

CONTROL SYMBOLS			
	AIRFLOW MEASURING STATION		PRESSURE CONTROLLER (PNEUMATIC)
	AIR FLOW SWITCH		PRESSURE SENSOR (ELECTRONIC)
	ALARM		PRESSURE TRANSMITTER (PNEUMATIC)
	BACKDRAFT DAMPER (GRAVITY)		RETURN AIR DAMPER
	CO2 SENSOR		SMOKE DETECTOR
	CURRENT SENSOR		START/STOP SWITCH
	DISCONNECT SWITCH - FUSED		STEAM FLOW MEASUREMENT ORIFICE PLATE & MASS FLOW COMPUTER
	DUCT DETECTOR		SUPPLY AIR DAMPER
	CURRENT TO PNEUMATIC TRANSDUCER		OVERRIDE TIMER
	DAMPER END SWITCH (BINARY)		TEMPERATURE CONTROLLER (PNEUMATIC)
	DIFFERENTIAL PRESSURE SENSOR (ANALOG)		TEMPERATURE SENSOR (ELECTRONIC)
	DIFFERENTIAL PRESSURE SWITCH (BINARY)		TEMPERATURE TRANSMITTER (PNEUMATIC)
	DIFFERENTIAL PRESSURE SWITCH (HIGH LIMIT) (BINARY)		TERMINAL BOX ACTUATOR
	(LOW LIMIT) (BINARY)		TURBINE METER
	ELECTRO-PNEUMATIC SWITCH		VARIABLE FREQUENCY CONTROLLER FOR MOTOR
	ELECTRO-PNEUMATIC TRANSDUCER		UNINTERRUPTED POWER (SUPPLY)
	EXHAUST AIR DAMPER		VELOCITY PRESSURE SENSOR (ELECTRONIC)
	FACE & BY-PASS DAMPER		VIBRATION SWITCH
	FAN DISCHARGE DAMPER		VOLUME DAMPER (MANUAL)
	FAN INLET DAMPER		VORTEX SHEDDING AIR FLOW TRANSMITTER
	FLOW CONTROLLER		VORTEX SHEDDING FLOW METER
	FLOW SENSOR		20 PSIG MAIN CONTROL AIR
	HAND-OFF-AUTO SWITCH		PRESSURE GAUGE
	HUMIDITY CONTROLLER (PNEUMATIC)		HOLDING COIL
	HUMIDITY HIGH LIMIT (ELECTRONIC)		HUMIDISTAT (SPACE)
	HUMIDITY SENSOR (ELECTRONIC)		SWITCH (PNEUMATIC)
	HUMIDITY TRANSMITTER (PNEUMATIC)		THERMOSTAT OR TEMPERATURE SENSOR (SPACE)
	LOW LIMIT SAFETY THERMOSTAT		NIGHT CYCLE
	MAGNETIC INDUCTIVE FLOW METER SENSOR		0-30 PSIG AIR GAUGE
	MANUAL SWITCH (ELECTRIC)		LOCAL AREA NETWORK
	MODEM		TEMPERATURE CONTROL CONTRACTOR
	MOTOR CONTROL CENTER		ENVIRONMENTAL CONTROL CONTRACTOR
	MOTOR STARTER		ELECTRICAL CONTRACTOR
	OCCUPANCY SENSOR		PNEUMATIC
	OPERATOR'S WORK STATION		ELECTRICAL WIRING
	OUTDOOR AIR DAMPER		ANALOG INPUT (DDC CONTROLLER)
	OUTDOOR AIR VOLUME PROBE, TRANSDUCER AND MONITOR		ANALOG OUTPUT (DDC CONTROLLER)
	PNEUMATIC RELAY		BINARY INPUT (DDC CONTROLLER)
	PRESSURE-ELECTRIC SWITCH		BINARY OUTPUT (DDC CONTROLLER)

GENERAL SYMBOLS	
	REVISION NUMBER - SHOWN ON PLANS
	POINT WHERE NEW CONNECTS TO EXISTING
	NUMBER OF DETAIL ON SHEET
	NUMBER OF SHEET WHERE DETAIL APPEARS
	PLAN NOTE
	CONTINUATION SYMBOL
	ROOM NAME AND NUMBER
	ITEM TO BE DEMOLISHED
	AREA NOT IN CONTRACT
	GENERAL NOTE
	1. PLAN NOTE LIST
	T THERMOSTAT
	H HUMIDISTAT
	M PRESSURE MONITOR (FLOOR PLAN)
	Co2 Co2 MONITOR (FLOOR PLAN)
LINE SYMBOLS	
	LIGHT (SCREENED SOLID OR DASHED LINES INDICATE EXISTING TO REMAIN)
	HEAVY DASHED LINES INDICATE EXISTING TO BE REMOVED
	HEAVY CONTINUOUS LINES INDICATE NEW WORK
	LIGHT DOT LINES INDICATE FUTURE WORK

ABBREVIATIONS	
Ø	ROUND
ABV	ABOVE
AC	AIR CONDITIONING
AD	AREA DRAIN
ADD	ADDENDUM
AF	ABOVE FINISHED FLOOR
AFUE	ANNUAL FUEL UTILIZATION EFFICIENCY
ALT	ALTERNATE
AP	ACCESS PANEL
ARCH	ARCHITECT/ARCHITECTURAL
BFF	BELOW FINISHED FLOOR
BLW	BELOW
BTU	BRITISH THERMAL UNITS
BTUH	BRITISH THERMAL UNITS PER HOUR
CB	CATCH BASIN
CFM	CUBIC FEET PER MINUTE
CLG	CEILING
CO	CLEAN OUT
CW	COLD WATER
D	DEGREE
DB	DRY BULB
DIA	DIAMETER
DN	DOWN
DW	DISTILLED WATER
E	EACH
EAT	ENTERING AIR TEMPERATURE
ELEC	ELECTRICAL
EQUIP	EQUIPMENT
EWC	ELECTRIC WATER COOLER
EWT	ENTERING WATER TEMPERATURE
FA	FIRE ALARM
EXIST	EXISTING
F	DEGREES FAHRENHEIT
FCO	FLOOR CLEAN OUT
FD	FLOOR DRAIN
FDV	FIRE DEPARTMENT VALVE
FL	FLOOR
FO	FUEL OIL
FOV	FUEL OIL VENT
FOR	FUEL OIL RETURN
FOS	FUEL OIL SUPPLY
FRM	FEET PER MINUTE
FS	FLOOR SINK
FT	FOOT/FEET
FTR	FIN TUBE RADIATION
GAL	GALLON
GC	GENERAL CONTRACTOR
GPM	GALLONS PER MINUTE
GW	GREASE WASTE
HB	HOSE BIB
HP	HORSE POWER
HTG	HEATING
HTR	HEATER
HW	HOT WATER
HYD	HYDRO
ID	INDIRECT
IN	INCH
INV	INVERT
LB	POUND
LBHR	POUNDS PER HOUR
LAT	LEAVING AIR TEMPERATURE
LP	LOW PRESSURE
LPG	LIQUEFIED PETROLEUM GAS
LVR	LEAVING WATER TEMPERATURE
MIA	MIXED AIR
MBH	MAXIMUM
MBH	ONE THOUSAND BTU PER HOUR
MCF	ONE THOUSAND CUBIC FEET
MD	MECHANICAL
MFR	MANUFACTURER
MN	MINIMUM
MISC	MISCELLANEOUS
MTR	MOTOR
MUA	MAKE-UP AIR
NC	NOISE CRITERIA
NC	NORMALLY CLOSED
NC	NOT IN CONTRACT
NO	NORMALLY OPEN
NTS	NOT TO SCALE
O	OUTSIDE AIR
ORD	OVERFLOW ROOF DRAIN
PD	PRESSURE DROP
PV	POST INDICATOR VALVE
PLBG	PLUMBING
PR	PRESSURE
PRV	PRESSURE REDUCING VALVE
PSI	POUNDS PER SQUARE INCH
PSIG	POUNDS PER SQUARE INCH GAUGE
PWR	POWER
R	DUCT RISER
RIA	RETURN AIR
RLA	RELIEF AIR
RM	ROOM
RRM	REVOLUTIONS PER MINUTE
RW	RAIN WATER
SF	SQUARE FOOT
SAN	SANITARY
SAN	SANITARY
SF	SQUARE FOOT
SD	SMOKE DAMPER
SM	SURFACE MOUNT
SP	STANDPIPE
SP	STATIC PRESSURE
STM	STEAM
T	THERMOSTAT
TD	TEMPERATURE DROP
TDR	TRENCH DRAIN
TEMP	TEMPERATURE
TYP	TYPICAL
US	UNDERGROUND
VAC	VACUUM
V	VENT
VAV	VARIABLE AIR VOLUME
VENT	VENTILATION
VTR	VENT THROUGH ROOF
W	WASTE
WB	WET BULB
WCO	WALL CLEAN OUT
WH	WALL HYDRANT

EQUIPMENT ABBREVIATIONS	
AC	AIR CONDITIONING UNIT
ACCU	AIR COOLING CONDENSING UNIT
AHU	AIR HANDLING UNIT
AS	AIR SEPARATOR
B	BOILER
CH	CHILLER
CT	COOLING TOWER
CUH	CABINET UNIT HEATER
CHWP	CHILLED WATER PUMP
DBP	DOMESTIC WATER BOOSTER PUMP
DC	DUCT MOUNTED COIL
DCP	DOMESTIC WATER CIRCULATING PUMP
EF	EXHAUST FAN
EDC	ELECTRIC DUCT COIL
ET	EXPANSION TANK
EW	ELECTRIC WATER HEATER
FCU	FAN COIL UNIT
FP	FIRE PUMP
GI	GREASE INTERCEPTOR
GRV	GRAVITY ROOF VENTILATOR
HWP	HEATING WATER PUMP
HRU	HEAT RECOVERY UNIT
HRV	HEAT RECOVERY VENTILATOR
RE	RETURN/EXHAUST FAN
RTU	ROOFTOP UNIT
SP	SUMP PUMP
UH	UNIT HEATER
WH	WATER HEATER

HVAC SYMBOLS	
	SQUARE DUCT SIZE TAG (WIDTH X HEIGHT)
	OVAL DUCT SIZE TAG (WIDTH / HEIGHT)
	ROUND DUCT SIZE TAG (DIAMETER)
	EXISTING DUCT TAG
	DUCT BEING DEMOLISHED
	SUPPLY AIR
	CONDITIONED OUTSIDE AIR
	OUTSIDE AIR
	RETURN AIR
	TRANSFER AIR
	EXHAUST AIR
	RELIEF AIR
	GREASE EXHAUST AIR
	SMOKE EXHAUST AIR
	EXHAUST GAS FLUE
	COMBUSTION AIR
	FLEX DUCT
	RECTANGULAR SUPPLY/OUTSIDE AIR DUCT RISE
	ROUND SUPPLY/OUTSIDE AIR DUCT RISE
	RECTANGULAR RETURN/TRANSFER AIR DUCT RISE
	ROUND RETURN/TRANSFER AIR DUCT RISE
	RECTANGULAR EXHAUST/RELIEF AIR DUCT RISE
	ROUND EXHAUST/RELIEF AIR DUCT RISE
	AIR FLOW MEASURING STATION
	INSULATED METAL PANEL
	BOOT TAP OR 45 DEGREE ENTRY FITTING
	R (RISE), D (DROP) ARROW IN DIRECTION OF AIR FLOW
	ACCESS DOOR
	90 DEGREE ELBOW WITH TURNING VANES
	STANDARD RADIUS ELBOW WITH CENTER RADIUS EQUAL TO 1-1/2 TIMES WIDTH OF DUCT.
	FLAT OVAL DUCT (WIDTH X HEIGHT)
	ROUND DUCT
	RECTANGULAR DUCT (WIDTH X HEIGHT)
	SPIN-IN FITTING WITH MANUAL VOLUME DAMPER AND FLEXIBLE DUCT CONNECTION
	CONICAL FITTING
	SQUARE-TO-ROUND TRANSITION

GRILLES, REGISTERS & DIFFUSERS TAGS	
	TYPE (SEE SCHEDULE) SDX - 300
	ROUND DIFFUSER
	TYPE (SEE SCHEDULE) SWGX - 300
	SIDE WALL GRILLE
	TYPE (SEE SCHEDULE) SLBX - 300
	LINEAR BAR GRILLE
	TYPE (SEE SCHEDULE) LSD1 - 200
	LINEAR SLOT DIFFUSER

MECHANICAL EQUIPMENT TAGS	
	IDENTITY
	AIRFLOW
	EXISTING EQUIPMENT TO REMAIN
	EXISTING RELOCATED EQUIPMENT
	EXISTING EQUIPMENT TO REMAIN
	EXISTING RELOCATED EQUIPMENT

DAMPER TAGS	
	FIRE DAMPER
	SMOKE DAMPER
	MOTORIZED DAMPER
	MANUAL BALANCING DAMPER
	BACKDRAFT DAMPER
	COMBINATION FIRE/SMOKE DAMPER

MECHANICAL PIPING SYMBOLS	
	CHILLED WATER RETURN
	CHILLED WATER SUPPLY
	CONDENSATE DRAINAGE
	CONDENSER WATER RETURN
	CONDENSER WATER SUPPLY
	GEOTHERMAL WATER RETURN
	GEOTHERMAL WATER SUPPLY
	HEATING WATER RETURN
	HEATING WATER SUPPLY
	NATURAL GAS
	PROPANE GAS
	REFRIGERANT LIQUID
	REFRIGERANT SUCTION
	REFRIGERANT HOT GAS
	STEAM
	CONDENSATE RETURN
	PIPE SIZE TAG (DIAMETER)
	ABOVE GROUND PIPING
	PIPE SLOPE TAG
	BELOW GROUND PIPING
	PIPE INVERT ELEVATION TAG
	EXISTING PIPE TAG
	PIPING BEING DEMOLISHED
	PIPE DROP
	PIPE RISE
	PIPE TEE
	CAP
	PLUG
	REDUCING 45 DEGREE TEE
	45 DEGREE TEE
	2" DOM. WM DOMESTIC WATER METER
	2" M-CNTRL MOTORIZED CONTROL VALVE
	2" BALANCING BALANCING VALVE
	2" 3-WAY CNTRL 3-WAY MOTORIZED CONTROL VALVE
	2" SHUTOFF 1/4 TURN BALL VALVE
	2" PRV PRESSURE REDUCING VALVE
	2" CHECK CHECK VALVE
	2" SOLENOID REFRIGERANT SOLENOID VALVE
	2" TMV 3-WAY MIXING VALVE
	2" BUTTERFLY BUTTERFLY VALVE

VALVE AND FITTING SYMBOLS	
PLAN VIEW	DETAIL VIEW
	PIPING FLEXIBLE CONNECTION
	PIPE TURNED UP (UNLESS NOTED OTHERWISE)
	PIPE TURNED DOWN
	PIPE OUT TOP
	PIPE OUT BOTTOM
	THREADED NIPPLE W/CAP
	PIPE WITH BLIND FLANGE
	CONCENTRIC REDUCER
	ECCENTRIC REDUCER
	ISOLATION VALVE
	CHECK VALVE
	UNION
	GATE VALVE
	BALL VALVE
	GLOBE VALVE
	BUTTERFLY VALVE
	TEMPERATURE CONTROL - 2-WAY MODULATING VALVE
	TEMPERATURE CONTROL - 2-WAY, 2-POSITION ISOLATION VALVE
	TEMPERATURE CONTROL - 3-WAY MODULATING VALVE
	TEMPERATURE CONTROL - 3-WAY, 2-POSITION ISOLATION VALVE
	CALIBRATED BALANCE VALVE
	SAFETY RELIEF VALVE
	STRAINER
	FLOW METER
	FLOOR DRAIN BY PC
	2-WAY MOTORIZED CONTROL VALVE
	3-WAY MOTORIZED CONTROL VALVE
	FLOW MEASURING DEVICE
	MANUAL AIR RELIEF VENT
	AUTOMATIC AIR RELIEF VENT
	LUBRICATED PLUG VALVE
	PRESSURE REDUCING VALVE (STEAM, GAS, COMPRESSED AIR)
	ANGLE VALVE
	REFRIGERANT HOT GAS BY-PASS VALVE
	SHUT-OFF COCK (HYDRONICS)
	SOLENOID VALVE
	REFRIGERANT EXPANSION VALVE
	WATER PRESSURE REDUCING/REGULATING VALVE
	STEAM TRAP
	DIELECTRIC FITTING BETWEEN STEEL AND COPPER
	STRAINER (STEAM)
	PRESSURE AND TEMPERATURE PLUG
	REFRIGERANT SIGHT GLASS
	THERMOMETER
	(P=PRESS V=VAC T=TEMP) GAUGE
	SENSOR (T-TEMP H-HUMIDITY)
	FLOW SWITCH
	CLEAN OUT
	INDICATED EXPANSION LOOP (COLD SPRING)
	ANCHOR
	GUIDE
	REFRIGERANT SHUT-OFF VALVE
	EXPANSION JOINT
	DIFFERENTIAL SWITCH
	REDUCED PRESSURE BACKFLOW PREVENTER

PROJECT GENERAL NOTES	
A	REMOVE ALL UNUSED PIPING, DUCTWORK AND ACCESSORIES.
B	THE MECHANICAL CONTRACTOR SHALL PERFORM SERVICE AND REPAIR ON THE EXISTING EQUIPMENT AND ITS ACCESSORIES AS FOLLOWS: CLEAN ALL COILS, REPLACE THE FILTERS AND BELTS, INSPECT, REPAIR, OR REPLACE THE ECONOMIZERS, DRIVES AND FAN BEARINGS, MOTORS, CONTROL COMPONENTS, VALVES AND ANY OTHER ITEM NECESSARY FOR A COMPLETE AND PROPER OPERATING SYSTEM. THIS CONTRACTOR SHALL ALSO VISIT THE SITE, PRIOR TO FINAL BIDDING, AND VERIFY ALL EXISTING SITE CONDITIONS. PROVIDE ALL MATERIAL AND COMPONENTS AS NEEDED TO BRING THE UNITS TO FULL COMPLIANCE OF THE LANDLORD'S CRITERIA AND LOCAL AUTHORITY HAVING JURISDICTION.
C	WHERE FLOOR DRAINS OCCUR WITHIN THE LIMITS OF CONSTRUCTION, PREVENT CONSTRUCTION DEBRIS FROM ENTERING DRAIN BODY BY SEALING DRAIN OPENING PRIOR TO START OF WORK. UNSAFE DRAINS AT COMPLETION OF CONSTRUCTION.
D	COORDINATE INSTALLATION OF PIPING, DUCTWORK, CONDUIT, LIGHTS, CABLE TRAY, STRUCTURE, AND EQUIPMENT TO PREVENT CONFLICTS.
E	THE CONTRACTOR SHALL BE FAMILIAR WITH ALL THE CONDITIONS BOTH EXISTING AND THOSE ILLUSTRATED BY THESE DOCUMENTS AS WELL AS THOSE WHICH CAN BE REASONABLY ANTICIPATED INCLUDING, BUT NOT LIMITED TO ARCHITECTURAL, ELECTRICAL, VENTILATION, PLUMBING, AND OTHER SYSTEMS INVOLVED ON THIS PROJECT.
F	FINAL PRODUCT SHALL BE A COMPLETE AND FUNCTIONING SYSTEM, AND SHALL CONFORM TO ALL REQUIREMENTS OF APPLICABLE FEDERAL, STATE, AND LOCAL CODES, INCLUDING BUT NOT LIMITED TO THE INTERNATIONAL BUILDING CODE AND INTERNATIONAL MECHANICAL CODE.
G	LOCATE EQUIPMENT REQUIRING ACCESS 2'-0" MAXIMUM ABOVE CEILING.
H	ALL ROOF MOUNTED EQUIPMENT SHALL BE A MINIMUM 10'-0" FROM EDGE OF ROOF.
I	LOCATE DUCTWORK, PIPING AND MECHANICAL EQUIPMENT AWAY FROM THE SPACE ABOVE ELECTRICAL PANELS, TRANSFORMERS AND OTHER ELECTRICAL EQUIPMENT.
J	FIRE SEAL AROUND DUCT AND PIPING PENETRATIONS OF FIRE RATED WALLS. REFER TO SPECIFICATION.
K	PROVIDE SLEEVES AND/OR OPENINGS TO RUN PIPES AND DUCTS THROUGH FOUNDATIONS, FLOORS, WALLS, AND ROOF.
L	ADJUST PIPING AND DUCTWORK SIZES TO PROPERLY CONNECT TO MECHANICAL EQUIPMENT.
M	REFER TO PLUMBING SERIES DRAWINGS FOR GAS AND A.C. CONDENSATE DRAIN PIPING. PIPE SIZES SHOWN SHALL BE CONTINUED IN THE DIRECTION OF FLOW UNTIL ANOTHER SIZE IS SHOWN.
O	FOR DETAILS, EQUIPMENT CONNECTIONS, AND PIPE SIZES NOT SHOWN ON THE SEGMENTS, REFER TO DETAILS, SCHEDULES, AND SPECIFICATIONS.
P	INSTALL ALL EQUIPMENT IN ACCORDANCE WITH THE RESPECTIVE MANUFACTURERS WRITTEN INSTALLATION INSTRUCTIONS, AT A LEVEL OF QUALITY AND WORKMANSHIP CONSISTENT WITH THE SPECIFICATIONS.
Q	LOCATIONS OF PIPING, DUCTWORK AND EQUIPMENT AS INDICATED ON THE DRAWING, ARE APPROXIMATE AND SUBJECT TO MINOR ADJUSTMENTS IN THE FIELD. WORK SHALL BE COORDINATED WITH ALL OTHER TRADES TO AVOID INTERFERENCE IN THE FIELD.
R	INSTALL EXPOSED PIPING AND DUCTWORK AS HIGH AS PRACTICAL IN ROOMS WITHOUT CEILINGS.

HVAC GENERAL NOTES	
A	SUPPLY AND RETURN PIPING TO COILS ARE THE SAME SIZE.
B	CONTRACTOR SHALL LOCATE THERMOSTATS AND TEMPERATURE SENSORS AT 4'-6" AFF, A MINIMUM OF 8" FROM LIGHT SWITCH.
C	REFER TO PIPING DRAWINGS FOR THERMOSTAT AND TEMPERATURE SENSOR LOCATIONS.
D	CONDENSATE DRAINS SHALL BE SUPPLIED FOR ALL COOLING EQUIPMENT; CONTRACTOR SHALL ENSURE PROPER INSTALLATION AND DRAINAGE AS REQUIRED BY FEDERAL, STATE, AND LOCAL CODES. CONDENSATE PIPING SHALL BE TYPE "C" COPPER.
E	PROVIDE 4" HOUSING/KEEPING PAD FOR EACH PIECE OF MECHANICAL EQUIPMENT. COORDINATE SIZES WITH MECHANICAL EQUIPMENT SELECTED.
F	ALL SUPPLY, RETURN, AND EXHAUST DUCTWORK SHALL BE RATED FOR PRESSURE CLASS OF 2" W.G. UNLESS NOTED OTHERWISE.
G	THIS CONTRACTOR SHALL BE REQUIRED TO REPLACE FILTERS ON HVAC EQUIPMENT AFTER ALL DUST PRODUCING CONSTRUCTION HAS BEEN COMPLETED AND PRIOR TO THE FINAL PUNCH.
H	NEW WORK SHOWN ON DRAWINGS IS CONCEPTUAL IN NATURE. CONTRACTOR IS RESPONSIBLE FOR THE COORDINATION OF ALL TRADES. CONTRACTOR SHALL PERFORM FINAL COORDINATION OF ALL TRADES AFTER CEILING REMOVAL ALLOWS FURTHER SITE INVESTIGATION ON EXISTING BUILDINGS.
I	UNLESS NOTED OTHERWISE ON THE DRAWINGS, REFER TO SCHEDULES FOR DIFFUSER, GRILLE, TERMINAL BOX, AND OTHER TERMINAL EQUIPMENT RUNOUT SIZES.
J	WHERE EQUIPMENT REQUIRING ACCESS MUST BE INSTALLED ABOVE GYPSUM CEILING, CONTRACTOR SHALL COORDINATE ADEQUATE ACCESS WITH ALL TRADES AND PROVIDE THE REQUIRED ACCESS PANELS.

HVAC DEMOLITION GENERAL NOTES	
A	CONTRACTOR IS TO COORDINATE ALL SERVICE INTERRUPTIONS WITH OWNER REPRESENTATIVES, MAINTENANCE STAFF, AND END-USERS. CONTRACTOR SHALL PROVIDE SUFFICIENT ADVANCE NOTICE TO ALL AFFECTED PARTIES OF IMPENDING OUTAGE.
B	SYSTEMS INDICATED BY DASHED LINE ARE TO BE REMOVED IN THEIR ENTIRETY (UNLESS OTHERWISE DESCRIBED BY PLAN NOTES). THIS INCLUDES THE REMOVAL OF EXISTING HANGERS, INSULATION, ASSOCIATED CONTROLS AND ACCESSORIES. ALL DUCTWORK AND PIPING TERMINATIONS NOT SHOWN AS BEING REUSED SHALL BE CAPPED. CONTRACTOR SHALL PATCH INSULATION DAMAGED DURING DEMOLITION AND WHERE PIPING IS SHOWN TO BE CAPPED.
C	THE MECHANICAL CONTRACTOR SHALL PERFORM SERVICE AND REPAIR ON THE EXISTING EQUIPMENT AND ITS ACCESSORIES AS FOLLOWS: CLEAN ALL COILS, REPLACE THE FILTERS AND BELTS, INSPECT, REPAIR, OR REPLACE THE ECONOMIZERS, DRIVES AND FAN BEARINGS, MOTORS, CONTROL COMPONENTS, VALVES AND ANY OTHER ITEM NECESSARY FOR A COMPLETE AND PROPER OPERATING SYSTEM. THIS CONTRACTOR SHALL ALSO VISIT THE SITE, PRIOR TO FINAL BIDDING, AND VERIFY ALL EXISTING SITE CONDITIONS. PROVIDE ALL MATERIAL AND COMPONENTS AS NEEDED TO BRING THE UNITS TO FULL COMPLIANCE OF THE LANDLORD'S CRITERIA AND LOCAL AUTHORITY HAVING JURISDICTION.

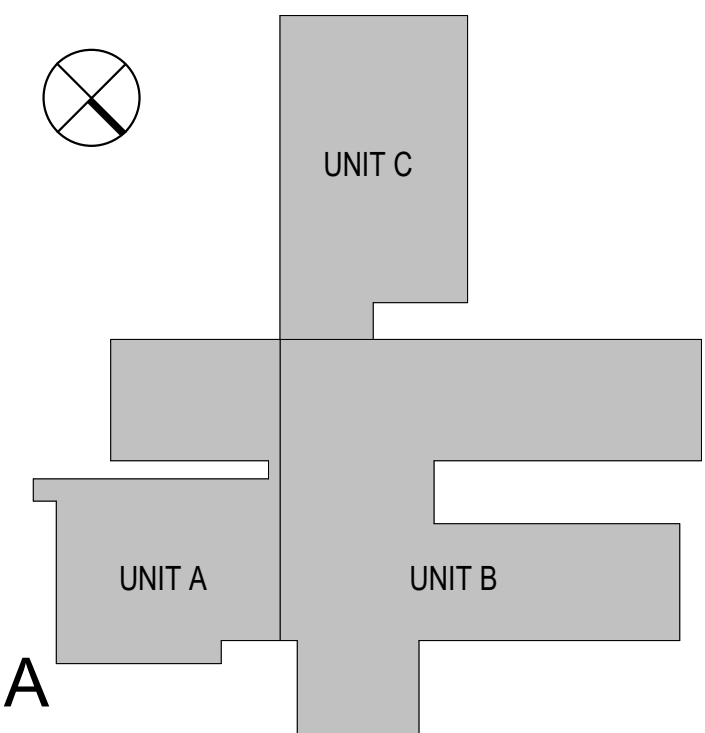
Project #: 17-700-078-2

Designed By: A.M.M

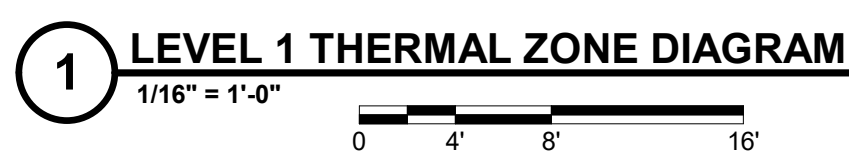
Drawn By: A.M.M

Checked By: D.E.B

Date: 04.12.2021

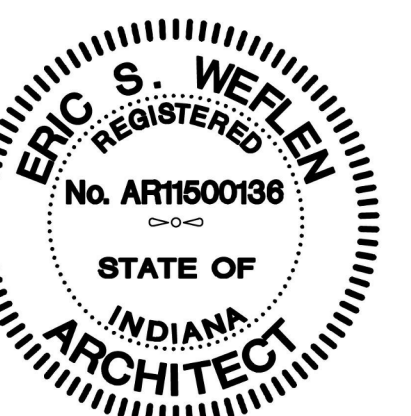


M012

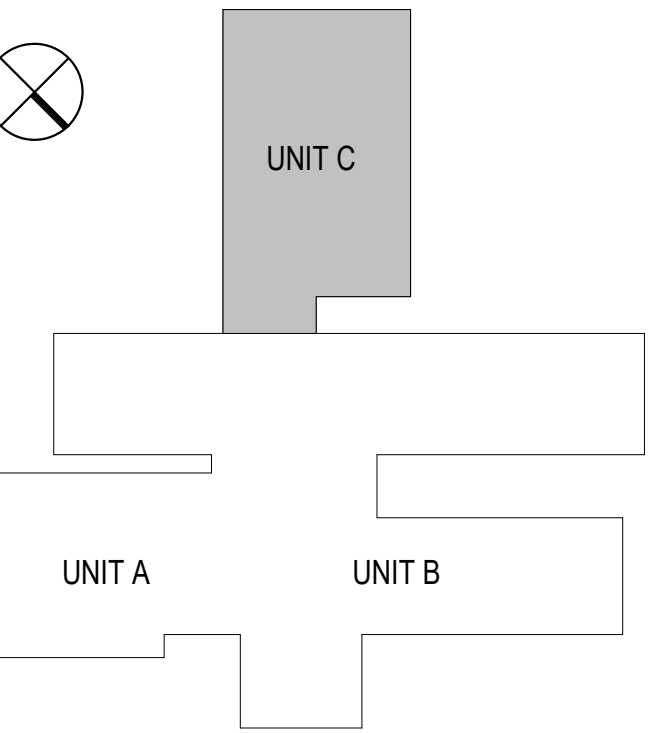


#	Revision	Date
---	----------	------

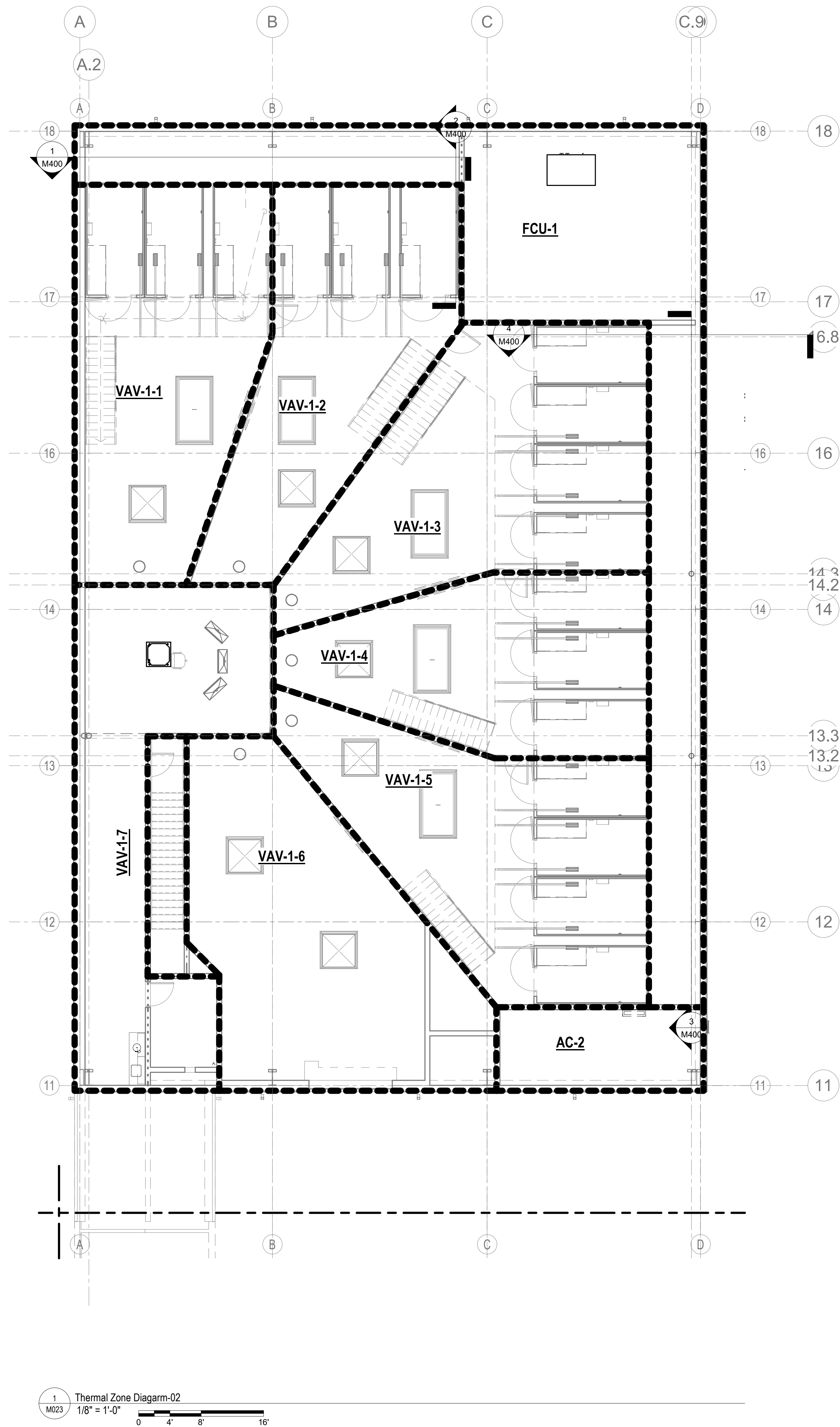
ate: 04.12.2021



These Drawings and Specifications, and all copies thereof are and shall remain the property and copyright of the Architect. They shall be used only with respect to this Project and are not to be used on any other Project or Work without prior written permission from the Architect.



M023



E

D

C

B

A

6

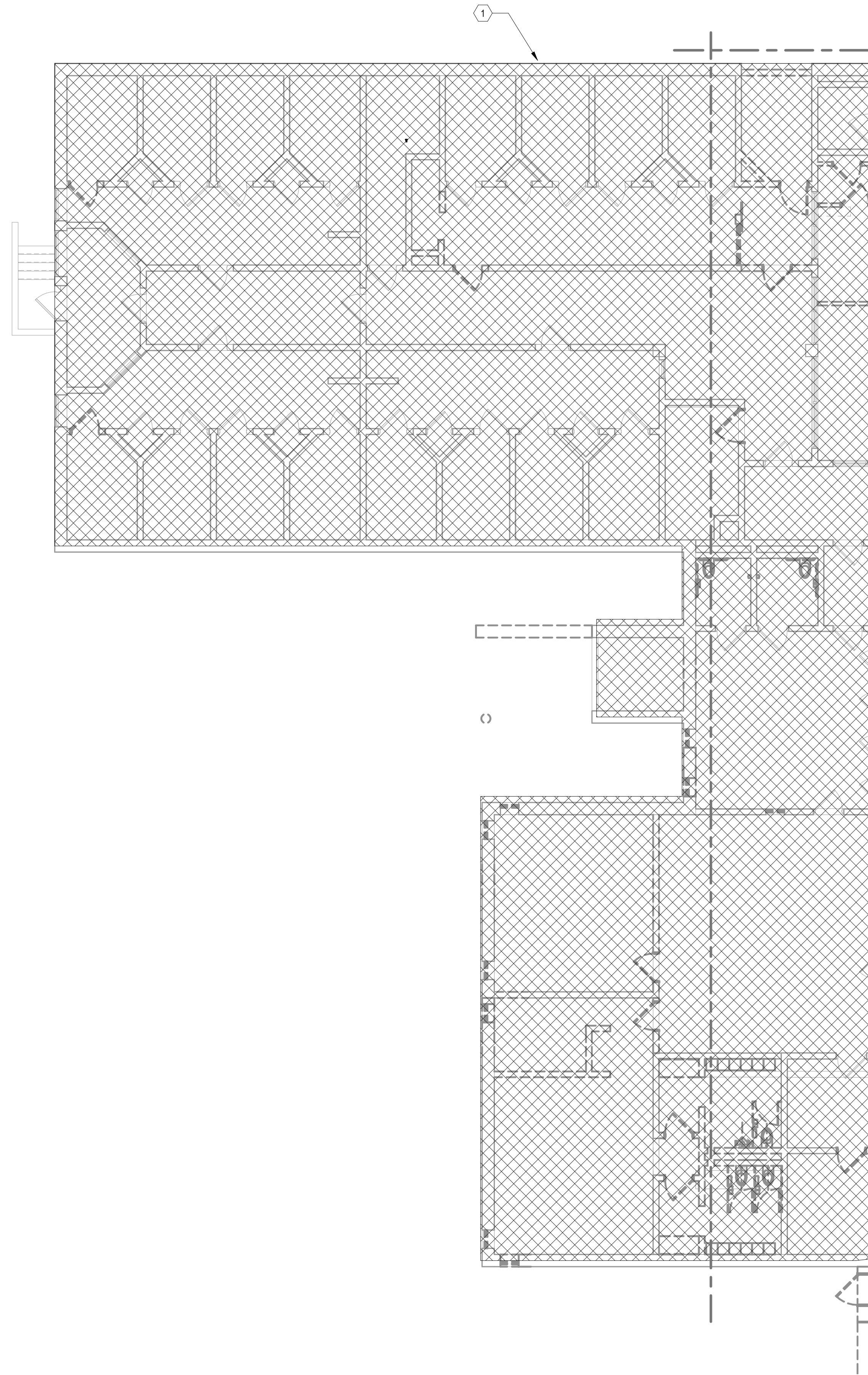
5

4

3

2

1



1 FIRST FLOOR DEMOLITION PLAN - UNIT A - MECHANICAL
1/8" = 1'-0"

0 4 8 16'

GENERAL NOTES:

- A. ALL DUCT DIMENSIONS GIVEN ARE INSIDE DIMENSIONED, UNLESS NOTED OTHERWISE.
- B. COORDINATE ALL OPENINGS IN FLOORS, ROOF AND WALLS WITH THE GENERAL AND STRUCTURAL CONTRACTORS.
- C. RUN ALL DUCTS AS HIGH AS POSSIBLE WITH RETURN AND EXHAUST ABOVE SUPPLY AS OFTEN AS POSSIBLE.
- D. PROVIDE TURNING VANES IN ALL SUPPLY AIR SQUARE THROATED ELBOWS.
- E. ALL RECTANGULAR SUPPLY AIR BRANCH CONNECTIONS SHALL BE 45° TAKE-OFF FITTINGS AND ROUND SUPPLY AIR BRANCH CONNECTIONS SHALL BE BELLMOUTH FITTINGS
- F. REFER TO DIFFUSER SCHEDULE FOR DUCT RUNOUT SIZE UNLESS NOTED OTHERWISE. ALL FLEXIBLE DUCTS TO DIFFUSERS MAXIMUM LENGTH 5'-0". FLEXIBLE DUCTS ARE NOT TO BE USED AS ELBOWS.
- G. ALL ABOVE FINISH FLOOR (A.F.F.) DIMENSIONS ARE TO BE MEASURED TO BOTTOM OF EQUIPMENT OR DUCTWORK UNLESS NOTED OTHERWISE.
- H. PROVIDE CEILING RADIATION DAMPERS (CRD) FOR ALL DIFFUSERS/GRILLES LOCATED IN RATED CEILINGS. COORDINATE TYPE OF CRD REQUIRED WITH ARCHITECTURALL CONSTRUCTION TYPE AS SHOWN ON ARCHITECTURAL PLANS.

PLAN NOTES	
#	NOTE
1	DEMOLISH ALL DIFFUSERS, DUCTWORK, PIPING, MECHANICAL EQUIPMENT, AND ALL ASSOCIATED WIRING, AND CONDUIT UNLESS OTHERWISE NOTED. PATCH PENETRATIONS TO MATCH EXISTING AS REQUIRED.

RQAW
ARCHITECTURE

100% CD Bid Set

VERMILLION COUNTY COMMISSIONERS
VERMILLION COUNTY JAIL
ADDITION & RENOVATION
1888 IN-43
HILLSDALE IN 47854

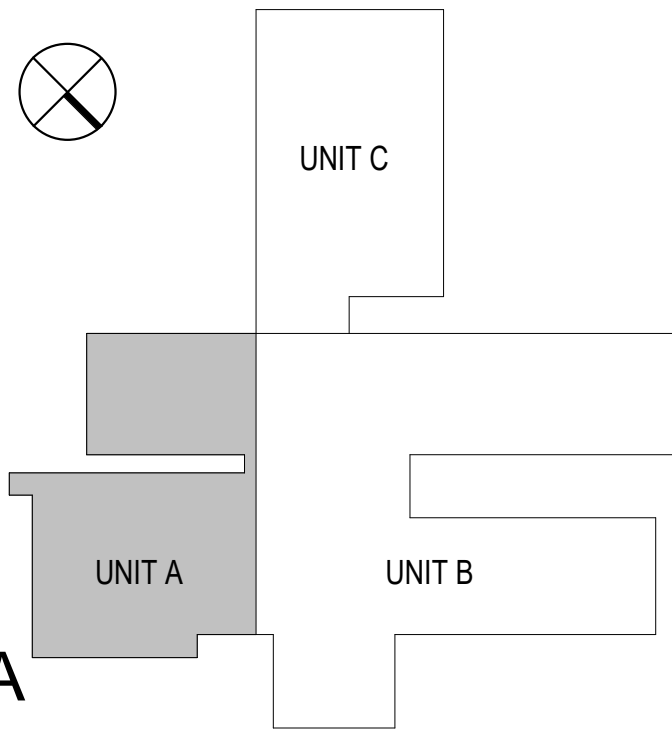
#	Revision	Date
---	----------	------

Project #: 17-700-078-2
Designed By: A.M.M
Drawn By: A.M.M
Checked By: D.E.B
Date: 04.12.2021



Eric Wepler

These Drawings and Specifications, and all copies thereof are and shall remain the property and copyright of the Architect. They shall be used only with respect to this Project and are not to be used on any other Project or Work without prior written permission from the Architect



FIRST FLOOR
DEMOLITION PLAN - UNIT
A - MECHANICAL

M110A

6

5

4

3

2

1

E

D

C

B

A

6

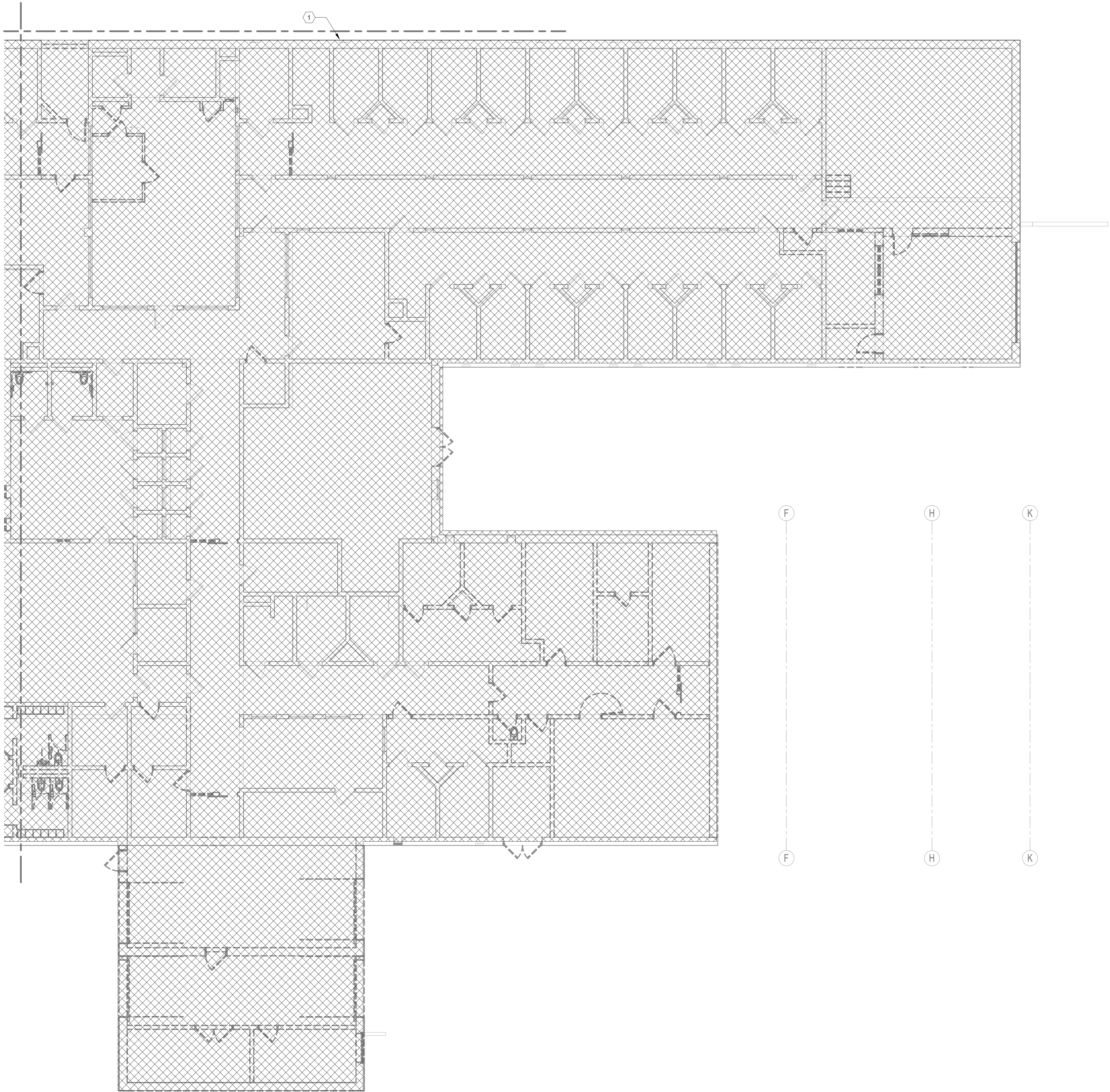
5

4

3

2

1



1 FIRST FLOOR DEMOLITION PLAN - UNIT B - MECHANICAL
1/8" = 1'-0"
0 4 8 16'

GENERAL NOTES:

- A. ALL DUCT DIMENSIONS GIVEN ARE INSIDE DIMENSIONED, UNLESS NOTED OTHERWISE.
- B. COORDINATE ALL OPENINGS IN FLOORS, ROOF AND WALLS WITH THE GENERAL AND STRUCTURAL CONTRACTORS.
- C. RUN ALL DUCTS AS HIGH AS POSSIBLE WITH RETURN AND EXHAUST ABOVE SUPPLY AS OFTEN AS POSSIBLE.
- D. PROVIDE TURNING VANES IN ALL SUPPLY AIR SQUARE THROATED ELBOWS.
- E. ALL RECTANGULAR SUPPLY AIR BRANCH CONNECTIONS SHALL BE 45° TAKE-OFF FITTINGS AND ROUND SUPPLY AIR BRANCH CONNECTIONS SHALL BE BELLMOUTH FITTINGS
- F. REFER TO DIFFUSER SCHEDULE FOR DUCT RUNOUT SIZE UNLESS NOTED OTHERWISE. ALL FLEXIBLE DUCTS TO DIFFUSERS MAXIMUM LENGTH 5'-0". FLEXIBLE DUCTS ARE NOT TO BE USED AS ELBOWS.
- G. ALL ABOVE FINISH FLOOR (A.F.F.) DIMENSIONS ARE TO BE MEASURED TO BOTTOM OF EQUIPMENT OR DUCTWORK UNLESS NOTED OTHERWISE.
- H. PROVIDE CEILING RADIATION DAMPERS (CRD) FOR ALL DIFFUSERS/GRILLES LOCATED IN RATED CEILINGS. COORDINATE TYPE OF CRD REQUIRED WITH ARCHITECTURALL CONSTRUCTION TYPE AS SHOWN ON ARCHITECTURAL PLANS.

PLAN NOTES	
#	NOTE
1	DEMOLISH ALL DIFFUSERS, DUCTWORK, PIPING, MECHANICAL EQUIPMENT, AND ALL ASSOCIATED WIRING, AND CONDUIT UNLESS OTHERWISE NOTED. PATCH PENETRATIONS TO MATCH EXISTING AS REQUIRED.

100% CD Bid Set

VERMILLION COUNTY COMMISSIONERS
**VERMILLION COUNTY JAIL
ADDITION & RENOVATION**
1888 IN-43
HILLSDALE IN 47854

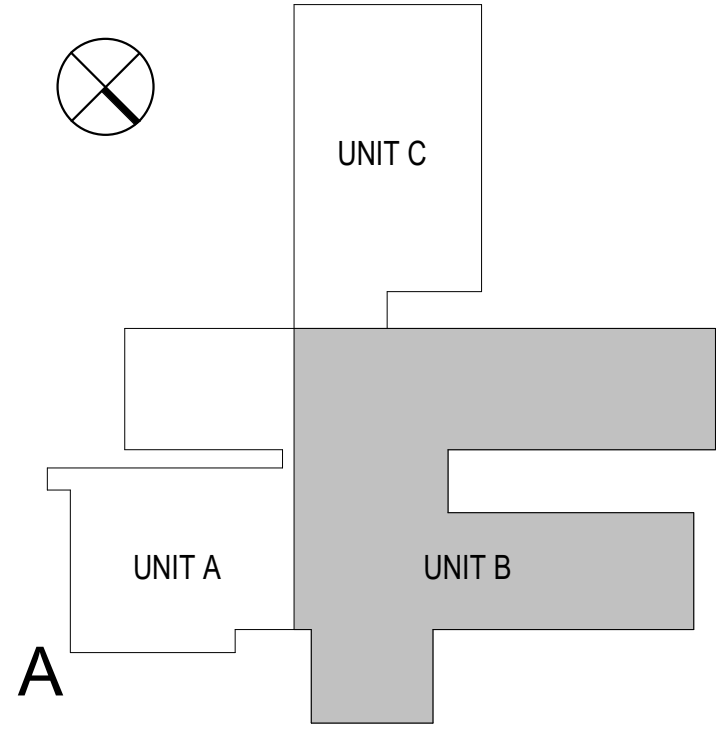
#	Revision	Date
---	----------	------

Project #: 17-700-078-2
Designed By: A.M.M
Drawn By: A.M.M
Checked By: D.E.B
Date: 04.12.2021



Eric Wefer

These Drawings and Specifications, and all copies thereof are and shall remain the property and copyright of the Architect. They shall be used only with respect to this Project and are not to be used on any other Project or Work without prior written permission from the Architect



FIRST FLOOR
DEMOLITION PLAN - UNIT
B - MECHANICAL

M110B

E

D

C

B

A

6

5

4

3

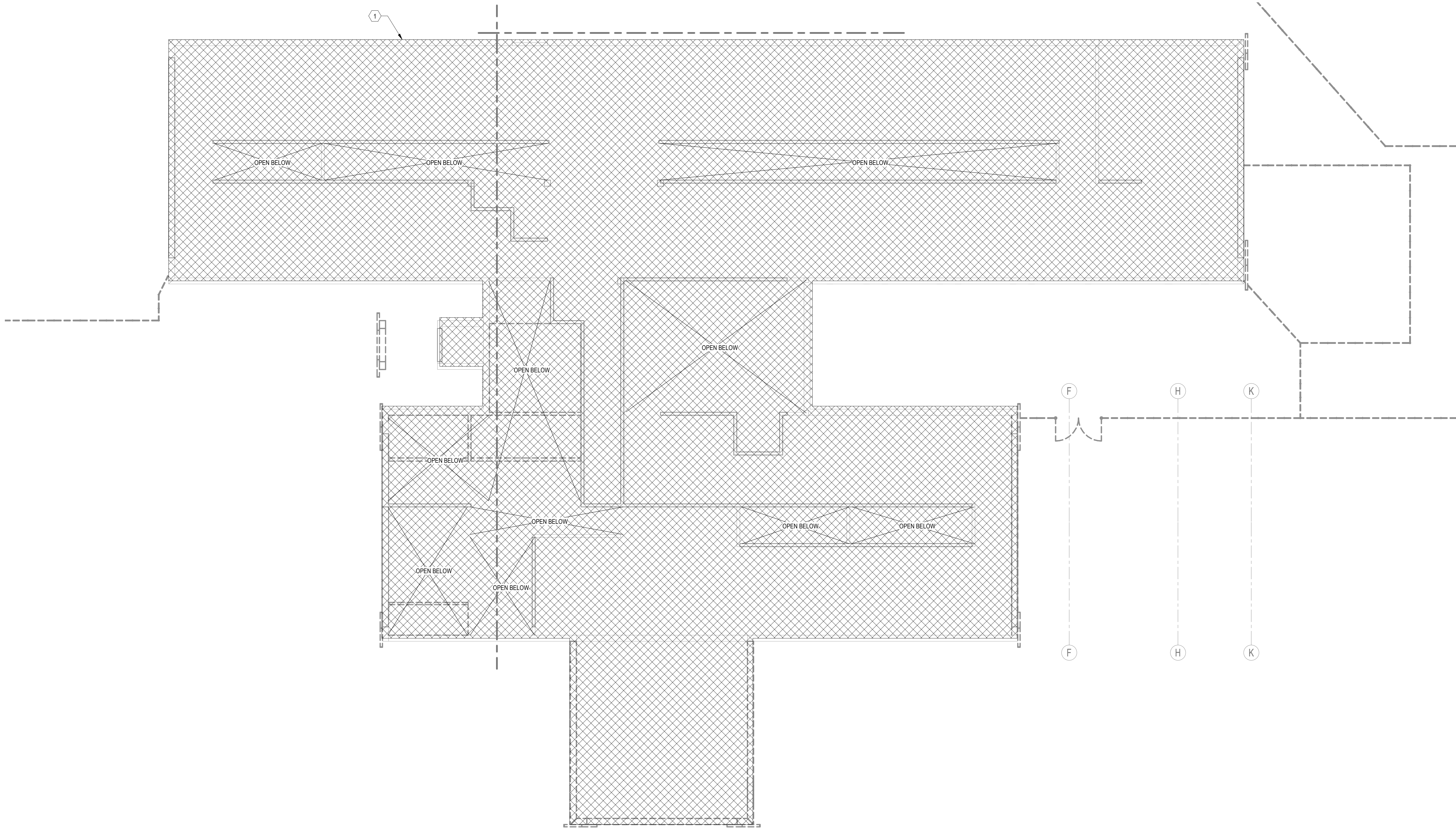
2

1

GENERAL NOTES:

- A. ALL DUCT DIMENSIONS GIVEN ARE INSIDE DIMENSIONED, UNLESS NOTED OTHERWISE.
- B. COORDINATE ALL OPENINGS IN FLOORS, ROOF AND WALLS WITH THE GENERAL AND STRUCTURAL CONTRACTORS.
- C. RUN ALL DUCTS AS HIGH AS POSSIBLE WITH RETURN AND EXHAUST ABOVE SUPPLY AS OFTEN AS POSSIBLE.
- D. PROVIDE TURNING VANES IN ALL SUPPLY AIR SQUARE THROATED ELBOWS.
- E. ALL RECTANGULAR SUPPLY AIR BRANCH CONNECTIONS SHALL BE 45° TAKE-OFF FITTINGS AND ROUND SUPPLY AIR BRANCH CONNECTIONS SHALL BE BELLMOUTH FITTINGS
- F. REFER TO DIFFUSER SCHEDULE FOR DUCT RUNOUT SIZE UNLESS NOTED OTHERWISE. ALL FLEXIBLE DUCTS TO DIFFUSERS MAXIMUM LENGTH 5'-0". FLEXIBLE DUCTS ARE NOT TO BE USED AS ELBOWS.
- G. ALL ABOVE FINISH FLOOR (A.F.F.) DIMENSIONS ARE TO BE MEASURED TO BOTTOM OF EQUIPMENT OR DUCTWORK UNLESS NOTED OTHERWISE.
- H. PROVIDE CEILING RADIATION DAMPERS (CRD) FOR ALL DIFFUSERS/GRILLES LOCATED IN RATED CEILINGS. COORDINATE TYPE OF CRD REQUIRED WITH ARCHITECTURALL CONSTRUCTION TYPE AS SHOWN ON ARCHITECTURAL PLANS.

PLAN NOTES	
#	NOTE
1	DEMOLISH ALL DIFFUSERS, DUCTWORK, PIPING, MECHANICAL EQUIPMENT, AND ALL ASSOCIATED WIRING, AND CONDUIT UNLESS OTHERWISE NOTED. PATCH PENETRATIONS TO MATCH EXISTING AS REQUIRED.



1 OVERALL ATTIC FLOOR DEMOLITION PLAN - MECHANICAL
3/32\" = 1'-0"

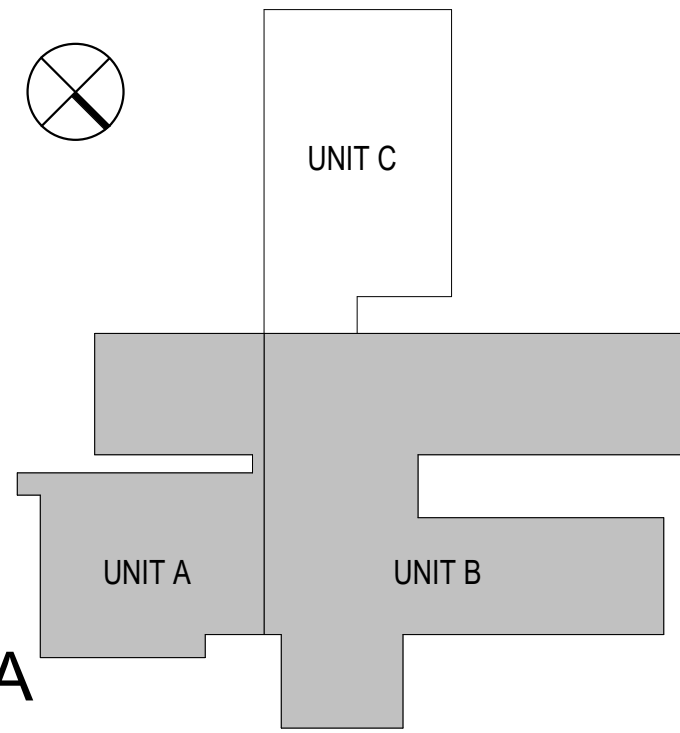
#	Revision	Date
---	----------	------

Project #: 17-700-078-2
Designed By: A.M.M
Drawn By: A.M.M
Checked By: D.E.B
Date: 04.12.2021



Eric Wepler

These Drawings and Specifications, and all copies thereof are and shall remain the property and copyright of the Architect. They shall be used only with respect to this Project and are not to be used on any other Project or Work without prior written permission from the Architect



E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

E

D

C

B

A

A

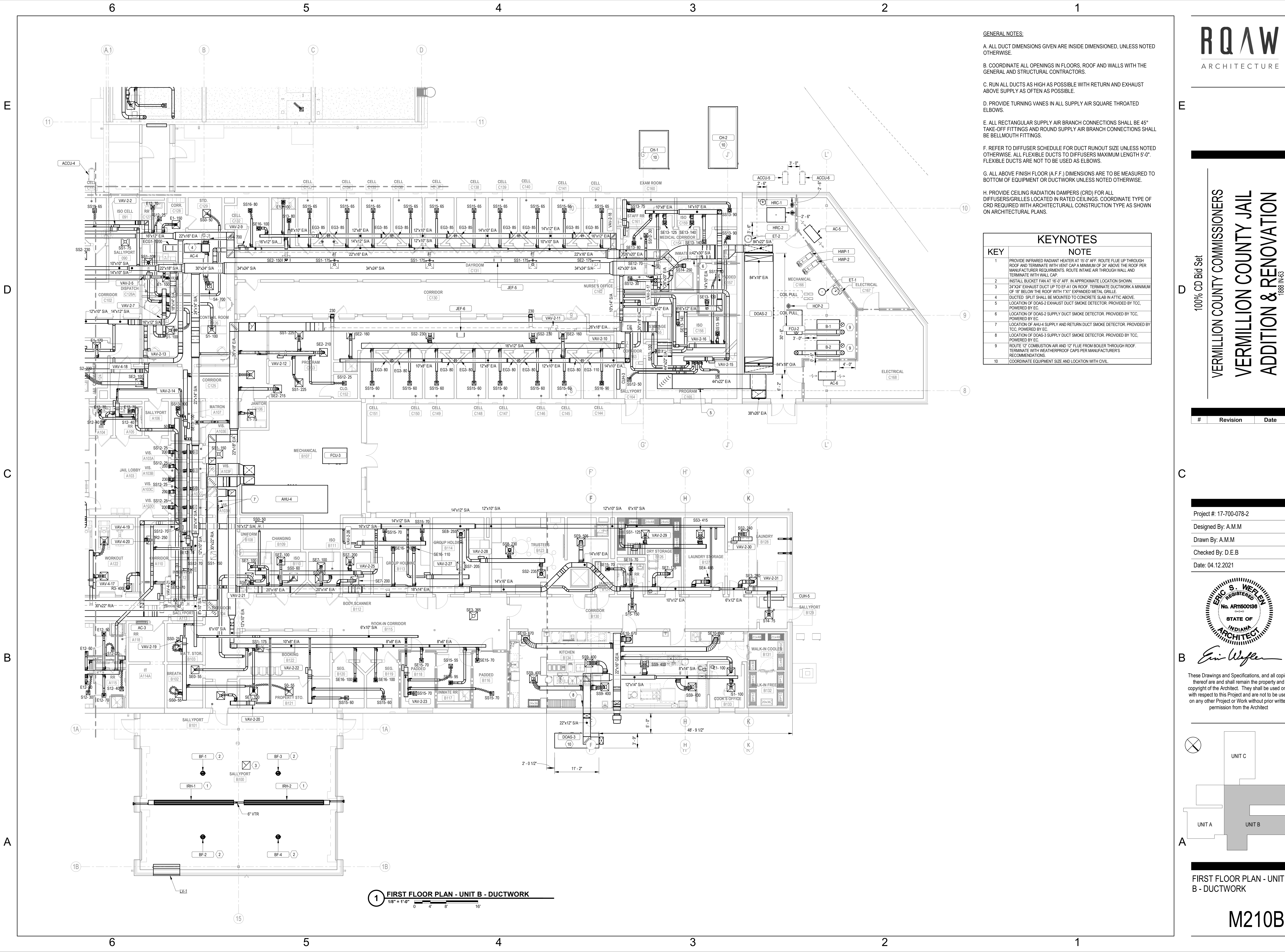
E

D

C

B

A



- GENERAL NOTES:**
- A. ALL DUCT DIMENSIONS GIVEN ARE INSIDE DIMENSIONED, UNLESS NOTED OTHERWISE.
- B. COORDINATE ALL OPENINGS IN FLOORS, ROOF AND WALLS WITH THE GENERAL AND STRUCTURAL CONTRACTORS.
- C. RUN ALL DUCTS AS HIGH AS POSSIBLE WITH RETURN AND EXHAUST ABOVE SUPPLY AS OFTEN AS POSSIBLE.
- D. PROVIDE TURNING VANES IN ALL SUPPLY AIR SQUARE THROATED ELBOWS.
- E. ALL RECTANGULAR SUPPLY AIR BRANCH CONNECTIONS SHALL BE 45° TAKE-OFF FITTINGS AND ROUND SUPPLY AIR BRANCH CONNECTIONS SHALL BE BELLMOUTH FITTINGS.
- F. REFER TO DIFFUSER SCHEDULE FOR DUCT RUNOUT SIZE UNLESS NOTED OTHERWISE. ALL FLEXIBLE DUCTS TO DIFFUSERS MAXIMUM LENGTH 5'-0". FLEXIBLE DUCTS ARE NOT TO BE USED AS ELBOWS.
- G. ALL ABOVE FINISH FLOOR (A.F.F.) DIMENSIONS ARE TO BE MEASURED TO BOTTOM OF EQUIPMENT OR DUCTWORK UNLESS NOTED OTHERWISE.
- H. PROVIDE CEILING RADIATION DAMPERS (CRD) FOR ALL DIFFUSERS/GRILLES LOCATED IN RATED CEILINGS. COORDINATE TYPE OF CRD REQUIRED WITH ARCHITECTURAL CONSTRUCTION TYPE AS SHOWN ON ARCHITECTURAL PLANS.

KEY	NOTE
1	PROVIDE INFRARED RADIANT HEATER AT 15'-0" AFF. ROUTE FLUE UP THROUGH ROOF AND TERMINATE WITH VENT CAP A MINIMUM OF 24" ABOVE THE ROOF PER MANUFACTURER REQUIREMENTS. ROUTE INTAKE AIR THROUGH WALL AND TERMINATE WITH WALL CAP.
2	INSTALL BUCKET FAN AT 16'-0" AFF. IN APPROXIMATE LOCATION SHOWN.
3	24"X24" EXHAUST DUCT UP TO EF-A1 ON ROOF. TERMINATE DUCTWORK A MINIMUM OF 18" BELOW THE ROOF WITH 1'X1' EXPANDED METAL GRILLE.
4	DUCTED SPLIT SHALL BE MOUNTED TO CONCRETE SLAB IN ATTIC ABOVE.
5	LOCATION OF DOAS-2 EXHAUST DUCT SMOKE DETECTOR. PROVIDED BY TCC, POWERED BY EC.
6	LOCATION OF DOAS-2 SUPPLY DUCT SMOKE DETECTOR. PROVIDED BY TCC, POWERED BY EC.
7	LOCATION OF AHU-4 SUPPLY AND RETURN DUCT SMOKE DETECTOR. PROVIDED BY TCC, POWERED BY EC.
8	LOCATION OF DOAS-3 SUPPLY DUCT SMOKE DETECTOR. PROVIDED BY TCC, POWERED BY EC.
9	ROUTE 12" COMBUSTION AIR AND 12" FLUE FROM BOILER THROUGH ROOF. TERMINATE WITH WEATHERPROOF CAPS PER MANUFACTURER'S RECOMMENDATIONS.
10	COORDINATE EQUIPMENT SIZE AND LOCATION WITH CIVIL.

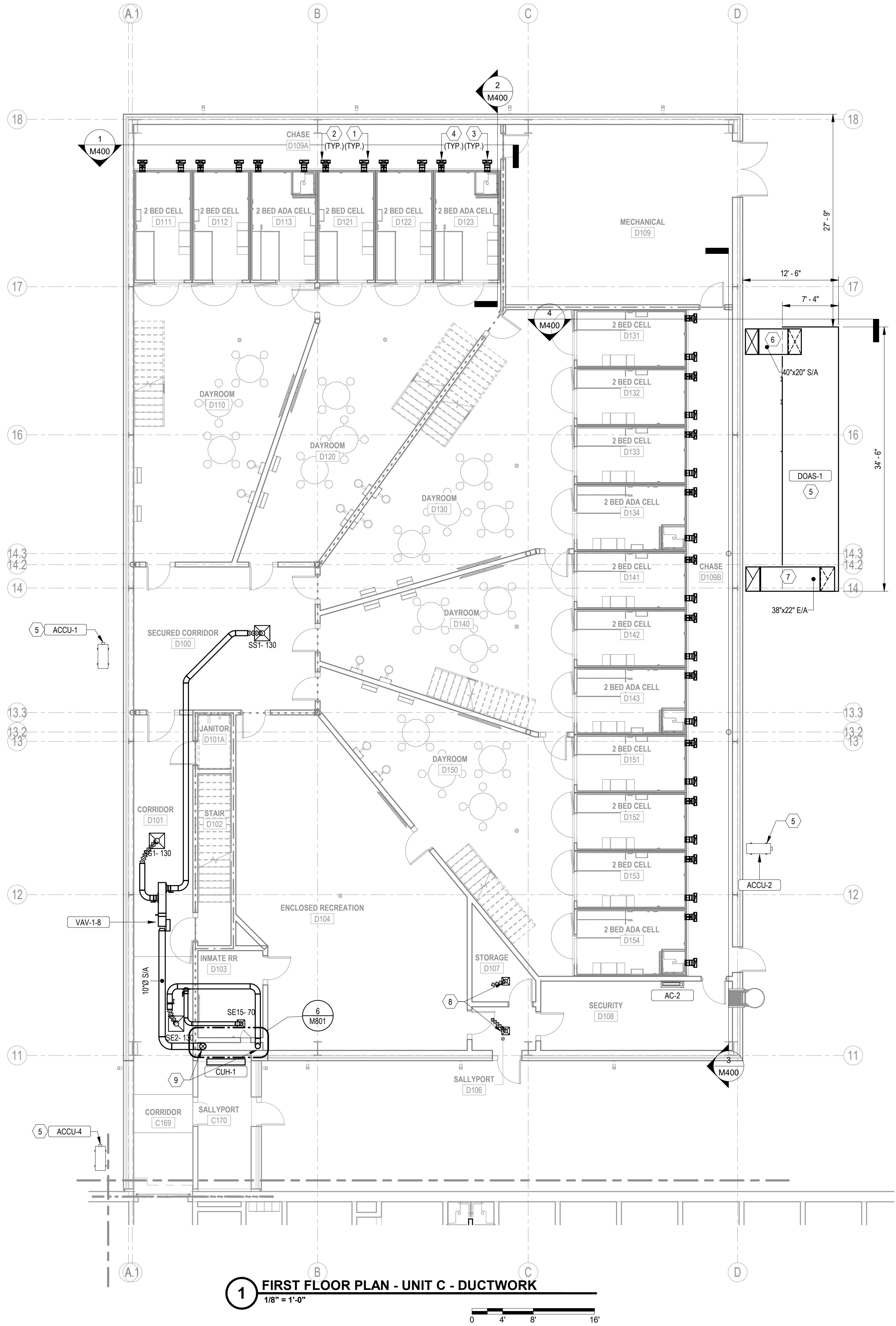
E

D

C

B

A



- GENERAL NOTES:**
- A. ALL DUCT DIMENSIONS GIVEN ARE INSIDE DIMENSIONED, UNLESS NOTED OTHERWISE.
- B. COORDINATE ALL OPENINGS IN FLOORS, ROOF AND WALLS WITH THE GENERAL AND STRUCTURAL CONTRACTORS.
- C. RUN ALL DUCTS AS HIGH AS POSSIBLE WITH RETURN AND EXHAUST ABOVE SUPPLY AS OFTEN AS POSSIBLE.
- D. PROVIDE TURNING VANES IN ALL SUPPLY AIR SQUARE THROATED ELBOWS.
- E. ALL RECTANGULAR SUPPLY AIR BRANCH CONNECTIONS SHALL BE 45° TAKE-OFF FITTINGS AND ROUND SUPPLY AIR BRANCH CONNECTIONS SHALL BE BELLMOUTH FITTINGS.
- F. REFER TO DIFFUSER SCHEDULE FOR DUCT RUNOUT SIZE UNLESS NOTED OTHERWISE. ALL FLEXIBLE DUCTS TO DIFFUSERS MAXIMUM LENGTH 5'-0". FLEXIBLE DUCTS ARE NOT TO BE USED AS ELBOWS.
- G. ALL ABOVE FINISH FLOOR (A.F.F.) DIMENSIONS ARE TO BE MEASURED TO BOTTOM OF EQUIPMENT OR DUCTWORK UNLESS NOTED OTHERWISE.
- H. PROVIDE CEILING RADIATION DAMPERS (CRD) FOR ALL DIFFUSERS/GRILLES LOCATED IN RATED CEILINGS. COORDINATE TYPE OF CRD REQUIRED WITH ARCHITECTURALL CONSTRUCTION TYPE AS SHOWN ON ARCHITECTURAL PLANS.

PLAN NOTES	
#	NOTE
1	CONNECT 8" DIA EXHAUST DUCT TO 2 MEN CELL EXHAUST CONNECTION. PROVIDE 8" ROUND MANUAL DAMPER AT CONNECTION AND BALANCE TO 105 CFM. ROUTE 4"X14" EXHAUST DUCT UP. REFER TO M220C FOR CONTINUATION.
2	CONNECT 8" DIA SUPPLY DUCT TO 2 MEN CELL SUPPLY CONNECTION. PROVIDE 8" ROUND MANUAL DAMPER AT CONNECTION AND BALANCE TO 85 CFM. ROUTE 4"X14" SUPPLY DUCT UP. REFER TO M220C FOR CONTINUATION.
3	CONNECT 8" DIA EXHAUST DUCT TO 2 MEN ADA CELL EXHAUST CONNECTION. PROVIDE 8" ROUND MANUAL DAMPER AT CONNECTION AND BALANCE TO 120 CFM. ROUTE 4"X14" EXHAUST DUCT UP. REFER TO M220C FOR CONTINUATION.
4	CONNECT 8" DIA SUPPLY DUCT TO 2 MEN ADA CELL SUPPLY CONNECTION. PROVIDE 8" ROUND MANUAL DAMPER AT CONNECTION AND BALANCE TO 105 CFM. ROUTE 4"X14" SUPPLY DUCT UP. REFER TO M220C FOR CONTINUATION.
5	COORDINATE EQUIPMENT SIZE AND LOCATION WITH CONCRETE PADS LOCATED ON CIVIL PLANS.
6	CONNECT 40"X20" SUPPLY DUCT TO DOAS-1. ROUTE DUCT UP EXTERIOR OF BUILDING. REFER TO M220C FOR CONTINUATION.
7	CONNECT 36"X22" EXHAUST DUCT TO DOAS-1. ROUTE DUCT UP EXTERIOR OF BUILDING. REFER TO M220C FOR CONTINUATION.
8	8" SUPPLY DUCT DOWN FROM ABOVE. REFER TO M220C FOR CONTINUATION.
9	DUCTS UP TO SECOND FLOOR. REFER TO M220C FOR CONTINUATION.

#	Revision	Date
---	----------	------

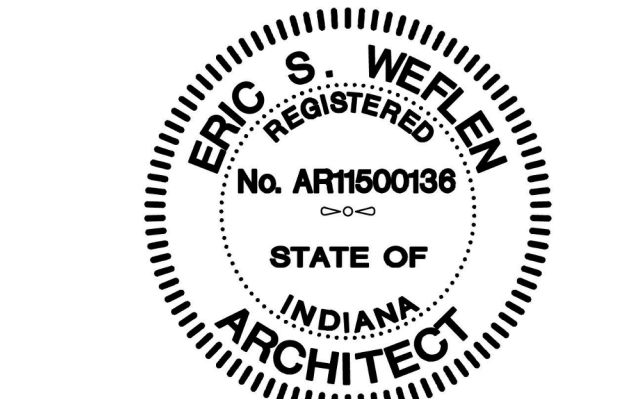
Project #: 17-700-078-2

Designed By: A.M.M

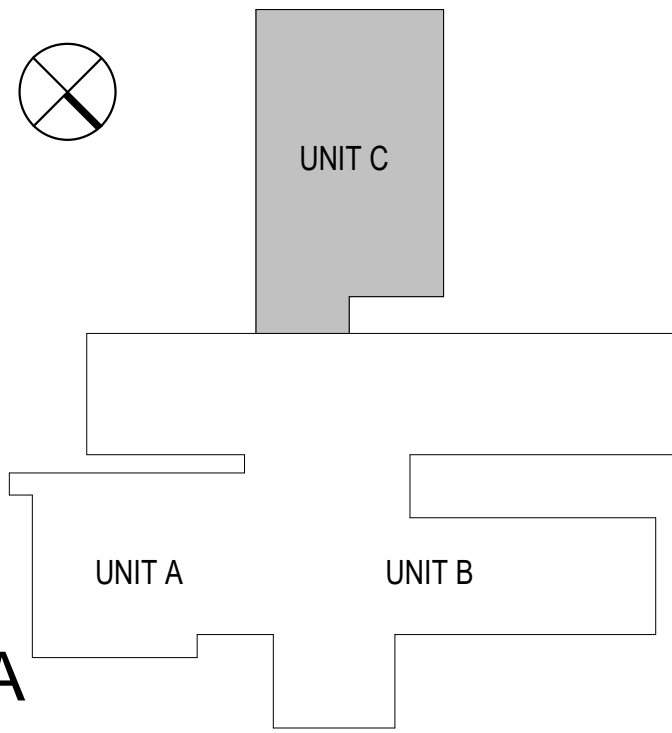
Drawn By: A.M.M

Checked By: D.E.B

Date: 04.12.2021

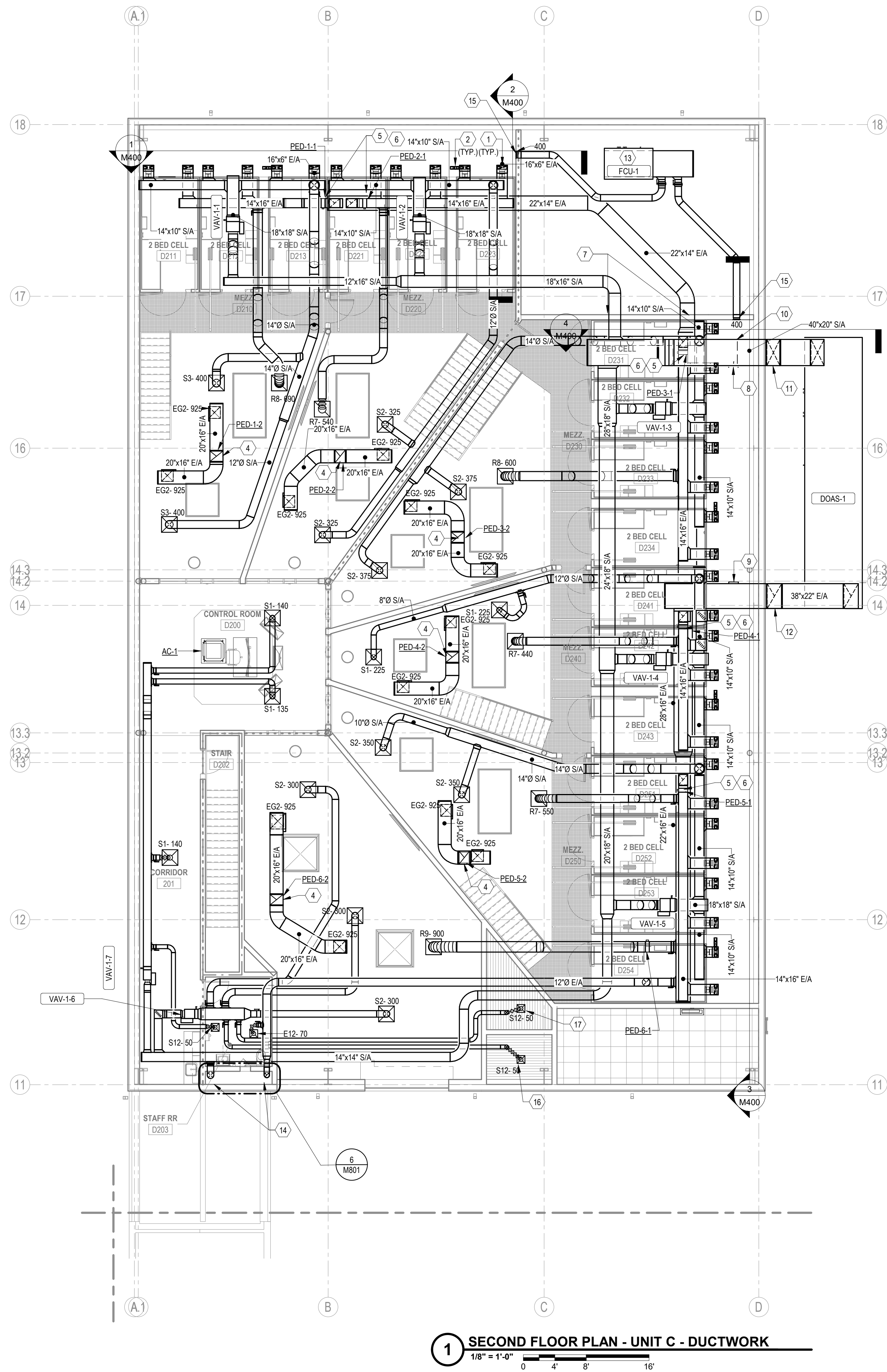


These Drawings and Specifications, and all copies thereof are and shall remain the property and copyright of the Architect. They shall be used only with respect to this Project and are not to be used on any other Project or Work without prior written permission from the Architect



FIRST FLOOR PLAN - UNIT C - DUCTWORK

M210C



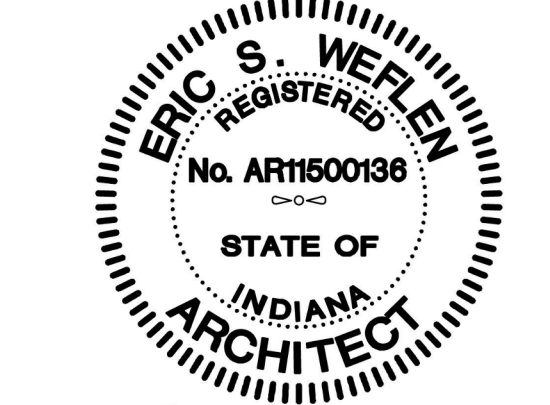
1 SECOND FLOOR PLAN - UNIT C - DUCTWORK
1/8" = 1'-0"

- GENERAL NOTES:
- A. ALL DUCT DIMENSIONS GIVEN ARE INSIDE DIMENSIONED, UNLESS NOTED OTHERWISE.
- B. COORDINATE ALL OPENINGS IN FLOORS, ROOF AND WALLS WITH THE GENERAL AND STRUCTURAL CONTRACTORS.
- C. RUN ALL DUCTS AS HIGH AS POSSIBLE WITH RETURN AND EXHAUST ABOVE SUPPLY AS OFTEN AS POSSIBLE.
- D. PROVIDE TURNING VANES IN ALL SUPPLY AIR SQUARE THROATED ELBOWS.
- E. ALL RECTANGULAR SUPPLY AIR BRANCH CONNECTIONS SHALL BE 45° TAKE-OFF FITTINGS AND ROUND SUPPLY AIR BRANCH CONNECTIONS SHALL BE BELLMOUTH FITTINGS
- F. REFER TO DIFFUSER SCHEDULE FOR DUCT RUNOUT SIZE UNLESS NOTED OTHERWISE. ALL FLEXIBLE DUCTS TO DIFFUSERS MAXIMUM LENGTH 5'-0". FLEXIBLE DUCTS ARE NOT TO BE USED AS ELBOWS.
- G. ALL ABOVE FINISH FLOOR (A.F.F.) DIMENSIONS ARE TO BE MEASURED TO BOTTOM OF EQUIPMENT OR DUCTWORK UNLESS NOTED OTHERWISE.
- H. PROVIDE CEILING RADIATION DAMPERS (CRD) FOR ALL DIFFUSERS/GRILLES LOCATED IN RATED CEILINGS. COORDINATE TYPE OF CRD REQUIRED WITH ARCHITECTURALL CONSTRUCTION TYPE AS SHOWN ON ARCHITECTURAL PLANS.

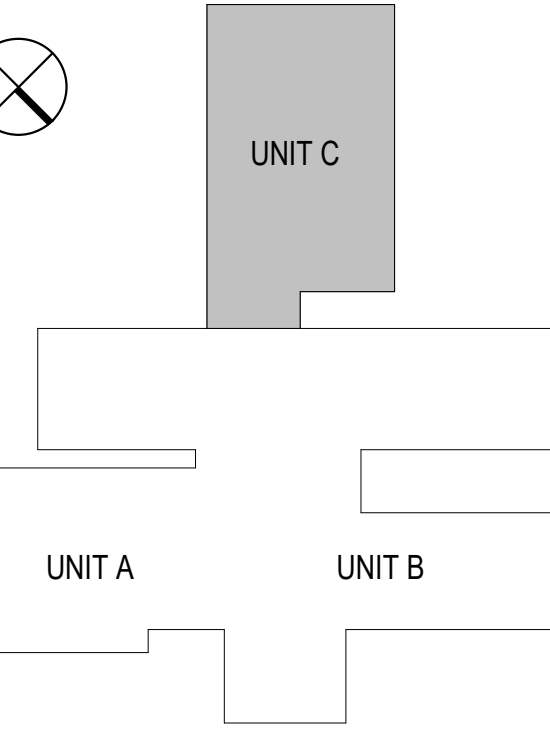
#	NOTE
1	CONNECT 8" DIA EXHAUST DUCT TO 2 MEN CELL EXHAUST CONNECTION. PROVIDE 8" ROUND MANUAL DAMPER AT CONNECTION AND BALANCE TO 105 CFM. ROUTE 4"x14" EXHAUST DUCT DOWN. REFER TO M210C FOR CONTINUATION.
2	CONNECT 8" DIA SUPPLY DUCT TO 2 MEN CELL SUPPLY CONNECTION. PROVIDE 8" ROUND MANUAL DAMPER AT CONNECTION AND BALANCE TO 85 CFM. ROUTE 4"x14" SUPPLY DUCT DOWN. REFER TO M210C FOR CONTINUATION.
4	18"x18" POD EXHAUST DUCT UP TO POD EXHAUST FAN ON ROOF. PROVIDE MOTORIZED DAMPER IN RISER.
5	PROVIDE TERMINAL BOX TEMPERATURE SENSOR IN EXHAUST DUCT FROM DAYROOM TO ZONE TEMPERATURE CONTROL.
6	PROVIDE MOTORIZED DAMPER FOR POD EXHAUST SYSTEM. REFER TO M-905 FOR SEQUENCE OF OPERATION.
7	PROVIDE SMOKE DAMPER AT PENETRATION IN WALL.
8	LOCATION OF DOAS-1 SUPPLY DUCT DETECTOR. PROVIDED BY AND INSTALLED BY EC.
9	LOCATION OF DOAS-1 RETURN DUCT DETECTOR. PROVIDED BY AND INSTALLED BY EC.
10	LOCATION OF DOAS-1 SUPPLY DUCT STATIC SENSOR.
11	ROUTE 40"x20" SA DUCT DOWN TO DOAS-1. REFER TO M210C FOR CONTINUATION.
12	ROUTE 38"x22" EA DUCT DOWN TO DOAS-1. REFER TO M210C FOR CONTINUATION.
13	INSTALL FOU AT APPROX. 19'-0" AFF. PROVIDE FULL SIZE 18" LONG RETURN PLENUM AND TERMINATE WITH 1'X1' EXPANDED METAL GRILLE.
14	ROUTE 10"O SUPPLY AND 6"O EXHAUST DUCT DOWN THRU PLUMBING CHASE TO FIRST FLOOR. COORDINATE DUCT LOCATION WITHIN CHASE WITH PLUMBING CONTRACTOR.
15	PROVIDE 10"x10" EXPANDED METAL GRILLE AT PENETRATION. BALANCE TO 400 CFM. PROVIDE RADIATION DAMPER IN 1 HR RATED WALL.
16	ROUTE 6"O SA DUCT DOWN TO DIFFUSER SERVING SALLYPORT BELOW.
17	ROUTE 6"O SA DUCT DOWN TO DIFFUSER SERVING STORAGE ROOM BELOW.

#	Revision	Date
---	----------	------

Project #: 17-700-078-2
Designed By: A.M.M
Drawn By: A.M.M
Checked By: D.E.B
Date: 04.12.2021



These Drawings and Specifications, and all copies thereof are and shall remain the property and copyright of the Architect. They shall be used only with respect to this Project and are not to be used on any other Project or Work without prior written permission from the Architect



SECOND FLOOR PLAN - UNIT C - DUCTWORK

E

D

C

B

A

6

5

4

3

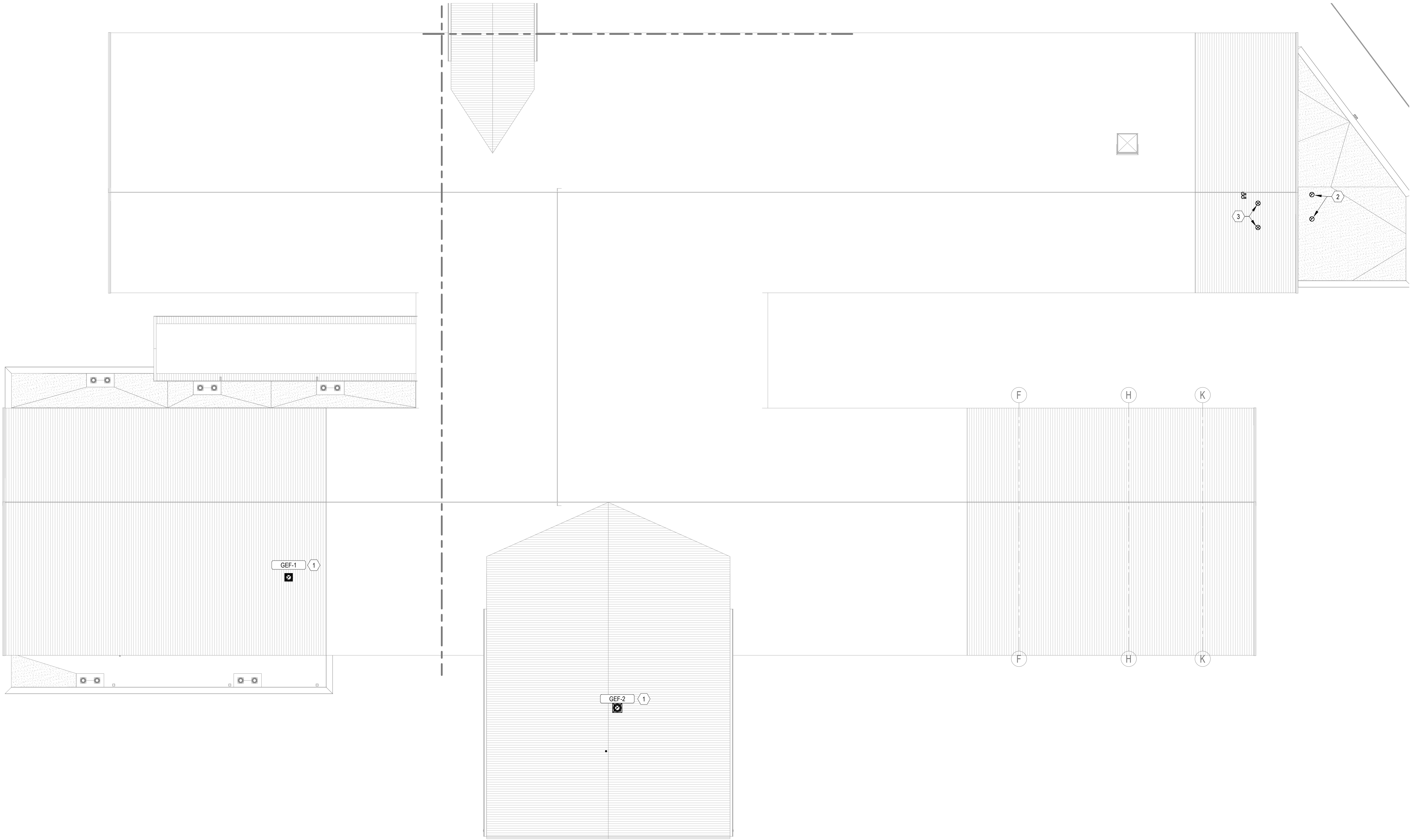
2

1

GENERAL NOTES:

- A. ALL DUCT DIMENSIONS GIVEN ARE INSIDE DIMENSIONED, UNLESS NOTED OTHERWISE.
- B. COORDINATE ALL OPENINGS IN FLOORS, ROOF AND WALLS WITH THE GENERAL AND STRUCTURAL CONTRACTORS.
- C. RUN ALL DUCTS AS HIGH AS POSSIBLE WITH RETURN AND EXHAUST ABOVE SUPPLY AS OFTEN AS POSSIBLE.
- D. PROVIDE TURNING VANES IN ALL SUPPLY AIR SQUARE THROATED ELBOWS.
- E. ALL RECTANGULAR SUPPLY AIR BRANCH CONNECTIONS SHALL BE 45° TAKE-OFF FITTINGS AND ROUND SUPPLY AIR BRANCH CONNECTIONS SHALL BE BELLMOUTH FITTINGS
- F. REFER TO DIFFUSER SCHEDULE FOR DUCT RUNOUT SIZE UNLESS NOTED OTHERWISE. ALL FLEXIBLE DUCTS TO DIFFUSERS MAXIMUM LENGTH 5'-0". FLEXIBLE DUCTS ARE NOT TO BE USED AS ELBOWS.
- G. ALL ABOVE FINISH FLOOR (A.F.F.) DIMENSIONS ARE TO BE MEASURED TO BOTTOM OF EQUIPMENT OR DUCTWORK UNLESS NOTED OTHERWISE.
- H. PROVIDE CEILING RADIATION DAMPERS (CRD) FOR ALL DIFFUSERS/GRILLES LOCATED IN RATED CEILINGS. COORDINATE TYPE OF CRD REQUIRED WITH ARCHITECTURALL CONSTRUCTION TYPE AS SHOWN ON ARCHITECTURAL PLANS.

PLAN NOTES	
#	NOTE
1	INSTALL EXHAUST FAN ON ROOF IN APPROXIMATE LOCATION SHOWN. ROUTE EXHAUST DUCT DOWN.
2	12" FLUE DUCT FROM BOILER THROUGH ROOF. TERMINATE WITH WEATHERPROOF CAPS PER MANUFACTURER'S RECOMMENDATIONS.
3	12" COMBUSTION AIR DUCT FROM BOILER THROUGH ROOF. TERMINATE WITH GOOSENECK PER MANUFACTURER'S RECOMMENDATIONS.



1 JAIL ROOF PLAN - MECHANICAL
3/32" = 1'-0"

E

D

C

B

A

100% CD Bid Set

VERMILLION COUNTY COMMISSIONERS

VERMILLION COUNTY JAIL

ADDITION & RENOVATION

1888 IN-43

HILLSDALE IN 47854

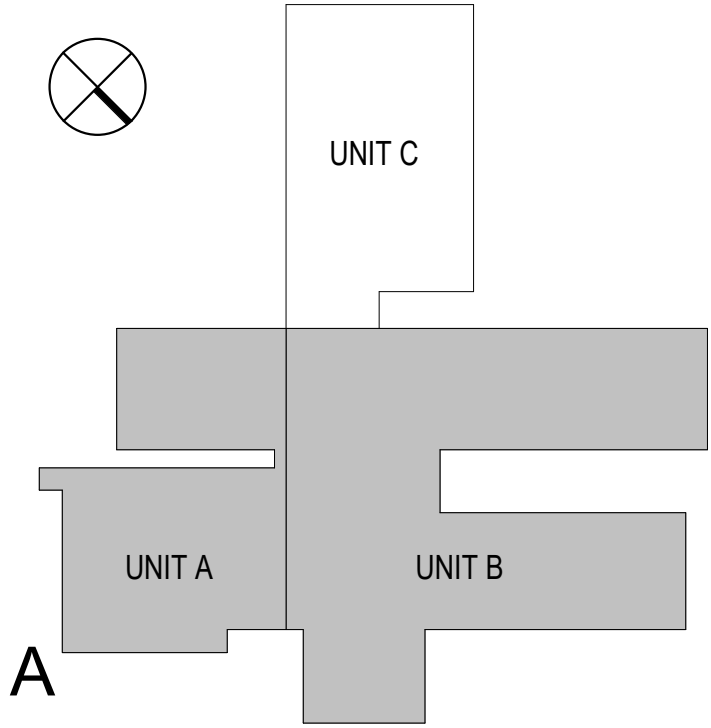
#	Revision	Date
---	----------	------

Project #: 17-700-078-2
Designed By: A.M.M
Drawn By: A.M.M
Checked By: D.E.B
Date: 04.12.2021



Eric Wepler

These Drawings and Specifications, and all copies thereof are and shall remain the property and copyright of the Architect. They shall be used only with respect to this Project and are not to be used on any other Project or Work without prior written permission from the Architect



JAIL ROOF PLAN - MECHANICAL

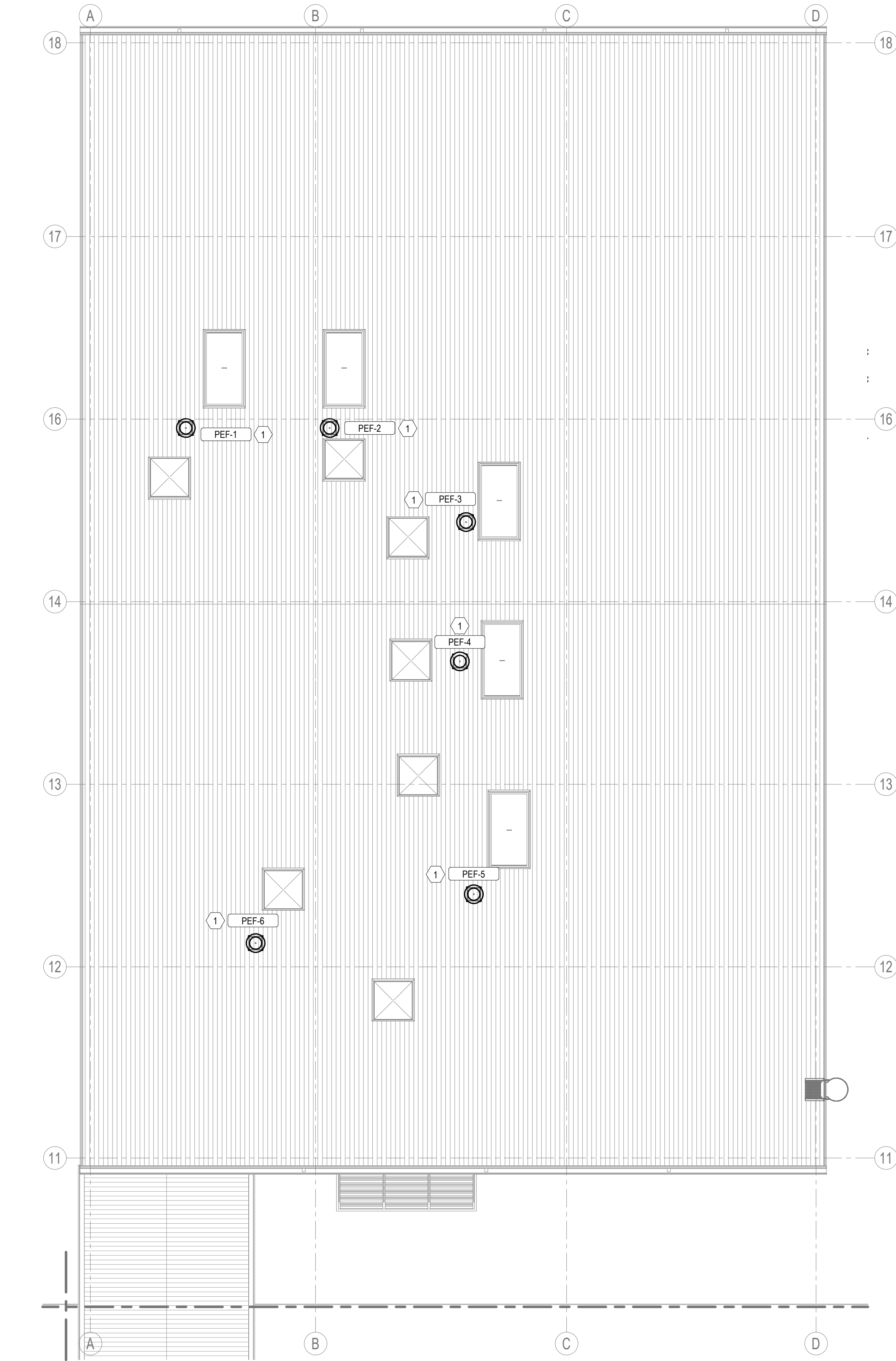
E

D

C

B

A



1 **POD ROOF PLAN - MECHANICAL**
1/8" = 1'-0"

GENERAL NOTES:

- A. ALL DUCT DIMENSIONS GIVEN ARE INSIDE DIMENSIONED, UNLESS NOTED OTHERWISE.
- B. COORDINATE ALL OPENINGS IN FLOORS, ROOF AND WALLS WITH THE GENERAL AND STRUCTURAL CONTRACTORS.
- C. RUN ALL DUCTS AS HIGH AS POSSIBLE WITH RETURN AND EXHAUST ABOVE SUPPLY AS OFTEN AS POSSIBLE.
- D. PROVIDE TURNING VANES IN ALL SUPPLY AIR SQUARE THROATED ELBOWS.
- E. ALL RECTANGULAR SUPPLY AIR BRANCH CONNECTIONS SHALL BE 45° TAKE-OFF FITTINGS AND ROUND SUPPLY AIR BRANCH CONNECTIONS SHALL BE BELLMOUTH FITTINGS
- F. REFER TO DIFFUSER SCHEDULE FOR DUCT RUNOUT SIZE UNLESS NOTED OTHERWISE. ALL FLEXIBLE DUCTS TO DIFFUSERS MAXIMUM LENGTH 5'-0". FLEXIBLE DUCTS ARE NOT TO BE USED AS ELBOWS.
- G. ALL ABOVE FINISH FLOOR (A.F.F.) DIMENSIONS ARE TO BE MEASURED TO BOTTOM OF EQUIPMENT OR DUCTWORK UNLESS NOTED OTHERWISE.
- H. PROVIDE CEILING RADIATION DAMPERS (CRD) FOR ALL DIFFUSERS/GRILLES LOCATED IN RATED CEILINGS. COORDINATE TYPE OF CRD REQUIRED WITH ARCHITECTURALL CONSTRUCTION TYPE AS SHOWN ON ARCHITECTURAL PLANS.

PLAN NOTES	
#	NOTE
1	INSTALL EXHAUST FAN ON ROOF IN APPROXIMATE LOCATION SHOWN. ROUTE EXHAUST DUCT DOWN.

E

D

C

B

A

100% CD Bid Set

VERMILLION COUNTY COMMISSIONERS
VERMILLION COUNTY JAIL
ADDITION & RENOVATION
1888 IN-63
HILLSDALE IN 47854

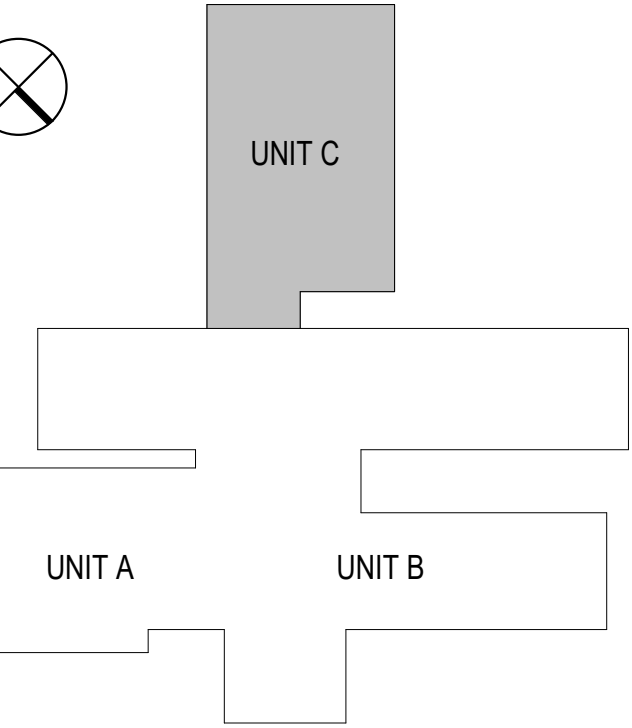
#	Revision	Date
---	----------	------

Project #: 17-700-078-2
Designed By: A.M.M
Drawn By: A.M.M
Checked By: D.E.B
Date: 04.12.2021



Eric Wepler

These Drawings and Specifications, and all copies thereof are and shall remain the property and copyright of the Architect. They shall be used only with respect to this Project and are not to be used on any other Project or Work without prior written permission from the Architect

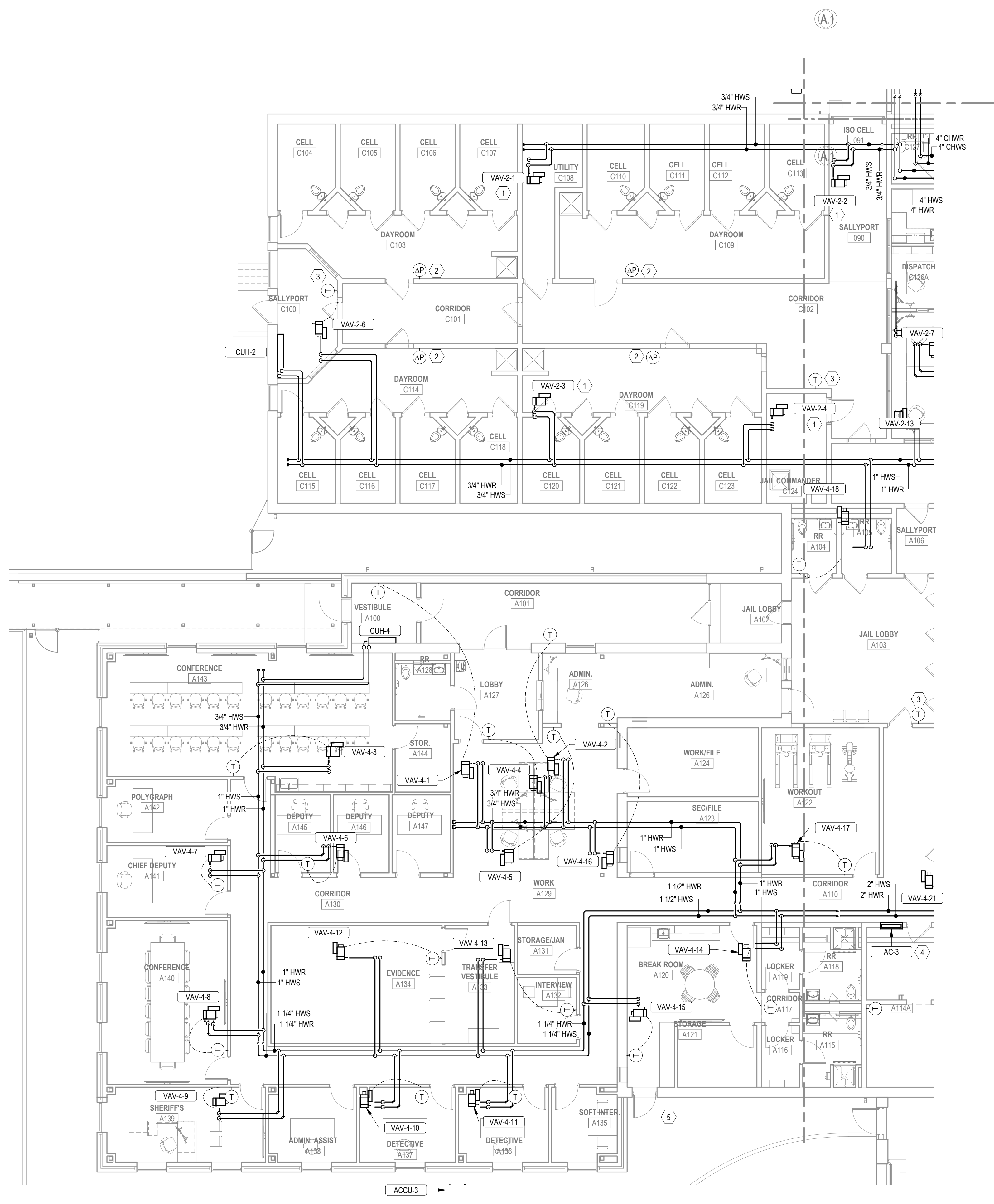


POD ROOF PLAN - MECHANICAL

M230C

E
D
C
B
A

6 5 4 3 2 1



1 FIRST FLOOR PLAN - UNIT A - PIPING
1/8" = 1'-0"

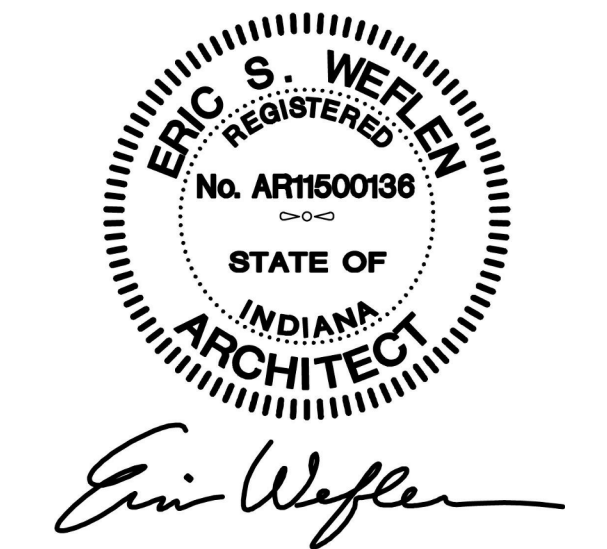
- GENERAL NOTES:
1. ALL HWS/R PIPING SHALL BE A MINIMUM OF 3/4" UNLESS NOTED OTHERWISE.
 2. ALL ABOVE FINISHED FLOOR (A.F.F.) DIMENSIONS ARE TO BE MEASURED TO BOTTOM OF EQUIPMENT, DUCTWORK OR PIPING UNLESS NOTED OTHERWISE.
 3. DETERMINE LINE LENGTH OF REFRIGERANT PIPING AND REFER TO MANUFACTURER'S INSTALLATION MANUAL FOR PREFERRED PIPE SIZING, INSULATION AND SPECIALTIES REQUIRED.

KEYNOTES	
KEY	NOTE
1	PROVIDE TEMPERATURE SENSOR IN EXHAUST DUCT FROM CELLS FOR TEMPERATURE CONTROL.
2	PROVIDE TAMPER PROOF DIFFERENTIAL PRESSURE SENSOR BETWEEN DAYROOM & CONTROL ROOM CORRIDOR. PROVIDED AND INSTALLED BY T.C.C.
3	THERMOSTAT TO HAVE LOCKBOX BUT CONTINUE TO HAVE LOCAL PROGRAMMING.
4	INSTALL AC UNIT AT 7'-6" AFF. ROUTE PUMPED CONDENSATE TO NEAREST FLOOR DRAIN. ASSOCIATED ACCU UNIT ON GRADE.
5	INSTALL ACCU UNIT IN APPROXIMATE LOCATION SHOWN. ROUTE REFRIGERANT PIPING AND CONTROL/POWER WIRING TO ASSOCIATED AC UNIT. FOR EXPOSED PIPING ON ROOF, INSULATE PIPING AND PROVIDE UV RESISTANT PVC JACKET. CONTROL/POWER WIRING SHALL BE INSTALLED IN CONDUIT.

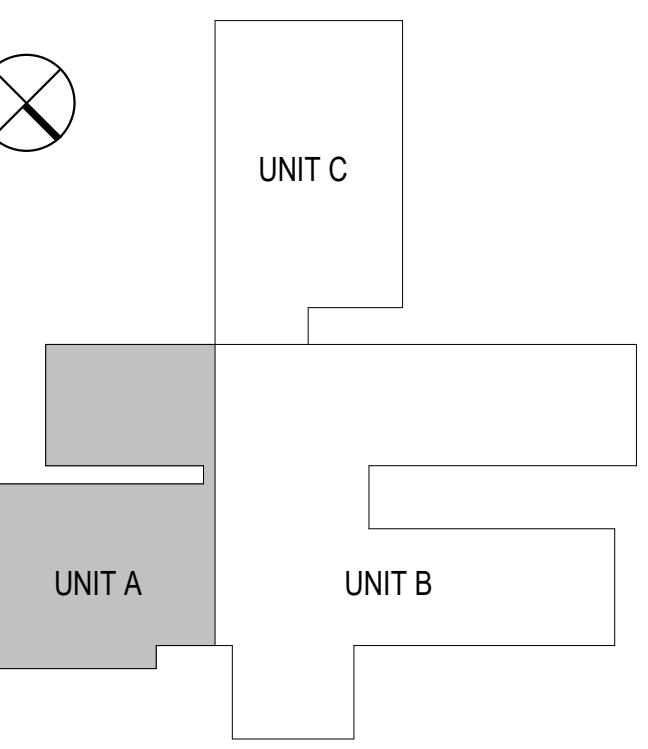
VERMILLION COUNTY COMMISSIONERS
VERMILLION COUNTY JAIL
ADDITION & RENOVATION
1888 IN-33
HILLSDALE IN 47854

#	Revision	Date
---	----------	------

Project #: 17-700-078-2
Designed By: A.M.M
Drawn By: A.M.M
Checked By: D.E.B
Date: 04.12.2021



These Drawings and Specifications, and all copies thereof are and shall remain the property and copyright of the Architect. They shall be used only with respect to this Project and are not to be used on any other Project or Work without prior written permission from the Architect



FIRST FLOOR PLAN - UNIT A - PIPING

- GENERAL NOTES:
1. ALL HWS/R PIPING SHALL BE A MINIMUM OF 3/4" UNLESS NOTED OTHERWISE.
 2. ALL ABOVE FINISHED FLOOR (A.F.F.) DIMENSIONS ARE TO BE MEASURED TO BOTTOM OF EQUIPMENT, DUCTWORK OR PIPING UNLESS NOTED OTHERWISE.
 3. DETERMINE LINE LENGTH OF REFRIGERANT PIPING AND REFER TO MANUFACTURER'S INSTALLATION MANUAL FOR PREFERRED PIPE SIZING, INSULATION AND SPECIALTIES REQUIRED.

KEY	NOTE
1	PROVIDE TEMPERATURE SENSOR IN EXHAUST DUCT FROM CELLS FOR TEMPERATURE CONTROL.
2	PROVIDE TERMINAL BOX TRANSFORMERS AT PRECISE LOCATION INDICATED ON ELECTRICAL DRAWINGS. TRANSFORMER POWERED BY EC WIRING TO TERMINAL BOXES BY TCC.
3	PROVIDE TAMPER PROOF DIFFERENTIAL PRESSURE SENSOR BETWEEN DAYROOM & CONTROL ROOM CORRIDOR. PROVIDED AND INSTALLED BY T.C.C.
4	THERMOSTAT TO HAVE LOCKBOX BUT CONTINUE TO HAVE LOCAL PROGRAMMING.
5	INSTALL ACQU UNIT IN APPROXIMATE LOCATION SHOWN. ROUTE REFRIGERANT PIPING AND CONTROL POWER WIRING TO ASSOCIATED AC UNIT. FOR EXPOSED PIPING ON ROOF, INSULATE PIPING AND PROVIDE UV RESISTANT PVC JACKET. CONTROL POWER WIRING SHALL BE INSTALLED IN CONDUIT.
6	PIPE UP HEATING COIL CIRCULATION PUMP ACCORDING TO DETAIL XM80X.

100% CD Bid Set

VERMILION COUNTY COMMISSIONERS

VERMILION COUNTY JAIL

ADDITION & RENOVATION

1888 IN-53

HILLSDALE IN 47854

#	Revision	Date
---	----------	------

Project #: 17-700-078-2

Designed By: A.M.M

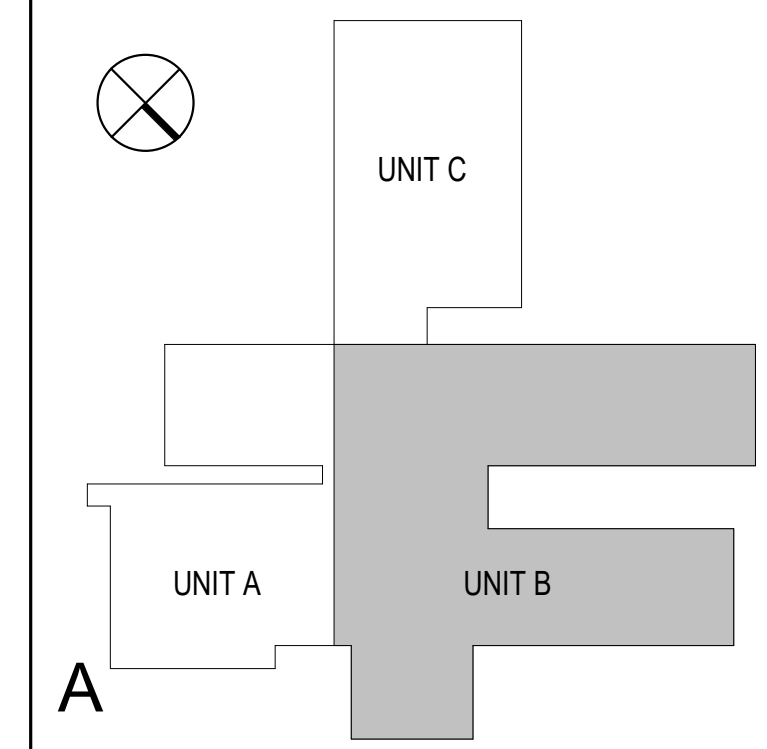
Drawn By: A.M.M

Checked By: D.E.B

Date: 04.12.2021

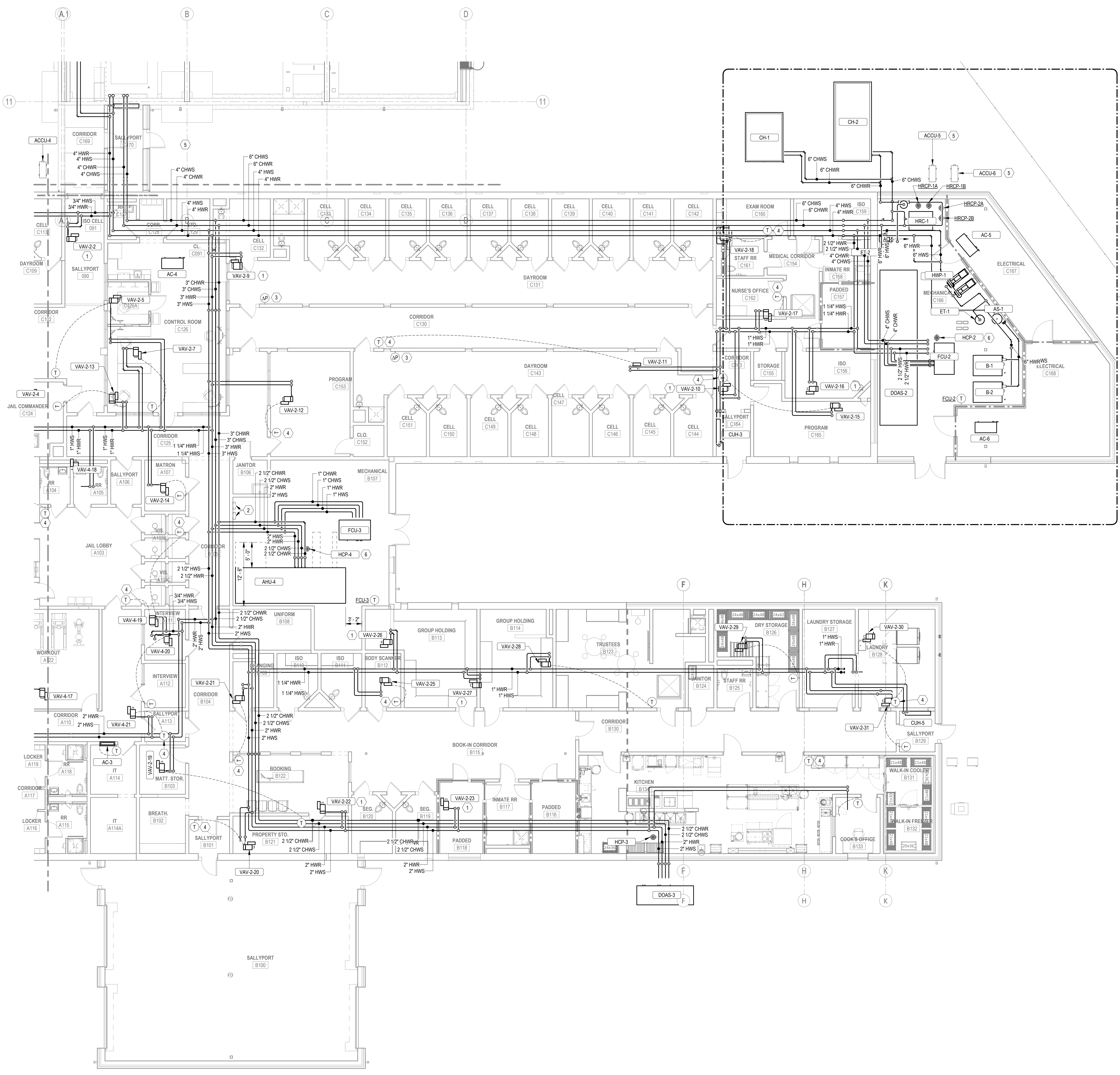


These Drawings and Specifications, and all copies thereof are and shall remain the property and copyright of the Architect. They shall be used only with respect to this Project and are not to be used on any other Project or Work without prior written permission from the Architect



FIRST FLOOR PLAN - UNIT B - PIPING

M310B



1 FIRST FLOOR PLAN - UNIT B - PIPING

1/8" = 1'-0"

0 4 8 16

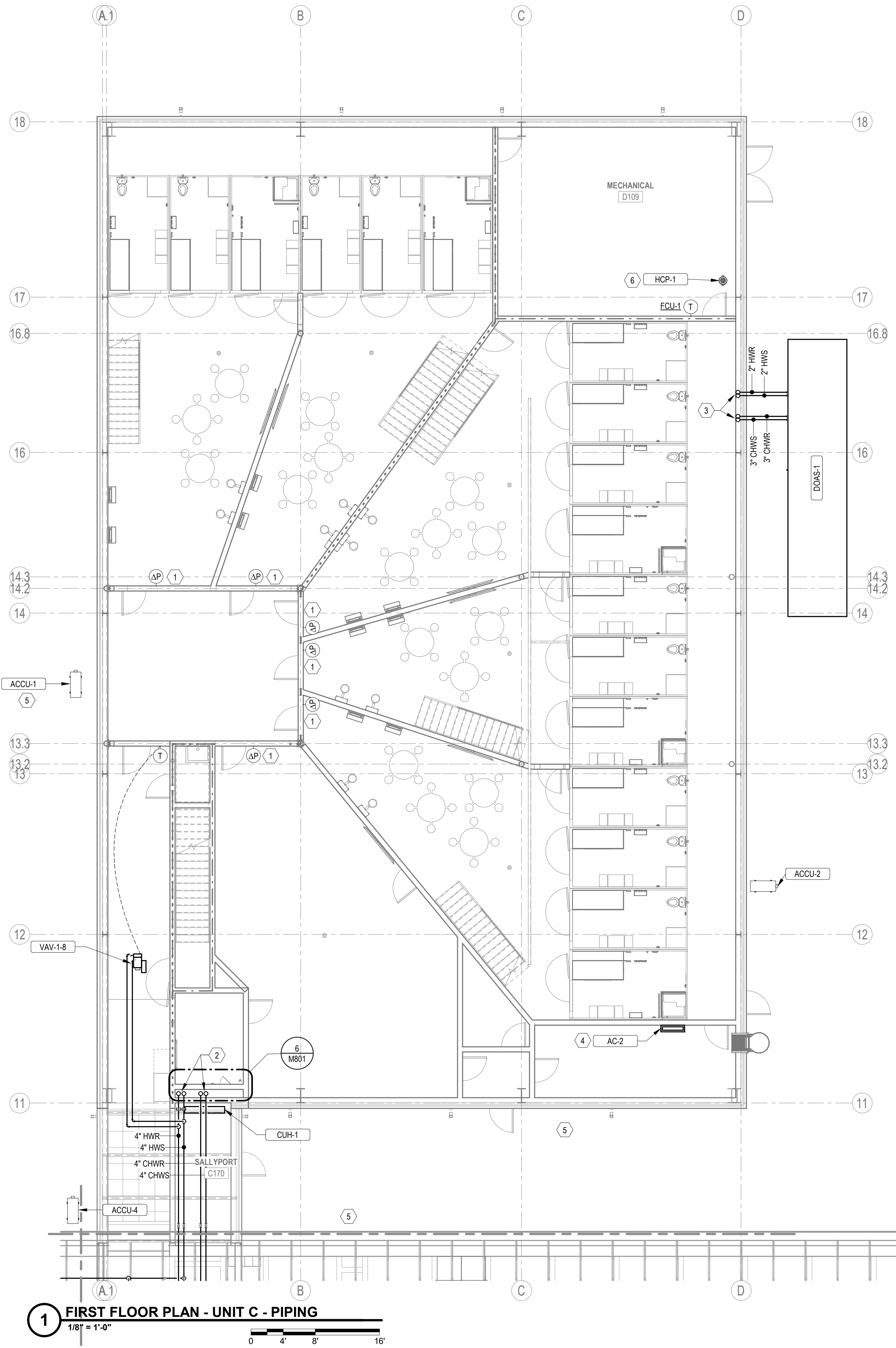
E

D

C

B

A



- GENERAL NOTES:
1. ALL HWS/R PIPING SHALL BE A MINIMUM OF 3/4" UNLESS NOTED OTHERWISE.
 2. ALL ABOVE FINISHED FLOOR (A.F.F.) DIMENSIONS ARE TO BE MEASURED TO BOTTOM OF EQUIPMENT, DUCTWORK OR PIPING UNLESS NOTED OTHERWISE.
 3. DETERMINE LINE LENGTH OF REFRIGERANT PIPING AND REFER TO MANUFACTURER'S INSTALLATION MANUAL FOR PREFERRED PIPE SIZING, INSULATION AND SPECIALTIES REQUIRED.

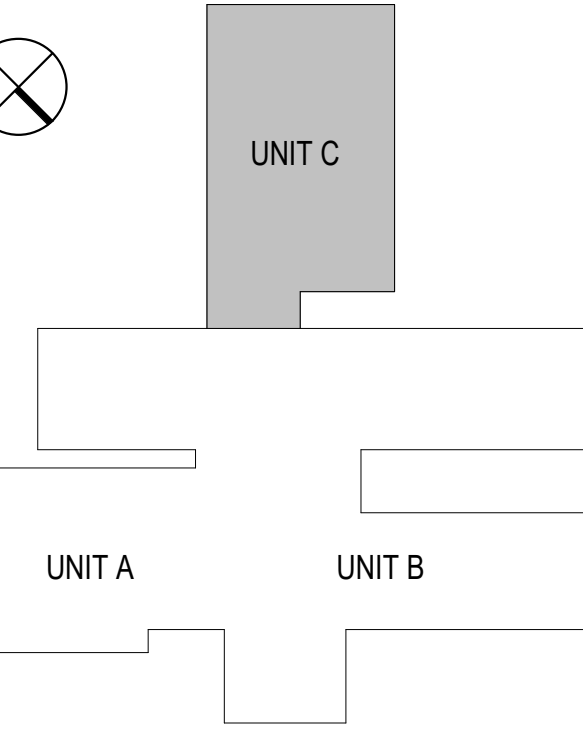
PLAN NOTES	
#	NOTE
1	PROVIDE TAMPER PROOF DIFFERENTIAL PRESSURE SENSOR BETWEEN DAYROOM AND SECURED CORRIDOR. MOUNT DAYROOM SIDE OF PRESSURE SENSOR HIGH ON WALL OUT OF REACH FROM INMATES. PROVIDED AND INSTALLED BY TCC.
2	4" CW/S&R AND 4" HWS&R UP. REFER TO M320C FOR CONTINUATION.
3	3" CW/S&R AND 2" HWS&R UP. REFER TO M320C FOR CONTINUATION.
4	INSTALL AC UNIT AT 7'-6" AFF. ROUTE PUMPED CONDENSATE TO NEAREST FLOOR DRAIN. ASSOCIATED ACCU UNIT ON GRADE.
5	INSTALL ACCU UNIT IN APPROXIMATE LOCATION SHOWN. ROUTE REFRIGERANT PIPING AND CONTROL POWER WIRING TO ASSOCIATED AC UNIT. FOR EXPOSED PIPING ON ROOF, INSULATE PIPING AND PROVIDE UV RESISTANT PVC JACKET. CONTROL POWER WIRING SHALL BE INSTALLED IN CONDUIT.
6	PIPE UP HEATING COIL CIRCULATION PUMP ACCORDING TO DETAIL 8/M805.

#	Revision	Date
---	----------	------

Project #: 17-700-078-2
Designed By: A.M.M
Drawn By: A.M.M
Checked By: D.E.B
Date: 04.12.2021



These Drawings and Specifications, and all copies thereof are and shall remain the property and copyright of the Architect. They shall be used only with respect to this Project and are not to be used on any other Project or Work without prior written permission from the Architect.



FIRST FLOOR PLAN - UNIT C - PIPING

M310C

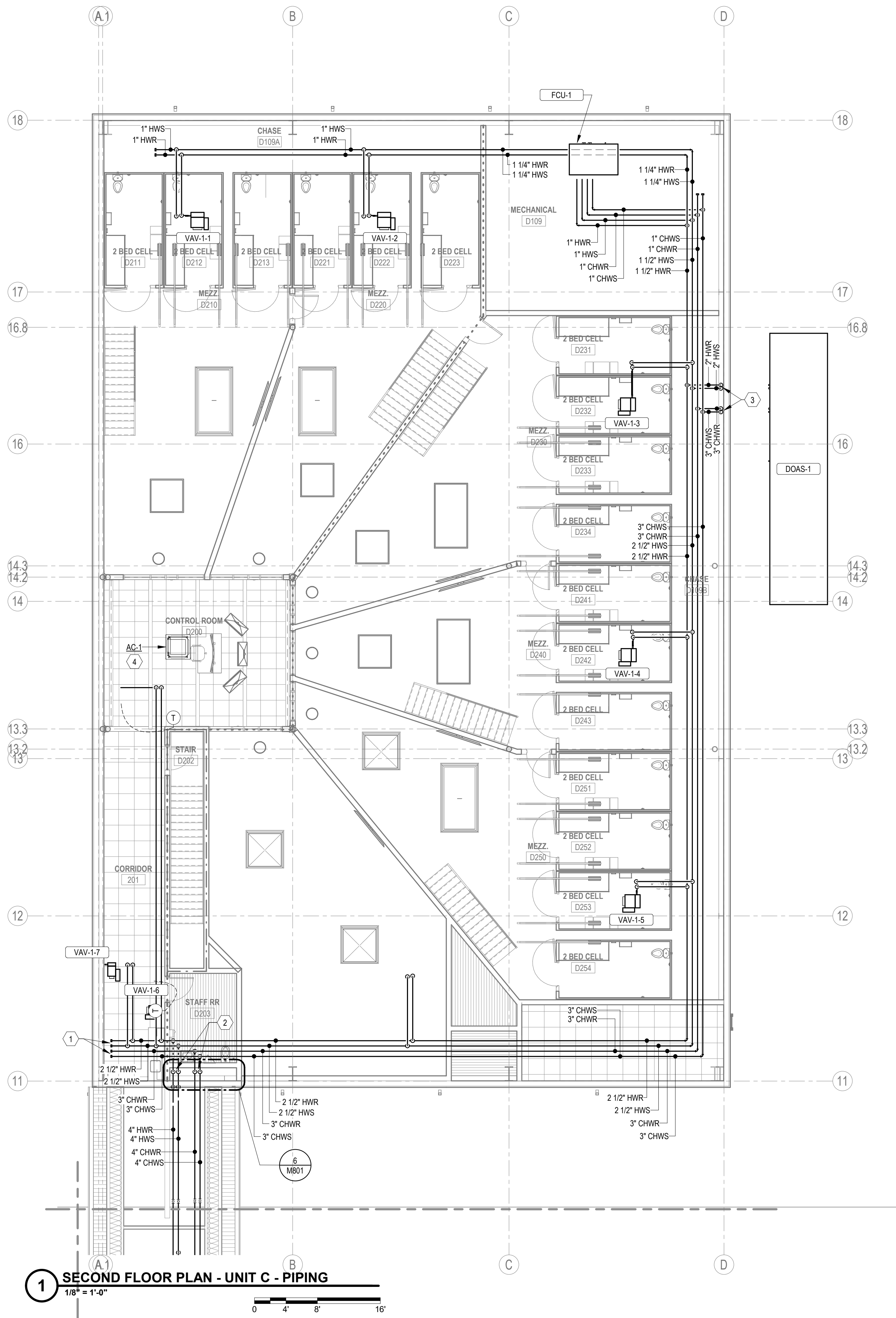
E

D

C

B

A



GENERAL NOTES:

1. ALL HWS/R PIPING SHALL BE A MINIMUM OF 3/4" UNLESS NOTED OTHERWISE.
2. ALL ABOVE FINISHED FLOOR (A.F.F.) DIMENSIONS ARE TO BE MEASURED TO BOTTOM OF EQUIPMENT, DUCTWORK OR PIPING UNLESS NOTED OTHERWISE.
3. DETERMINE LINE LENGTH OF REFRIGERANT PIPING AND REFER TO MANUFACTURER'S INSTALLATION MANUAL FOR PREFERRED PIPE SIZING, INSULATION AND SPECIALTIES REQUIRED.

KEYNOTES	
KEY	NOTE
1	CAP HWSR & CHWSR FOR FUTURE POD BUILDOUT.
2	4" CHWSR AND 4" HWSR DOWN. REFER TO M310C FOR CONTINUATION.
3	3" CHWSR AND 2" HWSR DOWN. REFER TO M310C FOR CONTINUATION.
4	INSTALL AC UNIT IN CEILING OR AT 9'-0" AFF IN LOCATION INDICATED. ROUTED PUMPED CONDENSATE TO NEAREST FLOOR DRAIN. ASSOCIATED ACCU UNIT ON GRADE.

E

D

C

B

A

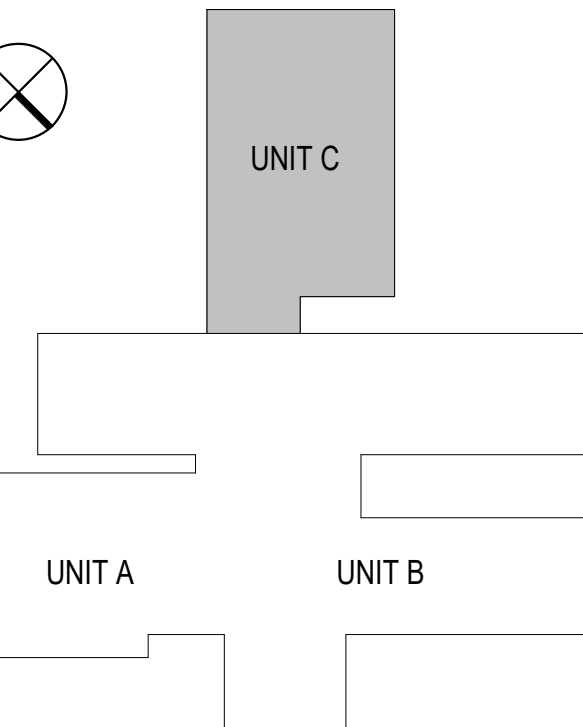
#	Revision	Date
---	----------	------

Project #: 17-700-078-2
Designed By: A.M.M
Drawn By: A.M.M
Checked By: D.E.B
Date: 04.12.2021



Eric Wepler

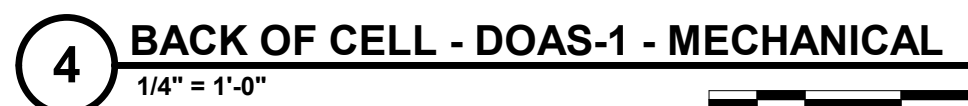
These Drawings and Specifications, and all copies thereof are and shall remain the property and copyright of the Architect. They shall be used only with respect to this Project and are not to be used on any other Project or Work without prior written permission from the Architect



SECOND FLOOR PLAN -
UNIT C - PIPING

M320C

PLAN NOTES	
#	NOTE



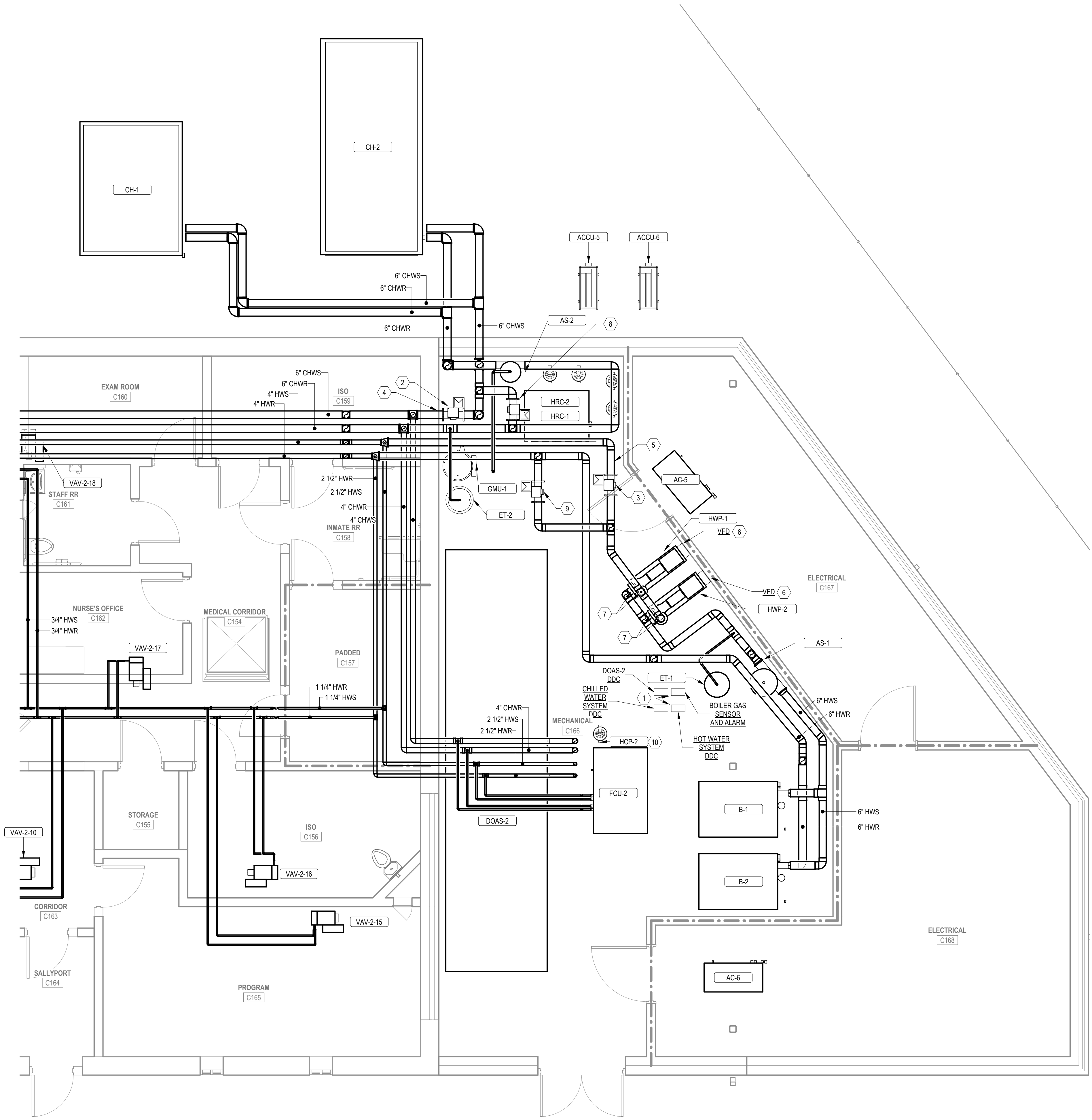
E

D

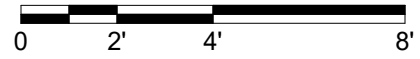
C

B

A



1 MECHANICAL ROOM - UNIT B - PIPING
1/4" = 1'-0"



GENERAL NOTES:

1. ALL HWS/R PIPING SHALL BE A MINIMUM OF 3/4" UNLESS NOTED OTHERWISE.
2. ALL ABOVE FINISHED FLOOR (A.F.F.) DIMENSIONS ARE TO BE MEASURED TO BOTTOM OF EQUIPMENT, DUCTWORK OR PIPING UNLESS NOTED OTHERWISE.
3. DETERMINE LINE LENGTH OF REFRIGERANT PIPING AND REFER TO MANUFACTURER'S INSTALLATION MANUAL FOR PREFERRED PIPE SIZING, INSULATION AND SPECIALTIES REQUIRED.

KEYNOTES	
KEY	NOTE
1	MOUNT PANELS ON UNISTRUT.
2	BUILDING CHILLED WATER CONTROL VALVE.
3	BUILDING HOT WATER CONTROL VALVE.
4	CHILLED WATER FLOW METER.
5	HOT WATER FLOW METER.
6	HOT WATER PUMP VFC.
7	6" HWS/R DOWN TO HOT WATER PUMP INLET AND OUTLET.
8	CHILLED WATER BYPASS.
9	HOT WATER BYPASS.
10	PIPE UP HEATING COIL CIRCULATION PUMP ACCORDING TO DETAIL 8/M805.

100% CD Bid Set

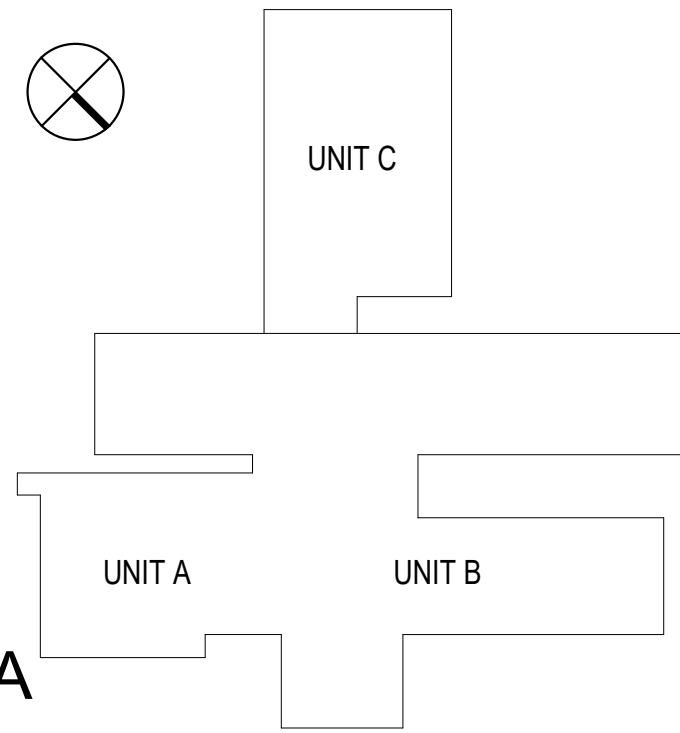
#	Revision	Date
---	----------	------

Project #: 17-700-078-2
Designed By: A.M.M
Drawn By: A.M.M
Checked By: D.E.B
Date: 04.12.2021



Eric Wepler

These Drawings and Specifications, and all copies thereof are and shall remain the property and copyright of the Architect. They shall be used only with respect to this Project and are not to be used on any other Project or Work without prior written permission from the Architect



AIR HANDLING UNIT SYSTEM INDEX SCHEDULE

1. AHU TO HAVE SINGLE POINT CONNECTION FOR 460/3 POWER AND A SEPARATE CONNECTION FOR 120/1.
2. MOUNT AHU ON MINIMUM 6" HIGH CONCRETE PAD WHICH EXTENDS 6" BEYOND PERIMETER OF AHU.
3. UNIT IS REQUIRED TO BE BUILT TO PRECISE OUTER DIMENSIONS AS NOTED ON SHEET M-607 AND FIT WITHIN DEMENSIONS SHOWN ON THE DUCTWORK DRAWINGS.
4. DOAS UNIT TO BE PROVIDED WITH FACTORY CONTROLS WITH BACKNET INTERFACE. SEE DOAS UNIT SPECIFICATIONS.

LOCATION					FAN																				DIMENSIONS (IN.)														DESIGN REFERENCE			
MARK	NAME	NO.	SYSTEM TYPE	AREA SERVED	SUPPLY FAN	RETURN FAN	TOTAL CFM	OA CFM	OA %	PREHEAT COIL	REHEAT COIL	COOLING COIL	DX HEAT RECOVERY COIL	FILTER(S)	SOUND ATTENUATOR	PLATE & FRAME HEAT EXCHANGER	OPERATING WEIGHT (LBS)	L	W	H	VOLTAGE	PHASE	Hz	FLA	MCA	MOCP	EMER. POWER	MANUFACTURER	MODEL	REMARKS												
DOAS-1	OUTSIDE POD	-	MULIZONE DOAS	POD	SF-1A/B	EF-1A/B	9000	9000	100.00%	PHC-1	-	CC-1	-	OAF-1/RF-1	SA-1A/B	HE-1	13,000	414"	88"	96"	460	3	60	32.4	34.8	40	YES	INNOVENT	ERU-SS-PL-9000-HW-CW-SL-SL-460	1,2,4												
DOAS-2	MECHANICAL	C166	MULIZONE DOAS	EXISTING JAIL/BOOKING	SF-2A/B	EF-2A/B	12000	12000	100.00%	PHC-2	-	CC-2	-	OAF-2/RF-2	SA-2A/B	HE-2	15,000	384"	88"	116"	460	3	60	43	46	50	YES	INNOVENT	ERU-OU-PL-12000-HW-CW-SL-SL-460	1,2,3,4												
DOAS-3	OUTSIDE KITCHEN	-	SINGLE ZONE DOAS	KITCHEN	SF-3	-	3000	3000	100.00%	-	RHC-3	CC-3	-	OAF-3	-	-	7,500	-	-	-	460	3	60	6	7.5	12	YES	PACE	PAQ-36X45	1,2,4												
AHU-4	MECHANICAL	B107	VAV	ADMIN	SF-4A/B	RF-4A/B	8000	2000	25.00%	PHC-3	-	CC-4	-	PF-4	SA-4A/B	-	2,500	148"	45"	36"	460	3	60	25.2	31.6	40	YES	PACE	PAI-42X84	1,2,3,4												

AHU SUPPLY FAN SCHEDULE

1. PROVIDE UNIT MOUNTED DISCONNECT SWITCH.
2. FAN SHALL HAVE OUTLET GUARD.
3. FAN SHALL HAVE BLANK-OFF PLATE PROVIDED BY MANUFACTURER FOR FAN ISOLATION

MARK	LOCATION	AIRFLOW	PRESSURE OF EACH FAN				WHEEL				FAN RPM	MAX. BHP/KW	VARIABLE CONTROL TYPE	MOTOR				DESIGN REFERENCE				REMARKS
			ESP (IN. W.G.)		TSP (IN. W.G.)		TYPE	CLASS	DIA	PH				MCA	MOCP	DIRECT DRIVE (YES/NO)	MANUFACTURER	MODEL				
SF-1A	DOAS-1	4500	2.00	5.55	PLENUM	II	SWSI	450	2493	5.6	VFD	7.5	1800	460	3	SINGLE POINT	YES	GREENHECK	-	-	1,2,3	
SF-1B	DOAS-1	4500	2.00	5.55	PLENUM	II	SWSI	450	2493	5.6	VFD	7.5	1800	460	3	SINGLE POINT	YES	GREENHECK	-	-	1,2,3	
SF-2A	DOAS-2	6000	2.00	5.97	PLENUM	II	SWSI	500	2314	8	VFD	10	1800	460	3	SINGLE POINT	YES	GREENHECK	-	-	1,2,3	
SF-2B	DOAS-2	6000	2.00	5.97	PLENUM	II	SWSI	500	2314	8	VFD	10	1800	460	3	SINGLE POINT	YES	GREENHECK	-	-	1,2,3	
SF-3	DOAS-3	3000	1.50	4.20	PLENUM	II	SWSI	135	3660	3.41	VFD	5	3600	460	3	SINGLE POINT	YES	LAU	DDPG2	-	1,2,3	
SF-4A	AHU-4	4000	2.00	3.85	PLENUM	II	SWSI	122	4243	4.84	VFD	5	3600	460	3	SINGLE POINT	YES	LAU	DDPG2	-	1,2,3	
SF-4B	AHU-4	4000	2.00	3.85	PLENUM	II	SWSI	122	4243	4.84	VFD	5	3600	460	3	SINGLE POINT	YES	LAU	DDPG2	-	1,2,3	

AHU RETURN/EXHAUST FAN SCHEDULE

1. PROVIDE UNIT MOUNTED DISCONNECT SWITCH.
2. FAN SHALL HAVE OUTLET GUARD.
3. FAN SHALL HAVE BLANK-OFF PLATE PROVIDED BY MANUFACTURER FOR FAN ISOLATION

													PRESSURE OF EACH FAN		WHEEL												MOTOR				DESIGN REFERENCE					
MARK	LOCATION	FAN DESCRIPTION	AIRFLOW (CFM)	ESP (IN. W.G.)	TSP (IN. W.G.)	TYPE	CLASS	TYPE	DIA	FAN RPM	MAX. BHP/KW	VARIABLE CONTROL TYPE	HP/KW	RPM	VOLT	PH	MCA	MOCP	DIRECT DRIVE (YES/NO)	MANUFACTURER	MODEL	SCHEDULE NOTES														
EF-1A	DOAS-1	EXHAUST FAN	4500	1.75	3.62	PLENUM	II	SWSI	450	2182	3.6	VFD	5	1800	460	3	SINGLE POINT		Yes	GREENHECK	-	1,2,3														
EF-1B	DOAS-1	EXHAUST FAN	4500	1.75	3.62	PLENUM	II	SWSI	450	2182	3.6	VFD	5	1800	460	3	SINGLE POINT		Yes	GREENHECK	-	1,2,3														
EF-2A	DOAS-2	EXHAUST FAN	6000	1.50	3.59	PLENUM	II	SWSI	500	1985	4.8	VFD	7.5	1800	460	3	SINGLE POINT		Yes	GREENHECK	-	1,2,3														
EF-2B	DOAS-2	EXHAUST FAN	6000	1.50	3.59	PLENUM	II	SWSI	500	1985	4.8	VFD	7.5	1800	460	3	SINGLE POINT		Yes	GREENHECK	-	1,2,3														
RF-4A	AHU-4	RETURN FAN	4000	1.75	2.48	AF	II	SWSI	150	2532	2.7	VFD	5	1800	460	3	SINGLE POINT		Yes	LUA	DDPG2	1,2,3														
RF-4B	AHU-4	RETURN FAN	4000	1.75	2.48	AF	II	SWSI	150	2532	2.7	VFD	5	1800	460	3	SINGLE POINT		Yes	LUA	DDPG2	1,2,3														

AHU CHILLED WATER COOLING COIL SCHEDULE

MARK	LOCATION	CFM	CAPACITY (MBH)		MAX. FACE VELOCITY (FPM)	MAX. APD (IN. W.G.)	# OF ROWS	# OF COILS	MAX. FINS PER INCH	ENTERING AIR TEMPERATURE		LEAVING AIR TEMPERATURE		FLUID	GPM	CIRCULATING FLUID			REMARKS
			TOTAL	SENSIBLE						DB (°F)	WB (°F)	DB (°F)	WB (°F)			ENTERING (°F)	LEAVING (°F)	MAX WPD (FT. HD.)	
CC-1	DOAS-1	9000	682	302	486	1.49	10	1	7	82.4	74.6	52.0	52.0	30% PG	96.4	40	55	23.5	
CC-2	DOAS-2	12000	923	409	500	0.20	2	1	8	82.6	74.7	51.8	51.8	30% PG	130.5	40	55	21.7	
CC-3	DOAS-3	3000	244	129	476	1.07	10	1	11	95.0	78.0	52.0	52.1	30% PG	48.3	40	51	16.6	
CC-4	AHU-4	8000	275	203	491	0.82	10	1	10	76.6	64.2	52.1	51.8	30% PG	39.6	40	55	16.5	

AHU HEATING WATER COIL SCHEDULE

MARK	LOCATION	TYPE	CFM	CAPACITY (MBH)		MAX. FACE VELOCITY (FPM)	MAX APD (IN. W.G.)	# OF ROWS	# OF COILS	MAX. FINS PER INCH	ENTERING AIR		LEAVING AIR		FLUID	GPM	CIRCULATING FLUID			REMARKS
				SENSIBLE							DB (°F)	WB (°F)	DB (°F)	WB (°F)			ENTERING (°F)	LEAVING (°F)	MAX. WPD (FT. HD.)	
PHC-1	DOAS-1	PRE-HEAT	9000	428	500	0.17	2	1	7	16.1	16.1	63.9	60.0	WATER	28.9	140	110	8.6		
PHC-2	DOAS-2	PRE-HEAT	12000	622	500	0.20	2	1	8	16.1	16.1	63.9	60.0	WATER	41.9	140	110	3.4		
PHC-3	DOAS-3	PRE-HEAT	3000	333	476	0.28	4	1	9	-10.0	-10.0	84.7	84.7	WATER	22.7	140	110	3.1		
PHC-4	AHU-4	PRE-HEAT	8000	312	491	0.11	2	1	8	36.8	36.8	71.7	71.7	WATER	21.2	140	110	1.5		

AHU PLATE AND FRAME HEAT EXCHANGER SCHEDULE

MARK	LOCATION	OA MIN FLOW	EXHAUST AIRFLOW	SUMMER DESIGN ENERGY RECOVERY										WINTER DESIGN ENERGY RECOVERY										EFFECTIVNESS		DESIGN REFERENCE		
				OUTSIDE AIR					EXHAUST AIR					OUTSIDE AIR					EXHAUST AIR									
				EAT		LAT		APD	EAT		LAT		APD	EAT		LAT		APD	EAT		LAT		APD	FROST CONTROL				
				DB (°F)	WB (°F)	DB (°F)	WB (°F)		DB (°F)	RH %	DB (°F)	RH %		DB (°F)	WB (°F)	DB (°F)	WB (°F)		DB (°F)	RH %	DB (°F)	RH %						
				DB (°F)	WB (°F)	DB (°F)	WB (°F)		DB (°F)	RH %	DB (°F)	RH %		DB (°F)	WB (°F)	DB (°F)	WB (°F)		DB (°F)	RH %	DB (°F)	RH %						
HE-1	DOAS-1	9000	9000	95.0	78.0	82.4	74.6	0.87	75.0	50.0	87.5	33.4	0.85	-10.0	-10.0	33.8	23.5	0.71	60.0	30.0	21.1	100.0	0.71	BYPASS @ 16.1 °F	INNOVENT	H-1-50B-1500		
HE-2	DOAS-2	12000	12000	95.0	78.0	82.6	74.7	1.09	75.0	50.0	87.3	33.6	1.07	-10.0	-10.9	33.2	23.1	0.91	60.0	30.0	21.4	100.0	0.98	BYPASS @ 16.1 °F	INNOVENT	H-1-50B-1650		

AHU AIR FILTER SCHEDULE

MARK	LOCATION	AIRFLOW (CFM)	EFFICIENCY MERV	MAX FACE VELOCITY (FPM)	MAX. A.P.D.		FILTER TYPE	HOUSING TYPE	CARTRIDGES		REMARKS
					INITIAL	FINAL			QUANTITY	SIZE (IN.)	
OAF-1	DOAS-1	9000	MERV-8	422	0.24	0.62	PLEATED	GALVANIZED	8	16X24	
RAF-1	DOAS-1	9000	MERV-8	422	0.24	0.62	PLEATED	GALVANIZED	8	16X24	
OAF-2	DOAS-2	12000	MERV-8	450	0.24	0.62	PLEATED	GALVANIZED	8	20X24	
RAF-2	DOAS-2	12000	MERV-8	450	0.24	0.62	PLEATED	GALVANIZED	8	20X24	
OAF-3	DOAS-3	3000	MERV-8	500	0.26	1	PLEATED	GALVANIZED	-	6.0 SF	
PF-4	AHU-4	8000	MERV-8	500	0.27	1	PLEATED	GALVANIZED	12	16.0 SF	

AHU SOUND ATTENUATOR SCHEDULE

MARK	LOCATION	SYSTEM	AIRFLOW (CFM)	MAX. VELOCITY (FPM)	DIMENSIONS			MAX. APD (IN. W.G.)	DYNAMIC INSERTION LOSS (IN DB) REQUIRED FOR GIVEN OCTAVE BAND								REMARKS
					WIDTH (IN.)	HEIGHT (IN.)	LENGTH (IN.)		63 Hz	125 Hz	250 Hz	500 Hz	1000 Hz	2000 Hz	4000 Hz	8000 Hz	
SA-1A	DOAS-1	SUPPLY	9000	341	47	85	24	0.1	6	9	15	21	17	14	14	12	
SA-1B	DOAS-1	RETURN	9000	341	47	85	24	0.1	6	9	15	21	17	14	14	12	
SA-2A	DOAS-2	SUPPLY	12000	393	83	53	24	0.1	6	9	15	21	17	14	14	12	
SA-2B	DOAS-2	RETURN	12000	393	83	53	24	0.1	6	9	15	21	17	14	14	12	
SA-4A	AHU-4	SUPPLY	8000	706	78	36	36	0.09	3.4	5.7	14.4	22.8	25.9	23.5	17.9	11	
SA-4B	AHU-4	RETURN	8000	706	78	36	36	0.09	2.5	5.7	13.5	22.1	25.2	23.5	16.9	13	

AIR COOLED CHILLER SCHEDULE

REMARKS:

1. FURNISH WITH SINGLE POINT ELECTRICAL CONNECTION WITH FACTORY MOUNTED AND WIRED FUSED DISCONNECT.
2. FURNISH WITH TEMPERATURE/PRESSURE SENSORS FOR CHILLED WATER ENTERING AND LEAVING TEMPERATURE/PRESSURE.
3. FURNISH WITH HIGH/LOW PRESSURE SAFETY SWITCHES.
4. PROVIDE INTEGRAL LEAD/LAG PUMPS AT THE CPL LISTED, 5HP, AND 140 FT HD (TOTAL).
5. PROVIDE BACNET CARD WITH READ/WRITE CAPABILITY THAT ALLOWS FOR THE IMPLEMENTATION OF THE SEQUENCE WRITTEN ON SHEET M801 FOR THE CHILLED WATER SYSTEM.
6. INTEGRAL CHILLED WATER PUMPS TO BE DUAL PUMPS WITH 100% BACK UP. PUMPS TO HAVE INTEGRAL VFD'S WITH SENSORLESS CONTROL. TECHNOLOGY. INTEGRAL PUMPS TO HAVE SEPARATE BACNET INTERFACE TO ALLOW FOR INDEPENDENT OPERATION FROM CHILLER.
7. DEDICATED CONSTANT VOLUME PUMPS TO BE PROVIDED BY CHILLER MANUFACTURER FOR FIELD INSTALLATION BY INSTALLING CONTRACTOR.

LOCATION					EVAPORATOR COOLING HEAT EXCHANGER													CONDENSER DATA					COMPRESSOR		UNIT ELECTRICAL DATA					ACCESSORIES					DESIGN REFERENCE							
																							REFRIGERANT																			
MARK	NAME	NO.	COOLING COP	HEATING COP	NPLV	NOMINAL CAPACITY (TON)	DESIGN COOLING CAPACITY (TON)	MAX FLOW (GPM)	DESIGN FLOW (GPM)	MIN FLOW (GPM)	FLUID	FOULING FACTOR	EWI (°F)	LWI (°F)	VOL (GAL)	WPD (FT. OF HEAD)	TOTAL WPD (FT. OF HEAD)	DESIGN FLOW (GPM)	FLUID	EWI (°F)	LWI (°F)	PD (FTH20)	Vol (GAL)	TYPE	TYPE	SUMMER AMBIENT DBT (°F)	FLA (A)	MCA (A)	MOCAP (A)	VOLTS PHASE HERTZ	EMER. POWER	UL LISTED (YES/NO)	SINGLE POINT ELEC. CONN. (YES/NO)	VIBRATION ISOLATORS (YES/NO)	WATER FLOW SWITCH (YES/NO)	UNIT MOUNTED VFD (YES/NO)	FACTORY AUTHORIZED STARTUP (YES/NO)	(BACNET) (YES/NO)	OPERATING WEIGHT (LBS)	MANUFACTURER	MODEL NO.	REMARKS
CH-1	MECHANICAL C166	OUTSIDE	-	-	15.29	85	77.0	400.1	128.1	99.9	30% PG	0.0001	55	40	8.8	4.2	6.8	-	-	-	-	-	-	SCROLL	R-410A	95.0	-	202.0	225.0	460/360	NO	Yes	Yes	Yes	Yes	Yes	Yes	5331	QUANTECH	QTC3085TSE46XFBSSXXX	1,2,3,4,5,6	
CH-2	MECHANICAL C168	OUTSIDE	-	-	15.15	140	134.0	624.5	222.3	149.9	30% PG	0.0001	55	40	13.2	6.5	10.8	-	-	-	-	-	-	SCROLL	R-410A	95.0	-	333.0	350.0	460/360	NO	Yes	Yes	Yes	Yes	Yes	Yes	7699	QUANTECH	QTC3140TSE46XFBSSXXX	1,2,3,4,5,6	
HRC-1	MECHANICAL C166	3.40	4.40	4.40	-	-	27.2	-	64.6	33.6	30% PG	0.0001	55	44	-	10.2	-	56.5	WATER	95	110	14.26	-	SCROLL	R-410A	-	-	68.0	100.0	460/360	NO	Yes	Yes	Yes	Yes	Yes	Yes	Yes	150	MULTISTACK	HSS520XNHCRAA	1,2,3,4,5,7
HRC-2	MECHANICAL C166	3.40	4.40	4.40	-	-	27.2	-	64.6	33.6	30% PG	0.0001	55	44	-	10.2	-	56.5	WATER	95	110	14.26	-	SCROLL	R-410A	-	-	68.0	100.0	460/360	NO	Yes	Yes	Yes	Yes	Yes	Yes	Yes	150	MULTISTACK	HSS520XNHCRAA	1,2,3,4,5,7

CONDENSING BOILER SCHEDULE

REMARKS

1. PROVIDE CONDENSATE TRAP AND CONDENSATION NEUTRALIZATION KIT. ROUTE 1" STAINLESS STEEL PIPE TO DRAIN. PROVIDE REMOVABLE UNION FITTINGS TO ALLOW FOR MAINTENANCE.
2. PROVIDE STAINLESS STEEL VENT AND COMBUSTION AIR KIT.
3. PROVIDE EMERGENCY STOP SWITCH FOR GAS AND ELECTRIC.

LOCATION			TYPE	GAS BURNER			WATERSIDE								THERMAL EFFICIENCY	DIMENSIONS (IN.)			Equipment Weight (LB)	FLA (A)	MCA (A)	MOCP (A)	VOLTS/PHASE/HERTZ	EMERGENCY POWER	DESIGN REFERENCE			
MARK	NAME	NO.		CAPACITY (MBH)	TYPE	FUEL		FLUID	FLOW (GPM)	EWT (°F)	LWT (°F)	TURNDOWN RATIO	MAX WPD (FT. OF HD.)	VOL (GAL)		HW CONNECTION	L	W							H	MANUFACTURER	MODEL NO.	REMARKS
						PRESS	AVAIL (PSI)																					
B-1	MECHANICAL	C166	CONDENSING	3000	NG	2.0	3500	WATER	220.0	110	140	10:1	0.5	198.0	5"	95.2%	68.4	48.4	91.2	5230	4.0	-	20.0	460/3/60	YES	CLEAVER BROOKS	CFC-E 3500	1.2
B-2	MECHANICAL	C166	CONDENSING	3000	NG	2.0	3500	WATER	220.0	110	140	10:1	0.5	198.0	5"	95.2%	68.4	48.4	91.2	5230	4.0	-	20.0	460/3/60	YES	CLEAVER BROOKS	CFC-E 3500	1.2

PUMP SCHEDULE

REMARKS:

1. PROVIDE WITH CLOSED IMPELLER, VORTEX SUPPRESSOR AND INLET STRAINER.
2. PROVIDE VFD BY PUMP TCC.
3. STARTER PROVIDED BY EC.

MARK	LOCATION		SYSTEM	FLOW (GPM)	MIN FLOW (GPM)	DESIGN HEAD (FT.HD.)	MAX SHUT OFF HEAD (FT.HD.)	EFFICIENCY (%)	PUMP TYPE	FLUID TYPE	TEMP (°F)	MOTOR					DESIGN REFERENCE				REMARKS
	NAME	NO.										HP	RPM	VOLT	PH	FLA	EMER. POWER	VSC	MANUFACTURER	MODEL NO.	
HWP-1	MECHANICAL	C166	HOT WATER	220	52	125	138	67.6	BASE MOUNTED END SUCTION	WATER	140	15	1800	460	3	-	YES	YES	B&G	E-1510	1,2
HWP-2	MECHANICAL	C166	HOT WATER	220	52	125	138	67.6	BASE MOUNTED END SUCTION	WATER	140	15	1800	460	3	-	YES	YES	B&G	E-1510	1,2
HCP-1	MECHANICAL	B107	DOAS-1	11	-	30	32	-	INLINE	WATER	140	1/6	3300	115	1	2.1	YES	NO	B&G	PL-36	3
HCP-2	MECHANICAL	C166	DOAS-2	11	-	30	32	-	INLINE	WATER	140	2/5	3250	115	1	4.7	YES	NO	B&G	PL-55	3
HCP-3	KITCHEN	B134	DOAS-3	11	-	30	32	-	INLINE	WATER	140	1/6	3300	115	1	2.1	YES	NO	B&G	PL-36	3
HCP-4	MECHANICAL	B107	AHU-4	11	-	30	32	-	INLINE	WATER	140	1/6	3300	115	1	2.1	YES	NO	B&G	PL-36	3
HRCP-1A	MECHANICAL	C166	HEAT RECOVERY CHILLER CHILLED WATER SIDE	65	-	10	20	-	INLINE	WATER 30%PG	55	2/5	3200	115	1	4.8	NO	NO	B&G	PL-130B/2"	3
HRCP-1B	MECHANICAL	C166	HEAT RECOVERY CHILLER HOT WATER SIDE	57	-	14	20	-	INLINE	WATER	140	2/5	3250	115	1	4.8	NO	NO	B&G	PL-100B	3
HRCP-2A	MECHANICAL	C166	HEAT RECOVERY CHILLER CHILLED WATER SIDE	65	-	10	20	-	INLINE	WATER 30%PG	55	2/5	3200	115	1	4.8	NO	NO	B&G	PL-130B/2"	3
HRCP-2B	MECHANICAL	C166	HEAT RECOVERY CHILLER HOT WATER SIDE	57	-	14	20	-	INLINE	WATER	140	2/5	3250	115	1	4.8	NO	NO	B&G	PL-100B	3

EXPANSION TANK SCHEDULE

REMARKS:

1. FILL PRESSURE SHALL BE FIELD ADJUSTABLE AND SET EQUAL TO DOMESTIC WATER MAKE UP PRESSURE REDUCING VALVE

[illegible]

AIR SEPARATOR SCHEDULE

REMARKS:

1. ASME CERTIFIED.

LOCATION			SYSTEM SERVED	CAPACITY (GPM)	CONNECTION SIZE	MAX W.P.D. (FT. HD.)	BUILT-IN STRAINER REQUIRED	DESIGN REFERENCE		REMARKS
MARK	NAME	NO.						MANUFACTURER	MODEL NO.	
AS-1	MECHANICAL	171	HEATING WATER					B&G		1
AS-2	MECHANICAL	171	CHILLED WATER					B&G		1

GLYCOL MAKE-UP UNIT SCHEDULE

MARK	LOCATION		PUMP			MOTOR			UNIT VOL	UNIT WEIGHT	FLA	MCA	MCCP	VOLT	PH	DESIGN REFERENCE		REMARKS
	NAME	NO.	FLOW	HEAD	DISCHARGE PRESS	QTY	POWER	RPM								MANUFACTURER	MODEL NO.	
GMU-1	MECHANICAL	171								13.8	17.3	25	120	1		BELL & GOSSETT		



100% CD Bid Set

VERMILLION COUNTY COMMISSIONERS

VERMILLION COUNTY JAIL

ADDITION & RENOVATION

100% CD Bid Set

1888 IN-63
HILL DALE IN 47854

#	Revision	Date
---	----------	------

Project #: 17-700-078-2

Designed By: A.M.M

Drawn By: A.M.M

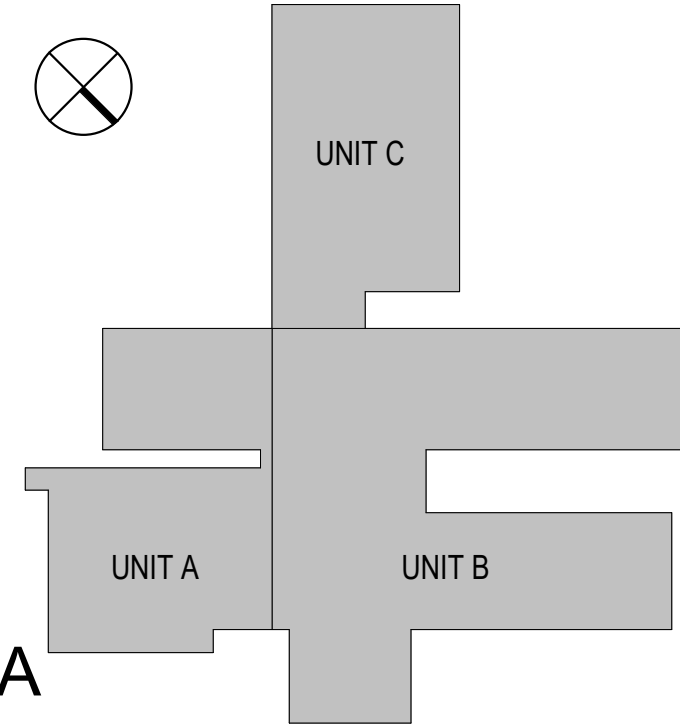
Checked By: D.E.B

Date: 04.12.2021



Erin Wefler

These Drawings and Specifications, and all copies thereof are and shall remain the property and copyright of the Architect. They shall be used only with respect to this Project and are not to be used on any other Project or Work without prior written permission from the Architect.



SCHEDULES - MECHANICAL

M601

E

D

C

B

A

E

D

C

B

A

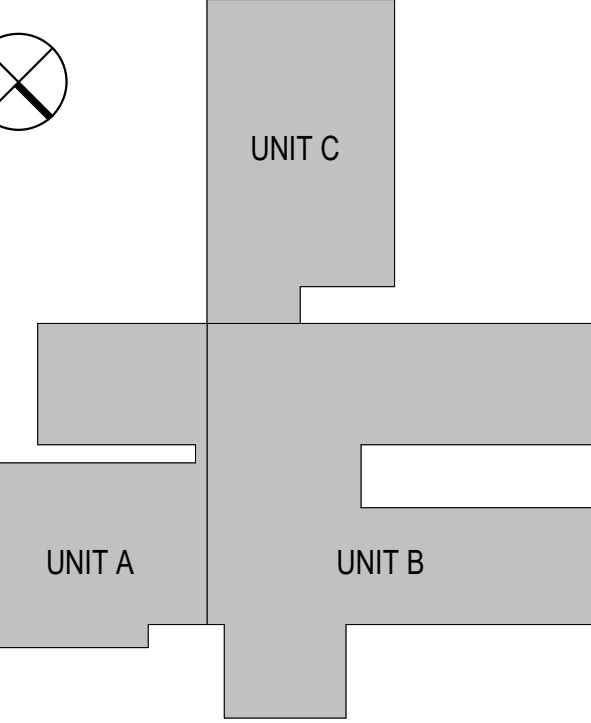
#	Revision	Date
---	----------	------

Project #: 17-700-078-2
Designed By: A.M.M
Drawn By: A.M.M
Checked By: D.E.B
Date: 04.12.2021



Erik Weisenborn

These Drawings and Specifications, and all copies thereof are and shall remain the property and copyright of the Architect. They shall be used only with respect to this Project and are not to be used on any other Project or Work without prior written permission from the Architect



UNIT HEATER SCHEDULE

REMARKS:
1. PROVIDE INTEGRAL THERMOSTAT.
2. PROVIDE KEY LOCKS FOR ACCESS DOORS AND FRONT PANEL.
3. PROVIDE HIGH CAPACITY 2-ROW COIL WITH MAX WATER PRESS DROP OF 5 FT HD.

		LOCATION			ENTERING AIR TEMPERATURE (°F)	LEAVING AIR TEMPERATURE (°F)	AIRFLOW (CFM)	CIRCULATING FLUID					ELECTRICAL DATA					MANUFACTURER	MODEL NO.	REMARKS
MARK	DESCRIPTION	NAME	NO.	ARRANGEMENT				PIPE RUNOUT SIZE	FLOW	EWT (°F)	LWT (°F)	MAX W.P.D (FT HD)	HP	VOLT	PH	FLA	FACTORY DISCONNECT	FUSED/ NON-FUSE D	EMER. POWER	
CUH-1	CABINET UNIT HEATER	SALLYPORT	C170	VERTICAL SLOPE TOP	62	107	560	3/4	1.9	140	110	2.05	1/20	115	1	0.75	YES	FUSED	NO	KRUEGER KVF506 1 THRU 3
CUH-2	CABINET UNIT HEATER	SALLYPORT	C100	VERTICAL SLOPE TOP	62	107	560	3/4	1.9	140	110	2.05	1/20	115	1	0.75	YES	FUSED	NO	KRUEGER KVF506 1 THRU 3
CUH-3	CABINET UNIT HEATER	SALLYPORT	C164	VERTICAL SLOPE TOP	62	107	560	3/4	1.9	140	110	2.05	1/20	115	1	0.75	YES	FUSED	NO	KRUEGER KVF506 1 THRU 3
CUH-4	CABINET UNIT HEATER	VESTIBULE	A100	VERTICAL RECESSED	62	107	560	3/4	1.9	140	110	2.05	1/20	115	1	0.75	YES	FUSED	NO	KRUEGER KVFH06 1 THRU 3
CUH-5	CABINET UNIT HEATER	SALLYPORT	B129	VERTICAL SLOPE TOP	62	107	560	3/4	1.9	140	110	2.05	1/20	115	1	0.75	YES	FUSED	NO	KRUEGER KVF506 1 THRU 3

DX SPLIT SCHEDULE

MARK	AREA SERVED		MANUFACTURER	INDOOR UNIT				OUTDOOR UNIT							REMARKS
	ROOM NAME	ROOM NUMBER		MARK	MODEL	MAX CFM	COOLING MBH	MARK	MODEL	SEER/EER	HSPF	MCA	MOCP	VOLTS/ PHASE	
AC-1	CONTROL ROOM	D200	LG	AC-1	LCN128HV4	875	12.0	ACCU-1	LUU1278HV	20.0 / 11.0	11.5	9	15	208 / 1	1 THRU 7
AC-2	SECURITY	D108	LG	AC-2	LSN240HEV2	875	24.0	ACCU-2	LSU240HEV2	20.0 / 11.0	11.5	15	20	208 / 1	1 THRU 7
AC-3	IT	A114	LG	AC-3	LSN240HEV2	875	24.0	ACCU-3	LSU240HEV2	20.0 / 11.0	11.5	15	20	208 / 1	1 THRU 7
AC-4	CONTROL ROOM	C126	LG	AC-4	ARNU543NKA4	1200	58.0	ACCU-4	ARUN060GSS4	20.0 / 11.0	11.5	25.4	40	208 / 1	1 THRU 7
AC-5	ELECTRICAL	C167	LG	AC-4	ARNU543NKA4	1200	58.0	ACCU-5	ARUN060GSS4	20.0 / 11.0	11.5	25.4	40	208 / 1	1 THRU 7
AC-6	ELECTRICAL	C167	LG	AC-4	ARNU543NKA4	1200	58.0	ACCU-6	ARUN060GSS4	20.0 / 11.0	11.5	25.4	40	208 / 1	1 THRU 7

- NOTES:
- PROVIDE WIRELESS REMOTE CONTROLLER KIT FOR EACH INDOOR UNIT.
 - OUTDOOR UNIT SHALL BE RATED FOR LOW AMBIENT CONTROL DOWN TO -5 F. PROVIDE RATED CAPACITIES AS LISTED FOR COOLING AT 47F AND HEATING AT 17F.
 - PROVIDE INDOOR UNITS WITH INTEGRAL CONDENSATE PUMP AND ALARM. FOR HIGH WALL STYLE UNITS PROVIDE CONDENSATE PUMP SIMILAR TO REFCO G0B1 II.
 - CONTRACTOR SHALL SIZE ALL REFRIGERANT PIPING SO AS NOT TO REDUCE RATED CAPACITY. SUBMIT REFRIGERANT PIPING DETAIL WITH SUBMITTAL.
 - INDOOR UNITS SHALL BE POWERED THROUGH OUTDOOR UNIT.
 - MOUNT OUTDOOR UNITS ON 12" HIGH ROOF RAILS OR STAND FOR GRADE MOUNTED UNITS.
 - ECC TO PROVIDE TEMPERATURE SENSOR IN ELECTRICAL / IT ROOM FOR MONITORING ONLY.

LOUVER SCHEDULE

MARK	MANUFACTURER	MODEL	SERVICE	LOCATION	AIRFLOW CFM	WIDTH IN.	HEIGHT IN.	MINIMUM FREE AREA SF	MAXIMUM VELOCITY FPM	MAX APD IN.WG.	FRAME DEPTH	REMARKS
LV-1	RUSKIN	ELF6375DXD	AHU-2 OA	MECHANICAL - 171							REFER TO ARCH PLANS	1,2,3
LV-2	RUSKIN	ELF6375DXD	AHU-2 EXHAUST	MECHANICAL - 171							REFER TO ARCH PLANS	1,2,3
LV-3	RUSKIN	ELF6375DXD	AHU-3 OA	MECHANICAL - 71							REFER TO ARCH PLANS	1,2,3
LV-4	RUSKIN	ELF6375DXD	AHU-3 EXHAUST	MECHANICAL - 71							REFER TO ARCH PLANS	1,2,3
LV-5	RUSKIN	ELF6375DXD	AHU-4 OA	KITCHEN ATTIC							REFER TO ARCH PLANS	1,2,3
LV-6	RUSKIN	ELF6375DXD	AHU-4 EXHAUST	KITCHEN ATTIC							REFER TO ARCH PLANS	1,2,3
LV-7	RUSKIN	ELF6375DXD	SALLYPORT EXHAUST	SALLYPORT - 12							REFER TO ARCH PLANS	1,2,4

- REMARKS:
- FURNISH WITH ALUMINUM BIRD SCREEN.
 - FURNISH WITH KYNAR PAINT FINISH OF STANDARD COLOR SELECTION BY ARCHITECT.
 - MC SHALL FURNISH WITH GRAVITY BACKDRAFT DAMPER.
 - ECC SHALL FURNISH WITH MOTORIZED DAMPER INTERLOCKED WITH GAS MONITORING SYSTEM.

GRAVITY HOOD SCHEDULE

LOCATION			SYSTEM NAME	QUANTITY	CFM	MAX APD (IN-WG)	THROAT VELOCITY (FT/MIN)	NO. OF LOUVERS	DIMENSIONS				GRAVITY BACKDRAFT DAMPER	MIN. CURB HEIGHT (IN.)	DESIGN REFERENCE		REMARKS
MARK	NAME	NO.							HOOD		THROAT				MANUFACTURER	MODEL NO.	
									WIDTH	LENGTH	HEIGHT	WIDTH					
OALP-1	POD ROOF	-	OUTSIDE AIR POD AHU'S									YES	18	GREENHECK	WIH		
EALP-1	POD ROOF	-	EXHAUST AIR POD AHU'S									YES	18	GREENHECK	WRH		

FAN COIL UNIT SCHEDULE

1. PROVIDE WITH FACTORY MOUNTED AND WIRED NON-FUSED DISCONNECT SWITCH. CONTROLS TO BE FIELD INSTALLED.
2. PROVIDE WITH 2" MERV 8 FILTER.
3. PROVIDE 3-SPEED EC MOTOR W/ POTENTIOMETER.
4. FCU HEATING COILS SHALL BE SUBMITTED FOR REVIEW TO ENGINEER WITH PERFORMANCE AT BOTH 140 EAT AND 110 EAT TO EVALUATE OPERATION DURING HHW TEMPERATURE RESET.
5. PROVIDE STAINLESS STEEL DRAIN PAN, AND CONDENSATE OVERFLOW SWITCH.

LOCATION			SUPPLY FAN DATA					COOLING COIL										HEATING COIL										UNIT ELECTRICAL DATA					DESIGN REFERENCE		
MARK	NAME	NO.	SUPPLY CFM		OA CFM	ESP	TOTAL MBH	SENSIBLE MBH	GPM	EWT	LWT	ENTERING AIR		LEAVING AIR		ROWS	WPD FT HD	SENSIBLE MBH	GPM	EWT	LWT	EAT	LAT	ROWS	WPD FT HD	HP	MCA	VOLTS/ PHASE	EMER. POWER	DISCONNECT	MANUFACTURER	MODEL	REMARKS		
			MAX	MIN								DB (°F)	WB (°F)	DB (°F)	WB (°F)																				
FCU-1	MECHANICAL	D109	1500	400	-	0.20	29.9	27.4	4.3	40.0	55.0	75.0	63.0	58.3	56.1	4	1.7	77.9	5.2	140.0	110.0	50.0	98.1	2	2.8	(2) 1/4	12.00	120/160	YES	FACTORY FUSED	KRUEGER	KHGP-16	1,2,3,4,5		
FCU-2	MECHANICAL	C166	1500	400	-	0.20	29.9	27.4	4.3	40.0	55.0	75.0	63.0	58.3	56.1	4	1.7	77.9	5.2	140.0	110.0	50.0	98.1	2	2.8	(2) 1/4	12.00	120/160	YES	FACTORY FUSED	KRUEGER	KHGP-16	1,2,3,4,5		
FCU-3	MECHANICAL	B107	1500	400	-	0.20	29.9	27.4	4.3	40.0	55.0	75.0	63.0	58.3	56.1	4	1.7	77.9	5.2	140.0	110.0	50.0	98.1	2	2.8	(2) 1/4	12.00	120/160	YES	FACTORY FUSED	KRUEGER	KHGP-16	1,2,3,4,5		

EXHAUST FAN SCHEDULE

- REMARKS:
1. 24V MOTORIZED DAMPER PROVIDED BY ECC. REFER TO DAMPER SCHEDULE.
2. PROVIDE WITH 16" HIGH INSULATED ROOF CURB.
3. PROVIDE WITH VARIABLE SPEED CONTROLLER AND SAFETY CABLE.
4. EXHAUST FAN SHALL BE UL LISTED FOR KITCHEN EXHAUST
5. EXHAUST FAN SHALL BE EXTERIOR WALL MOUNTED
6. EXHAUST FAN SHALL HAVE INTEGRAL GREASTRAP.

MARK	LOCATION		FAN SERVICE	Fan Type	CFM	ESP (IN-WG)	WHEEL		MOTOR				ACCESS DOOR	BELT DRIVE	BIRD SCREEN	GRAVITY BDD	MOTORIZED DAMPER	UNIT MTD. DISC....	DESIGN REFERENCE		REMARKS
	NAME	AREA					TYPE	FAN RPM	HP	BHP	VOLTS/PHASE/HERTZ	FLA							MANUFACTURER	MODEL NO.	
KEF-1	WALL	KITCHEN	KITCHEN EXHAUST	CENTRIFUGAL DOWNBLAST	3000	0.50	BI	1055	1.00	0.77	208/3/60	4.6	YES	NO	YES	YES	NO	YES	GREENHECK	CUBE-180-10	1,3,4,5,6
GEF-1	ROOF	ADMIN	GENERAL EXHAUST	CENTRIFUGAL DOWNBLAST	1000	0.75		1630	0.50	0.24	115/1/60	6.4	YES	NO	YES	YES	NO	YES	GREENHECK	G-103-VG	2
GEF-2	ROOF	SALLYPORT	GENERAL EXHAUST	CENTRIFUGAL DOWNBLAST	3350	0.50		1323	2.00	0.87	208/1/60	12.0	YES	NO	YES	YES	NO	YES	GREENHECK	CUE-161-VG	2
PEF-1	ROOF	POD	POD EXHAUST	CENTRIFUGAL UPBLAST	1400	0.56		1417	0.50	0.25	115/1/60	6.4	YES	YES	YES	NO	YES	YES	GREENHECK	CUE-121-VG	1,2
PEF-2	ROOF	POD	POD EXHAUST	CENTRIFUGAL UPBLAST	1300	0.50		1228	0.50	0.21	115/1/60	6.4	YES	YES	YES	NO	YES	YES	GREENHECK	CUE-131-VG	1,2
PEF-3	ROOF	POD	POD EXHAUST	CENTRIFUGAL UPBLAST	1500	0.50		1491	0.50	0.29	115/1/60	6.4	YES	YES	YES	NO	YES	YES	GREENHECK	CUE-121-VG	1,2
PEF-4	ROOF	POD	POD EXHAUST	CENTRIFUGAL UPBLAST	1100	0.50		1221	0.50	0.16	115/1/60	6.4	YES	YES	YES	NO	YES	YES	GREENHECK	CUE-121-VG	1,2
PEF-5	ROOF	POD	POD EXHAUST	CENTRIFUGAL UPBLAST	1350	0.50		1255	0.50	0.22	115/1/60	6.4	YES	YES	YES	NO	YES	YES	GREENHECK	CUE-131-VG	1,2
PEF-6	ROOF	POD	POD EXHAUST	CENTRIFUGAL UPBLAST	2050	0.50		956	0.50	0.33	115/1/60	6.4	YES	YES	YES	NO	YES	YES	GREENHECK	CUE-161-VG	1,2
JEF-1	ATTIC	JAIL	POD EXHAUST	CENTRIFUGAL UPBLAST	700	0.50		1583	0.50	0.21	115/1/60	6.4	YES	YES	YES	NO	YES	YES	GREENHECK	SQ-99-VG	1,2
JEF-2	ATTIC	JAIL	POD EXHAUST	CENTRIFUGAL UPBLAST	705	0.50		1589	0.50	0.25	115/1/60	6.4	YES	YES	YES	NO	YES	YES	GREENHECK	SQ-99-VG	1,2
JEF-3	ATTIC	JAIL	POD EXHAUST	CENTRIFUGAL UPBLAST	680	0.50		1559	0.50	0.20	115/1/60	6.4	YES	YES	YES	NO	YES	YES	GREENHECK	SQ-99-VG	1,2
JEF-4	ATTIC	JAIL	POD EXHAUST	CENTRIFUGAL UPBLAST	570	0.50		1431	0.50	0.15	115/1/60	6.4	YES	YES	YES	NO	YES	YES	GREENHECK	SQ-99-VG	1,2
JEF-5	ATTIC	JAIL	POD EXHAUST	CENTRIFUGAL UPBLAST	1355	0.50		1474	0.50	0.28	115/1/60	6.4	YES	YES	YES	NO	YES	YES	GREENHECK	SQ-120-VG	1,2
JEF-6	ATTIC	JAIL	POD EXHAUST	CENTRIFUGAL UPBLAST	1070	0.50		1584	0.50	0.19	115/1/60	6.4	YES	YES	YES	NO	YES	YES	GREENHECK	SQ-100-VG	1,2
BF-1	SALLYPORT	SALLYPORT	DESTRATIFICATION	BUCKET FAN	1055	-	-	1675	94 WATTS	-	115/1/60	-	NO	NO	NO	NO	NO	YES	VENTS-US	1055	3

E

D

C

B

A

E

D

C

B

A

DIFFUSER / GRILLE SCHEDULE													
REMARKS:													
1. BRANCH DUCTWORK TO THE DIFFUSER SHALL BE THE SAME SIZE AS THE NECK UNLESS OTHERWISE NOTED.													
2. PROVIDE FRAME STYLE APPROPRIATE FOR CEILING TYPE (I.E. LAY IN, SURFACE MOUNT).													
3. PROVIDE FACE OPERABLE DAMPLER IN NECK.													
4. PROVIDE WITH 13/16" SQUARE PERFORATIONS 3/16" SPACING.													
5. PROVIDE SQUARE TO ROUND TRANSITION FOR 8" ROUND RUN OUTS.													
6. PROVIDE SQUARE TO ROUND TRANSITION FOR 10" ROUND RUN OUTS.													
7. PROVIDE SQUARE TO ROUND TRANSITION FOR 12" ROUND RUN OUTS.													
8. PROVIDE CLIPS TO SECURE DIFFUSERGRILLE TO CEILING GRID.													
9. PROVIDE SLEEVE AND SECURITY BARS. SLEEVE LENGTH AS REQUIRED FOR INSTALLATION.													
10. PROVIDE CEILING RADIATION DAMPER TO MEET RATING OF CEILING CONSTRUCTION.													
11. TERMINAL BOX DOWNSTREAM DUCT RUNOUT SHALL BE SIZE OF TERMINAL BOX DISCHARGE. COORDINATE WITH FINAL TERMINAL BOX SUBMITTAL.													
TAG	NECK SIZE	FACE LENGTH	FACE WIDTH	MATERIAL	FINISH	Min CFM	Max CFM	MAX NC	MAX THROW (FT)	MAX TOTAL APD (IN WG)	MANUFACTURER	MODEL	NOTES
E1	8"Ø	24"	24"	STEEL	WHITE	0	225	15	6	0.05	PRICE	SPD	
E6	8"Ø	24"	24"	ALUMINUM	WHITE	0	70	15	6	0.05	PRICE	ASPD	
E7	8"Ø	24"	24"	ALUMINUM	WHITE	71	225	15	6	0.05	PRICE	ASPD	
E8	10"Ø	24"	24"	ALUMINUM	WHITE	226	380	20	10	0.09	PRICE	ASPD	
E12	6"Ø	12"	12"	ALUMINUM	WHITE	0	70	15	6	0.05	PRICE	ASPD	
E13	8"Ø	12"	12"	ALUMINUM	WHITE	71	225	15	6	0.05	PRICE	ASPD	
EG2	18"X18"	18"	18"	STEEL	WHITE	800	1600	15	6	0.05	PRICE	PDR	
EG3	10"X6"	10"	6"	STEEL	WHITE	0	200	15	6	0.09	PRICE	PDR	
R1	8"Ø	24"	24"	STEEL	WHITE	71	225	15	6	0.05	PRICE	SPD	
R2	10"Ø	24"	24"	STEEL	WHITE	226	380	20	10	0.09	PRICE	SPD	
R3	12"Ø	24"	24"	STEEL	WHITE	381	550	22	12	0.13	PRICE	SPD	
R4	14"Ø	24"	24"	STEEL	WHITE	551	750	24	14	0.18	PRICE	SPD	
R7	12"Ø	24"	24"	STEEL	WHITE	381	550	22	12	0.13	PRICE	PDR	
R8	14"Ø	24"	24"	STEEL	WHITE	551	750	24	14	0.18	PRICE	PDR	
R9	16"Ø	24"	24"	STEEL	WHITE	751	900	25	15	0.20	PRICE	PDR	
S0	8"Ø	24"	24"	STEEL	WHITE	0	70	15	6	0.05	PRICE	SPD	
S1	8"Ø	24"	24"	STEEL	WHITE	71	225	15	6	0.05	PRICE	SPD	
S2	10"Ø	24"	24"	STEEL	WHITE	226	380	20	10	0.09	PRICE	SPD	
S3	12"Ø	24"	24"	STEEL	WHITE	381	550	22	12	0.13	PRICE	SPD	
S4	14"Ø	24"	24"	STEEL	WHITE	551	750	24	14	0.18	PRICE	SPD	
S6	6"Ø	24"	24"	ALUMINUM	WHITE	0	70	15	6	0.05	PRICE	ASPD	
S12	6"Ø	12"	12"	ALUMINUM	WHITE	0	70	15	6	0.05	PRICE	ASPD	
S13	8"Ø	12"	12"	ALUMINUM	WHITE	71	225	15	6	0.05	PRICE	ASPD	
S14	6"Ø	24"	24"	STEEL	WHITE	0	85	15	6	0.05	PRICE	PDS	
S15	8"Ø	24"	24"	STEEL	WHITE	86	275	15	6	0.05	PRICE	PDS	
SE0	8"Ø	24"	24"	STEEL	WHITE	0	70	15	6	0.05	AJ MFR	SDR	
SE2	8"Ø	24"	24"	STEEL	WHITE	71	225	15	6	0.05	AJ MFR	SDR	
SE3	10"Ø	24"	24"	STEEL	WHITE	226	380	20	10	0.09	AJ MFR	SDR	
SE4	12"Ø	24"	24"	STEEL	WHITE	381	550	22	12	0.13	AJ MFR	SDR	
SE7	8"Ø	24"	24"	ALUMINUM	WHITE	71	225	15	6	0.05	AJ MFR	SDRA	
SE8	10"Ø	24"	24"	ALUMINUM	WHITE	226	380	20	10	0.09	AJ MFR	SDRA	
SE9	12"Ø	24"	24"	ALUMINUM	WHITE	381	550	22	12	0.13	AJ MFR	SDRA	
SE10	14"Ø	24"	24"	ALUMINUM	WHITE	551	750	24	14	0.18	AJ MFR	SDRA	
SE12	6"Ø	12"	12"	ALUMINUM	WHITE	0	70	15	6	0.05	PRICE	MSRRP	
SE13	8"Ø	12"	12"	ALUMINUM	WHITE	71	225	15	6	0.05	PRICE	MSRRP	
SE15	6"Ø	12"	12"	ALUMINUM	WHITE	0	70	15	6	0.05	PRICE	MSRRP	
SE16	8"Ø	12"	12"	ALUMINUM	WHITE	71	225	15	6	0.05	PRICE	MSRRP	
SG1	10"X6"	10"	6"	ALUMINUM	WHITE	0	200	15	6	0.05	PRICE	600	
SS0	6"Ø	24"	24"	ALUMINUM	WHITE	0	70	15	6	0.05	AJ MFR	SDL	
SS1	8"Ø	24"	24"	ALUMINUM	WHITE	71	225	15	6	0.05	AJ MFR	SDL	
SS2	10"Ø	24"	24"	ALUMINUM	WHITE	226	380	20	10	0.09	AJ MFR	SDL	
SS3	12"Ø	24"	24"	ALUMINUM	WHITE	381	550	22	12	0.13	AJ MFR	SDL	
SS6	6"Ø	24"	24"	ALUMINUM	WHITE	0	70	15	6	0.05	AJ MFR	ASPD	
SS7	8"Ø	24"	24"	ALUMINUM	WHITE	71	225	15	6	0.05	AJ MFR	ASPD	
SS8	10"Ø	24"	24"	ALUMINUM	WHITE	226	380	20	10	0.09	AJ MFR	ASPD	
SS9	12"Ø	24"	24"	ALUMINUM	WHITE	381	550	22	12	0.13	AJ MFR	ASPD	
SS12	6"Ø	12"	12"	STEEL	WHITE	0	70	15	6	0.05	PRICE	PDS	
SS13	8"Ø	12"	12"	STEEL	WHITE	71	225	15	6	0.05	PRICE	PDS	
SS14	10"Ø	12"	12"	STEEL	WHITE	226	380	20	10	0.09	PRICE	PDS	
SS15	6"Ø	12"	12"	ALUMINUM	WHITE	0	70	15	6	0.05	PRICE	MSRRP	
SS16	8"Ø	12"	12"	ALUMINUM	WHITE	71	225	15	6	0.05	PRICE	MSRRP	

SLOT DIFFUSER SCHEDULE														
TAG	DUCT CONNECTION SIZE	SLOT LENGTH	SLOT WIDTH	# OF SLOTS	CFM/SF	CFM RANGE	MAX. NC	MAX THROW (FT)	MATERIAL	FINISH	MAX. TOTAL PD (IN. WG.)	MANUFACTURER	MODEL	REMARKS
SL-1	8	48"	1/2"	2	40	0-160	22	16	STEEL	WHITE	0.075	PRICE	SDA50	1,2
SL-2	10	48"	1/2"	4	60	160-240	18.0	18.0	STEEL	WHITE	0.043	PRICE	SDA50	1,2
REMARKS:														
1. BRANCH DUCTWORK TO THE DIFFUSER SHALL BE THE SAME SIZE AS THE NECK UNLESS OTHERWISE NOTED.														
2. PROVIDE FRAME STYLE APPROPRIATE FOR CEILING TYPE (I.E. LAY IN, SURFACE MOUNT)														

SIDEWALL GRILLE SCHEDULE										
TAG	NECK SIZE	FACE SIZE	MATERIAL	FINISH	CFM RANGE	MAX. NC	MAX. TOTAL PD (IN. WG.)	MANUFACTURER	MODEL	REMARKS
SRG-1	10"X10"	12"X12"	STEEL	WHITE	0 - 150	20	0.01	PRICE	MSRRP	1, 2, 3, 4
SWG-1	16"X8"	18"X10"	STAINLESS STEEL	WHITE	0 - 400	20	0.10	PRICE	MSRRP	3
SWG-2	24"X8"	18"X10"	STAINLESS STEEL	WHITE	400 - 750	20	0.10	PRICE	MSRRP	3
SWG-3	24"X16"	26"X18"	STEEL	WHITE	0 - 2000	45	0.30	TITUS	3300RL	3
SWG-4	30"X16"	32"X18"	STEEL	WHITE	0 - 2000	45	0.30	TITUS	3300RL	3
SWG-5	12"X6"	14"X8"	STEEL	WHITE	0-350	18	0.02	PRICE	SDG	
SWG-6	14"X10"	16"X12"	STEEL	WHITE	0-750	20	0.02	PRICE	SDG	
REMARKS:										
1. BRANCH DUCTWORK TO THE DIFFUSER SHALL BE THE SAME SIZE AS THE NECK UNLESS OTHERWISE NOTED.										
2. PROVIDE FRAME STYLE APPROPRIATE FOR CEILING TYPE (I.E. LAY IN, SURFACE MOUNT)										
3. PROVIDE SLEEVE AND SECURITY BARS. SLEEVE LENGTH AS REQUIRED FOR INSTALLATION.										
4. PROVIDE CEILING RADIATION DAMPER TO MEET RATING OF CEILING CONSTRUCTION.										

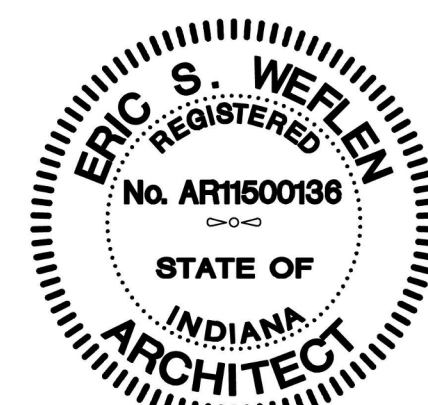
MOTORIZED DAMPER SCHEDULE							
TAG	DIMENSIONS	VOLTAGE	NORMALLY OPEN OR CLOSED	QUANTITY	DESIGNER REFERENCE		REMARKS
JED-1-1	18"X14"	24V	NO	1	POTTORFF	SD-151	1,3,4
JED-2-1	18"X14"	24V	NO	1	POTTORFF	SD-151	1,3,4
JED-3-1	18"X14"	24V	NO	1	POTTORFF	SD-151	1,3,4
JED-4-1	18"X14"	24V	NO	1	POTTORFF	SD-151	1,3,4
JED-5-1	18"X14"	24V	NO	1	POTTORFF	SD-151	1,3,4
JED-6-1	18"X14"	24V	NO	1	POTTORFF	SD-151	1,3,4
JED-1-2	18"X14"	24V	NC	1	POTTORFF	SD-151	1,2,4
JED-2-2	18"X14"	24V	NC	1	POTTORFF	SD-151	1,2,4
JED-3-2	18"X14"	24V	NC	1	POTTORFF	SD-151	1,2,4
JED-4-2	18"X14"	24V	NC	1	POTTORFF	SD-151	1,2,4
JED-5-2	18"X14"	24V	NC	1	POTTORFF	SD-151	1,2,4
JED-6-2	18"X14"	24V	NC	1	POTTORFF	SD-151	1,2,4
PED-1-1	20"X16"	24V	NO	1	POTTORFF	SD-151	1,3,4
PED-2-1	20"X16"	24V	NO	1	POTTORFF	SD-151	1,3,4
PED-3-1	20"X16"	24V	NO	1	POTTORFF	SD-151	1,3,4
PED-4-1	20"X16"	24V	NO	1	POTTORFF	SD-151	1,3,4
PED-5-1	20"X16"	24V	NO	1	POTTORFF	SD-151	1,3,4
PED-6-1	20"X16"	24V	NO	1	POTTORFF	SD-151	1,3,4
PED-1-2	18"X18"	24V	NC	1	POTTORFF	SD-151	1,2,4
PED-2-2	18"X18"	24V	NC	1	POTTORFF	SD-151	1,2,4
PED-3-2	18"X18"	24V	NC	1	POTTORFF	SD-151	1,2,4
PED-4-2	18"X18"	24V	NC	1	POTTORFF	SD-151	1,2,4
PED-5-2	18"X18"	24V	NC	1	POTTORFF	SD-151	1,2,4
PED-6-2	18"X18"	24V	NC	1	POTTORFF	SD-151	1,2,4
REMARKS:							
1. VERIFY LOCATION AND QUANTITY WITH TAG LOCATIONS ON DUCTWORK PLANS. VERIFY SIZE WITH MECHANICAL CONTRACTOR.							
2. MOTORIZED DAMPER FOR POD/JAIL EXHAUST FANS.							
3. OTORIZED DAMPER FOR POD/JAIL ZONE RETURN MAINS.							
4. PROVIDED BY TCC.							

VERMILLION COUNTY COMMISSIONERS
VERMILLION COUNTY JAIL
ADDITION & RENOVATION
1888 IN-53
HILLSDALE IN 47854

100% CD Bid Set

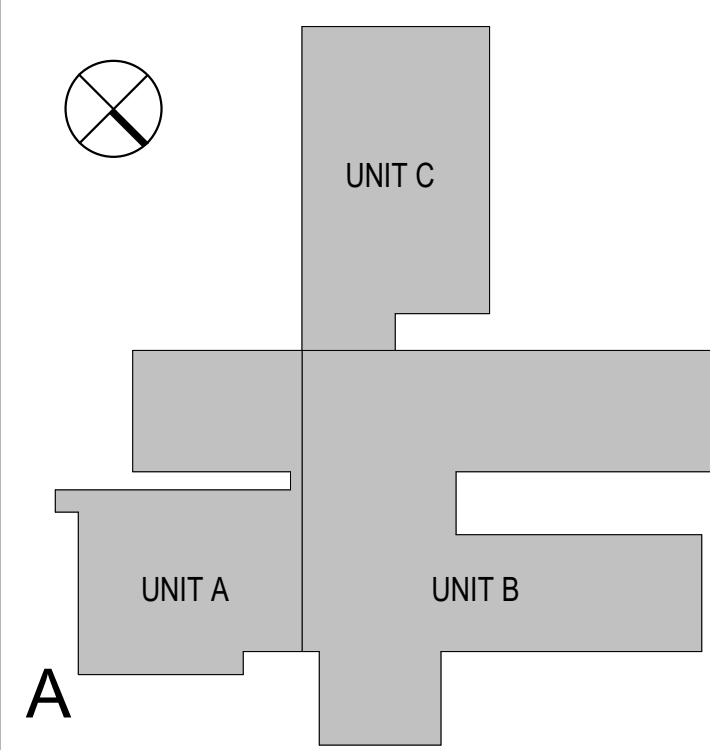
#	Revision	Date
---	----------	------

Project #: 17-700-078-2
Designed By: A.M.M
Drawn By: A.M.M
Checked By: D.E.B
Date: 04.12.2021



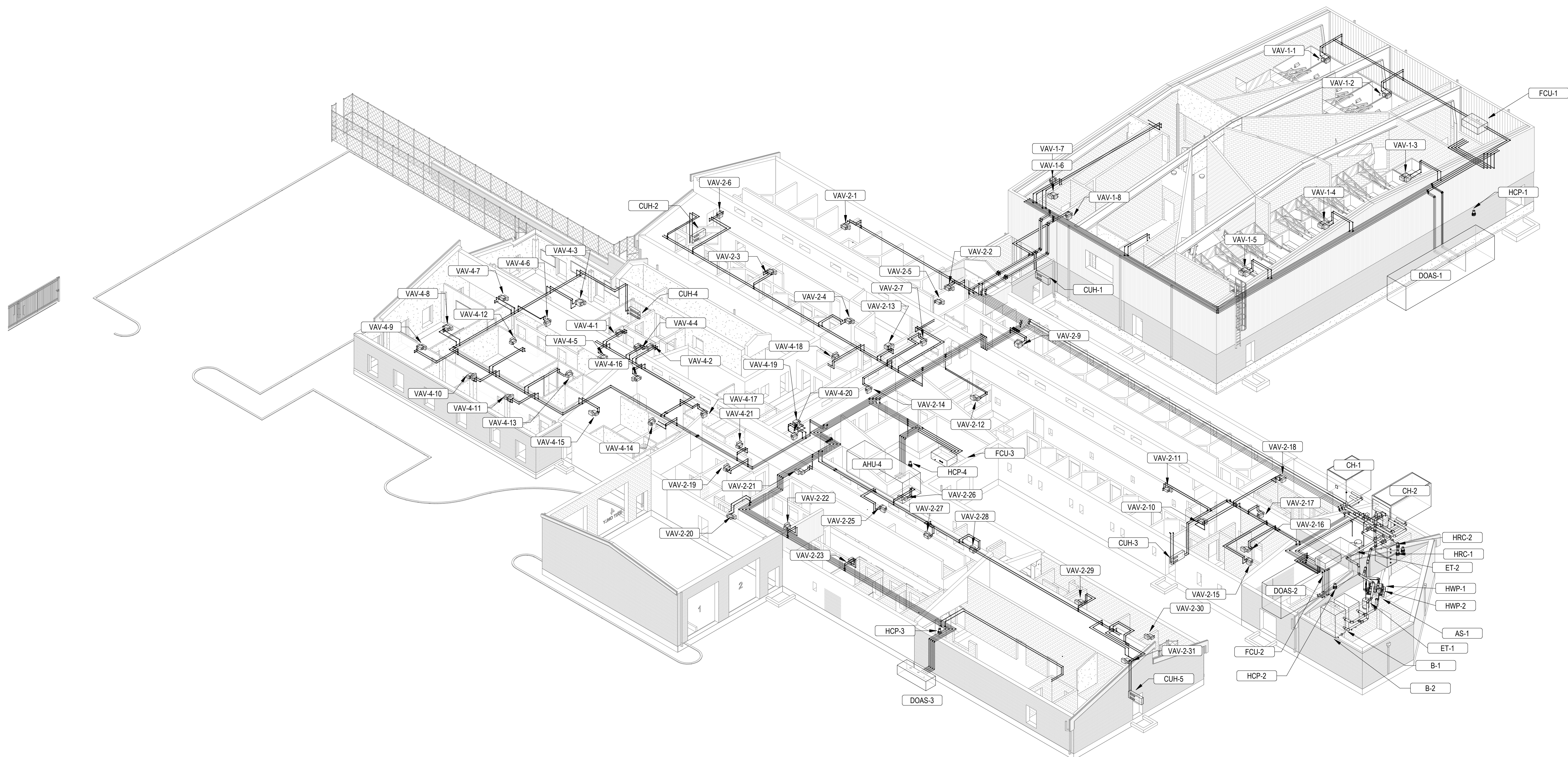
Eric Weylan

These Drawings and Specifications, and all copies thereof are and shall remain the property and copyright of the Architect. They shall be used only with respect to this Project and are not to be used on any other Project or Work without prior written permission from the Architect



HYDRONIC RISER
DIAGRAMS

M701



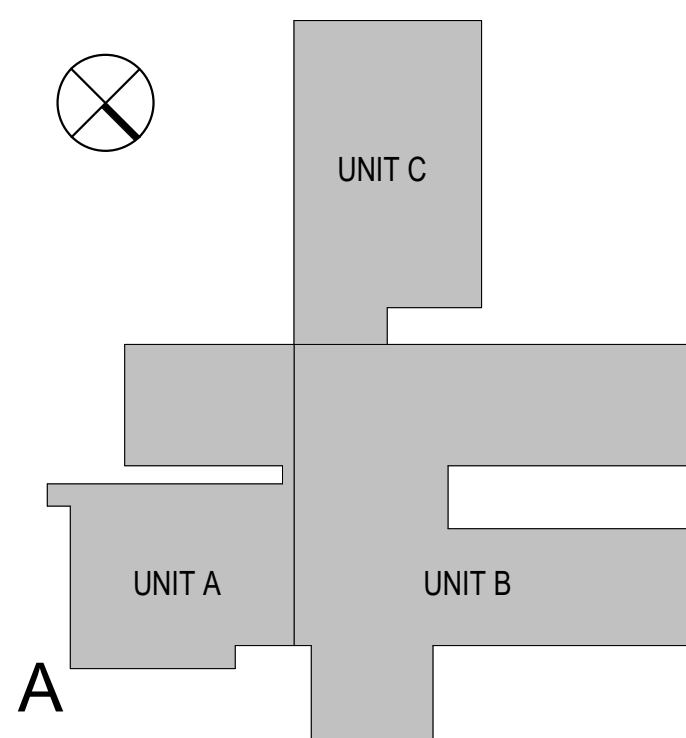
1 CHILLED AND HOT WATER ISOMETRIC

#	Revision	Date
---	----------	------

Project #: 17-700-078-2
Designed By: A.M.M
Drawn By: A.M.M
Checked By: D.E.B
Date: 04.12.2021

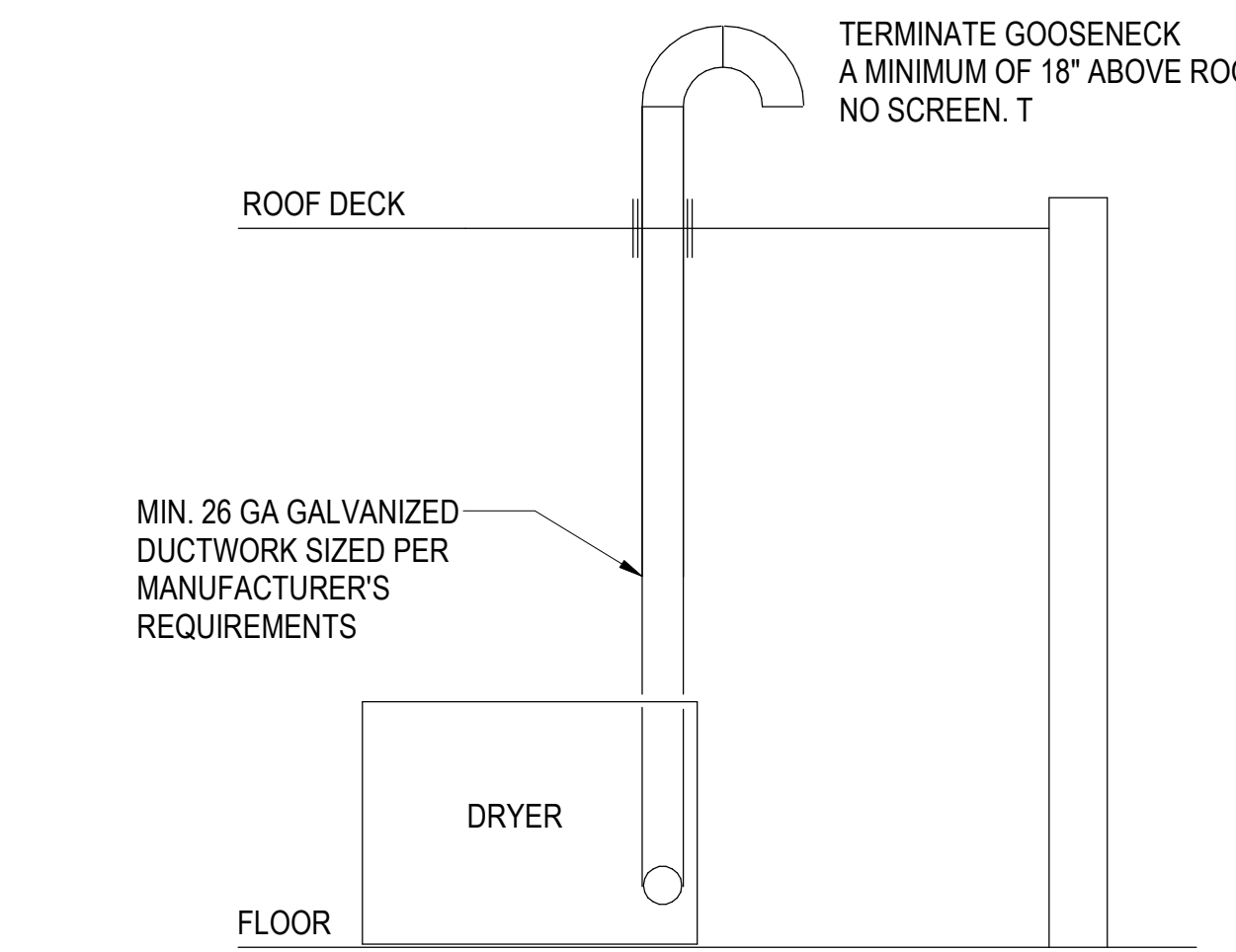


These Drawings and Specifications, and all copies thereof are and shall remain the property and copyright of the Architect. They shall be used only with respect to this Project and are not to be used on any other Project or Work without prior written permission from the Architect

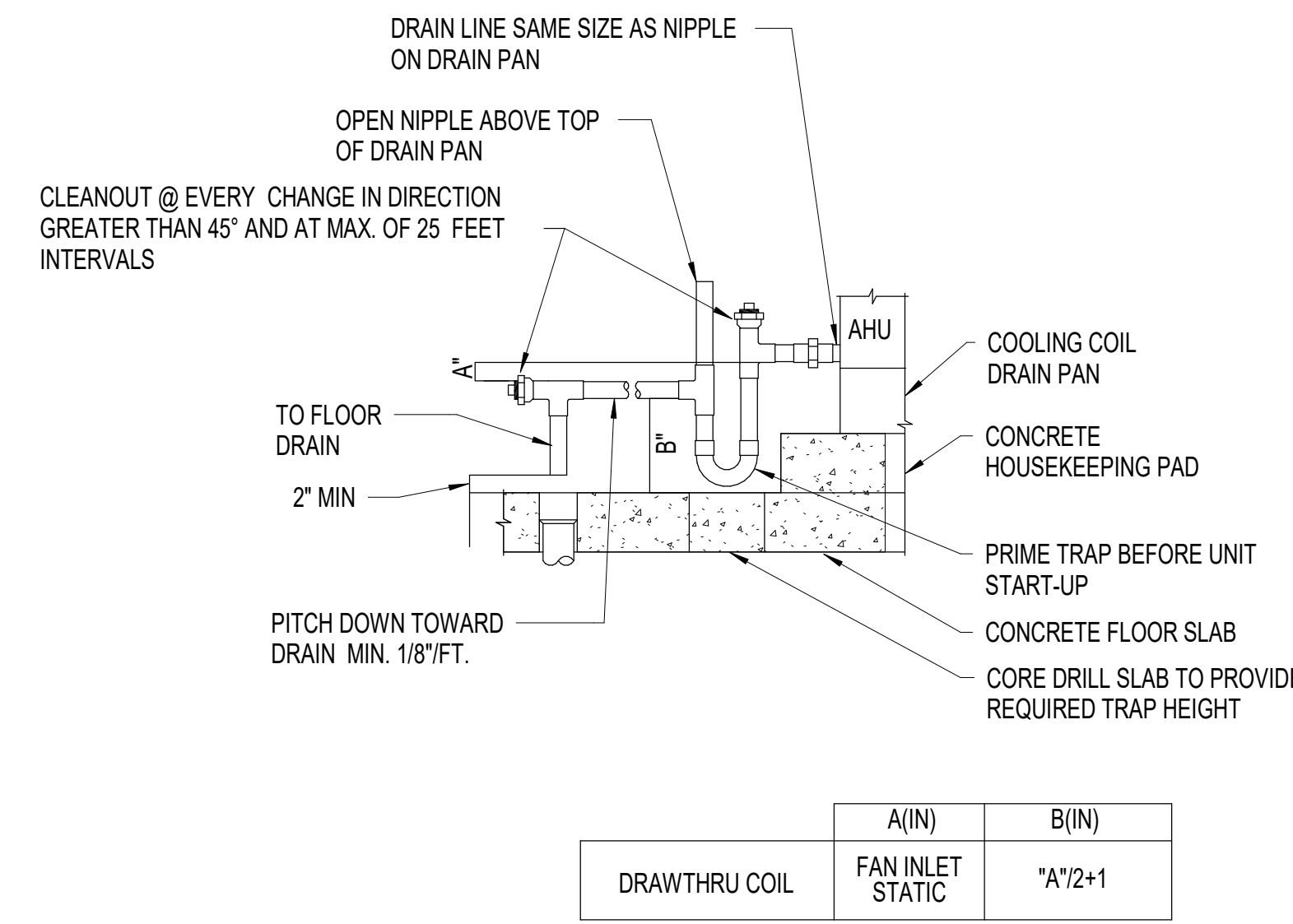


DETAILS - MECHANICAL

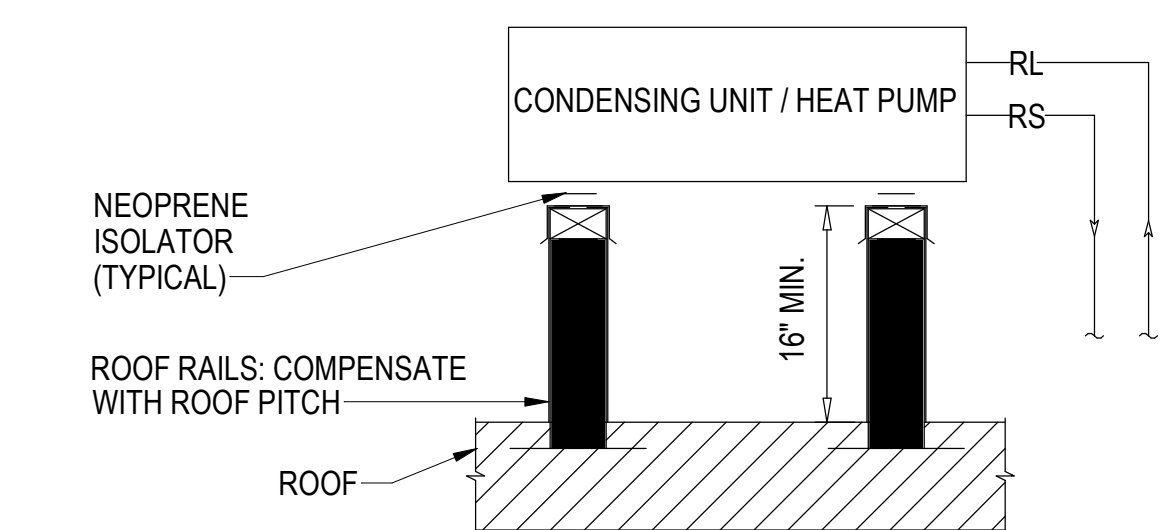
M801



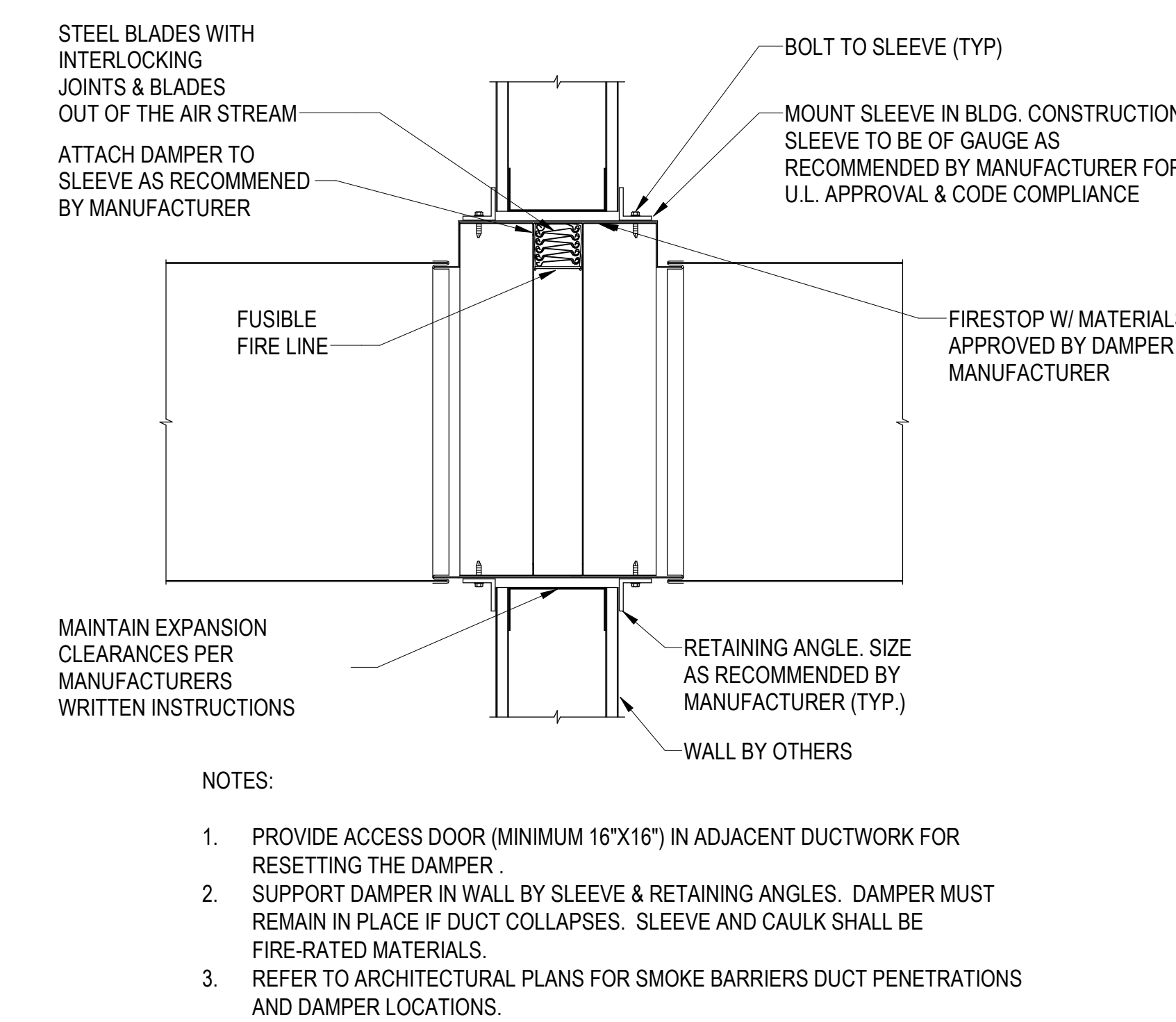
4 M - DETAIL - DRYER EXHAUST VENT1
1/8" = 1'-0"



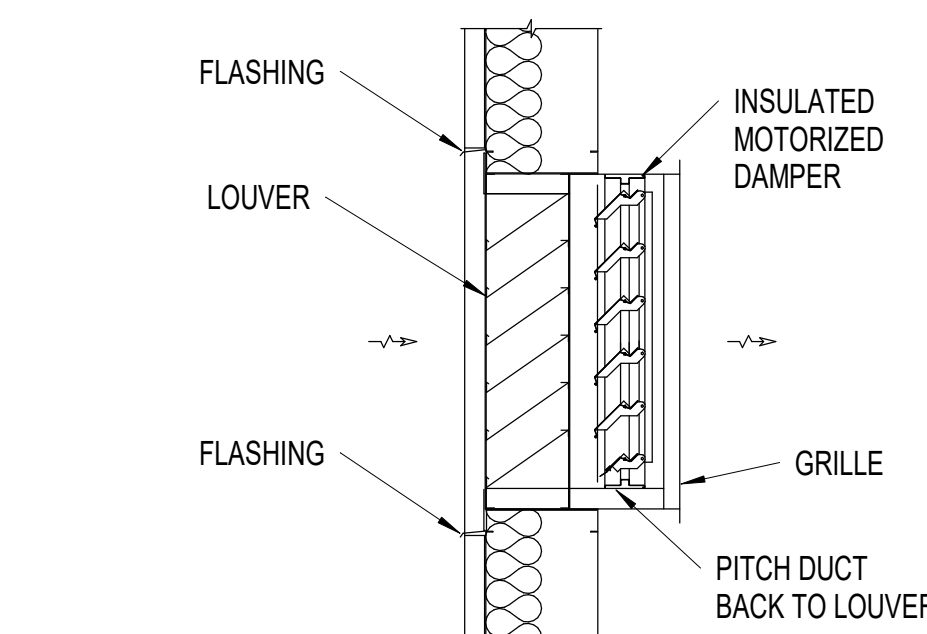
3 M - DETAIL - CONDENSATION DRAIN TRAP DETAIL
1/8" = 1'-0"



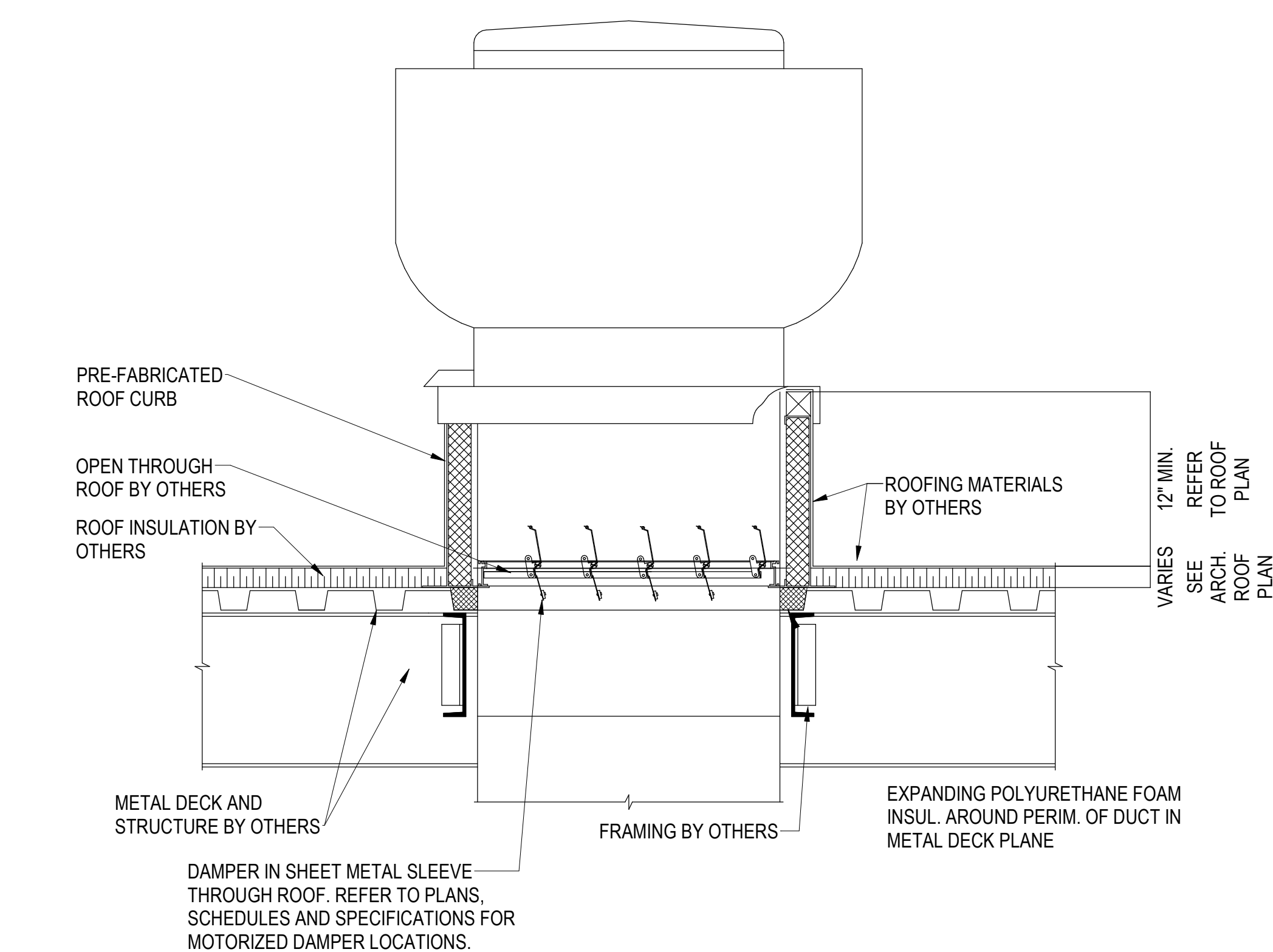
8 M - DETAIL - CONDENSING OR HEAT PUMP UNIT1
1/8" = 1'-0"



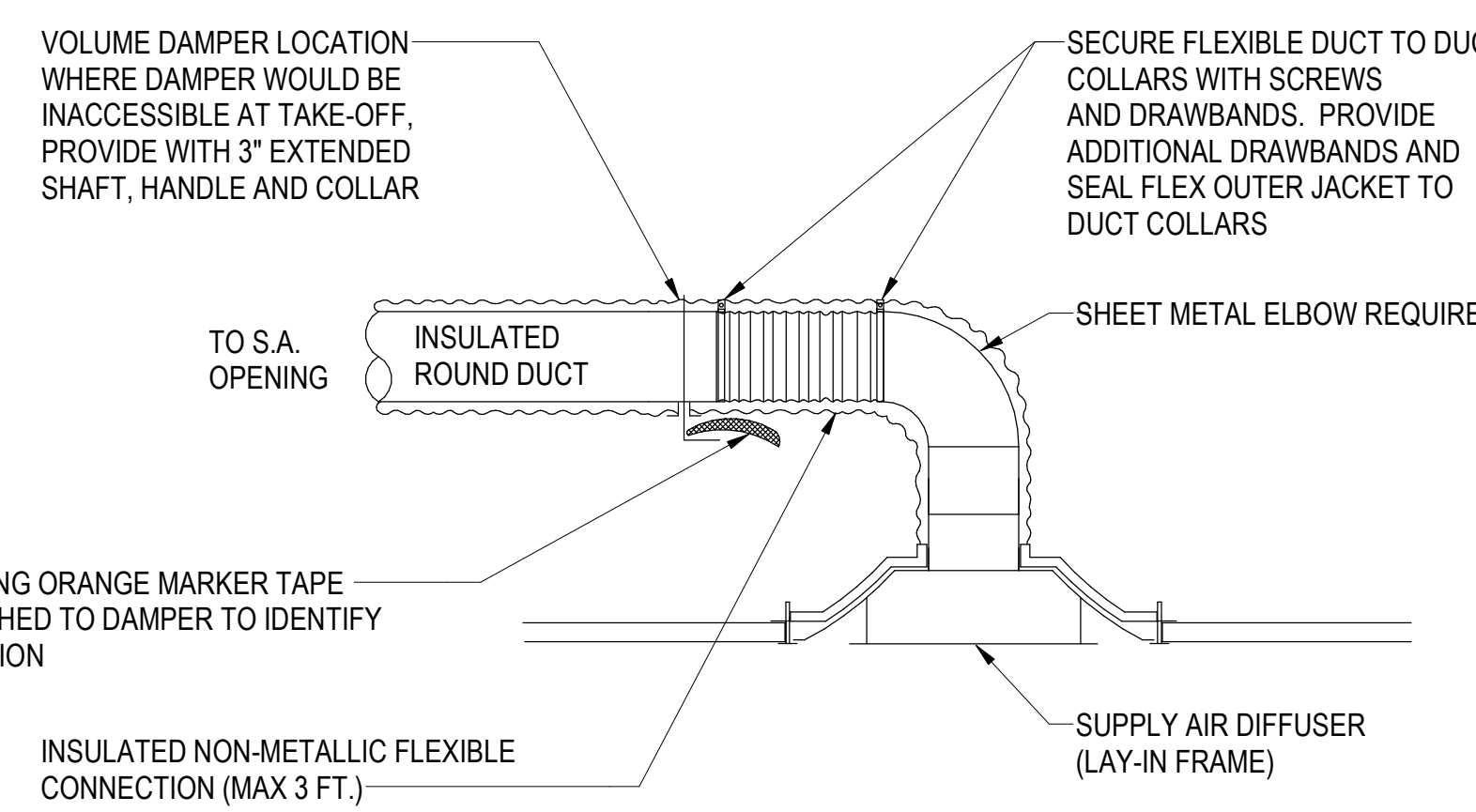
7 M - DETAIL - FIRE DAMPER INSTALLATION (STYLE B)1
1/8" = 1'-0"



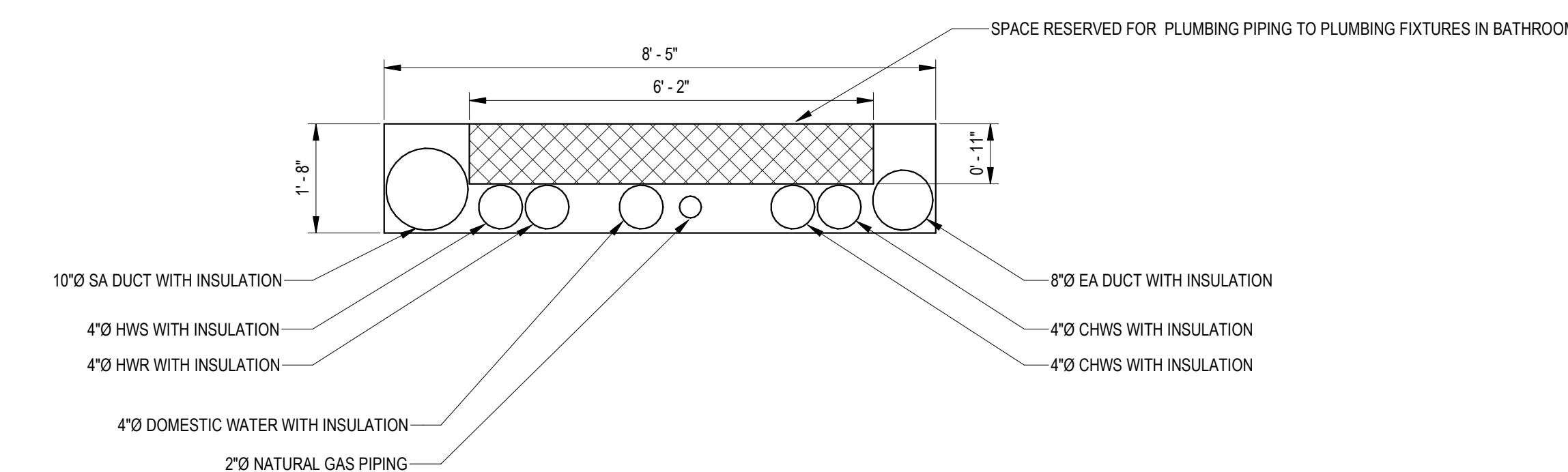
9 M - DETAIL - INTAKE LOUVER1
1/8" = 1'-0"



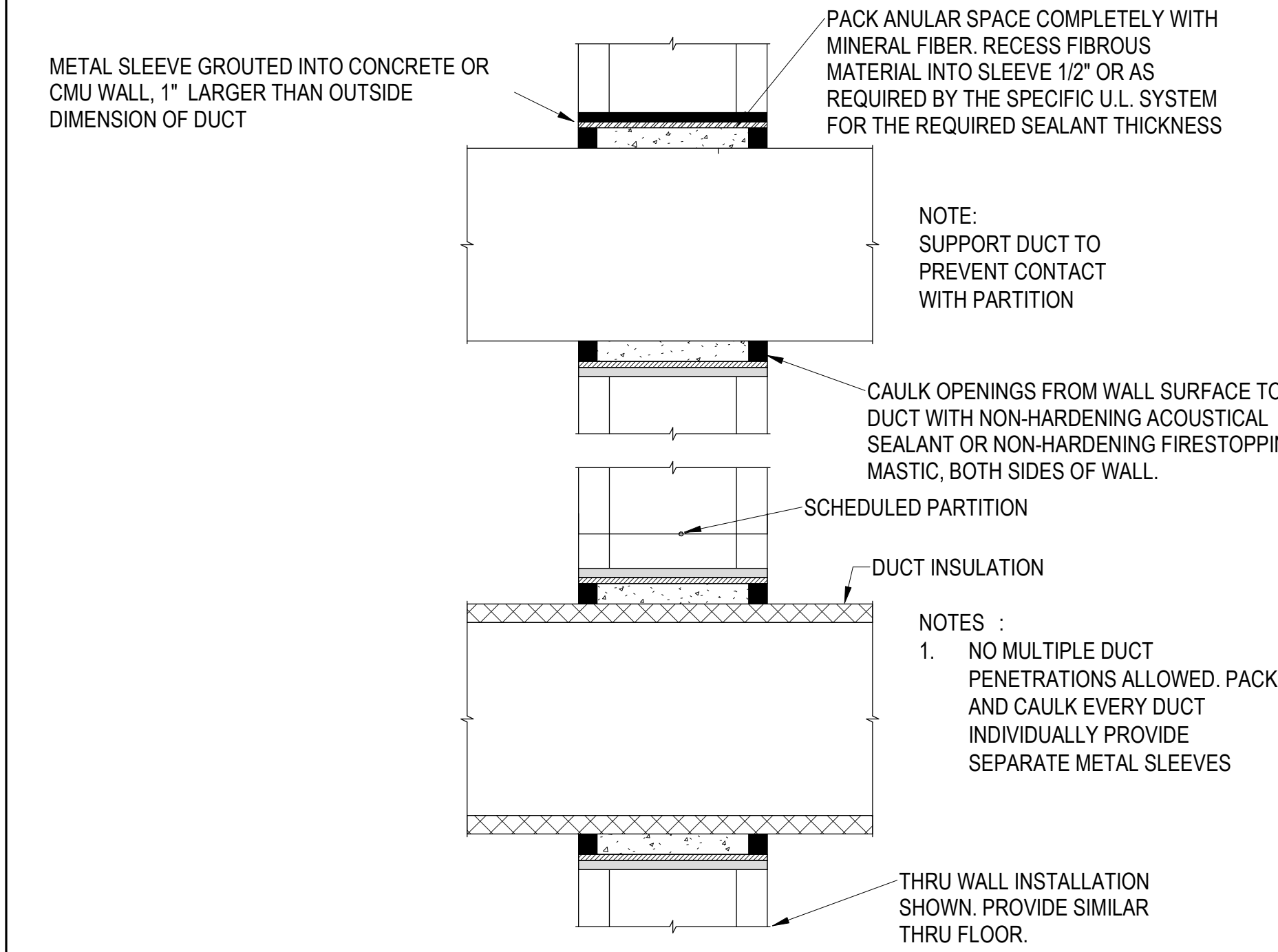
13 M - DETAIL - ROOF MOUNTED EXHAUST FAN1
1/8" = 1'-0"



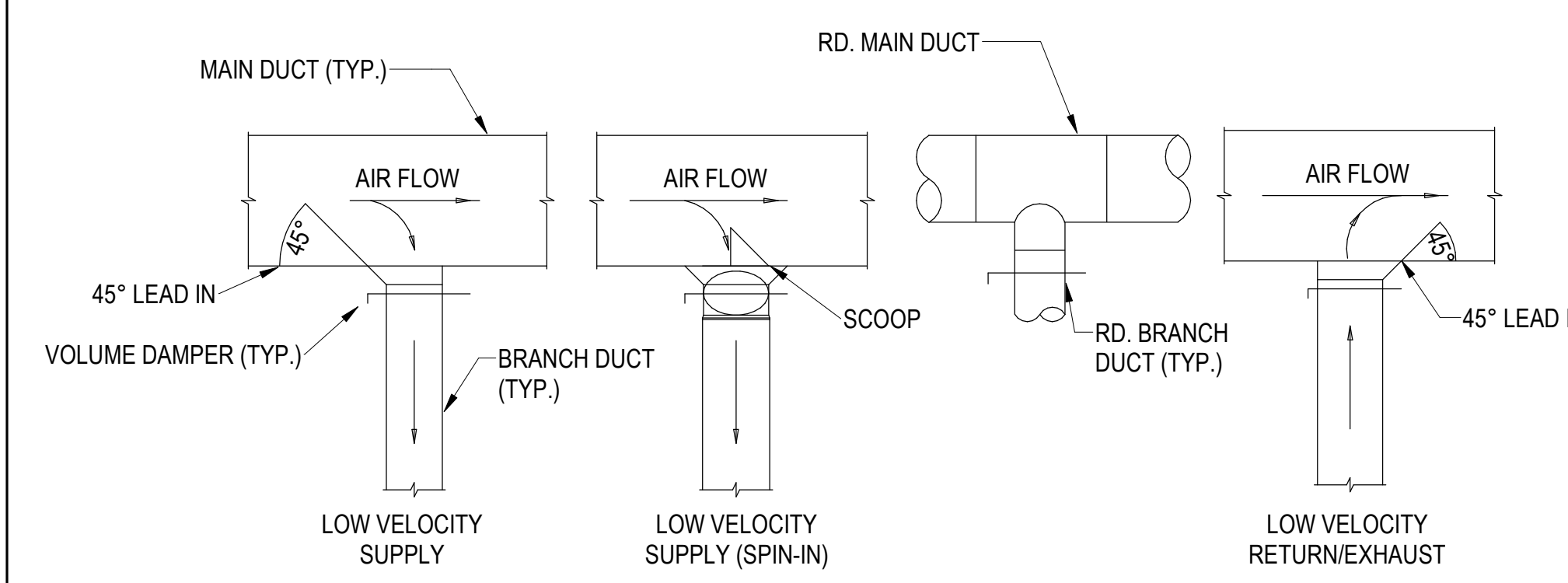
2 M - DETAIL - DIFFUSER MOUNTING1
1/8" = 1'-0"



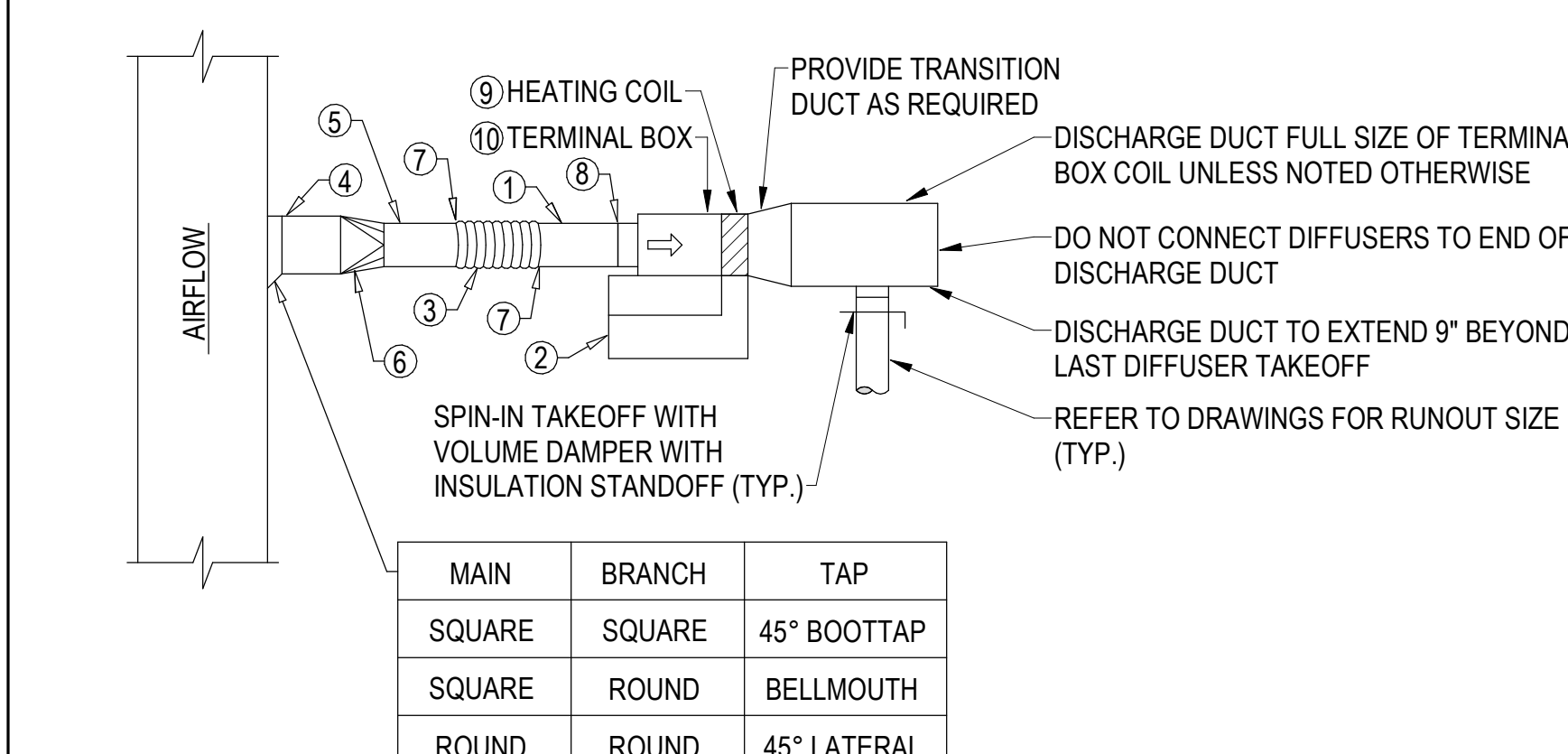
6 M - DETAIL - POD RESTROOM CHASE UTILITIES
1/2" = 1'-0"



1 M - DETAIL - DUCT PENETRATIONS1
1/8" = 1'-0"



5 M - DETAIL - TYPICAL DUCT CONNECTIONS1
1/8" = 1'-0"

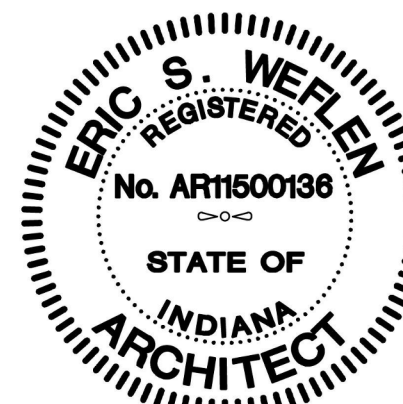


- DETAIL NOTES:
- A MINIMUM OF THREE (3) DUCT DIAMETERS (MAXIMUM OF 3'-0") OF STRAIGHT HARD METAL DUCT SHALL BE PROVIDED DIRECTLY UPSTREAM OF TERMINAL BOX INLET AFTER TRANSITION TO TERMINAL BOX INLET SIZE.
 - MAINTAIN A MINIMUM OF 24" CLEARANCE FOR SERVICE, CONTROLLER, AND HEATING COIL VALVING. PROVIDE 36" MINIMUM CLEARANCE WHERE VOLTAGE EXCEEDS 24V. COORDINATE ACCESS CLEARANCE FROM UNIT THROUGH CEILING BELOW WITH ALL TRADES.
 - OPTIONAL INSULATED FLEXIBLE DUCT CONNECTION TO THE TERMINAL BOX. HARD DUCT IS PERFECTLY ACCEPTABLE. FLEXIBLE DUCT SHALL NOT EXCEED ONE (1) FOOT IN LENGTH. FLEXIBLE DUCT SHALL BE INSTALLED STRAIGHT WITH NO OFFSETS. REFER TO DRAWINGS FOR RUNOUT SIZES. RUNOUT SIZE IS SAME AS TERMINAL UNIT INLET SIZE UNLESS NOTED OTHERWISE.
 - DUCT SIZE AFTER TRANSITION IS SAME SIZE AS SCHEDULED TERMINAL UNIT INLET SIZE AS MODIFIED BY COMPLETED SUBMITTALS.
 - TRANSITION, AS REQUIRED, IF RUNOUT SIZE DIFFERS FROM TERMINAL UNIT INLET SIZE.
 - TIE-WRAP CONNECT FLEX DUCT HELICAL COIL TO ROUND METAL DUCTWORK. PULL FLEX DUCT VAPOR BARRIER OVER LOWER TIE-WRAP AND INSTALL SECOND TIE-WRAP OVER THE VAPOR BARRIER. INSULATE METAL DUCT OVER TOP OF SECOND TIE-WRAP AND INSTALL INSULATOR'S TAPE ON EDGES.
 - INSULATE ALL EXPOSED METAL AT TERMINAL UNIT INLET. INSULATE AROUND FLOW SENSORS.
 - INSULATE ALL EXPOSED METAL AT COIL. INSULATE RETURN BENDS. SEAL CLEAT JOINTS. DUCTWORK INSULATION SHALL OVERLAP THE INTERNALLY INSULATED TERMINAL UNIT BY TWO INCHES. INSULATE COIL PIPING BETWEEN CONTROL VALVE AND COIL CONNECTION.
 - SUPPORT TERMINAL BOX SUCH THAT IT CAN HANG INDEPENDENTLY OF ANY DUCTWORK CONNECTIONS. ATTACH FOUR STRAP HANGERS TO UNIT. SEAL SCREW PENETRATIONS OF TERMINAL UNIT.

10 M - DETAIL - TERMINAL BOX1
1/8" = 1'-0"

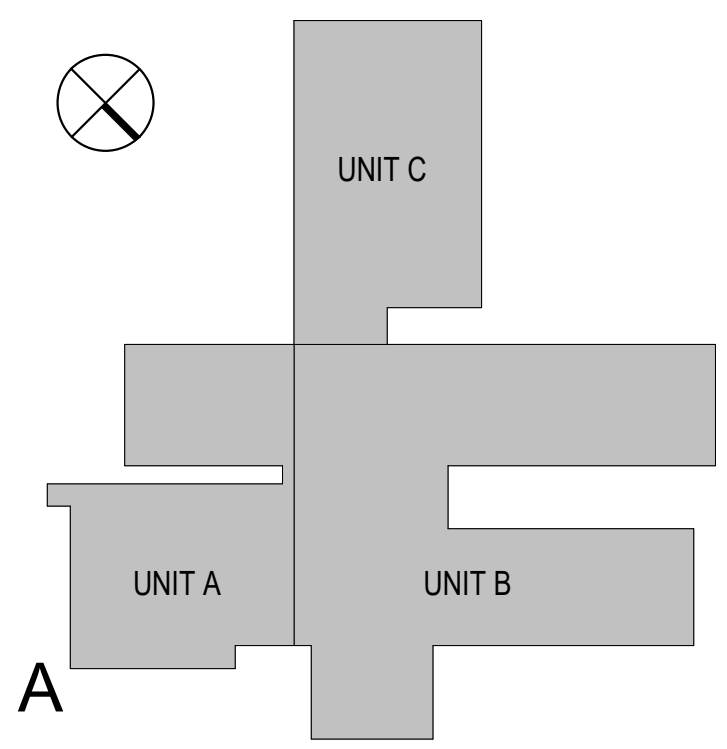
#	Revision	Date
---	----------	------

Project #: 17-700-078-2
Designed By: A.M.M
Drawn By: A.M.M
Checked By: D.E.B
Date: 04.12.2021



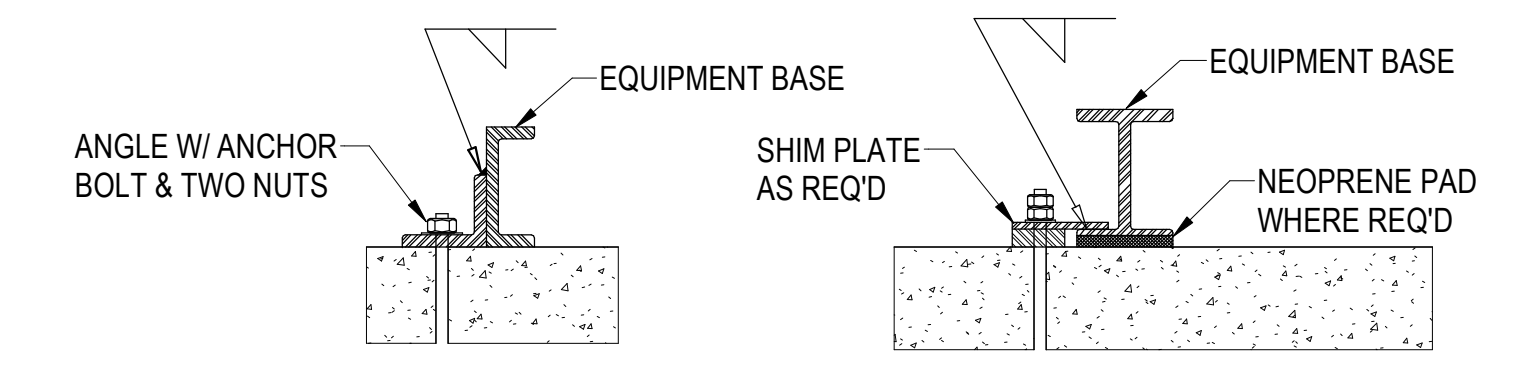
Eric Wepler

These Drawings and Specifications, and all copies thereof are and shall remain the property and copyright of the Architect. They shall be used only with respect to this Project and are not to be used on any other Project or Work without prior written permission from the Architect

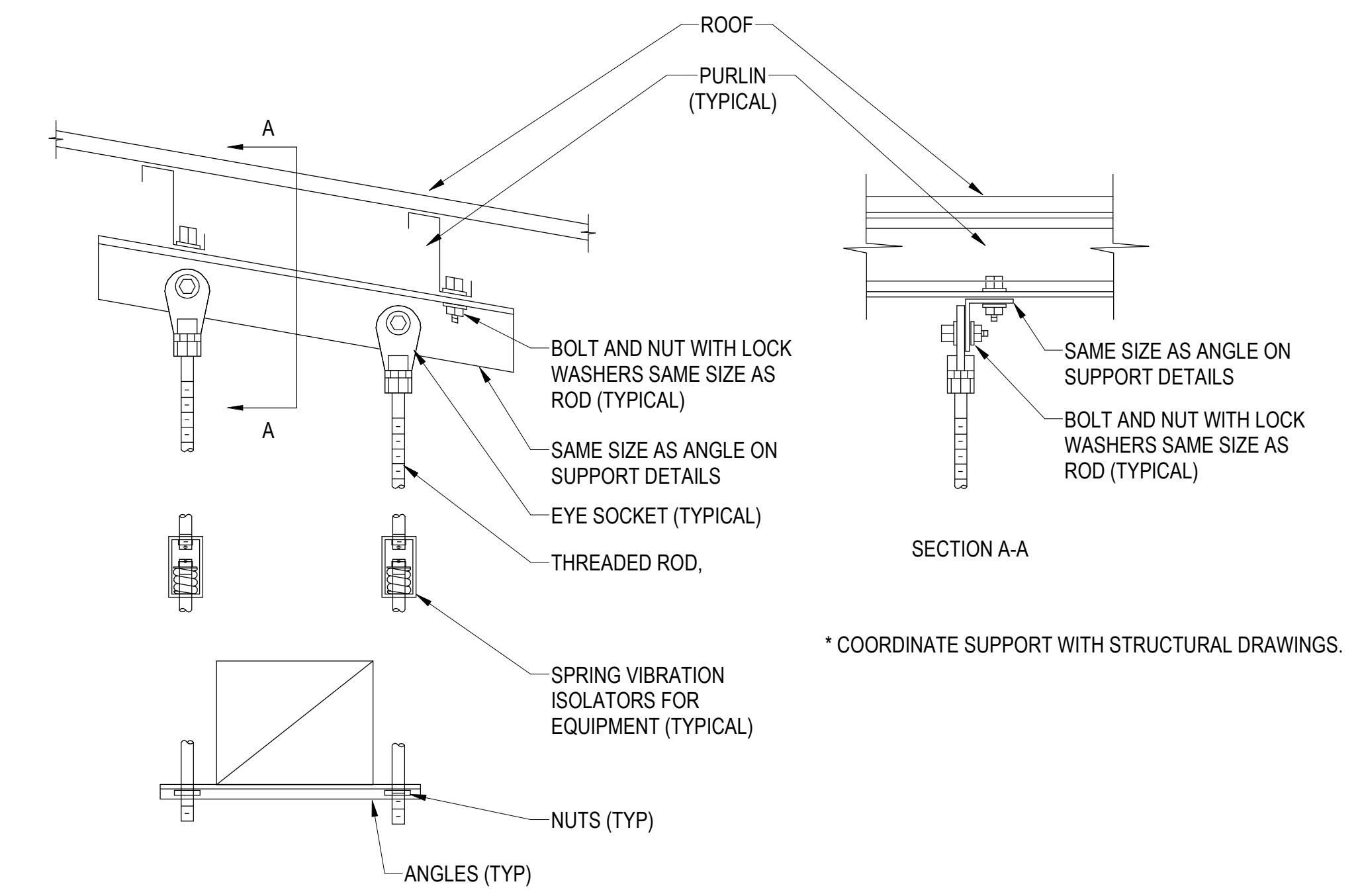


DETAILS - MECHANICAL

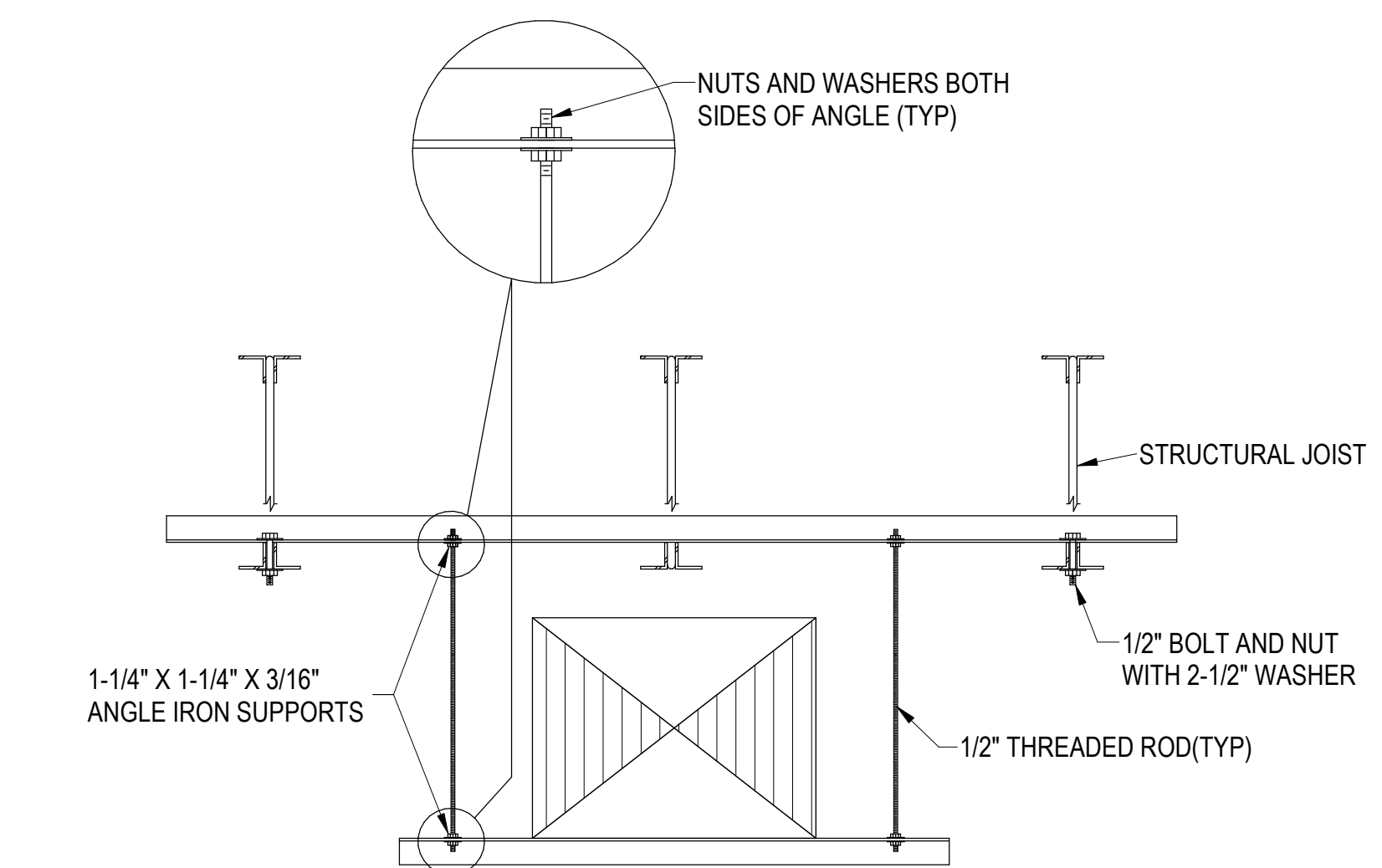
M802



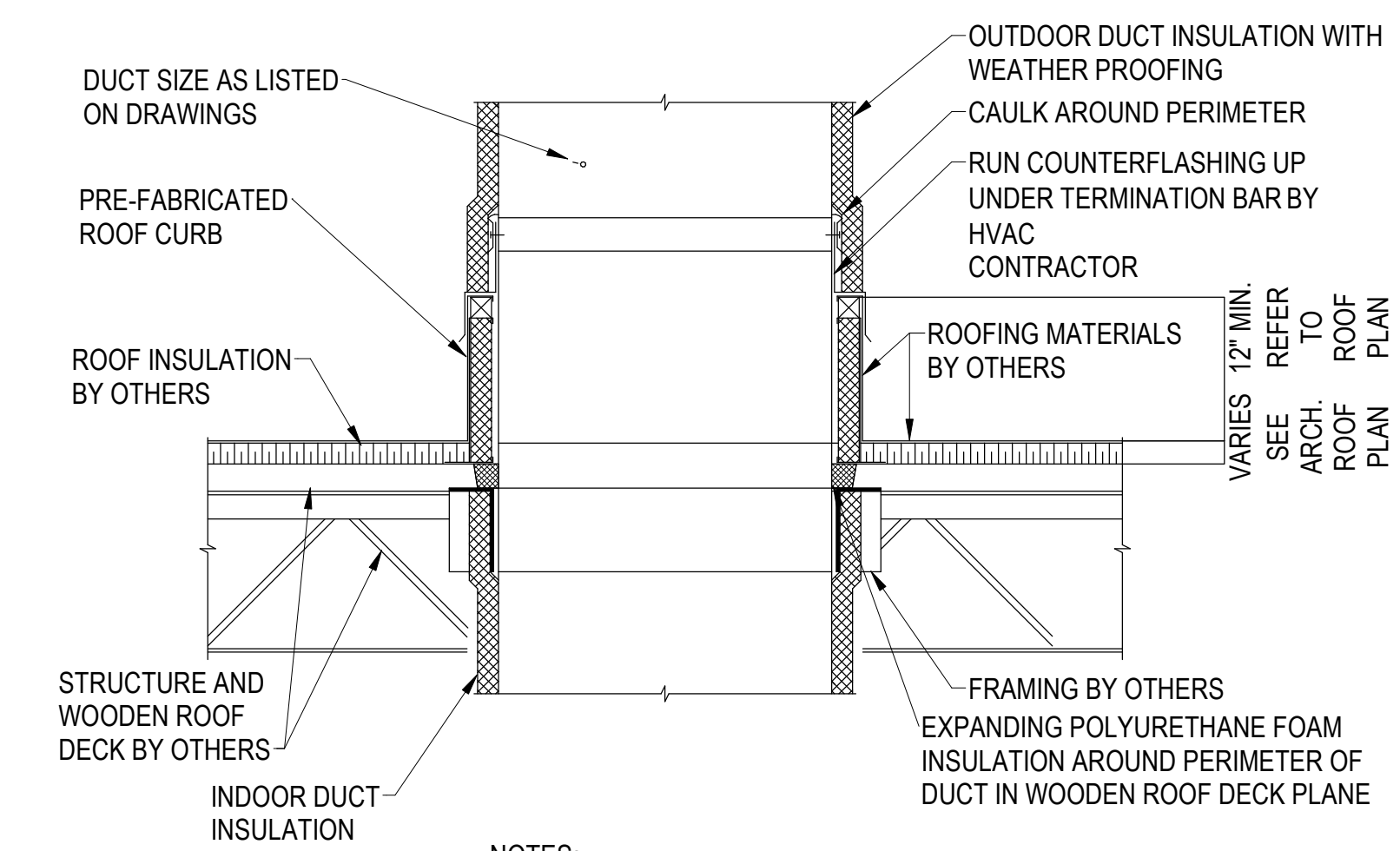
2 M - DETAIL - RIGID MOUNTED EQUIPMENT RESTRAINTS1
1/8" = 1'-0"



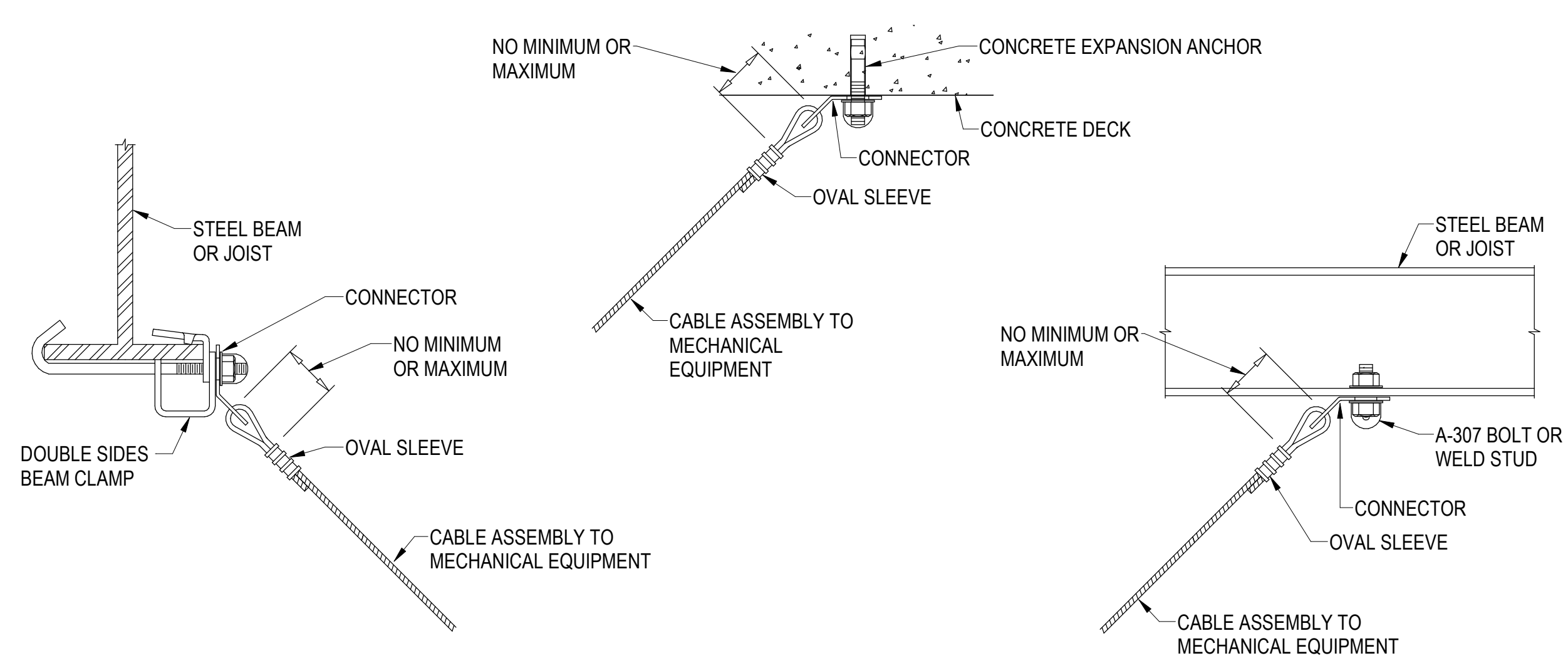
5 M - DETAIL - SUPPORT AND CONNECTION TO STRUCTURE1
1/8" = 1'-0"



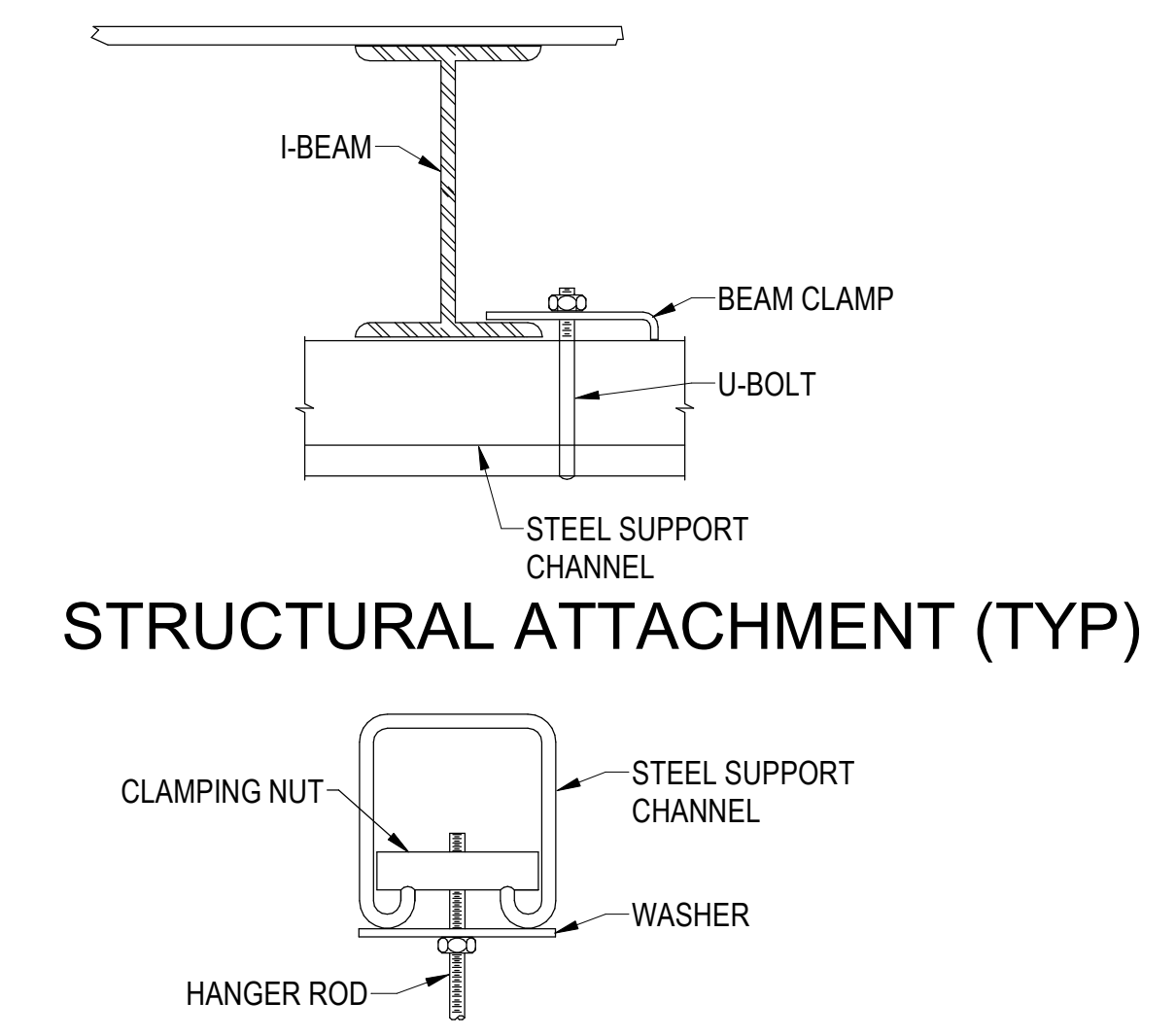
6 M - DETAIL - DUCT SUPPORT1
1/8" = 1'-0"



3 M - DETAIL - TYPICAL DUCTWORK THRU ROOF CURB1
1/8" = 1'-0"

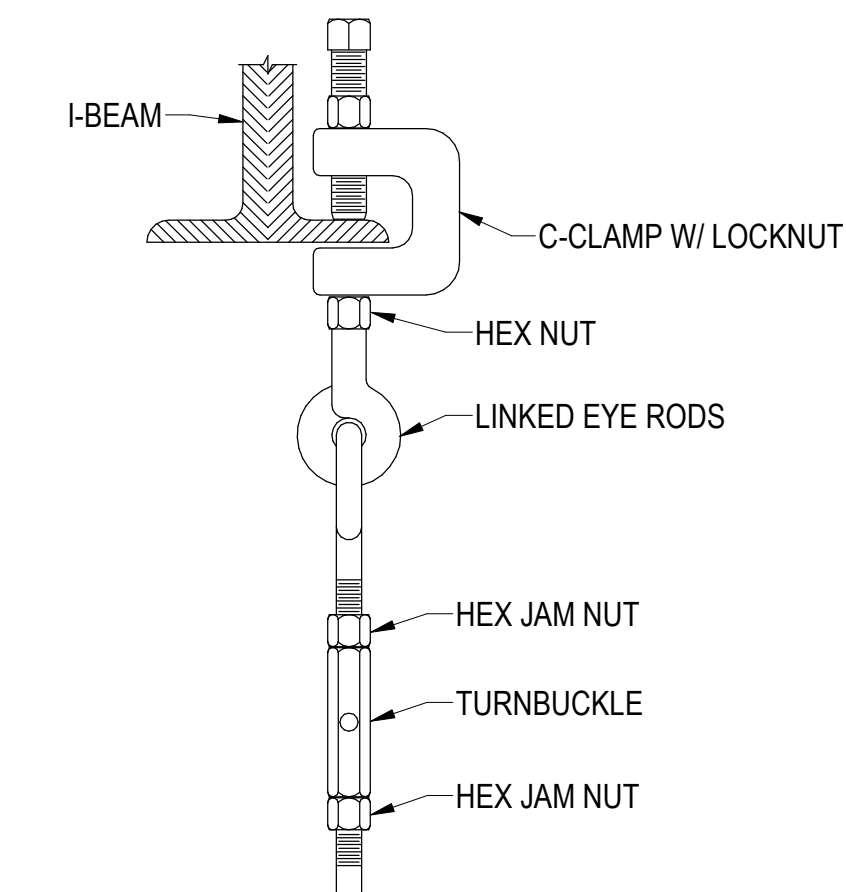


4 M - DETAIL - SUSPENDED EQUIPMENT SUPPORT1
1/8" = 1'-0"

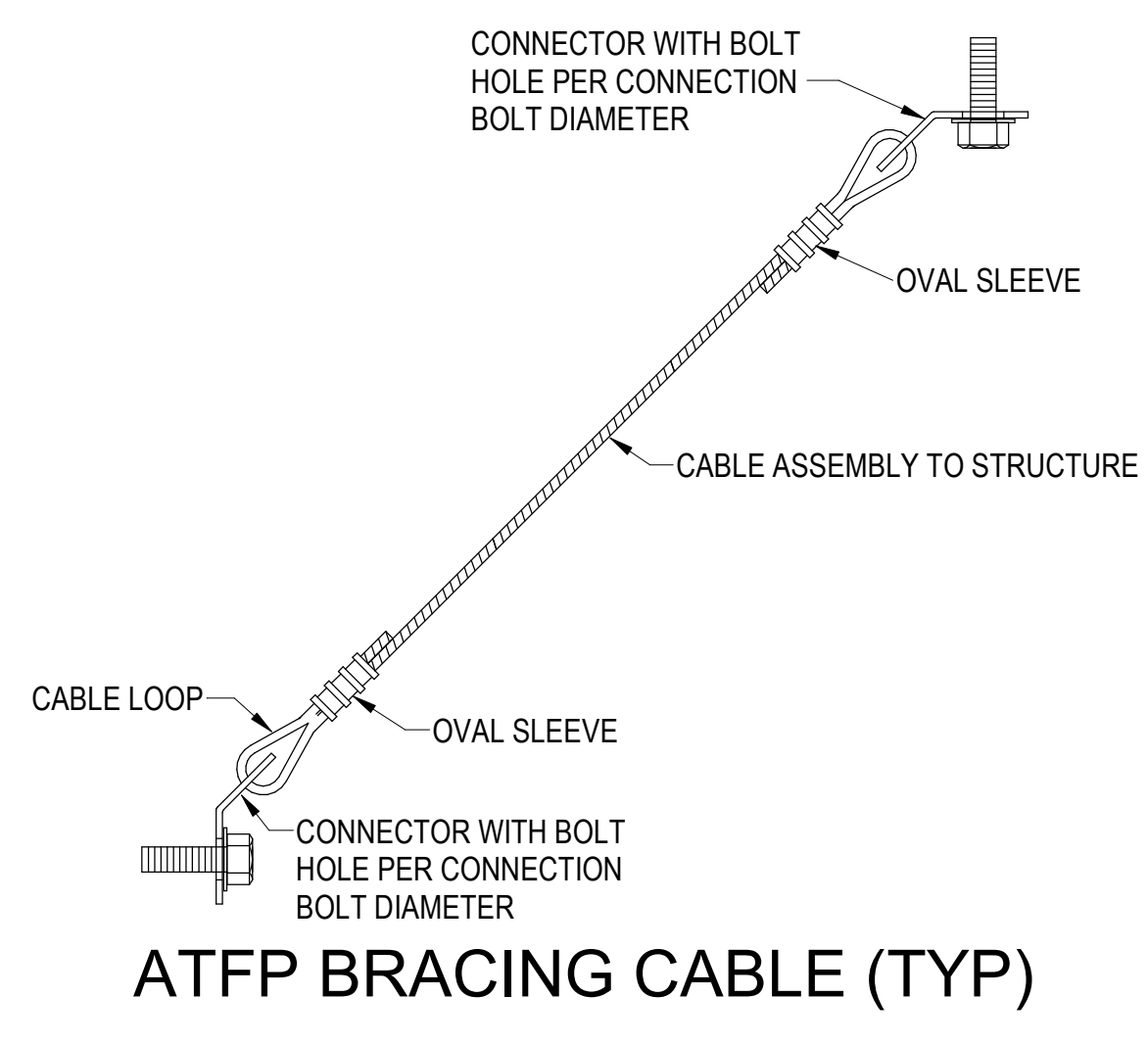


STRUCTURAL ATTACHMENT (TYP)

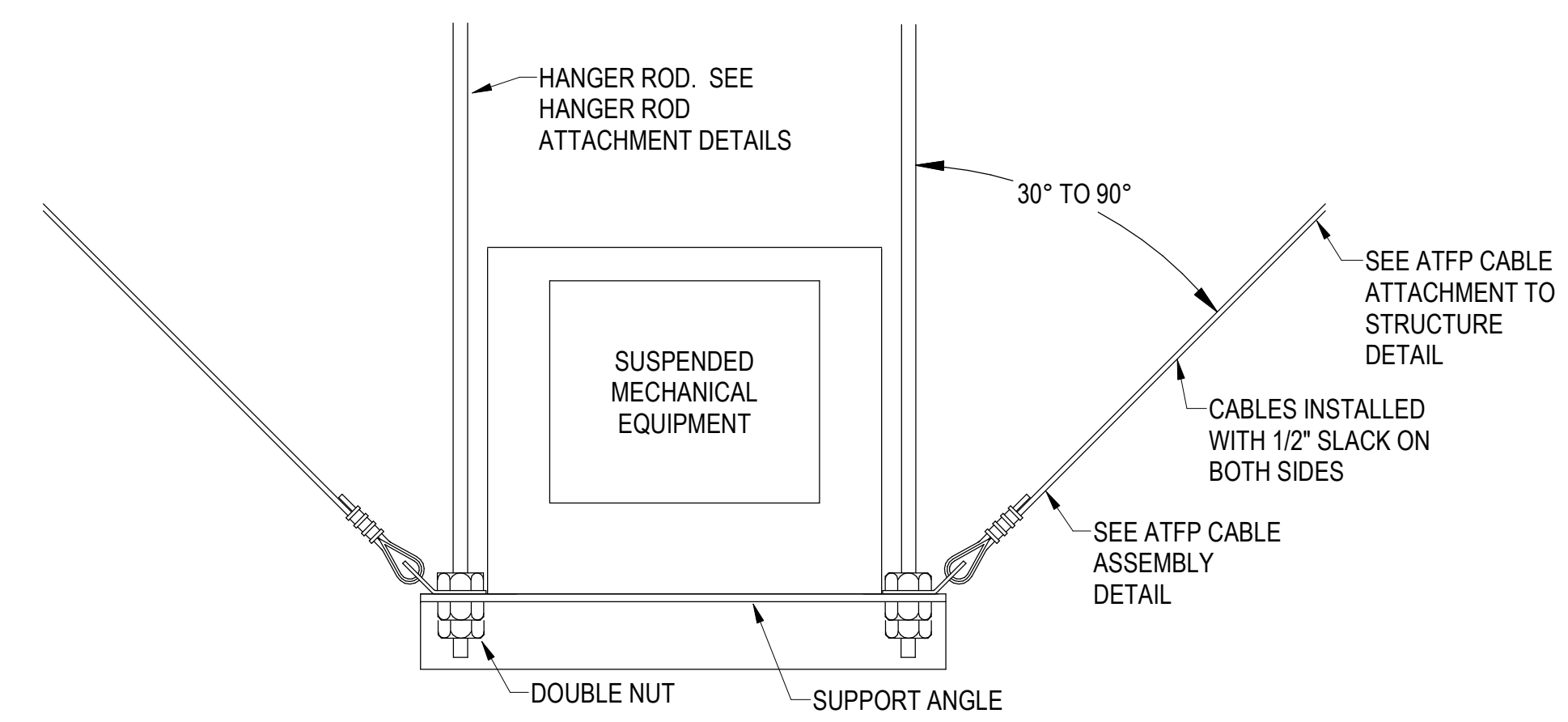
ATFP CABLE STRUCTURAL ATTACHMENT (TYP)



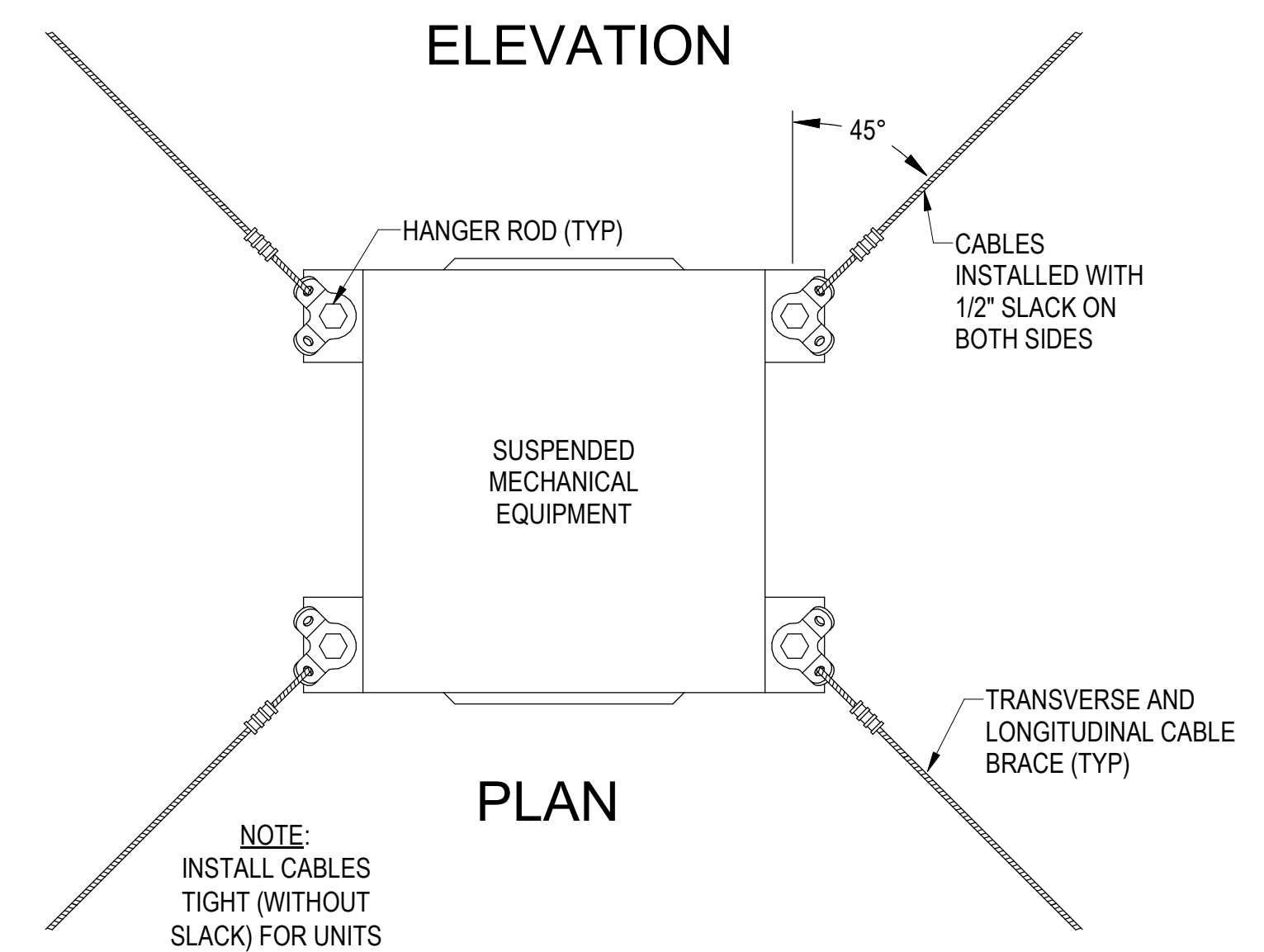
STRUCTURAL ATTACHMENT (TYP)



ATFP BRACING CABLE (TYP)



ELEVATION



PLAN

NOTE:
INSTALL CABLES
TIGHT (WITHOUT
SLACK) FOR UNITS
WITH INTERNAL
VIBRATION
ISOLATORS

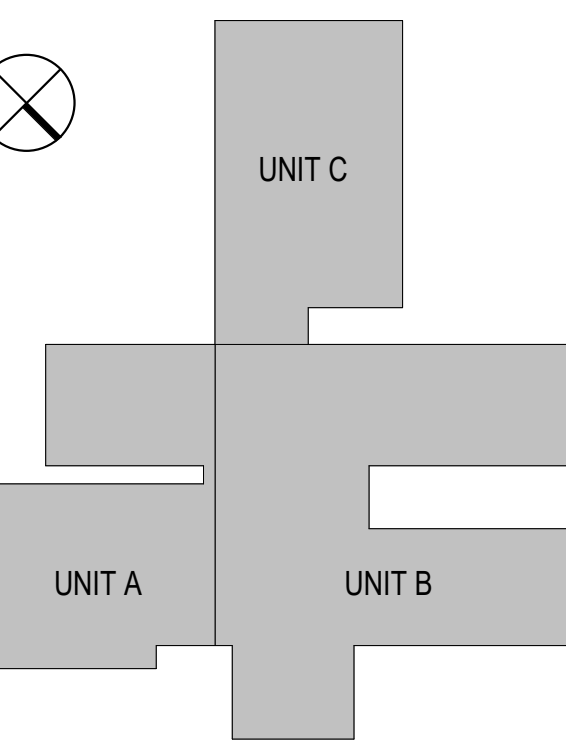
#	Revision	Date
---	----------	------

Project #: 17-700-078-2
Designed By: A.M.M
Drawn By: A.M.M
Checked By: D.E.B
Date: 04.12.2021



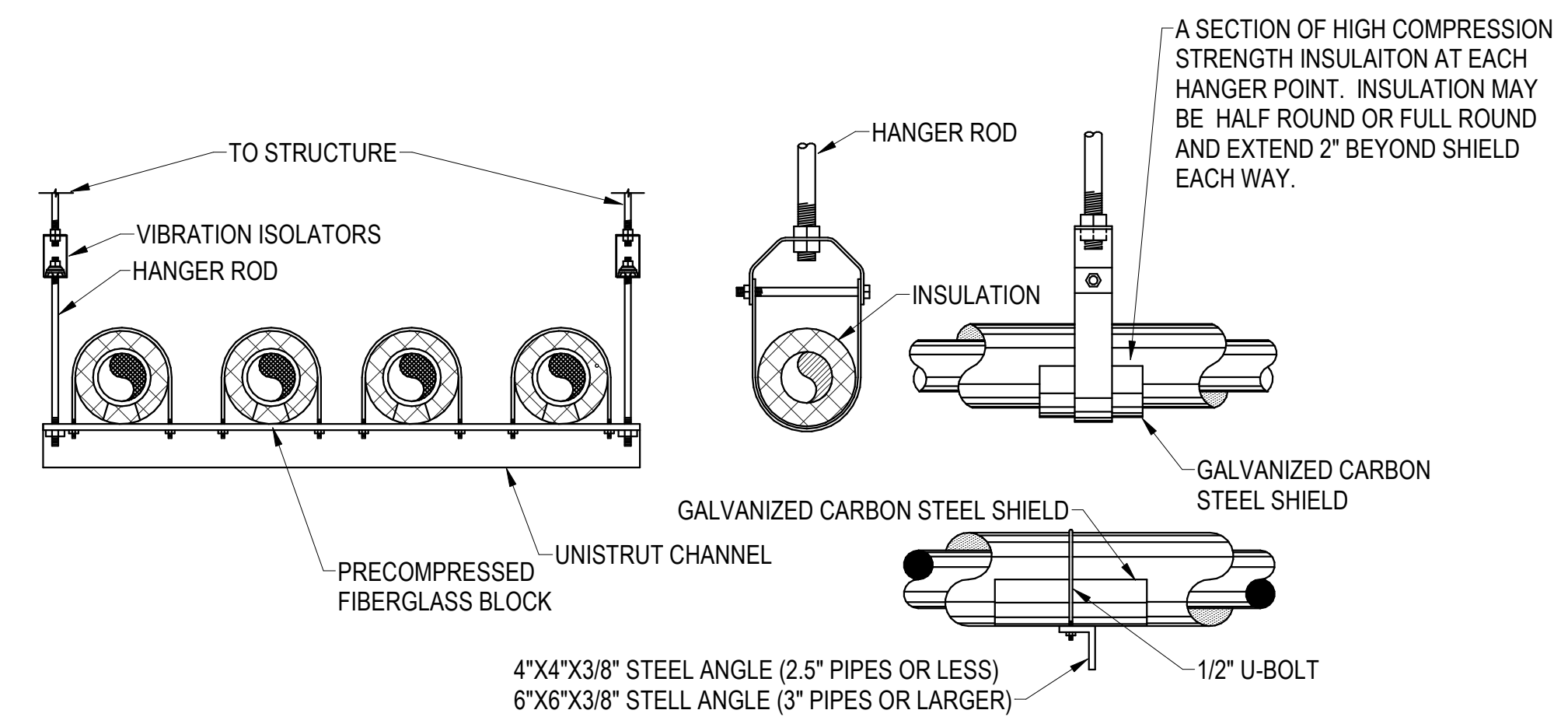
Eric Wepler

These Drawings and Specifications, and all copies thereof are and shall remain the property and copyright of the Architect. They shall be used only with respect to this Project and are not to be used on any other Project or Work without prior written permission from the Architect

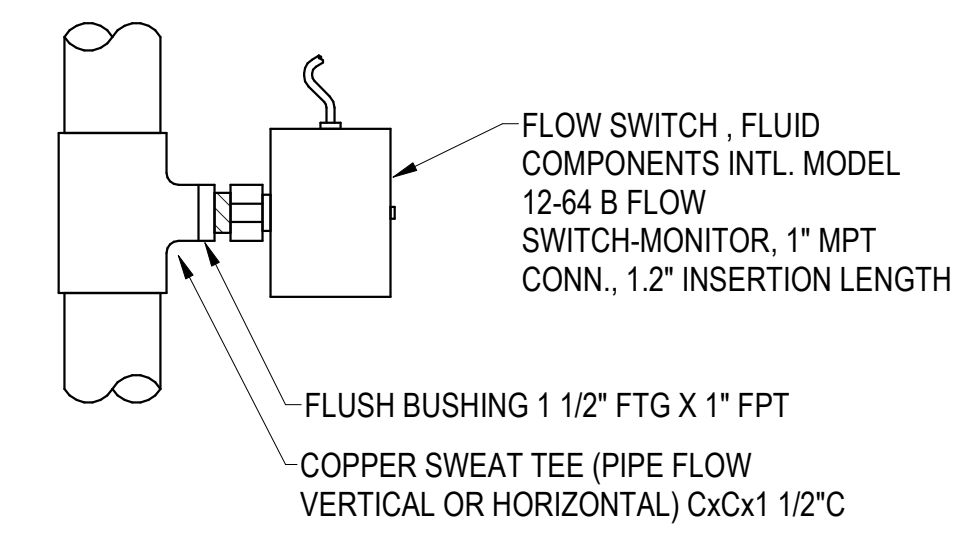


DETAILS - MECHANICAL

M803

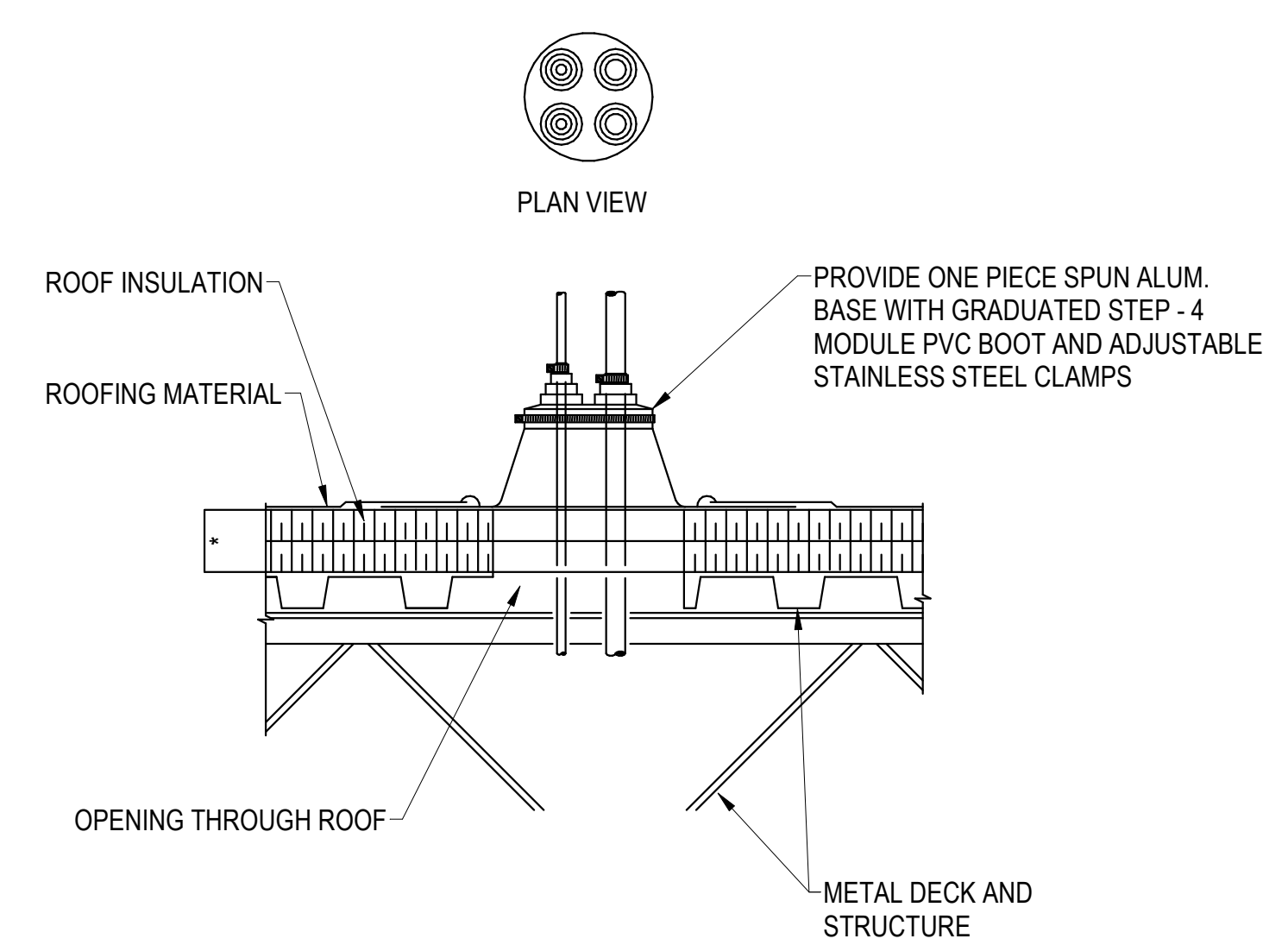


3 M - DETAIL - TYPICAL HANGER1
1/8" = 1'-0"

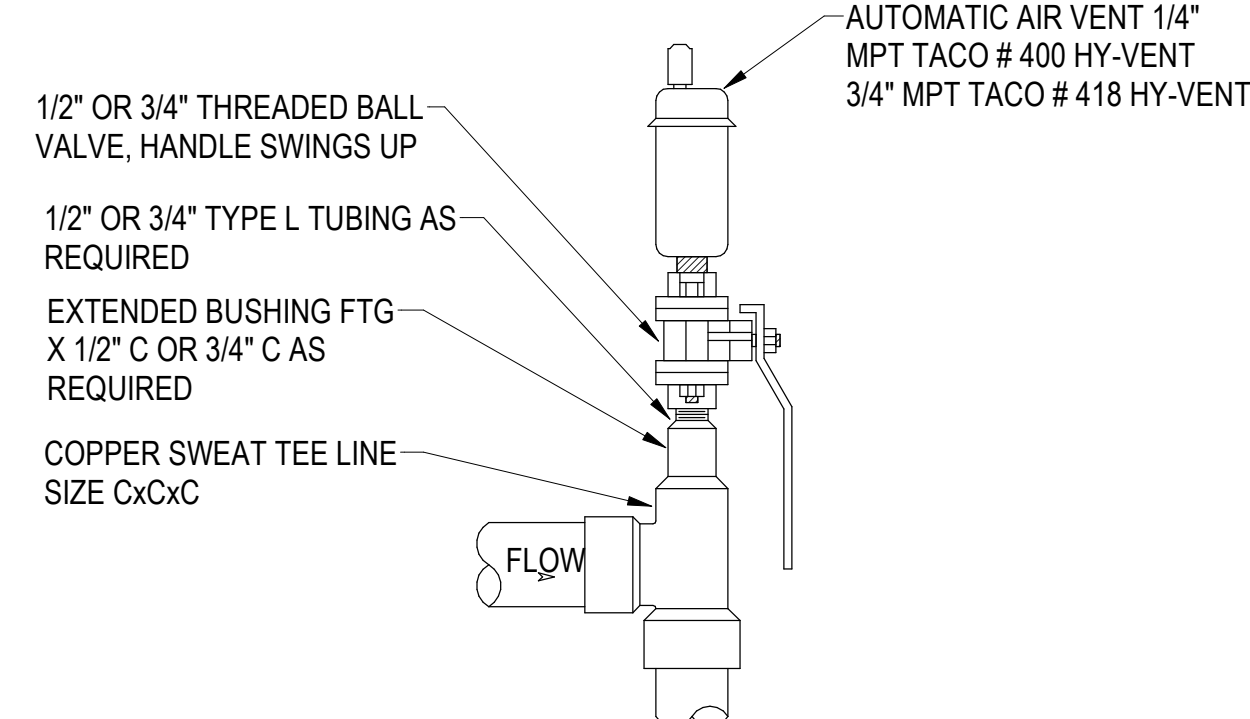


5 M - DETAIL - FLOW SWITCH MOUNTING1
1/8" = 1'-0"

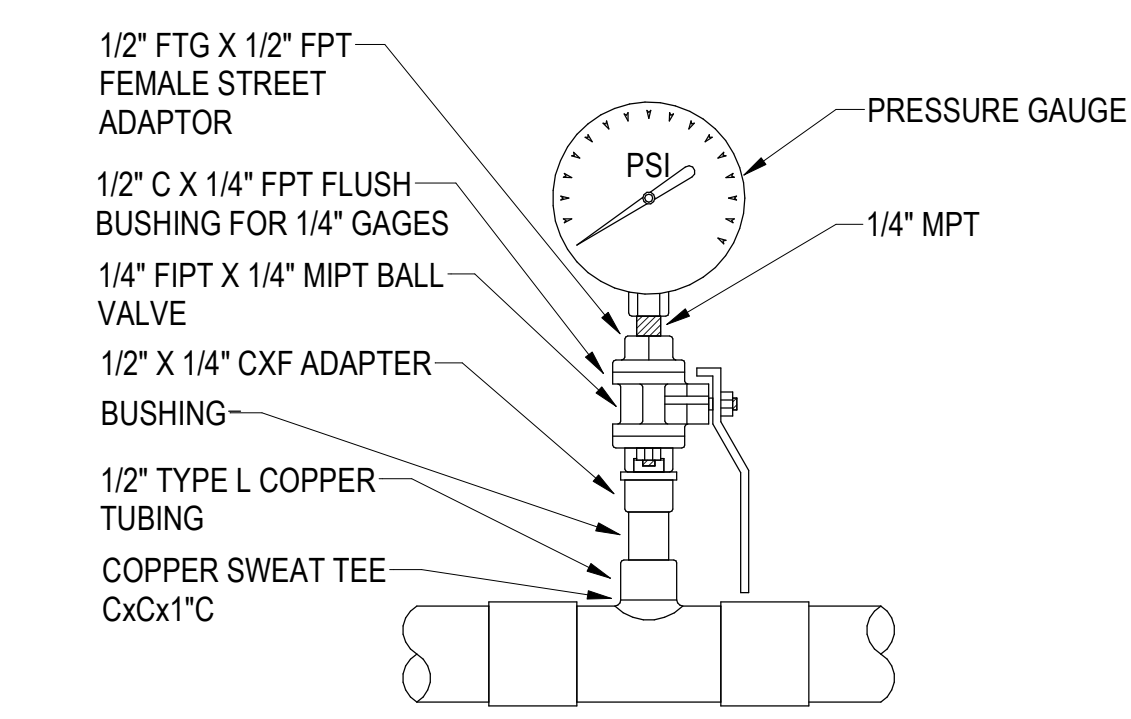
NOTES:
1. INSTALL PROBE HORIZONTALLY WITH "FLAT UP AND LEVEL" MARKING POSITIONED ON TOP



2 M - DETAIL - PIPE SEAL THRU ROOF1
1/8" = 1'-0"

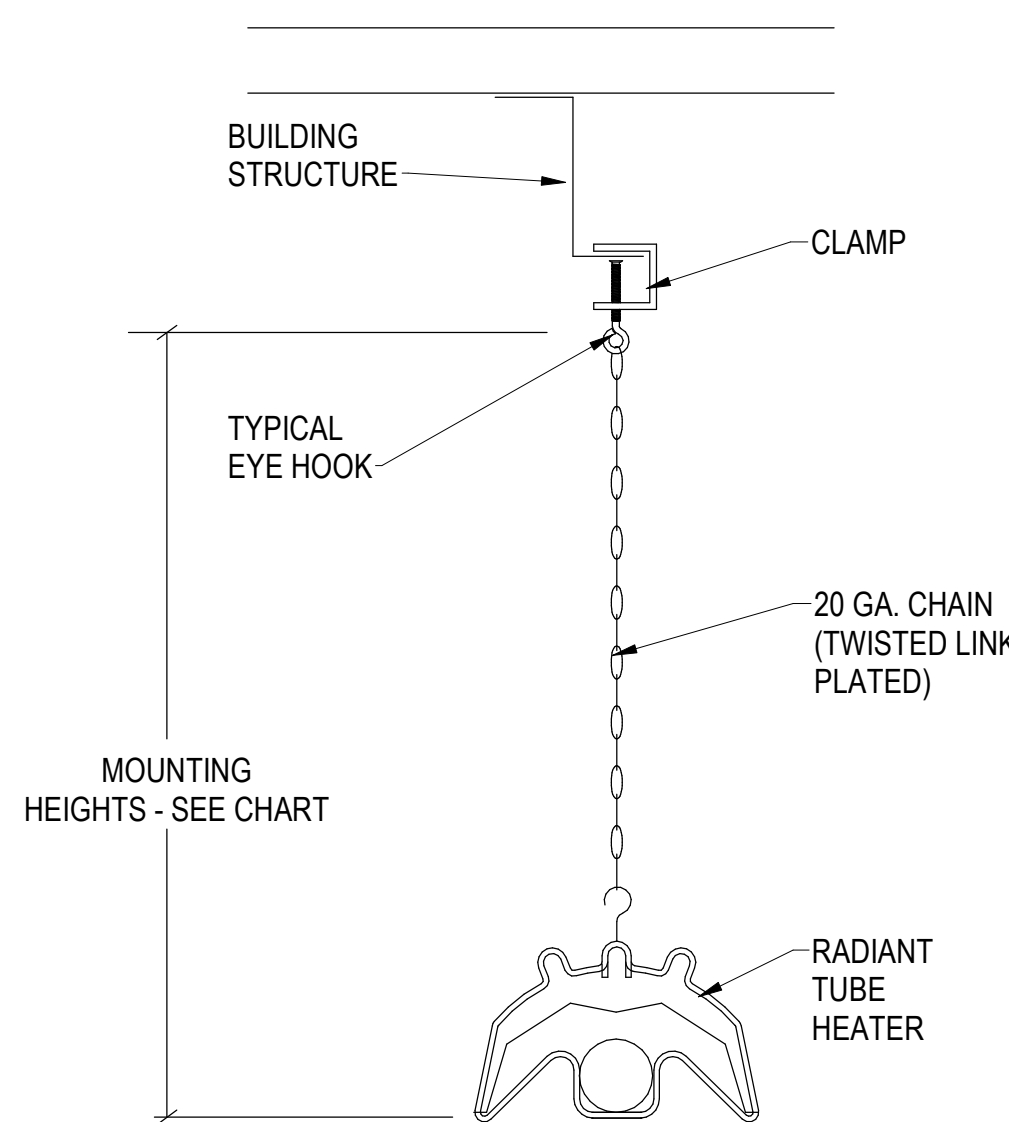


4 M - DETAIL - AUTOMATIC AIR RELEASE1
1/8" = 1'-0"

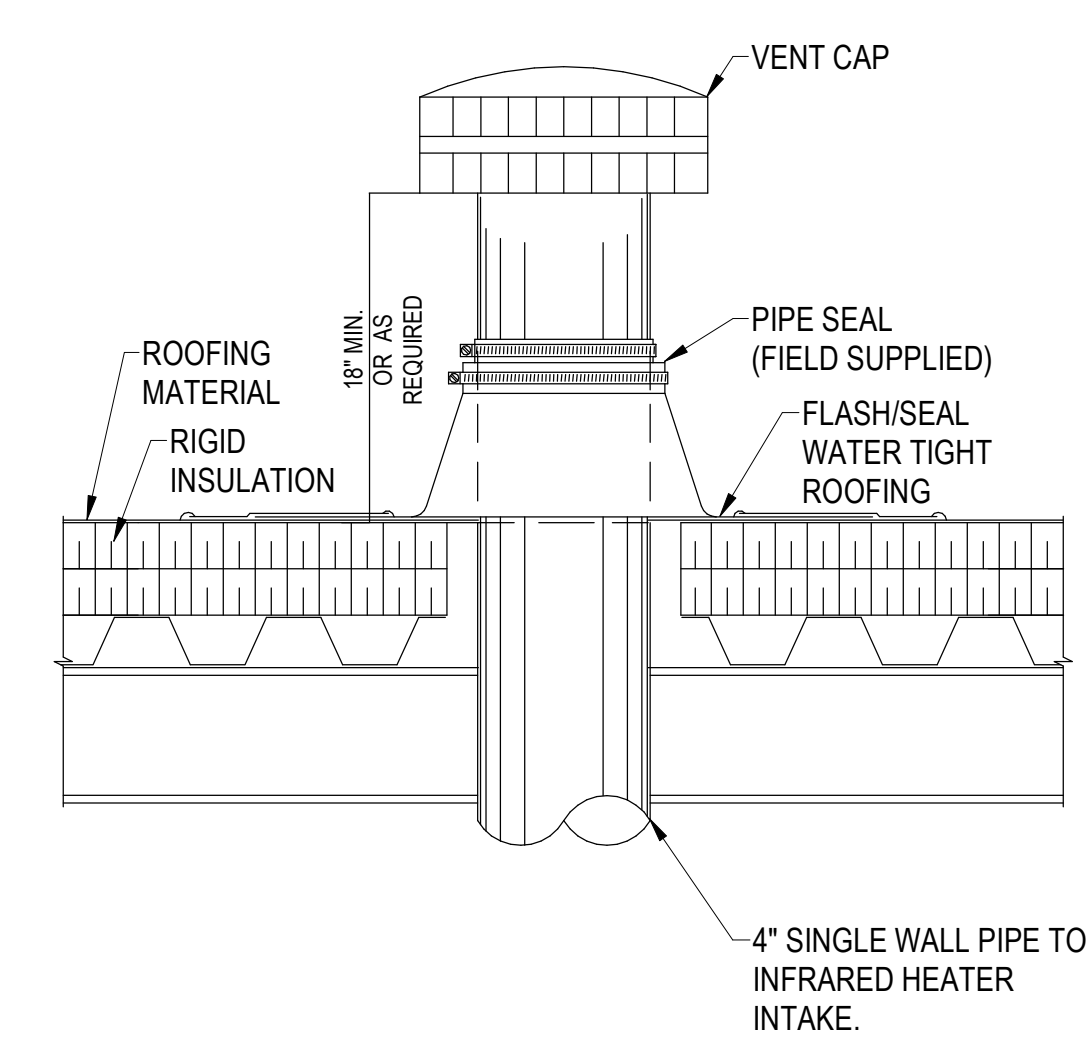


7 M - DETAIL - PRESSURE GAUGE MOUNTING1
1/8" = 1'-0"

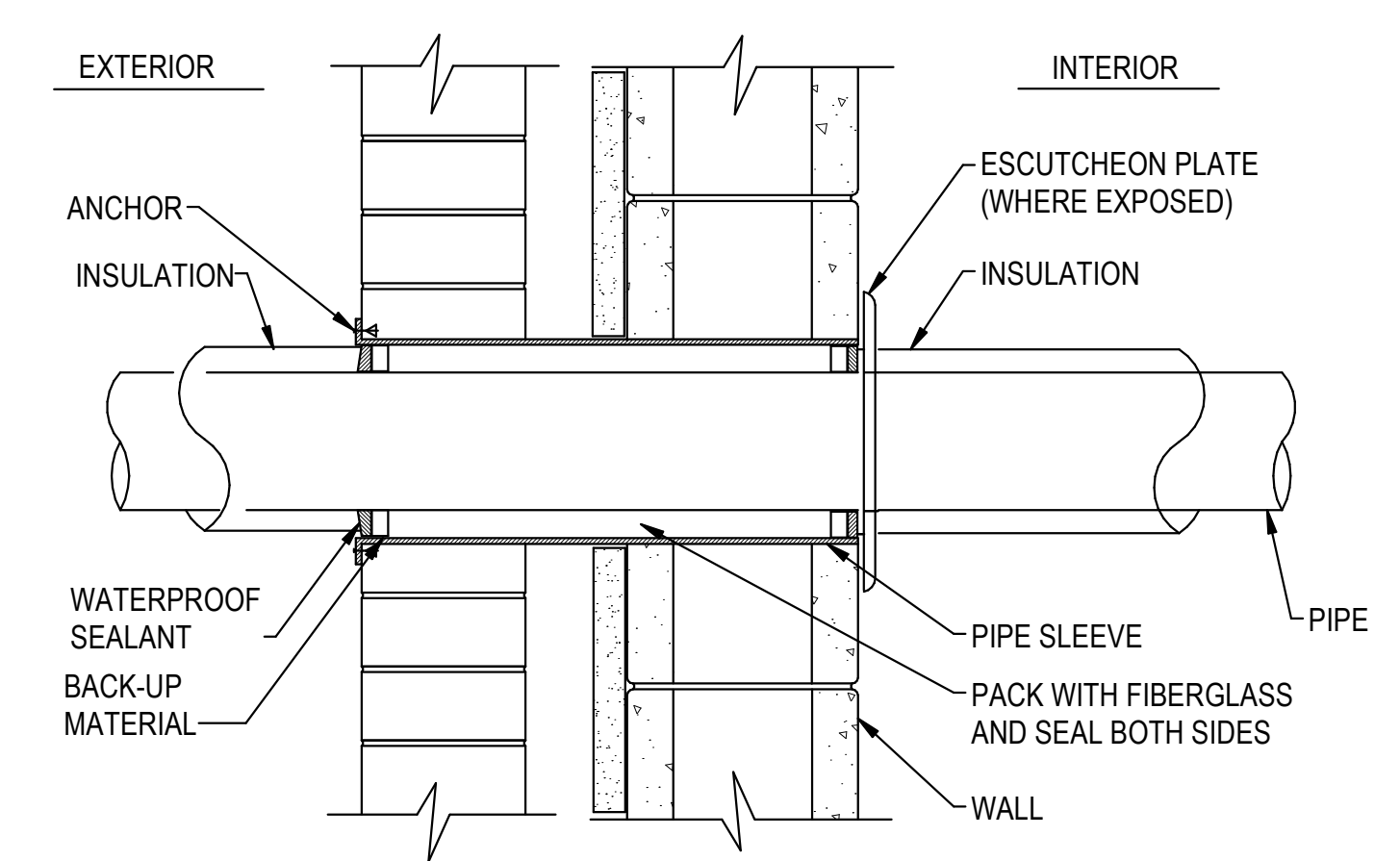
NOTES:
1. MOUNT GAUGE VERTICALLY FOR EASY READING. ON VERTICAL PIPE RUNS USE 90° SWEAT ELBOW AND 1/2" CU TUBING TO POSITION GAUGE VERTICALLY.



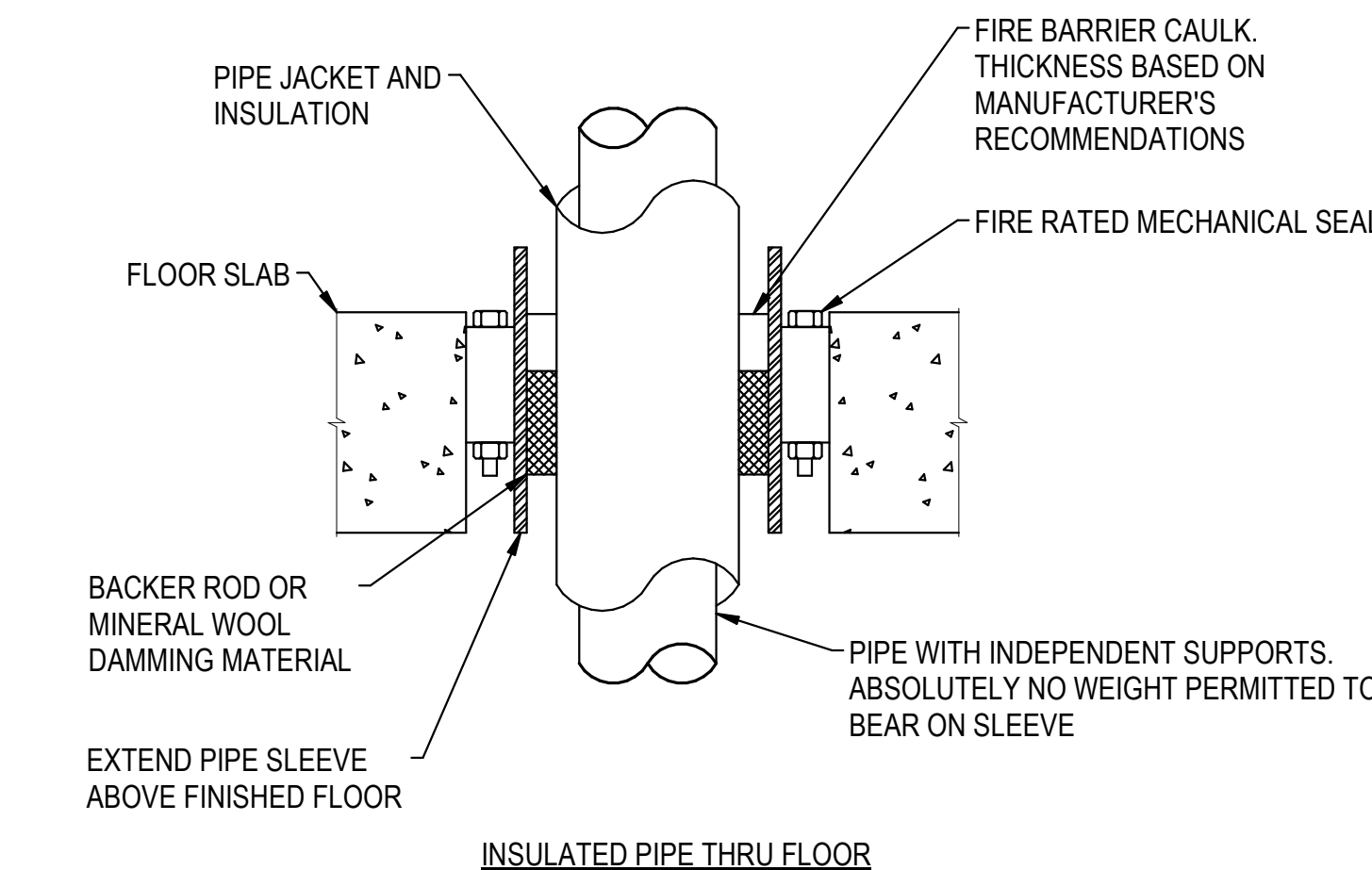
10 M - DETAIL - INFRARED HEATER INSTALLATION1
1/8" = 1'-0"



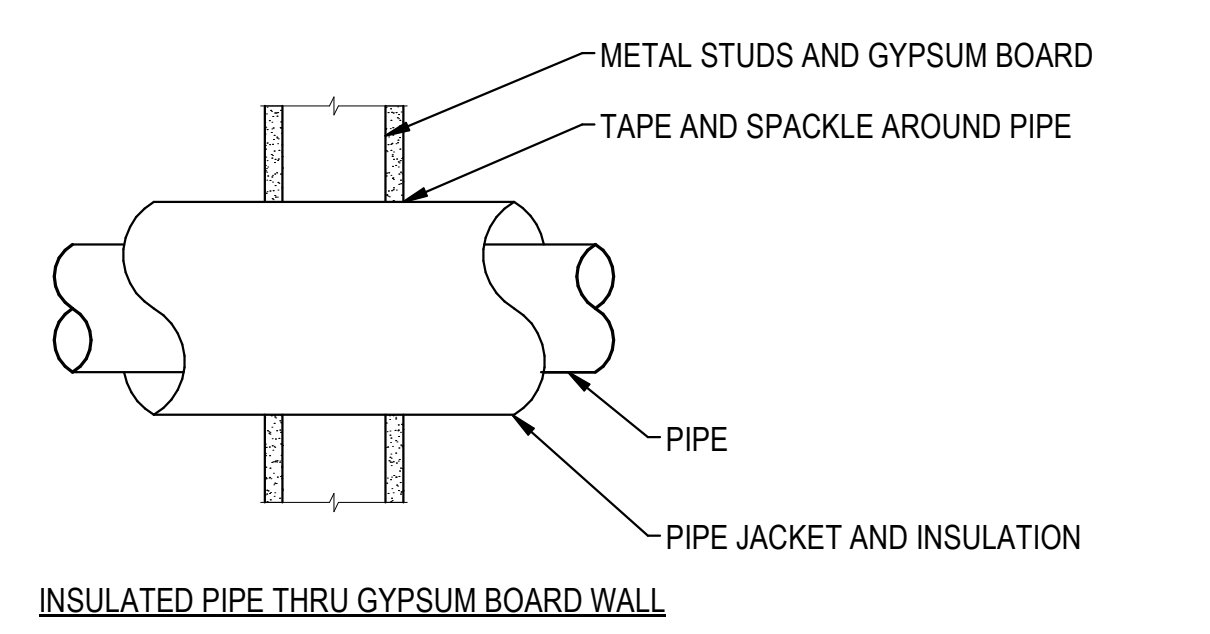
11 M - DETAIL - INFRARED HEATER INTAKE AND FLUE VENT1
1/8" = 1'-0"



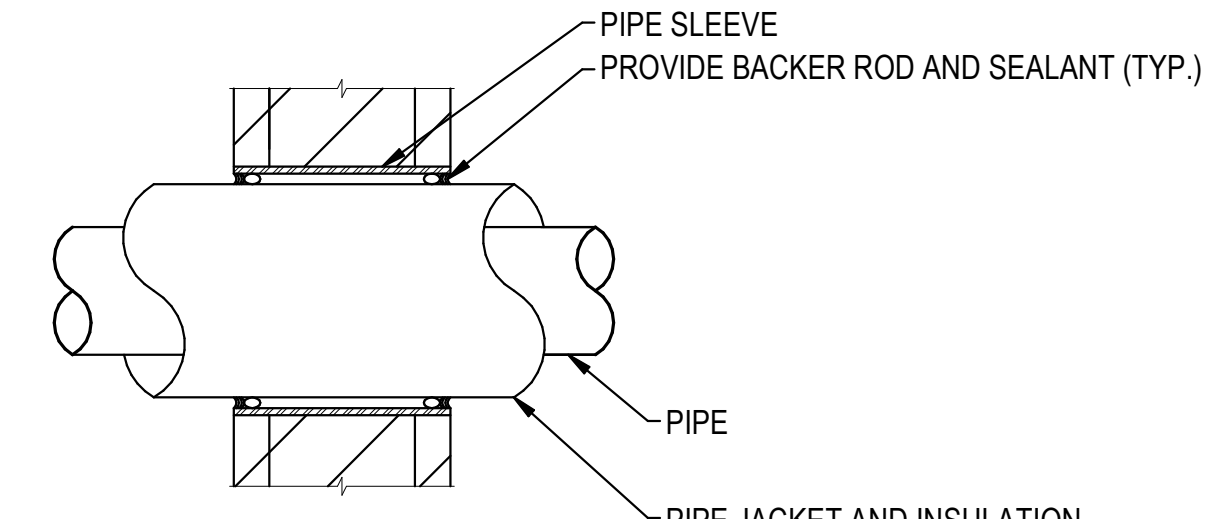
INSULATED PIPE WITH SLEEVE THRU EXTERIOR WALL



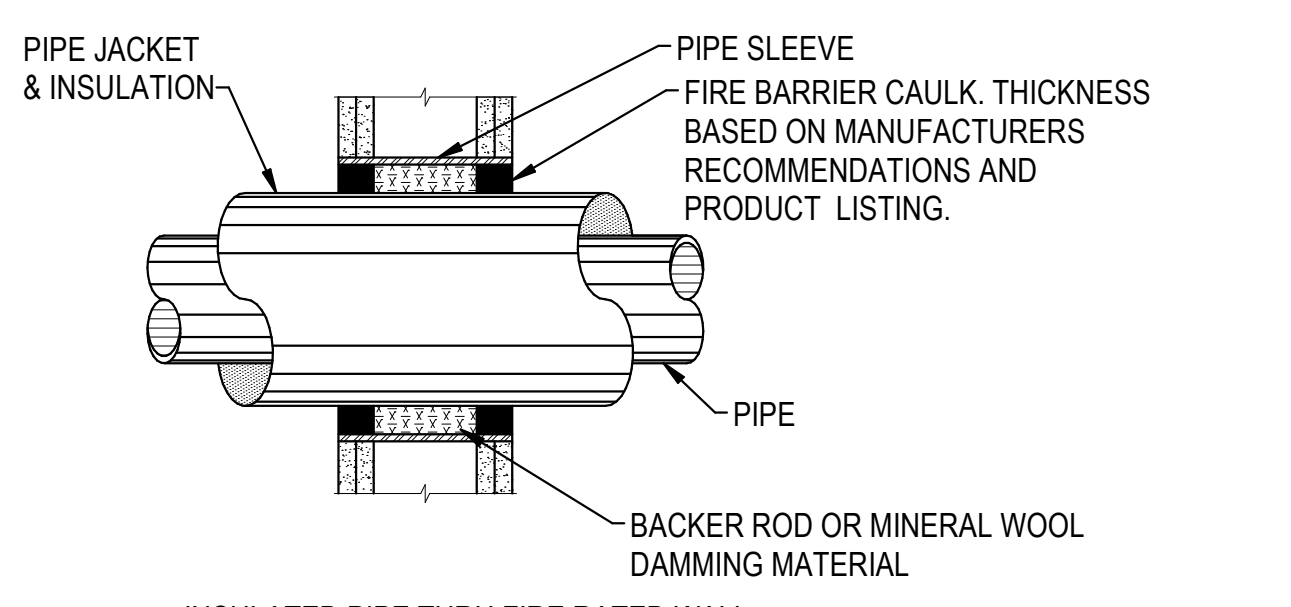
INSULATED PIPE THRU FLOOR



INSULATED PIPE THRU GYPSUM BOARD WALL



INSULATED PIPE THRU CMU WALL



INSULATED PIPE THRU FIRE RATED WALL

1 M - DETAIL - PIPE THROUGH WALL1
1/8" = 1'-0"

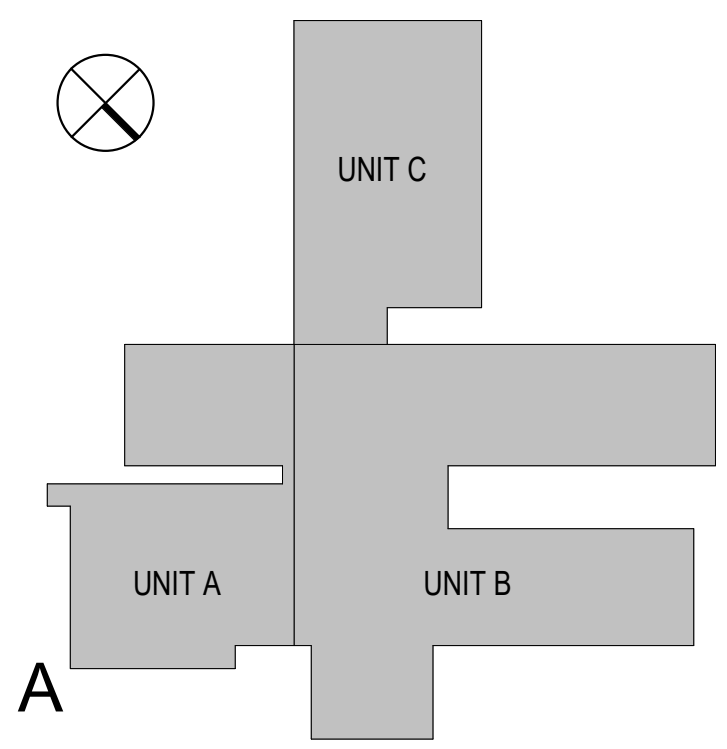
#	Revision	Date
---	----------	------

Project #: 17-700-078-2
Designed By: A.M.M
Drawn By: A.M.M
Checked By: D.E.B
Date: 04.12.2021



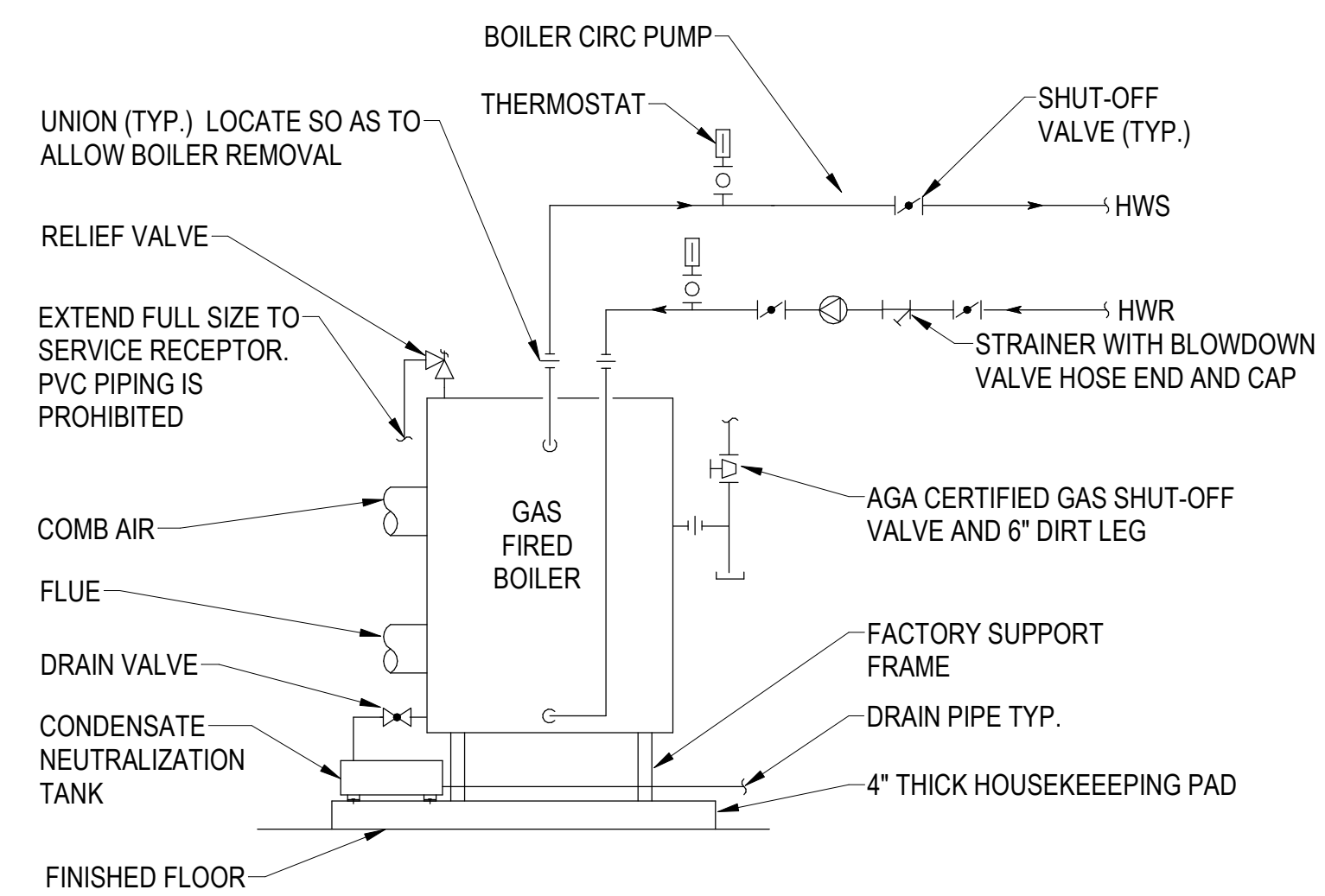
Eric Wepler

These Drawings and Specifications, and all copies thereof are and shall remain the property and copyright of the Architect. They shall be used only with respect to this Project and are not to be used on any other Project or Work without prior written permission from the Architect

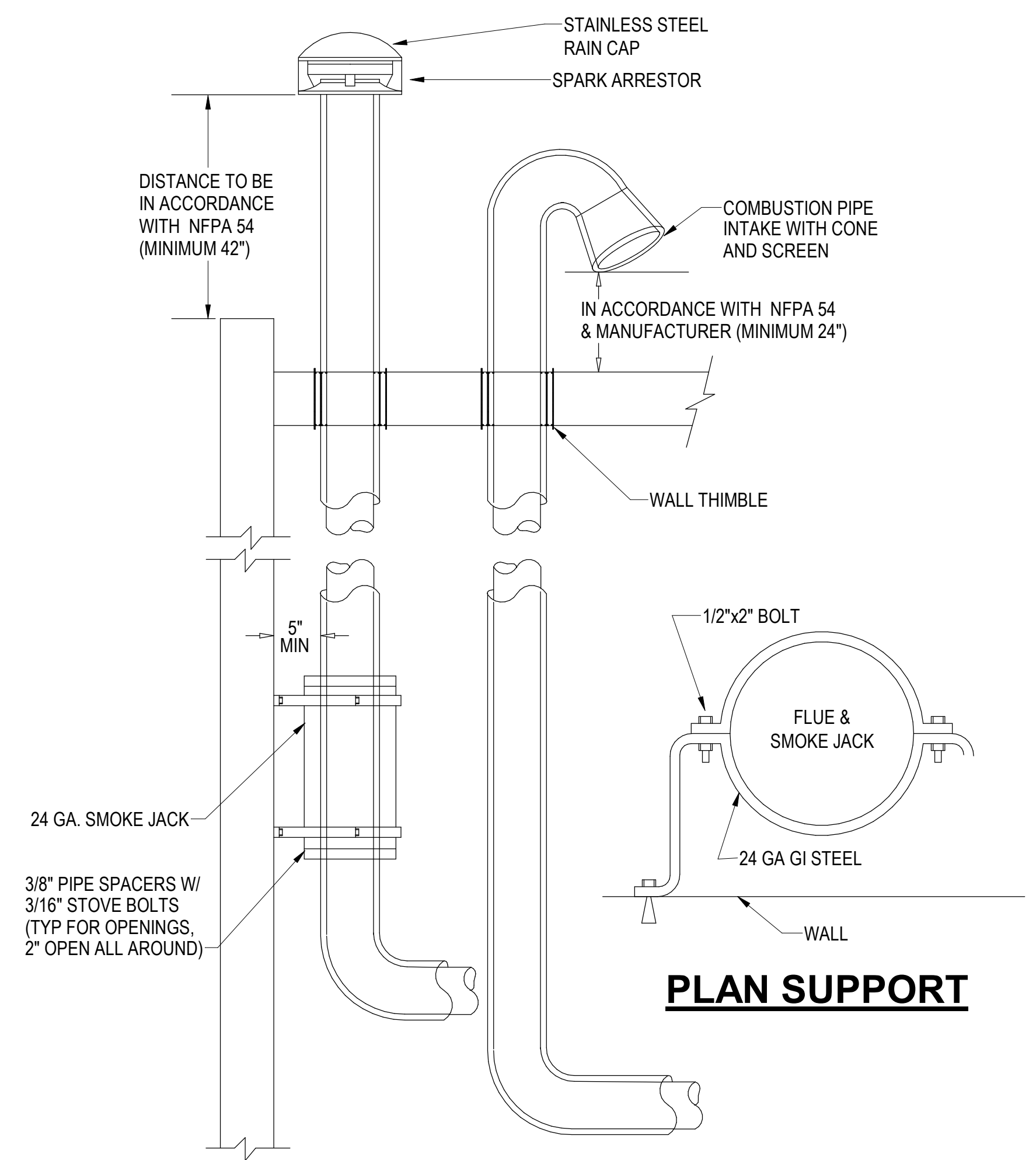


DETAILS - MECHANICAL

M804

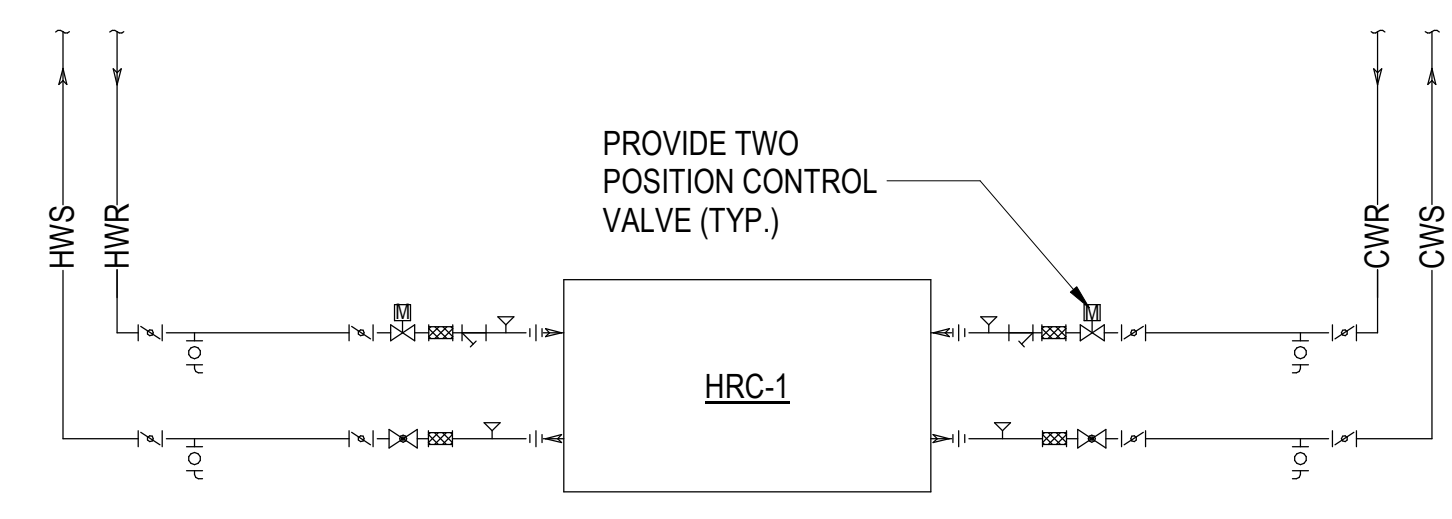


1 M804 M - DETAIL - BOILER INSTALLATION1
1/8" = 1'-0"

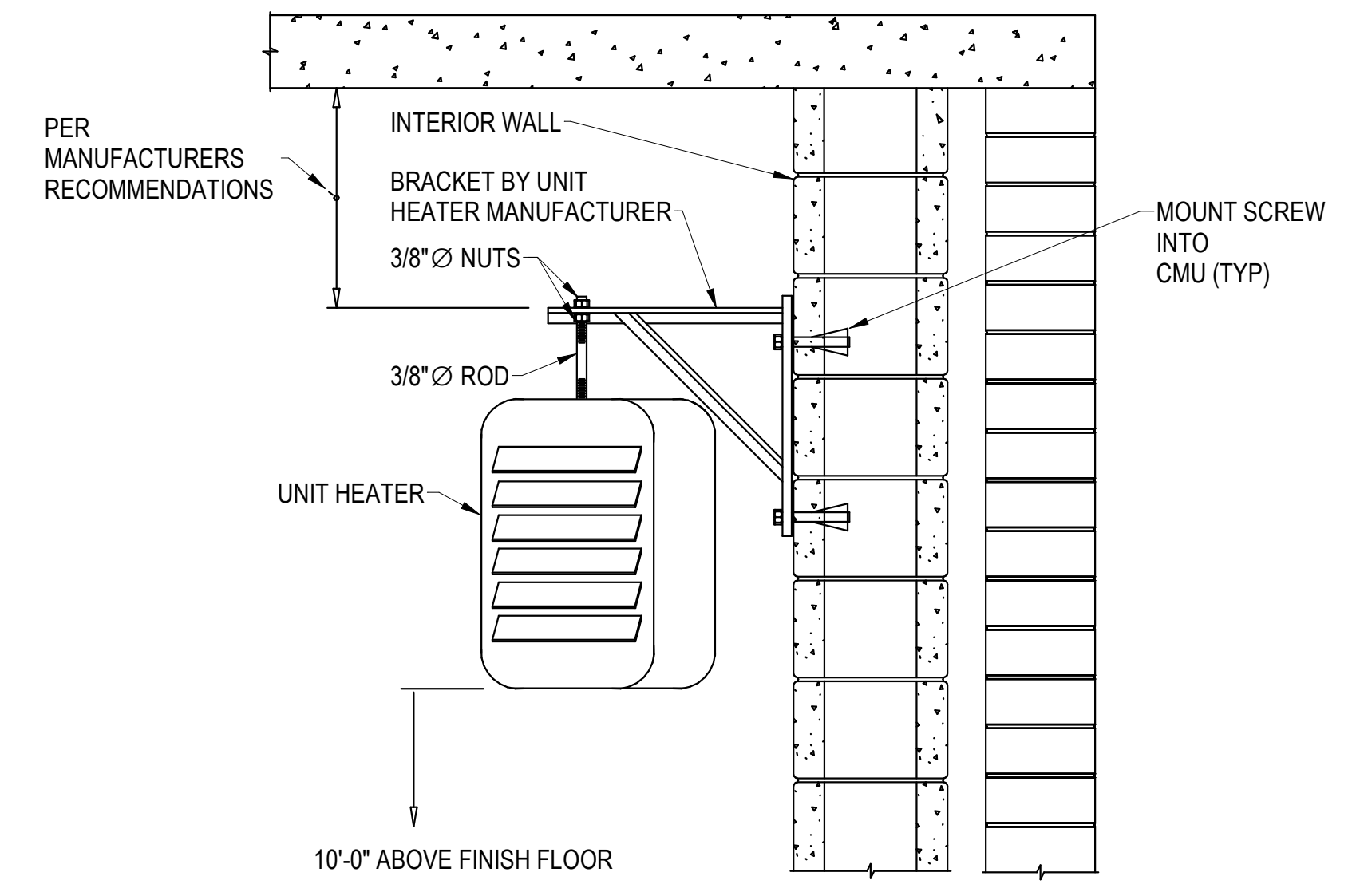


NOTES:
1. INSTALL PIPES IN ACCORDANCE WITH MANUFACTURER'S REQUIREMENTS FOR CLEARANCE BETWEEN INTAKE AND COMBUSTION FLUE PIPES. PIPES SHALL BE STAINLESS STEEL IN ACCORDANCE WITH MANUFACTURER'S REQUIREMENTS. INTAKE SHALL BE MINIMUM 24" ABOVE TOP OF ROOF MEMBRANE.

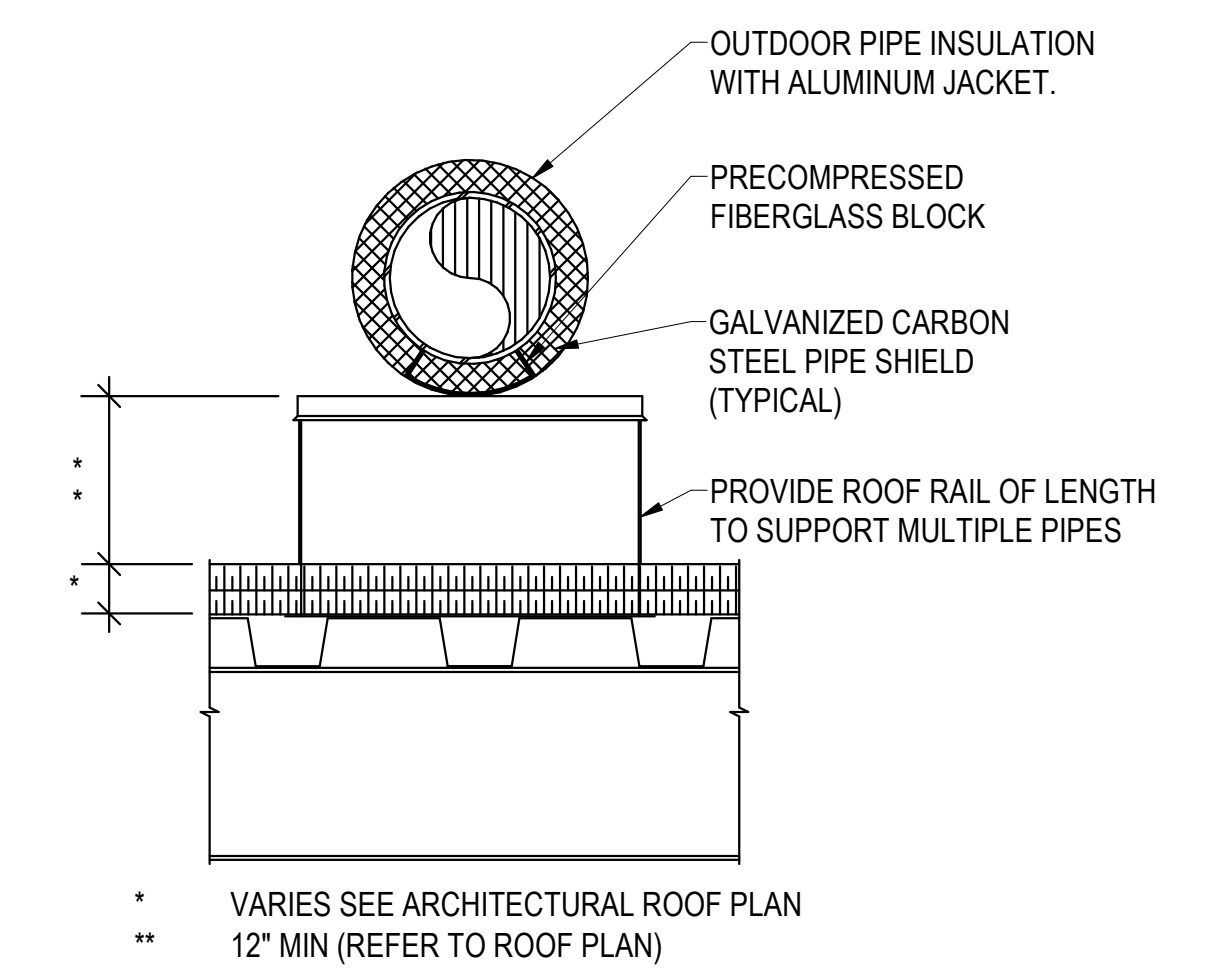
5 M804 M - DETAIL - FLUE STACK AND COMBUSTION AIR1
1/8" = 1'-0"



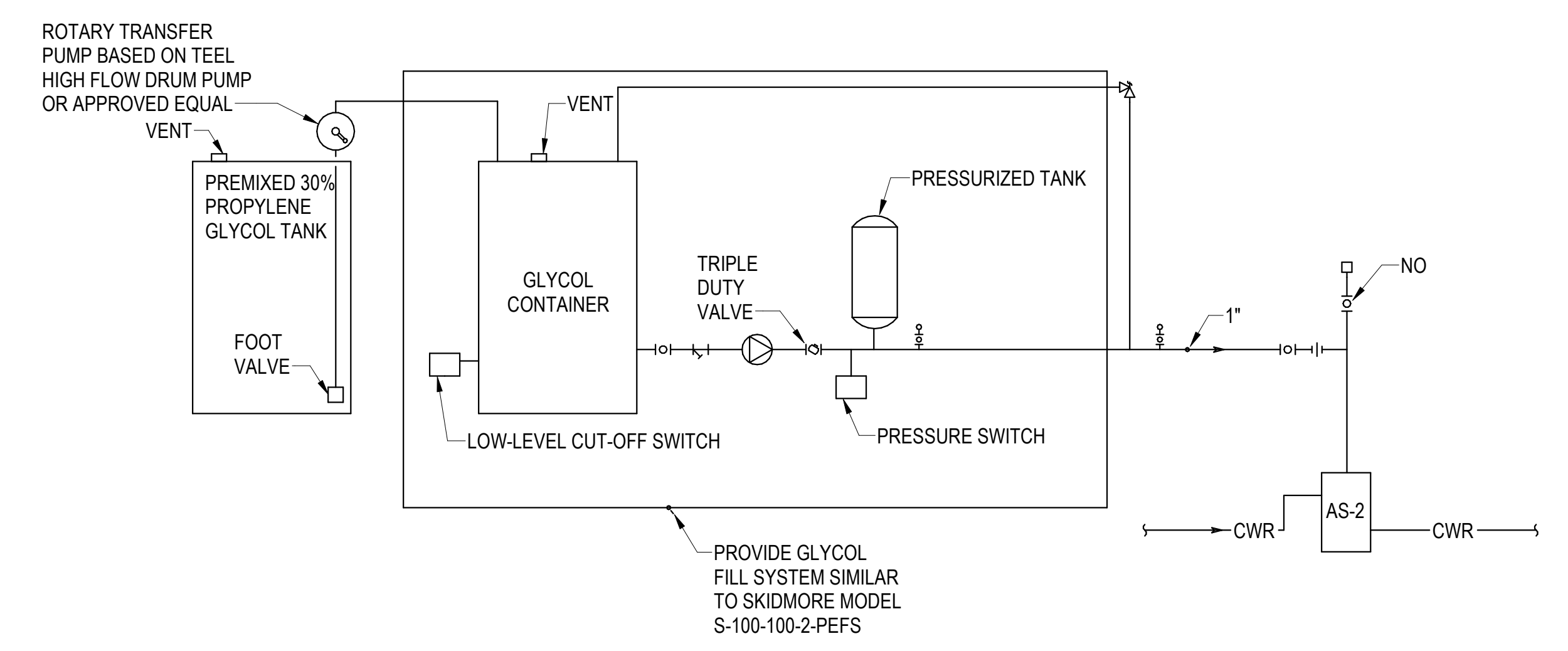
9 M804 M - DETAIL - HEAT RECOVERY CHILLER1
1/8" = 1'-0"



6 M804 M - DETAIL - UNIT HEATER SUPPORT1
1/8" = 1'-0"

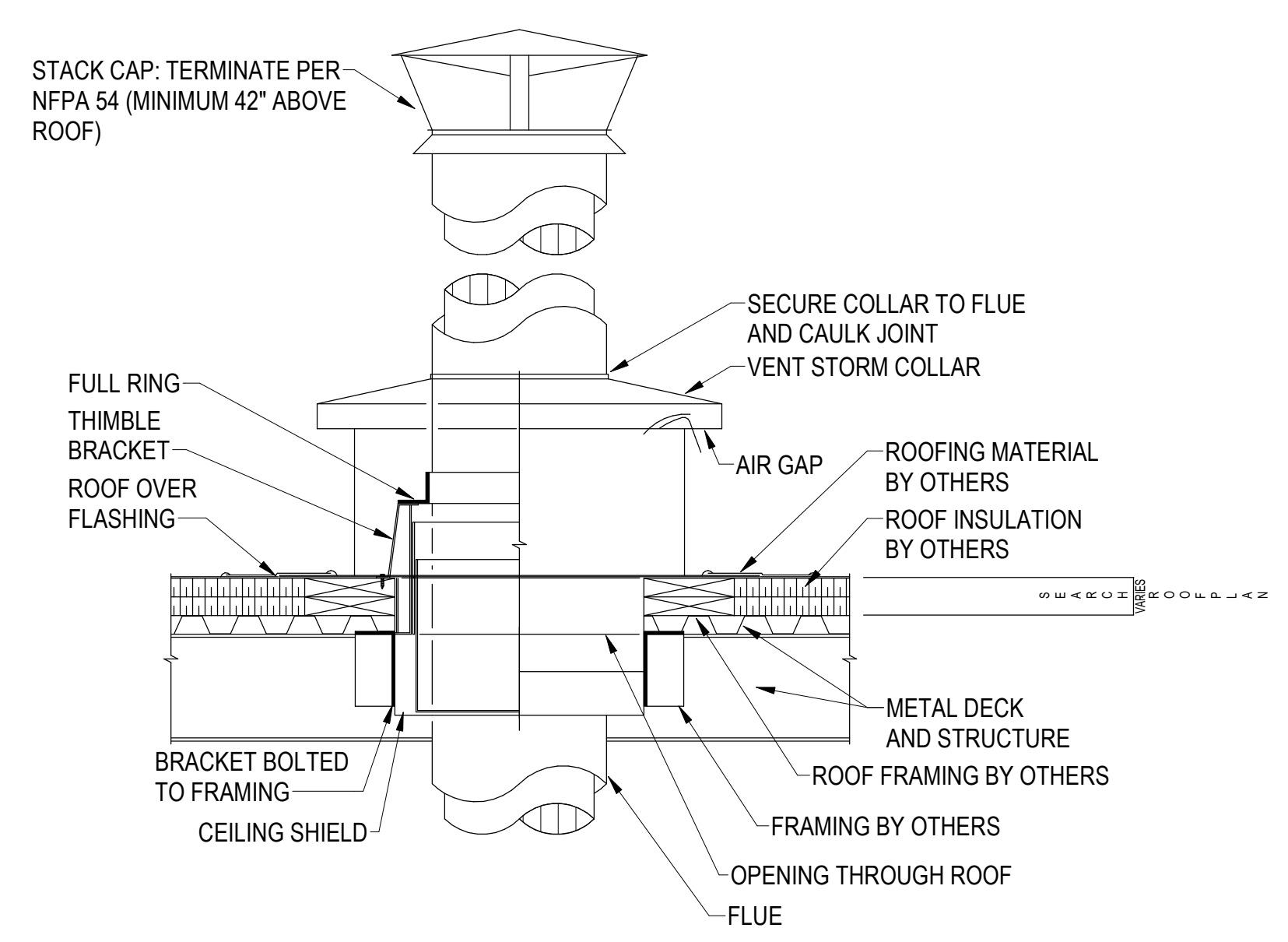


7 M804 M - DETAIL - PIPE ROOF SUPPORT1
1/8" = 1'-0"

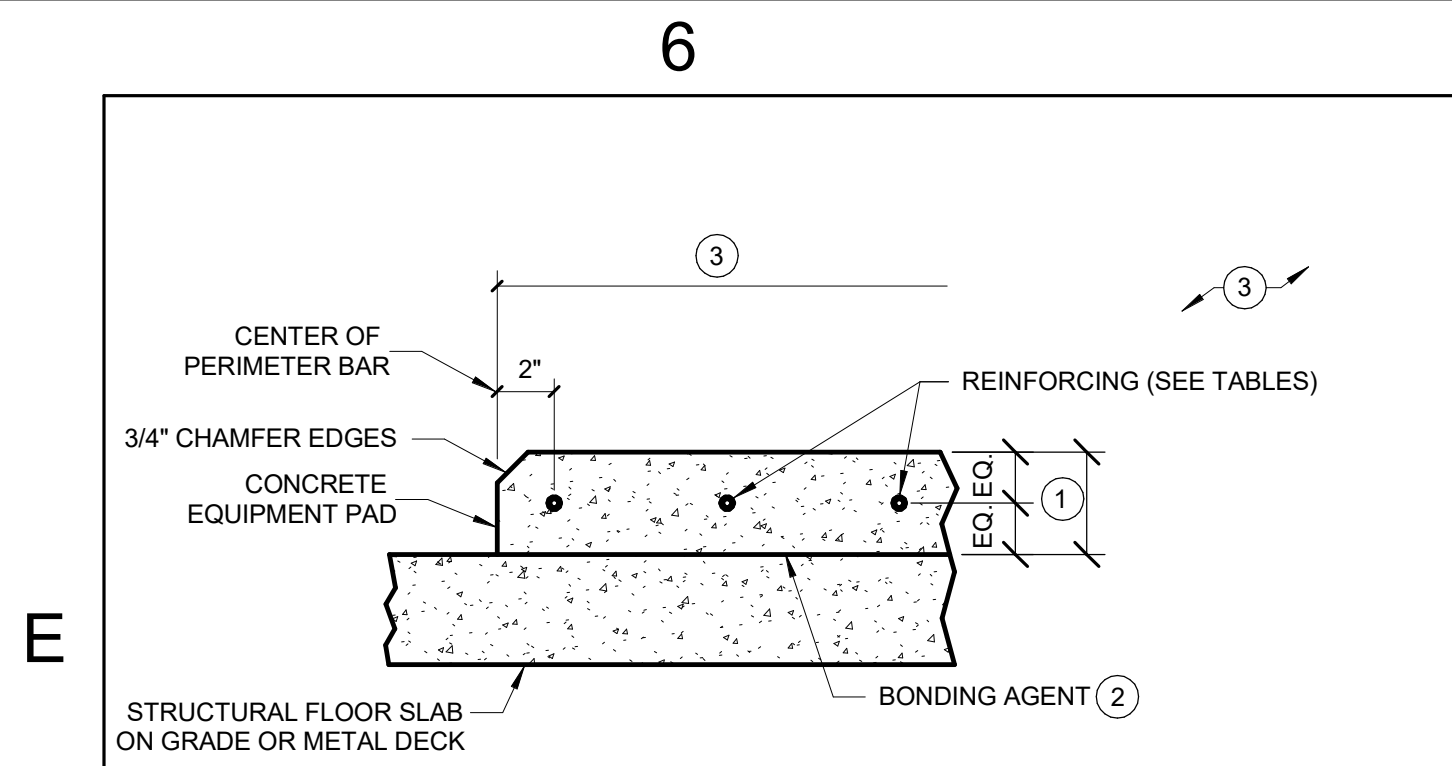


8 M804 M - DETAIL - AIR COOLED CHILLER1
1/8" = 1'-0"

11 M804 M - DETAIL - GLYCOL FILL SYSTEM - GLYCOL TANK AND EXPANSION TANK1
1/8" = 1'-0"



12 M804 M - DETAIL - BOILER FLUE ROOF PENETRATION1
1/8" = 1'-0"

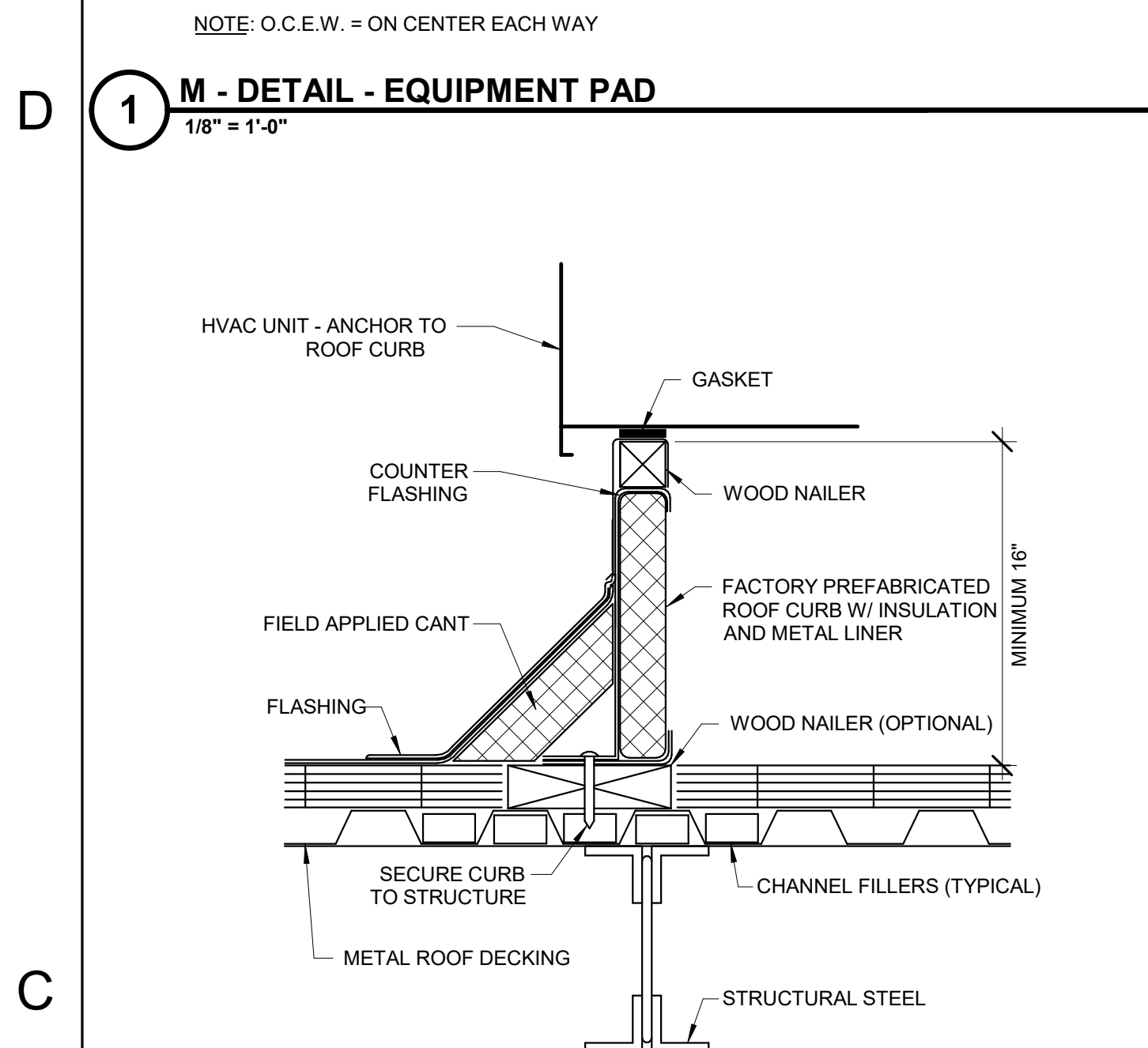


PLAN NOTES

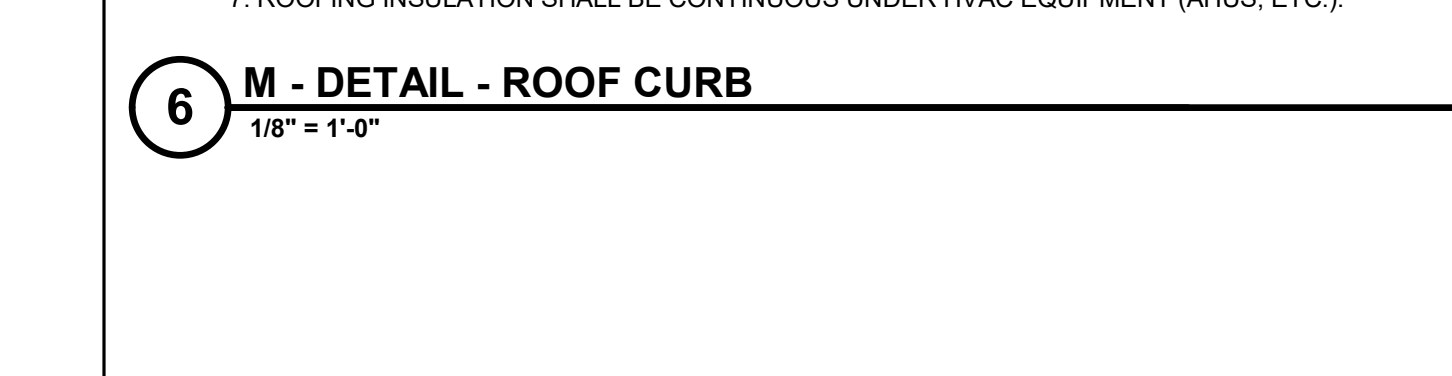
1. PROVIDE 3-1/2" HEIGHT OF EQUIPMENT PAD UNLESS INDICATED OTHERWISE ON THE DRAWINGS.
2. PROVIDE BONDING AGENT TO 100% OF THE CONTACT AREA BETWEEN THE EQUIPMENT PAD AND STRUCTURAL FLOOR. BONDING AGENT SHALL BE SBR LATEX AS MANUFACTURED BY THE EUCLID CHEMICAL COMPANY OR EQUAL. PREPARE SURFACES, MIX, APPLY BONDING COAT AND CURE PER MANUFACTURER'S WRITTEN INSTALLATION INSTRUCTIONS.
3. PAD SIZE TO BE A MINIMUM OF 6" LARGER THAN SIZE OF EQUIPMENT EACH DIRECTION (4" MINIMUM EACH SIDE OF EQUIPMENT TO EDGE OF PAD) WITH EQUIPMENT ANCHORS A MINIMUM OF 6" FROM EDGE OF PAD. VERIFY EDGE DISTANCE REQUIREMENTS FOR ACTUAL ANCHORS FROM MANUFACTURER'S WRITTEN LITERATURE.

AREA (FT²)	REINFORCING	MAX. LOAD (LB.)
UP TO 40	6 X 6 W2 9XW2 9 #3 ALONG PERIMETER	8,000
41 TO 100	#3 @ 12" O.C.E.W. #3 ALONG PERIMETER	15,000
101 TO 200	#4 @ 12" O.C.E.W. #4 ALONG PERIMETER	25,000
201 TO 400	#4 @ 12" O.C.E.W. #4 ALONG PERIMETER	50,000

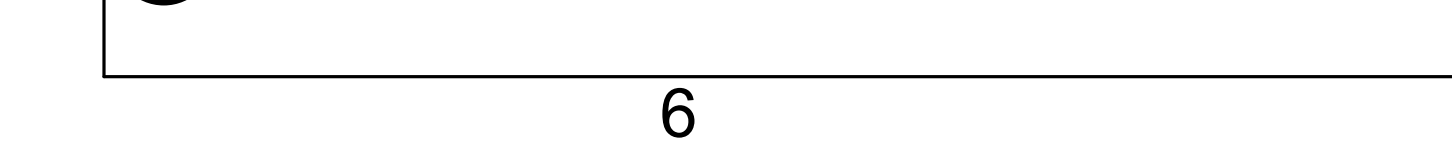
NOTE: O.C.E.W. = ON CENTER EACH WAY



- PLAN NOTES
1. ROOF CURBS SHALL BE FACTORY PREFABRICATED.
 2. ROOF CURBS SHALL BE PROVIDED WITH PROPERLY DESIGNED FABRICATED LEVELING DEVICES, SO THE HVAC EQUIPMENT IS LEVEL AND THE ROOF CURB INSTALLATION MATCHES THE ROOF SLOPE.
 3. PROVIDE PERIMETER ANGLE SUPPORT AT ALL SIDES BETWEEN STRUCTURAL STEEL. SIZE OF ANGLE SHALL BE BASED ON LOAD. MINIMUM SIZE SHALL BE 4"x4"x1/4".
 4. ROOF CURBS SHALL NOT REST ON ROOF INSULATION.
 5. RESLOPE ROOFING TO PROVIDE ADEQUATE DRAINAGE.
 6. COORDINATE LOCATION OF ROOF CURB WITH STRUCTURAL STEEL MEMBERS WHERE POSSIBLE.
 7. ROOFING INSULATION SHALL BE CONTINUOUS UNDER HVAC EQUIPMENT (AHUS, ETC.).



- PLAN NOTES
1. TANK CONNECTION LOCATIONS MAY VARY DEPENDING ON TANK SIZE AND MODEL. LOCATE TANK AS NOTED ON PLANS AND SEISMICALLY RESTRAIN AS REQUIRED.
 2. PROVIDE ANTI-THERMOSYPHON LOOP TO PREVENT GRAVITY HEATING OF TANK. 12" MIN. DROP.
 3. THE COLD WATER MAKE-UP. SEE PLUMBING DRAWINGS.
 4. "UV" STAMPED SAFETY VALVE WITH RESILIENT SEAT SUITABLE FOR 250°F WITH PACKED LEVER (LIQUID SERVICE ASME SECC VIII) AND STAINLESS SPRING (SIZED FOR 60 GPM OR 1200 PPH STEAM) AT 60°F (90% RATED AT 10% OVER PRESSURE). SET TO RELIEVE AT 125 PSIG OR AS REQUIRED BY ASME CODE. NO VALVES BETWEEN SAFETY VALVE AND VESSEL BEING PROTECTED. LOCATE SAFETY VALVE BELOW 10'-0" AFF.
 5. BYPASS TO BE SAME SIZE AS MAKE-UP LINE, 3/4" MINIMUM.
 6. 3/4" TOTALIZING FLOW METER (20 GPM CAPACITY) TO MEASURE MAKEUP WATER VOLUME. PROVIDE REMOTE-READING REGISTER FOR METER WHICH IS MORE THAN 8'-0" AFF.
 7. ROUTE DISCHARGE TO 6" ABOVE FLOOR DRAIN.
 8. ADJUSTMENT RANGE SHALL BE TANK PRECHARGE PLUS OR MINUS 10 PSIG.
 9. CHECK VALVE SHALL BE PROVIDED IN ADDITION TO THE BACKFLOW PREVENTOR ON THE PLUMBING SYSTEM.



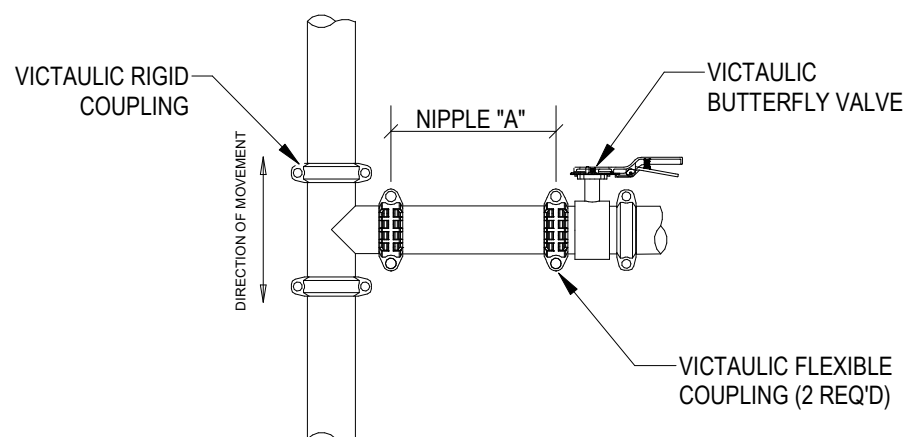
E

D

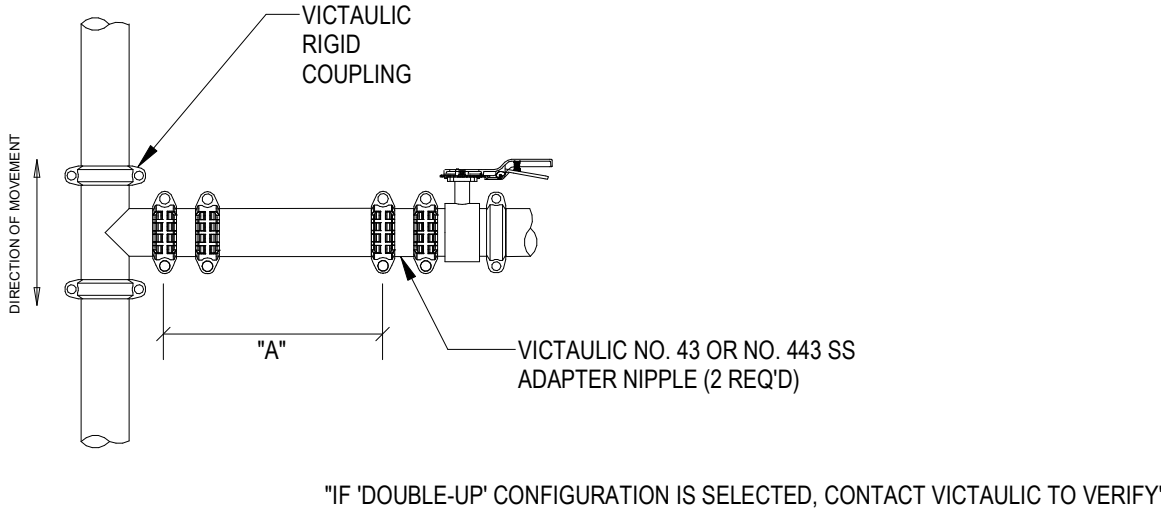
C

B

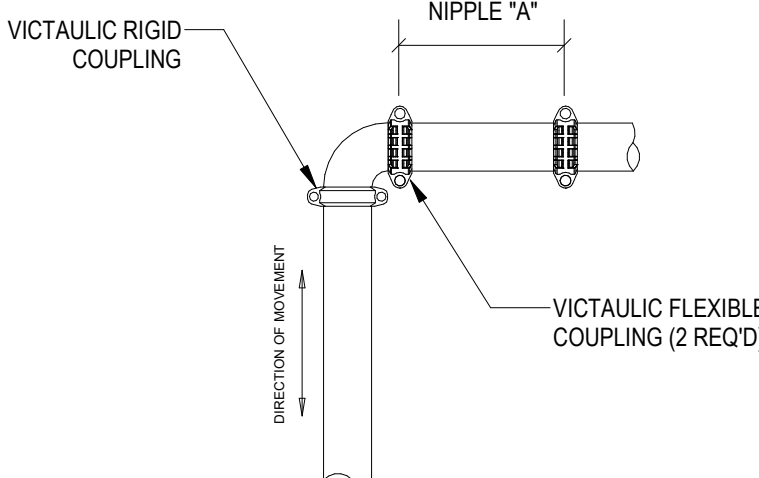
A



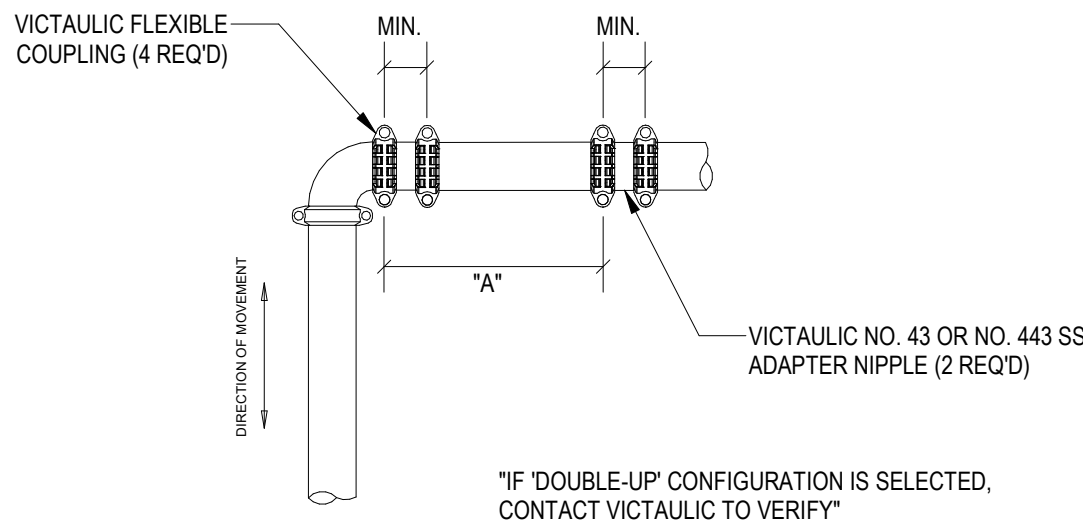
1 VICTAULIC FLEXIBLE BRANCH
1/8" = 1'-0"



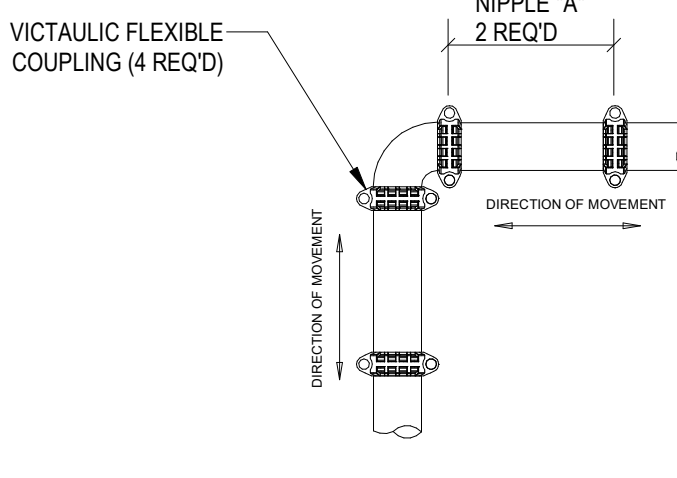
2 VICTAULIC "DOUBLED-UP" FLEXIBLE BRANCH
WHEN DIMENSION "A" IS TOO LONG PER DETAIL DH1
1/8" = 1'-0"



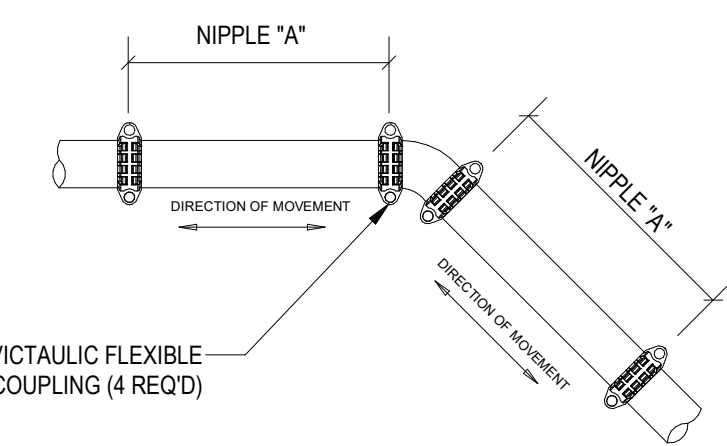
3 VICTAULIC FLEXIBLE OFFSET
1/8" = 1'-0"



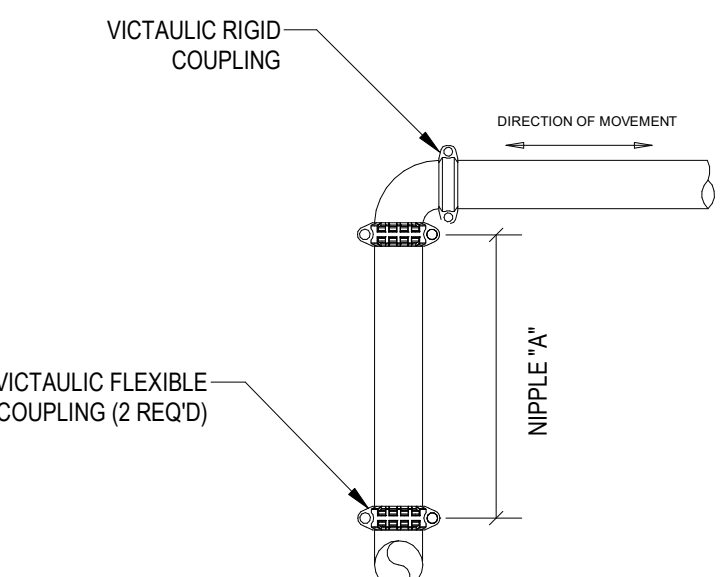
4 VICTAULIC "DOUBLED-UP" FLEXIBLE OFFSET
WHEN DIMENSION "A" IS TOO LONG PER DETAIL DH3
1/8" = 1'-0"



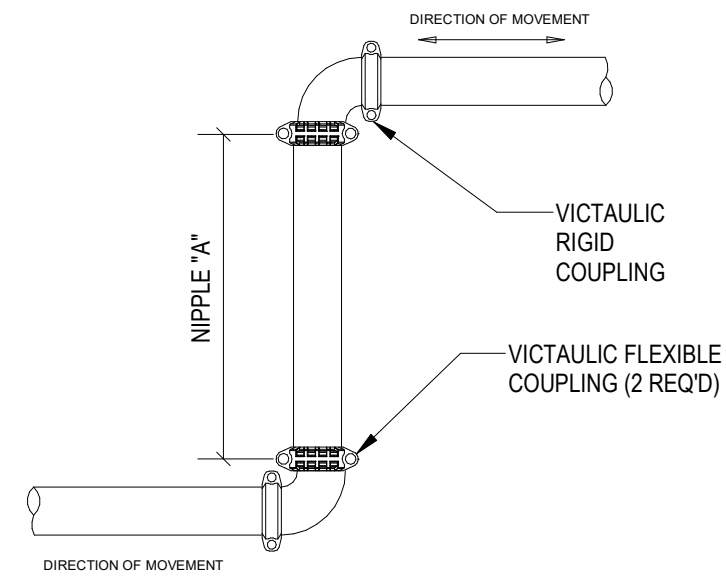
5 VICTAULIC FLEXIBLE OFFSET
BOTH SIDES OF ELBOW
1/8" = 1'-0"



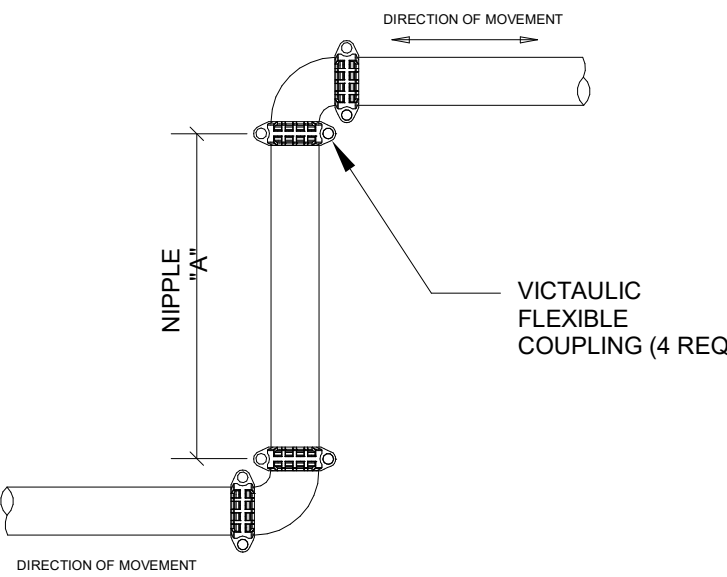
6 VICTAULIC FLEXIBLE DIRECTION CHANGE
- OBTUSE ANGLE
1/8" = 1'-0"



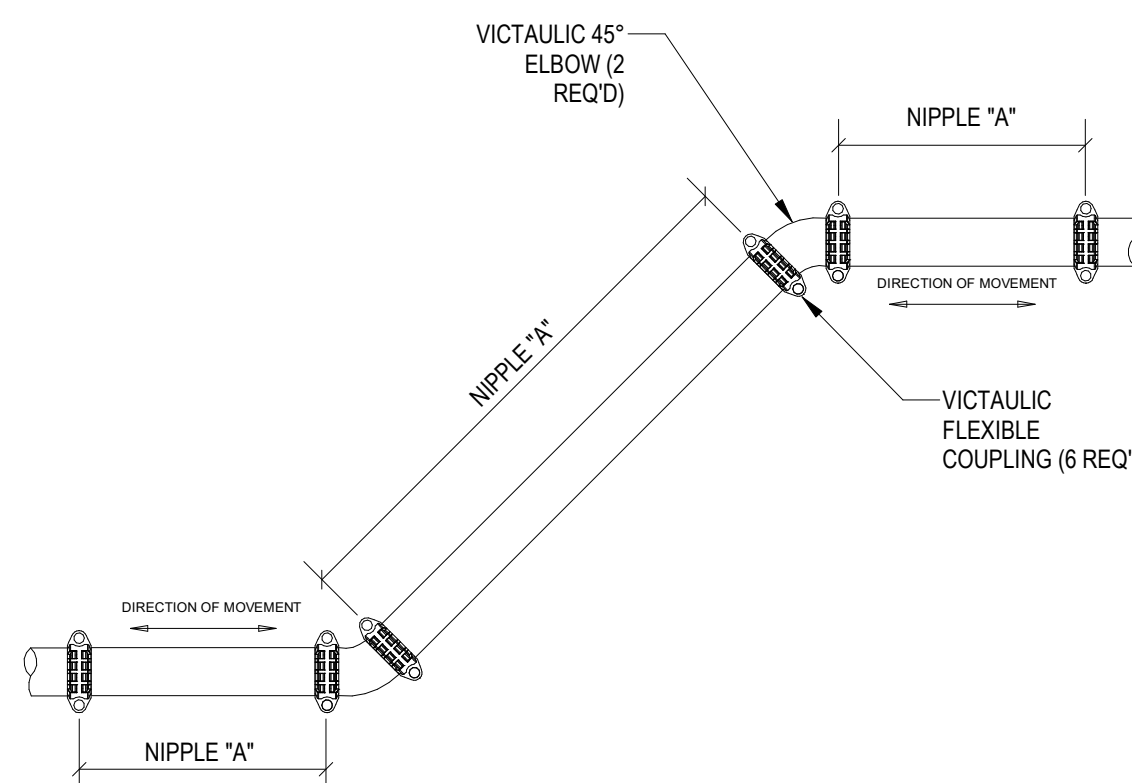
7 VICTAULIC CHANGE IN ELEVATION
1/8" = 1'-0"



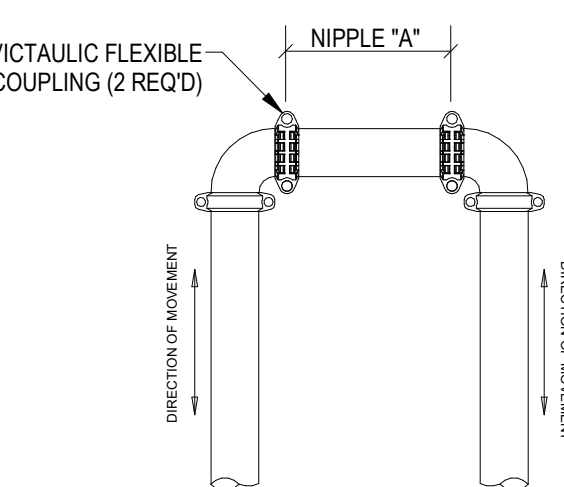
8 VICTAULIC "Z" TYPE CHANGE IN DIRECTION
1/8" = 1'-0"



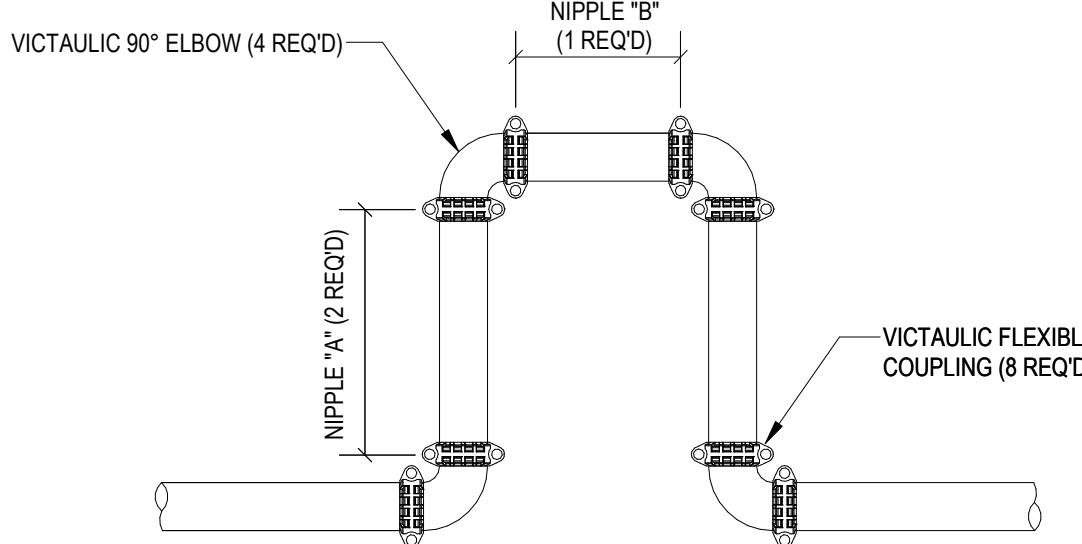
9 VICTAULIC "Z" TYPE CHANGE IN DIRECTION
1/8" = 1'-0"



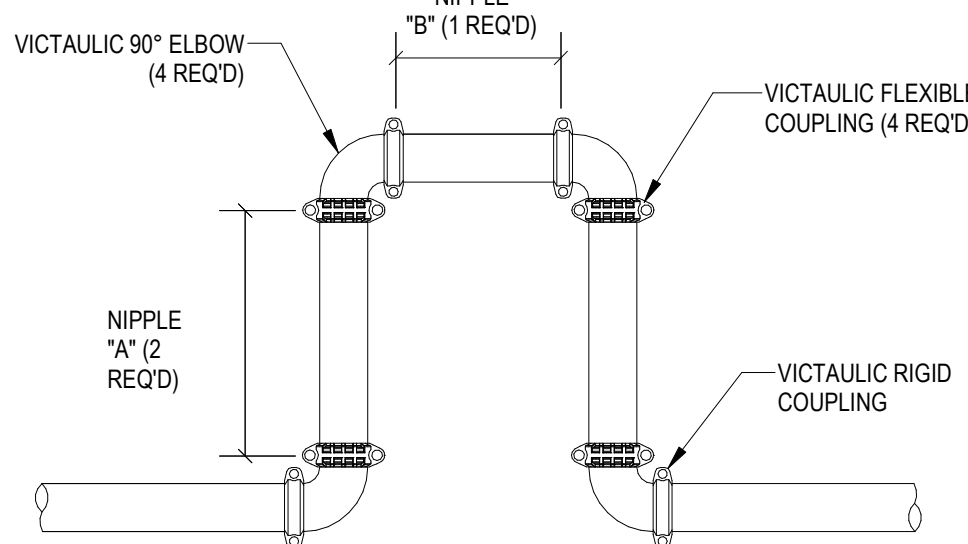
10 VICTAULIC 45° "Z" TYPE CHANGE IN DIRECTION
1/8" = 1'-0"



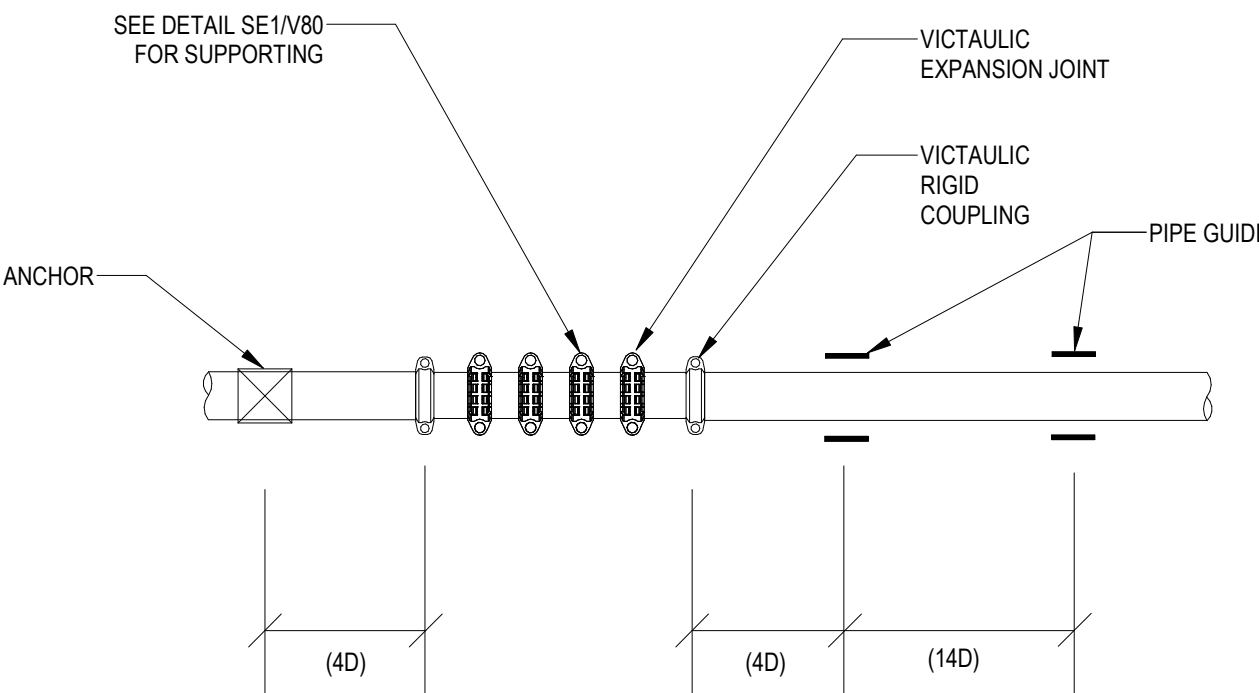
11 VICTAULIC FLEXIBLE ELBOW
1/8" = 1'-0"



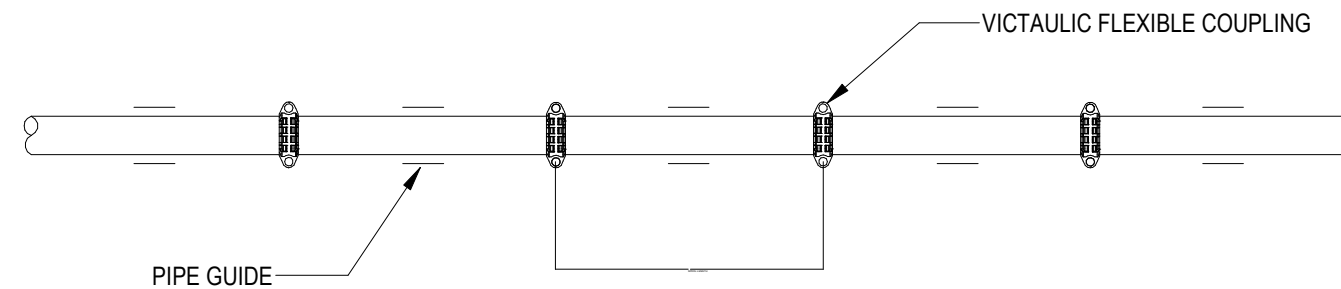
12 VICTAULIC FLEXIBLE EXPANSION LOOP 1
1/8" = 1'-0"



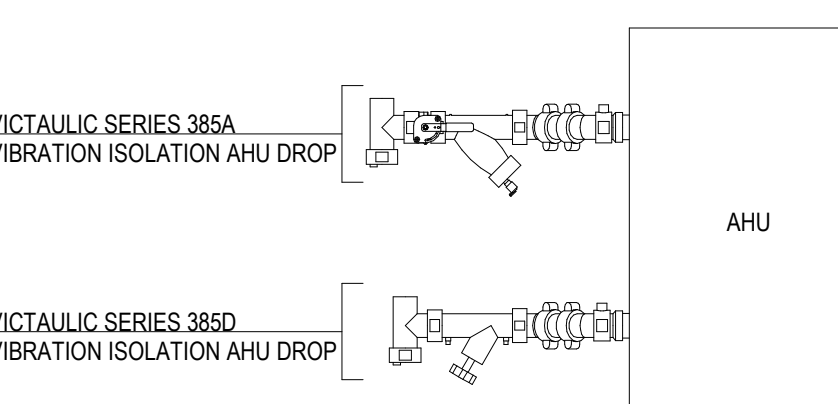
13 VICTAULIC FLEXIBLE EXPANSION LOOP 2
1/8" = 1'-0"



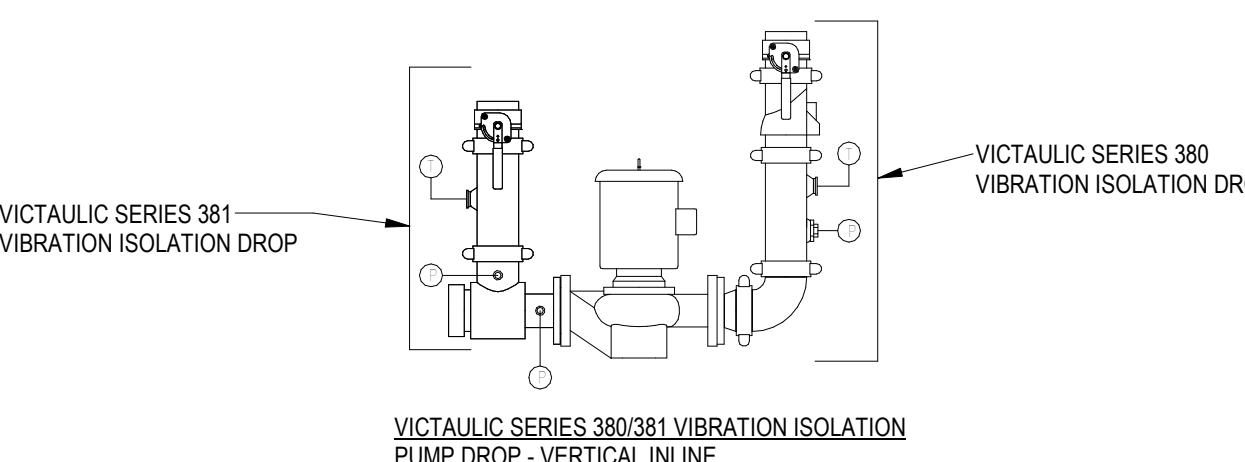
14 VICTAULIC EXPANSION JOINT GUIDING 1
1/8" = 1'-0"



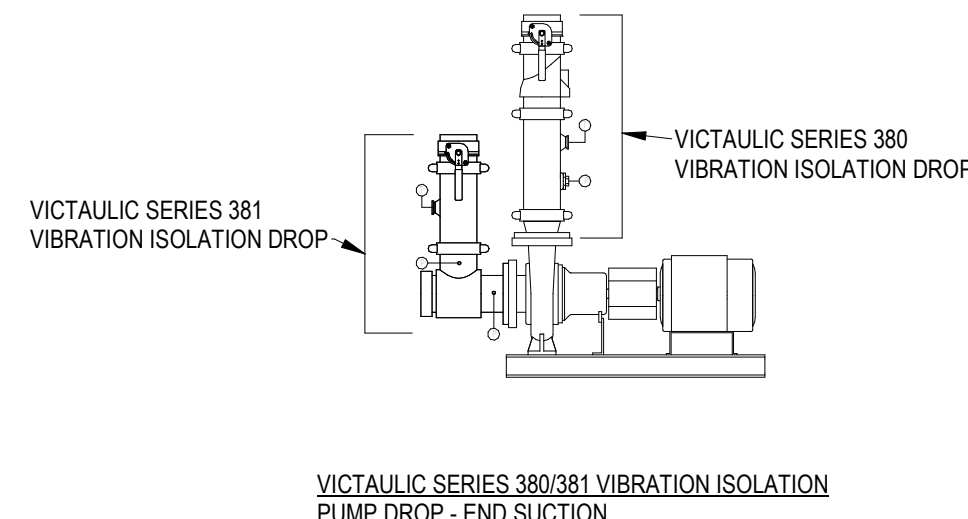
15 VICTAULIC FLEX COUPLINGS IN SERIES
1/8" = 1'-0"



16 AHU DROP SINGLE
1/8" = 1'-0"



17 PUMP DROP VERTICAL
1/8" = 1'-0"



18 PUMP DROP 380/381
1/8" = 1'-0"

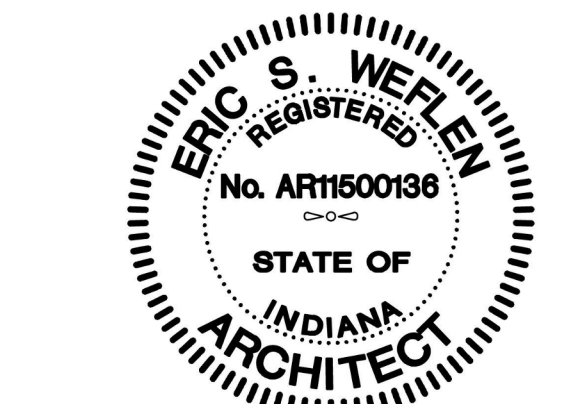
**FLEX COUPLING
SPACING AT OFFSETS
AND BRANCHES
(TYPICAL UNLESS
NOTED OTHERWISE)**

**HOT WATER S/R
INSTALL VICTAULIC
FLEXIBLE COUPLINGS
AS PER DETAIL AND
TABLE BELOW**

PIPE SIZE	"A" DIMENSION	STYLE
2" - 3"	3' - 0"	177N

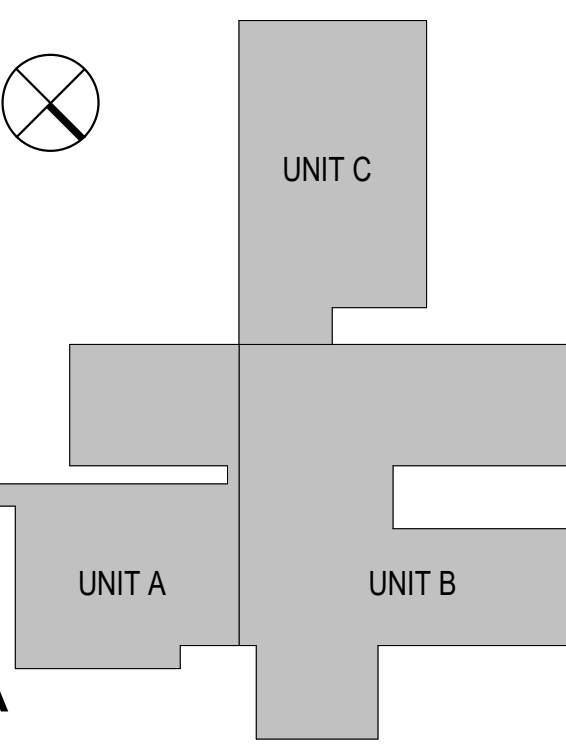
#	Revision	Date
---	----------	------

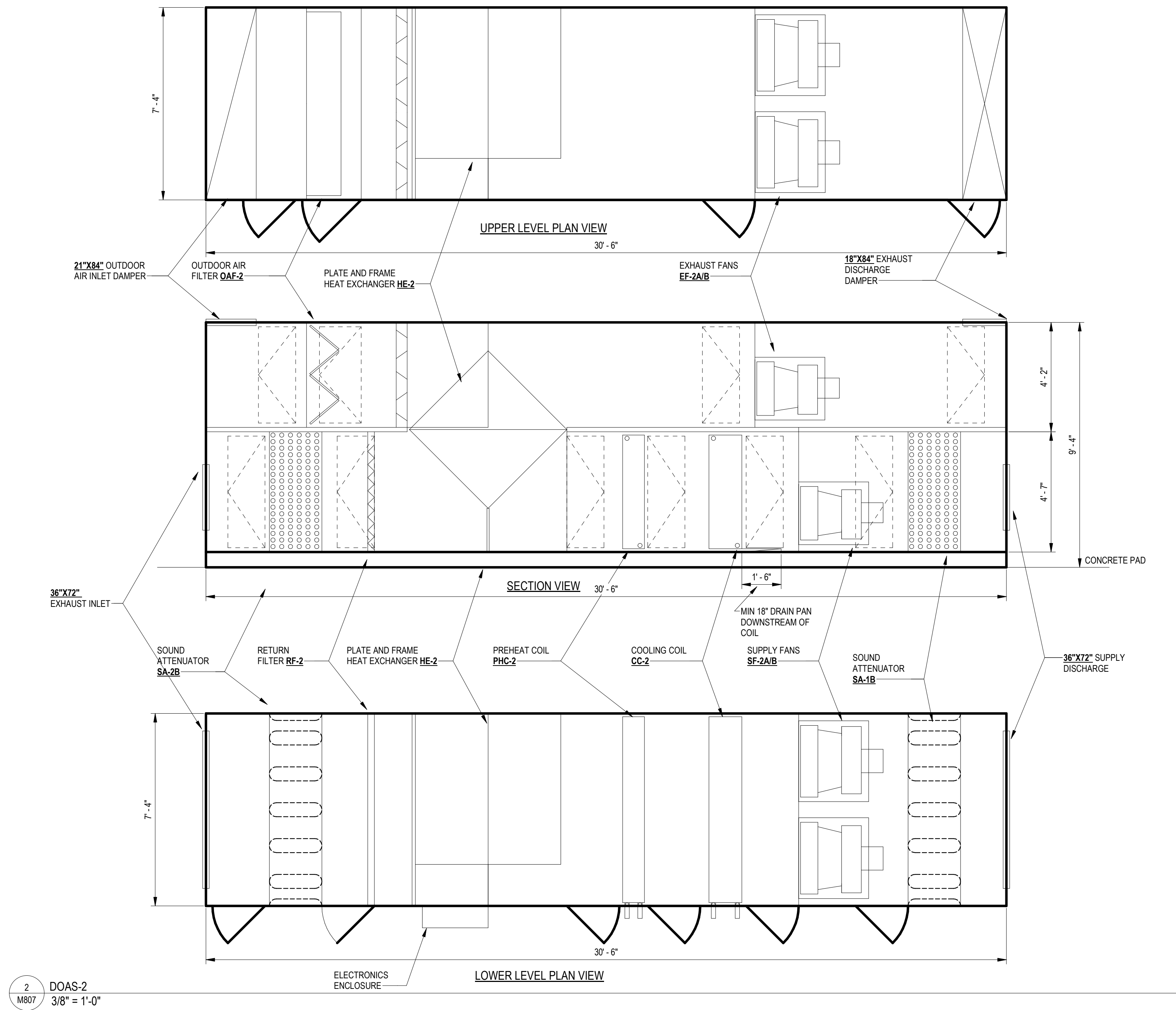
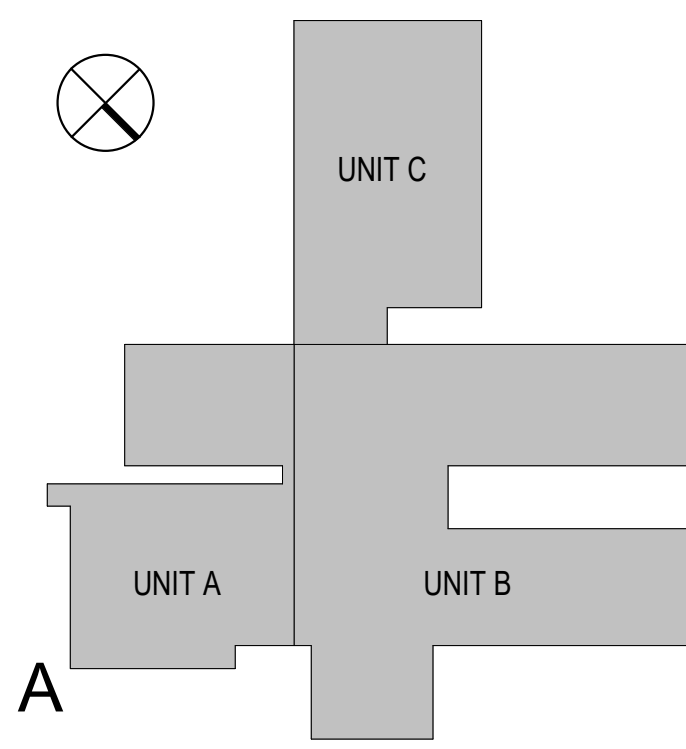
Project #: 17-700-078-2
Designed By: A.M.M
Drawn By: A.M.M
Checked By: D.E.B
Date: 04.12.2021



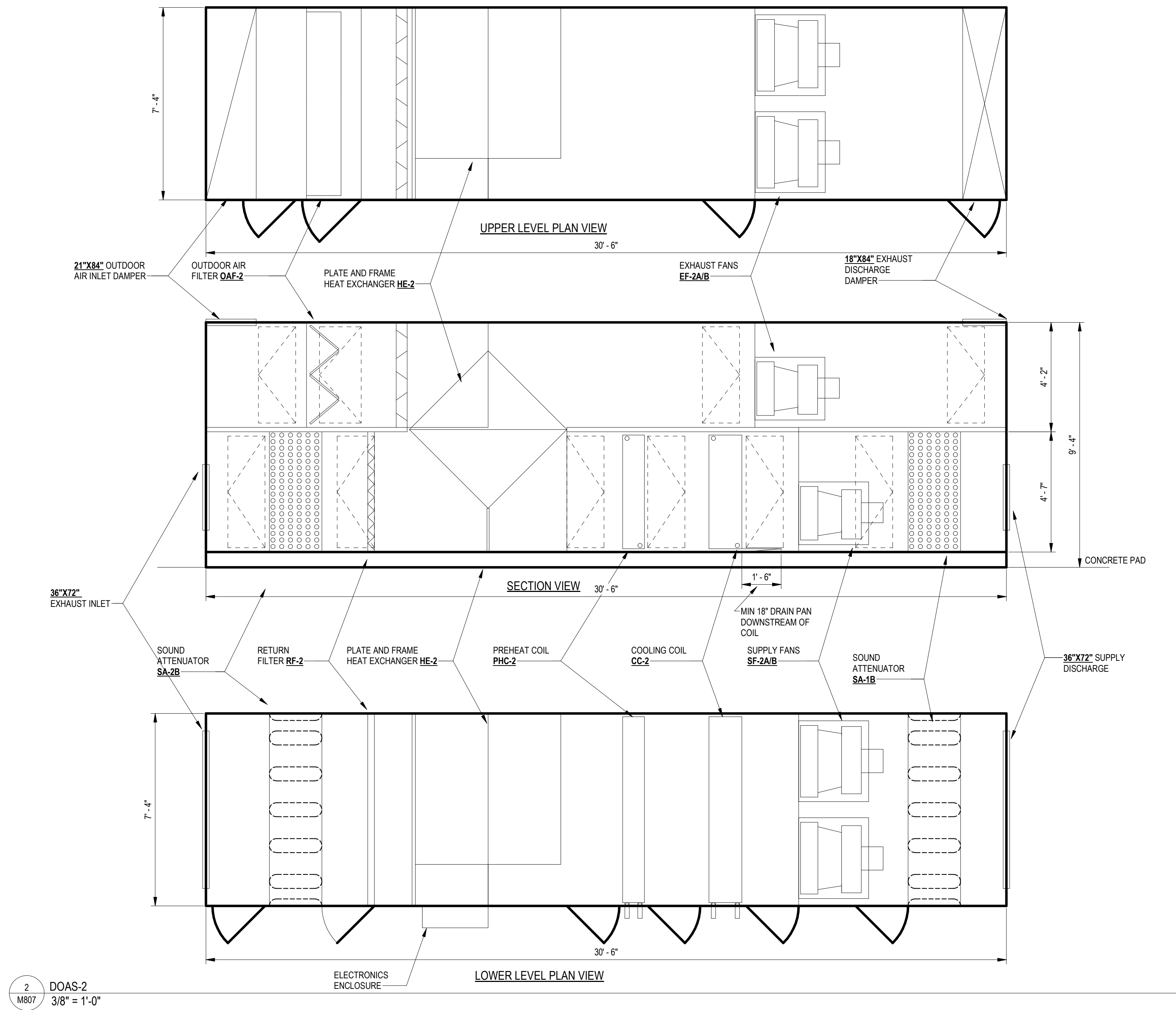
Eric Wefer

These Drawings and Specifications, and all copies thereof are and shall remain the property and copyright of the Architect. They shall be used only with respect to this Project and are not to be used on any other Project or Work without prior written permission from the Architect

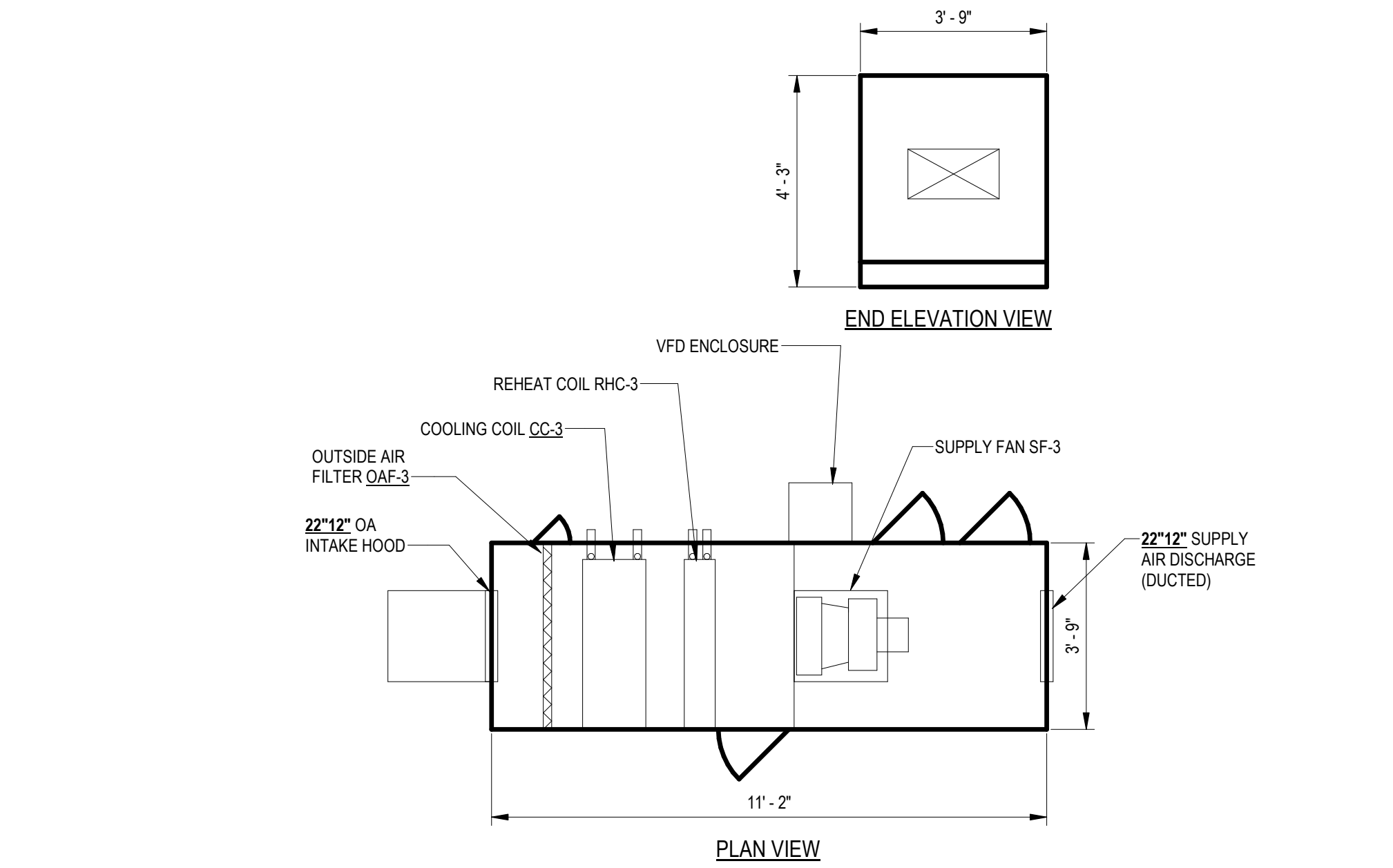




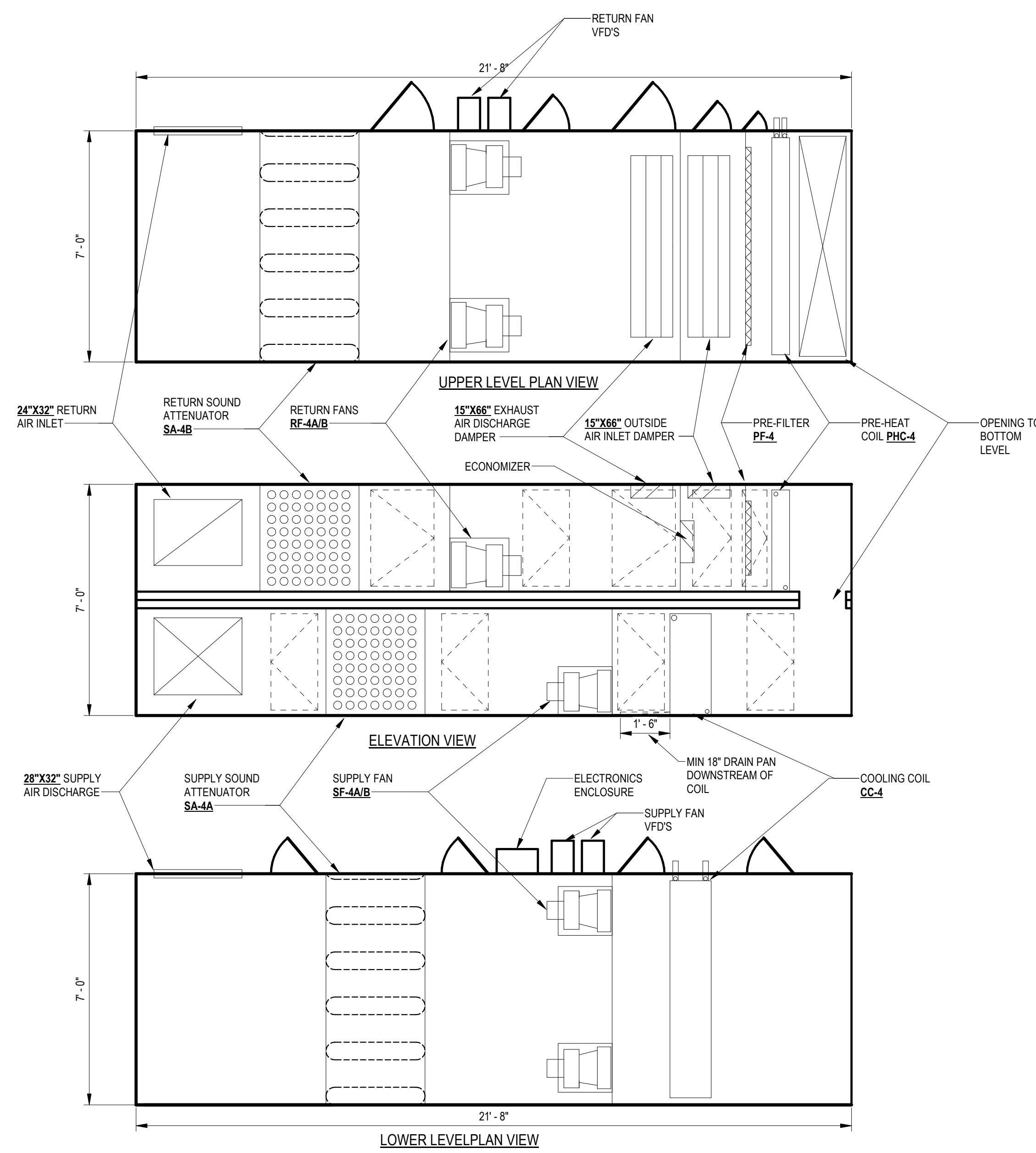
2 DOAS-2
3/8" = 1'-0"



1 DOAS-1
3/8" = 1'-0"



3 DOAS-3
3/8" = 1'-0"



4 AHU-1
3/8" = 1'-0"

1 DOAS-1
3/8" = 1'-0"

E

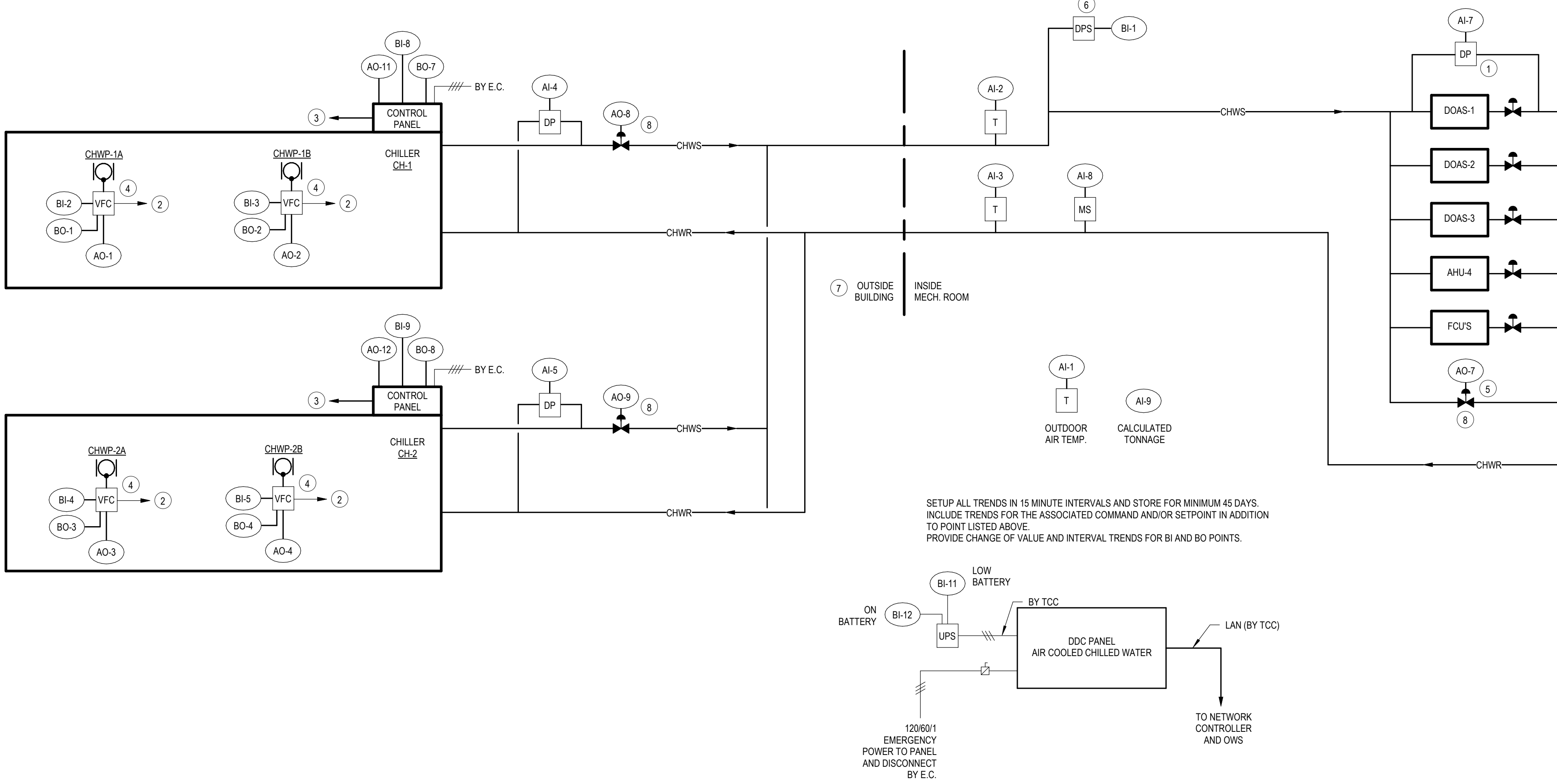
D

C

B

A

SCHEDULE OF DDC POINTS				
ID	DESCRIPTION	TREND	ALARM	GRAPHIC
AI-1	OUTDOOR AIR TEMPERATURE (GLOBAL)	X		X
AI-2	COMMON CHILLED WATER SUPPLY TEMPERATURE	X	X	X
AI-3	COMMON CHILLED WATER RETURN TEMPERATURE	X	X	X
AI-4	CHILLER CH-1 DIFFERENTIAL PRESSURE	X	X	X
AI-5	CHILLER CH-2 DIFFERENTIAL PRESSURE	X	X	X
AI-7	SYSTEM DIFFERENTIAL PRESSURE	X	X	X
AI-8	SYSTEM CHILLED WATER FLOW	X	X	X
AI-9	SYSTEM CHILLED WATER CALCULATED LOAD (VIRTUAL)	X	X	X
AO-1	CHILLED WATER PUMP CHWP-1A VFC	X		X
AO-2	CHILLED WATER PUMP CHWP-1B VFC	X		X
AO-3	CHILLED WATER PUMP CHWP-2A VFC	X		X
AO-4	CHILLED WATER PUMP CHWP-2B VFC	X		X
AO-7	CHILLED WATER SYSTEM PRESSURE BYPASS VALVE	X		X
AO-8	CHILLER CH-1 ISOLATION VALVE	X		X
AO-9	CHILLER CH-2 ISOLATION VALVE	X		X
AO-11	CHILLER CH-1 TEMPERATURE SETPOINT	X		X
AO-12	CHILLER CH-2 TEMPERATURE SETPOINT	X		X
BI-1	LOW LEVEL GLYCOL ALARM - CHILLED WATER		X	X
BI-2	CHILLED WATER PUMP CHWP-1A STATUS	X	X	X
BI-3	CHILLED WATER PUMP CHWP-1B STATUS	X	X	X
BI-4	CHILLED WATER PUMP CHWP-2A STATUS	X	X	X
BI-5	CHILLED WATER PUMP CHWP-2B STATUS	X	X	X
BI-8	CHILLER CH-1 GENERAL ALARM	X	X	X
BI-9	CHILLER CH-2 GENERAL ALARM	X	X	X
BI-11	UPS LOW BATTERY ALARM	X	X	X
BI-12	UPS ON BATTERY ALARM	X	X	X
BO-1	CHILLED WATER PUMP CHWP-1A START-STOP			X
BO-2	CHILLED WATER PUMP CHWP-1B START-STOP			X
BO-3	CHILLED WATER PUMP CHWP-2A START-STOP			X
BO-4	CHILLED WATER PUMP CHWP-2B START-STOP			X
BO-7	CHILLER CH-1 ENERGIZE/DEENERGIZE	X		X
BO-8	CHILLER CH-2 ENERGIZE/DEENERGIZE	X		X



1 AIR COOLED CHILLED WATER CONTROL SCHEMATIC
NOT TO SCALE

- PLAN NOTES
- LOCATED AT DOAS-2 HARDWIRE TO CHILLER DDC PANEL.
 - THRU THE LAN, VFC TO TRANSMIT TO BAS STATUS AND ALARMS OF ALL DATA AVAILABLE. VFC SUPPLIER TO FURNISH INTEGRAL COMMUNICATION CARD. TCC TO MAP ALL OWNER REQUESTED INFORMATION POINTS. NOTE THAT START/STOP SIGNAL, STATUS AND SPEED CONTROL ARE HARD WIRED TO DDC TO ENSURE OPERATION ON LOSS OF LAN.
 - THRU THE LAN, CHILLER TO TRANSMIT TO BAS STATUS AND ALARMS OF ALL DATA AVAILABLE. CHILLER SUPPLIER TO FURNISH INTEGRAL COMMUNICATION CARD. TCC TO MAP ALL OWNER REQUESTED INFORMATION POINTS. NOTE THAT ENERGIZABLE SIGNAL AND TEMPERATURE SET POINT ARE HARD WIRED TO DDC TO ENSURE OPERATION ON LOSS OF LAN.
 - VFC DRIVES FURNISHED, INSTALLED, WIRED AND COMMISSIONED BY MANUFACTURER. VFC SHALL MODULATE FROM THE CONTROLLED EQUIPMENT MANUFACTURERS REQUIRED SAFE OPERATING MAXIMUM AND MINIMUM SPEED.
 - CHILLED WATER BYPASS VALVE SIZED FOR XXX GPM AT XX PSID.
 - LOW GLYCOL ALARM SENSING POINT TO BE AT HIGHEST ELEVATION IN SYSTEM.
 - TCC TO NOTE THAT CHILLERS ARE AIR COOLED AND INSTALLED OUTDOORS. ANY TEMPERATURE CONTROLS INSTALLED OUTDOORS SHALL BE PROTECTED IN WEATHERPROOF ENCLOSURE.
 - VALVE TO FAIL LAST POSITION.

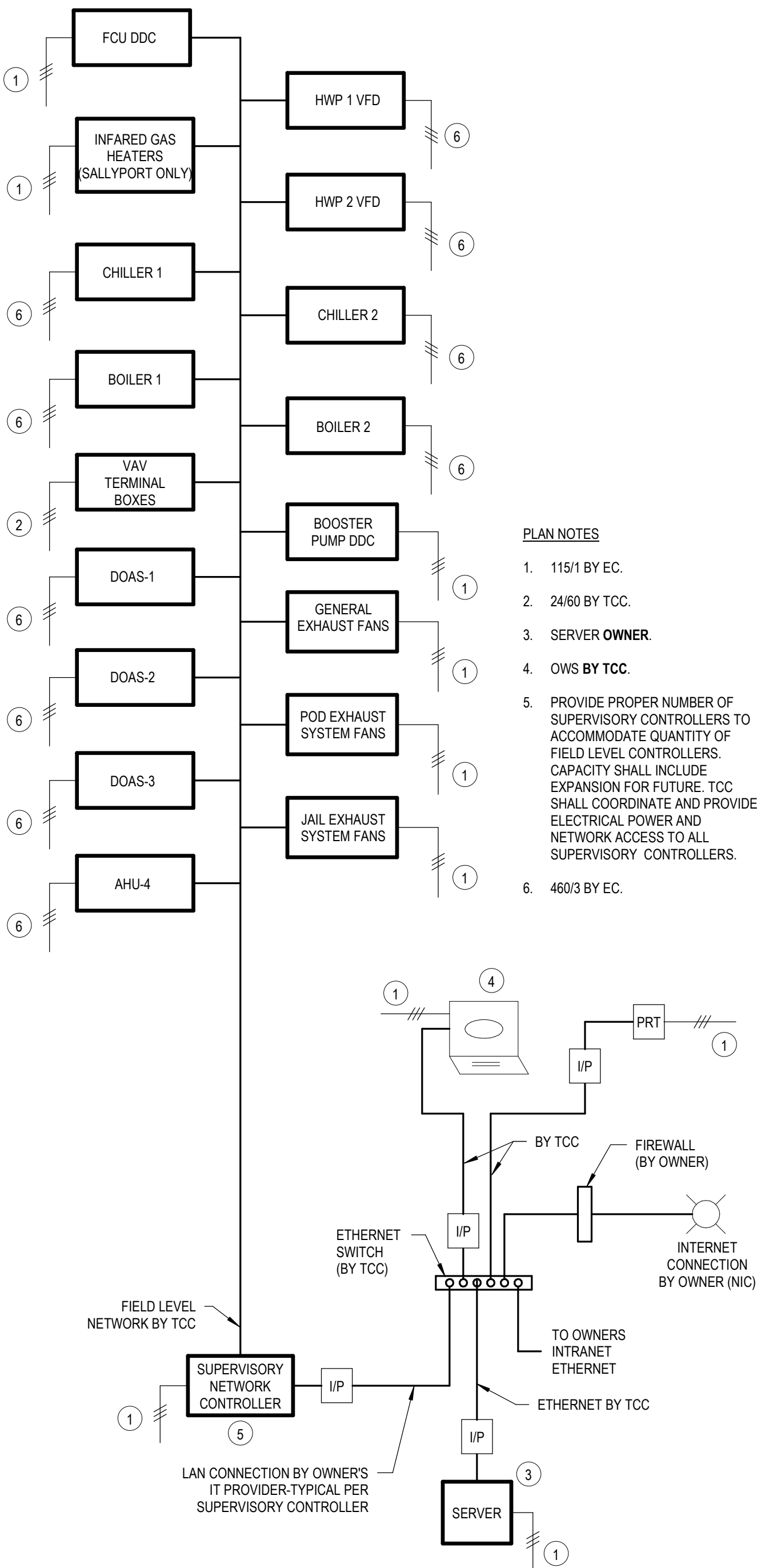
- SEQUENCE OF OPERATION:
- ALL SETPOINTS TO BE ADJUSTABLE. SETPOINTS TO BE EXPOSED ON GRAPHIC DISPLAY OR HIDDEN BASED ON OWNER REQUEST.
 - ENABLE: THE CHILLED WATER PUMPS SHALL BE OPERATIONAL WHENEVER THE OUTDOOR AIR TEMPERATURE IS GREATER THAN 50°F (ADJ).
 - DISABLE: THE CHILLED WATER SHALL SHUT DOWN WHENEVER THE OUTDOOR AIR TEMPERATURE FALLS BELOW 40°F (ADJ).
A. WHEN TEMPERATURE DROPS BELOW 10°F CHILLED WATER PUMP SHALL RUN AT 20Hz
 - CHILLERS:
A. CHILLERS ARE MANUALLY INDEXED FOR AUTOMATIC OPERATION THROUGH INTEGRAL ON-OFF SWITCHES.
B. CHILLERS SHALL ALTERNATE IN THE LEAD POSITION EVERY 750 HOURS (ADJ). FAILURE OF A CHILLER TO START WHEN COMMANDED BY THE BAS SHALL START THE NEXT CHILLER IN THE SERIES.
C. THE SYSTEM CHILLED WATER SET POINT SHALL BE 40°F (ADJ) ABOVE 75°F OAT (ADJ) AND 42°F OAT (ADJ) BELOW 70°F (ADJ).
D. WHEN THE CALCULATED LOAD EXCEEDS 90% (ADJ) OF GENERATING CAPACITY OF ONE CHILLER FOR 15 MINUTES (ADJ), THE FOLLOWING SHALL OCCUR:
i. THE CHILLED WATER SET POINT OF THE LEAD CHILLER IS RAISED 2°F (ADJ) TO ALLOW THE LAG CHILLER TO COME ONLINE.
ii. THE LAG CHILLER'S CHILLED WATER ISOLATION VALVE SHALL OPEN. THE CHILLED WATER ISOLATION VALVE SHALL OPEN AT A RATE SLOW ENOUGH SUCH THAT NO SHUTDOWN OF THE OPERATING CHILLER OCCURS DUE TO LOW FLOW (ECC SHALL COORDINATE WITH CHILLER MANUFACTURER AND TEST AND BALANCE CONTRACTOR TO DETERMINE THIS VALUE).
iii. WHEN FLOW THROUGH THE LAG CHILLER IS PROVED VIA THE CHILLER'S INTERNAL SAFETIES, THE CHILLER SHALL BE ENERGIZED TO PRODUCE CHILLED WATER AT THE SYSTEM SETPOINT.
iv. THE LEAD CHILLER'S CHILLED WATER SET POINT SHALL RETURN TO THE SYSTEM CHILLED WATER SET POINT AFTER THE LAG CHILLER HAS BEEN OPERATIONAL FOR 5 MINUTES (ADJ).
E. WHEN TWO CHILLERS ARE OPERATING AND THE CALCULATED LOAD FALLS BELOW 40% (ADJ) OF GENERATING CAPACITY OF BOTH CHILLERS FOR 15 MINUTES (ADJ), THE LAG CHILLER SHALL SHUT DOWN ACCORDING TO THE CHILLER MANUFACTURER'S REQUIREMENTS AND ITS ISOLATION VALVE SHALL CLOSE.
 - CHILLED WATER PUMPS:
A. EACH CHILLER HAS TWO INTEGRAL PUMPS, ONE LEAD AND ONE LAG.
B. CHILLED WATER PUMPS SHALL ROTATE AS THE LEAD PUMP FOR THEIR RESPECTIVE CHILLER EVERY 750 HOURS (ADJ). FAILURE OF ANY PUMP SHALL AUTOMATICALLY START THE NEXT PUMP IN THE OPERATING SEQUENCE.
C. WHEN A CHILLER IS ENABLED, THE NEXT PUMP IN SERVICE SHALL BE INDEXED TO START.
D. WHEN A CHILLER IS DISABLED, THE LAST PUMP IN SERVICE SHALL BE INDEXED TO STOP.
E. WHEN MULTIPLE PUMPS ARE OPERATING, PUMPS SHALL RUN AT THE SAME SPEED AND MODULATE FROM 18 TO 60 Hz (ADJ) TO MAINTAIN THE REQUIRED DIFFERENTIAL PRESSURE OF 15 PSID (ADJ). FINAL SET POINT DETERMINED BY TAB) 90% THROUGH THE CHILLED WATER PIPING SYSTEM. DIFFERENTIAL PRESSURE SET POINT SHALL BE RESET DOWNWARD BASED ON VALVE POSITION SUCH THAT THE WORST CASE CONTROL VALVE IS FULLY OPEN BASED ON A TRIM AND RESPOND LOOP OR SIMILAR LOGIC. THE MINIMUM DIFFERENTIAL PRESSURE SET POINT SHALL BE 8 PSID (ADJ).
 - CHILLED WATER BYPASS VALVE:
A. THE CHILLED WATER BYPASS VALVE SHALL MODULATE OPEN TO MAINTAIN THE SYSTEM FLOW RATE ABOVE THE REQUIRED MINIMUM LOW RATE BASED ON THE OPERATING CHILLER(S) AS MEASURED BY CHILLER DIFFERENTIAL PRESSURE.
 - ALARMS:
A. GENERATE ALARM ANYTIME SUPPLY TEMPERATURE DIFFERS FROM SET POINT BY GREATER THAN 8°F (ADJ) WHEN SYSTEM IS ENABLED.

- PLAN NOTES
- LOCATED AT 90% THROUGH HEATING HOT WATER SYSTEM WHERE SHOWN ON PLAN. HARD WIRED TO HEATING HOT WATER DDC PANEL.
 - THRU THE LAN, VFC TO TRANSMIT TO BAS STATUS AND ALARMS OF ALL DATA AVAILABLE. VFC SUPPLIER TO FURNISH INTEGRAL COMMUNICATION CARD. TCC TO MAP ALL OWNER REQUESTED INFORMATION POINTS. NOTE THAT START/STOP SIGNAL, STATUS AND SPEED CONTROL ARE HARD WIRED TO DDC PANEL TO ENSURE OPERATION ON LOSS OF LAN.
 - TCC TO FURNISH AND INSTALL ON WALL OUTSIDE BOILER ROOM AN EMERGENCY LOCKOUT SWITCH (RED MUSHROOM 2" DIA. WITH PROTECTIVE COVER) HARD WIRED TO ALL OF THE HOT WATER BOILERS PROVIDE MANUAL SILENCE SWITCH.
 - VFC DRIVES FURNISHED, INSTALLED, WIRED AND COMMISSIONED BY TCC. VFC SHALL MODULATE BETWEEN THE CONTROLLED EQUIPMENT MANUFACTURERS REQUIRED SAFE OPERATING MAXIMUM AND MINIMUM SPEED.
 - HEATING HOT WATER BYPASS VALVE SIZED AT MIN. BOILER GPM AT XX PSID.
 - CARBON MONOXIDE DETECTOR LOCATED IN BOILER ROOM.
 - VALVE ACTUATION TO FAIL LAST POSITION.

- SEQUENCE OF OPERATION:
- ALL SETPOINTS TO BE ADJUSTABLE. SETPOINTS TO BE EXPOSED ON GRAPHIC DISPLAY OR HIDDEN BASED ON OWNER REQUEST.
 - ENABLE: THE HEATING HOT WATER SYSTEM IS STARTED FROM DDC PANEL OR FROM COMMAND OF BUILDING MANAGEMENT SYSTEM. SYSTEM TO RUN CONTINUOUSLY.
 - DISABLE: WHEN SYSTEM IS DISABLED MANUALLY THROUGH THE BUILDING MANAGEMENT SYSTEM, BOILERS ARE DISABLED AND PUMPS STOP.
 - BOILERS:
A. BOILERS ARE MANUALLY INDEXED FOR AUTOMATIC OPERATION THROUGH INTEGRAL ON-OFF SWITCHES. ALL BOILERS ARE AVAILABLE FOR SERVICE AT ALL TIMES FOR NORMAL OPERATION.
B. A MINIMUM OF ONE BOILER SHALL ALWAYS BE ENABLED.
C. ANY BOILER TAKEN OUT OF SERVICE (MANUALLY DEENERGIZED) SHALL BE NOTED ON THE GRAPHICS AND IT SHALL BE REMOVED FROM THE CONTROL SEQUENCE.
D. QUANTITY OF BOILERS OPERATING AND THE FIRING RATE SHALL BE DETERMINED BY THE BUILDING AUTOMATION SYSTEM (BAS). MULTIPLE BOILERS SHALL RUN IN LOW FIRE MODE BEFORE MODULATING FIRING RATE TOWARD HIGHER FIRE TO MAINTAIN HIGHEST EFFICIENCY POSSIBLE. ALL OPERATING BOILERS SHALL NORMALLY RECEIVE THE SAME FIRING RATE.
E. ROTATE BOILER OPERATION WHEN DETERMINING WHICH BOILER TO ENABLE BY ENABLING THE UNIT WITH LEAST AMOUNT OF RUNTIME. WHEN ROTATING BOILERS OUT OF SEQUENCE DISABLE THE BOILER WITH THE MOST RUN-HOURS. DISPLAY BOILER RUN-HOURS AND ON/OFF RUN CYCLES ON THE GWS GRAPHICS.
F. THE SYSTEM SUPPLY HEATING HOT WATER SET POINT SHALL BE 140°F (ADJ). RESET HEATING HOT WATER FROM 140°F TO 110°F AS THE OUTDOOR AIR VARIES FROM 30°F TO 70°F.
G. LEAD BOILER ISOLATION VALVE SHALL BE OPEN AT ALL TIMES. LAG BOILER ISOLATION VALVE SHALL OPEN WHENEVER BOILER IS CALLED TO OPERATE. CLOSE THE ISOLATION VALVE WHEN THE BOILER IS TAKEN OUT OF SEQUENCE AFTER A 5 MIN. (ADJ) TIME DELAY.
 - HEATING HOT WATER PUMPS:
A. HEATING HOT WATER PUMPS SHALL OPERATE IN LEAD, LAG ARRANGEMENT. ONE (1) PUMP WILL HANDLE SYSTEM FLOW REQUIREMENTS.
B. HEATING HOT WATER PUMPS SHALL ROTATE AS THE LEAD PUMP EVERY 168 HOURS (ADJ). FAILURE OF ANY PUMP SHALL AUTOMATICALLY START THE NEXT PUMP IN THE OPERATING SEQUENCE.
C. LAG PUMP STARTS WHEN SYSTEM DIFFERENTIAL IS NOT MET BY A SINGLE PUMP FOR 10 MINUTES (ADJ). WHEN PRESSURE DROPS BELOW A 110 PSI (ADJ) LAG PUMP SHALL DE-ENERGIZE.
D. WHEN MULTIPLE PUMPS ARE OPERATING, PUMPS SHALL RUN AT THE SAME SPEED AND MODULATE FROM 18 TO 60 Hz (ADJ) TO MAINTAIN THE REQUIRED DIFFERENTIAL PRESSURE OF 10 PSID (ADJ). FINAL SET POINT DETERMINED BY TAB) 90% THROUGH THE HEATING HOT WATER PIPING SYSTEM. DIFFERENTIAL PRESSURE SET POINT SHALL BE RESET DOWNWARD BASED ON VALVE POSITION SUCH THAT THE WORST CASE CONTROL VALVE IS FULLY OPEN BASED ON A TRIM AND RESPOND LOOP OR SIMILAR LOGIC. THE MINIMUM DIFFERENTIAL PRESSURE SET POINT SHALL BE 5 PSID (ADJ).
 - HEATING HOT WATER BYPASS VALVE:
A. THE HEATING HOT WATER BYPASS VALVE SHALL MODULATE OPEN TO MAINTAIN THE SYSTEM FLOW RATE ABOVE THE REQUIRED MINIMUM FLOW RATE BASED ON THE OPERATING BOILER(S) AS MEASURED BY SYSTEM FLOW METER. XXX GPM SET POINT CONTROLLER.
B. SYSTEM BYPASS VALVE SHALL ALSO MODULATE OPEN TO MAINTAIN SYSTEM DIFFERENTIAL PRESSURE OF 2 PSID (ADJ) ABOVE EFFECTIVE SYSTEM DIFFERENTIAL SET POINT WHEN A SINGLE OPERABLE PUMP IS AT MINIMUM SPEED.
 - ALARMS:
A. GENERATE ALARM ANYTIME SUPPLY TEMPERATURE DIFFERS FROM SET POINT BY GREATER THAN 10°F (ADJ).
B. GENERATE ALARM ANYTIME CARBON MONOXIDE LEVEL IS ABOVE 25 PPM (ADJ) AND SHUT DOWN BOILERS.
C. MANUAL DEPRESSION OF EMERGENCY BOILER SHUTDOWN SWITCH SHALL SOUND HORN AND DISABLE ALL BOILERS.
D. GENERATE ALARM ANYTIME WATER IS DETECTED BY WATER DETECTOR.

2 HEATING HOT WATER CONTROL SCHEMATIC
NOT TO SCALE

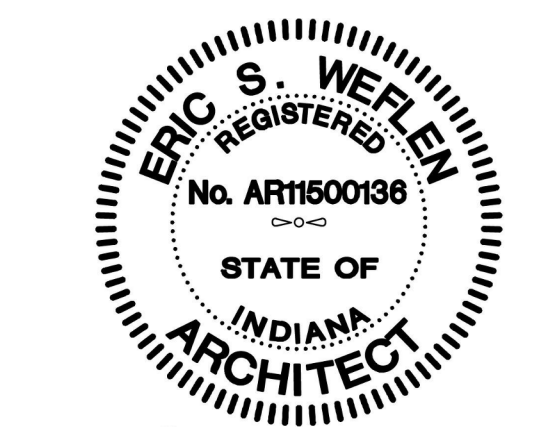
SCHEDULE OF DDC POINTS				
ID	DESCRIPTION	TREND	ALARM	GRAPHIC
AI-1	OUTDOOR TEMPERATURE (GLOBAL)	X		X
AI-2	SYSTEM HOT WATER SUPPLY TEMPERATURE	X	X	X
AI-3	SYSTEM HOT WATER RETURN TEMPERATURE	X	X	X
AI-4	SYSTEM HOT WATER FLOW	X		X
AI-5	SYSTEM HOT WATER CALCULATED LOAD	X		X
AI-6	SYSTEM DIFFERENTIAL PRESSURE	X	X	X
AI-7	BOILER B-1 SUPPLY TEMPERATURE	X	X	X
AI-8	BOILER B-2 SUPPLY TEMPERATURE	X	X	X
AI-10	CARBON MONOXIDE LEVEL	X	X	X
AO-1	HOT WATER PUMP HWP-1 VFC	X		X
AO-2	HOT WATER PUMP HWP-2 VFC	X		X
AO-4	BOILER B-1 FIRING RATE	X		X
AO-5	BOILER B-2 FIRING RATE	X		X
AO-7	SYSTEM BYPASS VALVE	X		X
BI-1	HOT WATER PUMP HWP-1 STATUS	X	X	X
BI-2	HOT WATER PUMP HWP-2 STATUS	X	X	X
BI-4	EMERGENCY BOILER SHUTDOWN	X	X	X
BI-5	BOILER B-1 GENERAL ALARM	X	X	X
BI-6	BOILER B-2 GENERAL ALARM	X	X	X
BI-8	BOILER ROOM WATER DETECTOR ALARM	X	X	X
BI-9	UPS LOW BATTERY ALARM	X	X	X
BI-10	UPS ON BATTERY ALARM	X	X	X
BO-1	HOT WATER PUMP HWP-1 START-STOP	X		X
BO-2	HOT WATER PUMP HWP-2 START-STOP	X		X
BO-4	BOILER B-1 START/STOP	X		X
BO-5	BOILER B-2 START/STOP	X		X
BO-7	BOILER B-1 HEATING WATER ISOLATION VALVE	X		X
BO-8	BOILER B-2 HEATING WATER ISOLATION VALVE	X		X



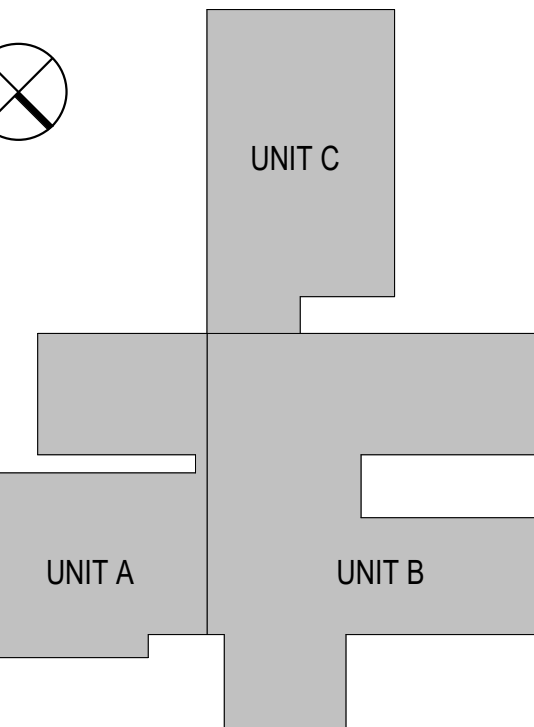
3 BAS SYSTEM ARCHITECTURE
NOT TO SCALE

#	Revision	Date
---	----------	------

Project #: 17-700-078-2
Designed By: A.M.M
Drawn By: A.M.M
Checked By: D.E.B
Date: 04.12.2021



These Drawings and Specifications, and all copies thereof are and shall remain the property and copyright of the Architect. They shall be used only with respect to this Project and are not to be used on any other Project or Work without prior written permission from the Architect



CHILLER AND HOT
WATER PLANTS
CONTROLS SCHEMATIC

#	Revision	Date
---	----------	------

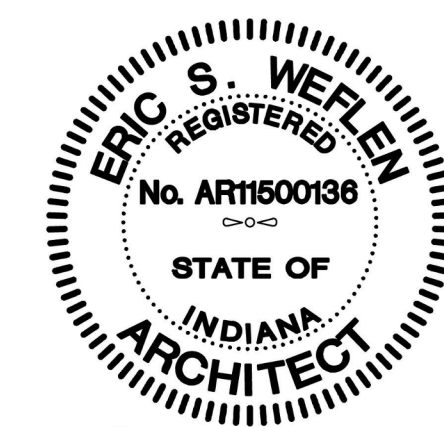
Project #: 17-700-078-2

Designed By: A.M.M

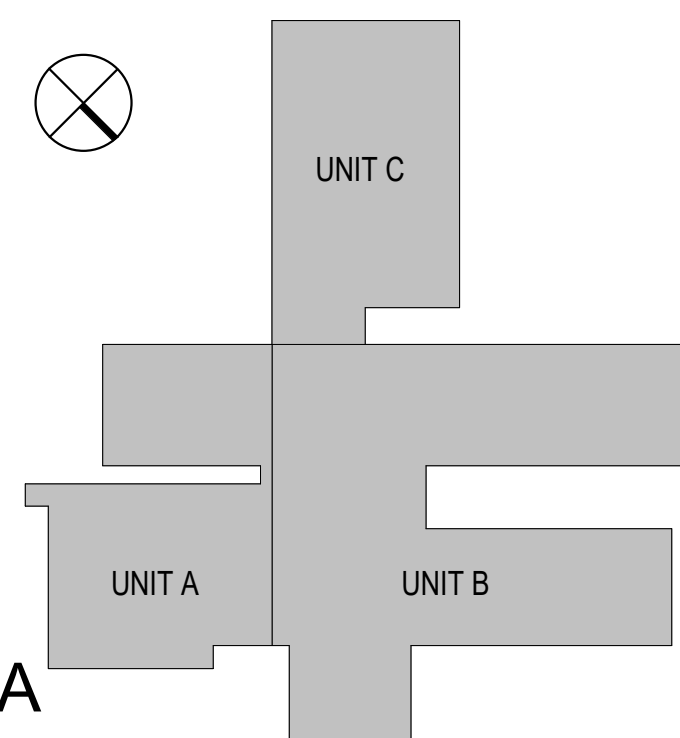
Drawn By: A.M.M

Checked By: D.E.B

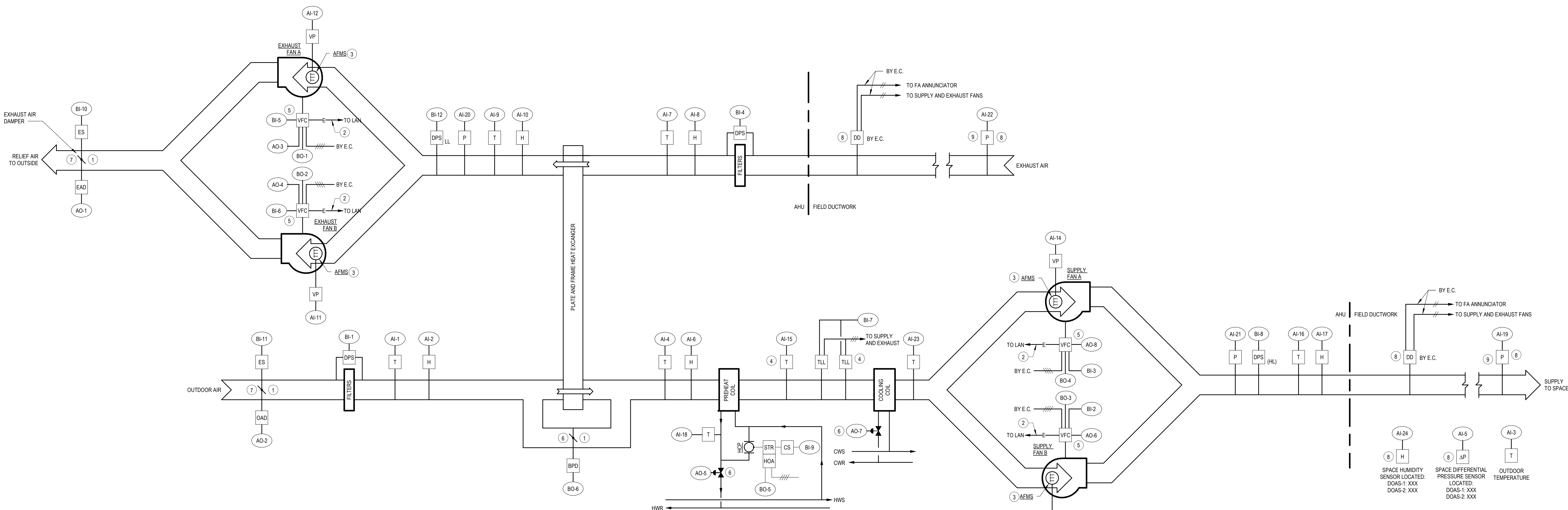
Date: 04.12.2021



These Drawings and Specifications, and all copies thereof are and shall remain the property and copyright of the Architect. They shall be used only with respect to this Project and are not to be used on any other Project or Work without prior written permission from the Architect

DOAS-1 & 2 CONTROLS
SCHEMATIC

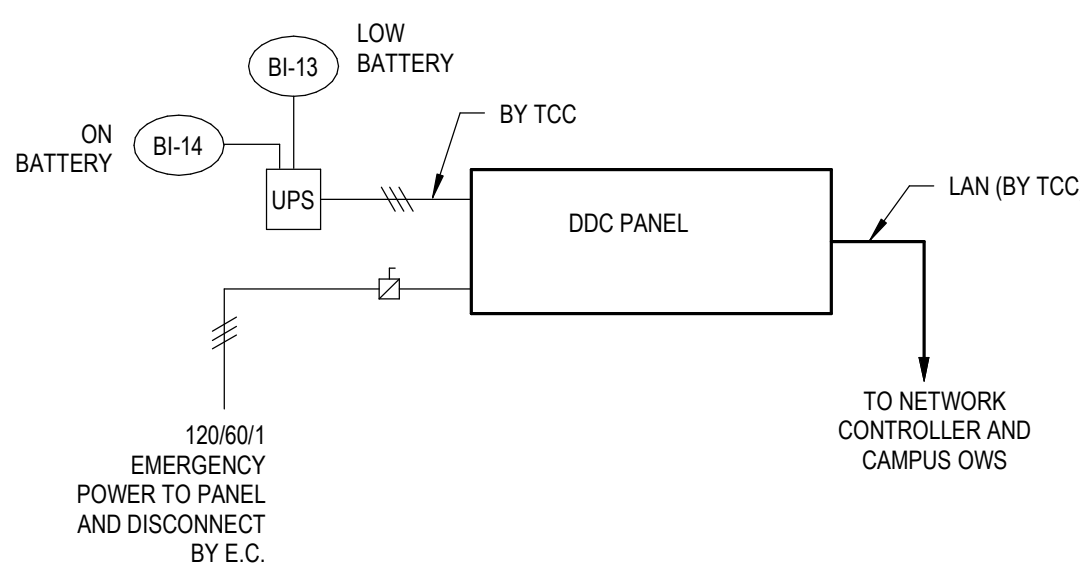
M902



SCHEDULE OF DDC POINTS

ID	DESCRIPTION	TREND	ALARM	GRAPHIC
AI-1	OUTDOOR AIR TEMPERATURE (ENTERING HX)	X		X
AI-2	OUTDOOR AIR HUMIDITY (ENTERING HX)	X		X
AI-3	OUTDOOR TEMPERATURE (GLOBAL)	X		X
AI-4	OUTDOOR AIR TEMPERATURE (LEAVING HX)	X		X
AI-5	SPACE DIFFERENTIAL PRESSURE	X		X
AI-6	OUTDOOR AIR HUMIDITY (LEAVING HX)	X		X
AI-7	EXHAUST AIR TEMPERATURE (ENTERING HX)	X		X
AI-8	EXHAUST AIR HUMIDITY (ENTERING HX)	X		X
AI-9	EXHAUST AIR TEMPERATURE (LEAVING HX)	X	X	
AI-10	EXHAUST AIR HUMIDITY (LEAVING HX)	X		X
AI-11	EXHAUST FAN AIR VOLUME	X		X
AI-12	EXHAUST FAN AIR VOLUME	X		X
AI-13	SUPPLY FAN AIR VOLUME	X		X
AI-14	SUPPLY FAN AIR VOLUME	X		X
AI-15	PREHEAT COIL DISCHARGE TEMPERATURE	X	X	X
AI-16	SUPPLY AIR TEMPERATURE	X	X	X
AI-17	SUPPLY AIR HUMIDITY	X		X
AI-18	PREHEAT COIL HWIR TEMPERATURE	X	X	X
AI-19	SUPPLY STATIC PRESSURE 90% THRU SYSTEM	X		X
AI-20	EXHAUST FAN INLET STATIC PRESSURE	X		X
AI-21	SUPPLY DISCHARGE STATIC PRESSURE	X		X
AI-22	EXHAUST STATIC PRESSURE 90% THRU SYSTEM	X		X
AI-23	COOLING COIL DISCHARGE TEMPERATURE	X	X	X
AI-24	SPACE HUMIDITY	X	X	X
AO-1	EXHAUST AIR DAMPER	X		X
AO-2	OUTDOOR AIR DAMPER	X		X
AO-3	EXHAUST FAN A SPEED	X		X
AO-4	EXHAUST FAN B SPEED	X		X
AO-5	PREHEAT COIL VALVE	X		X
AO-6	SUPPLY FAN B SPEED	X		X
AO-7	COOLING COIL VALVE	X		X
AO-8	SUPPLY FAN A SPEED	X		X
BI-1	OUTSIDE AIR FILTER STATUS	X	X	X
BI-2	SUPPLY FAN B STATUS	X	X	X
BI-3	SUPPLY FAN A STATUS	X	X	X
BI-4	EXHAUST AIR FILTER STATUS	X	X	X
BI-5	EXHAUST FAN A STATUS	X	X	X
BI-6	EXHAUST FAN B STATUS	X	X	X
BI-7	SAFETY LOW LIMIT THERMOSTAT ALARM	X	X	X
BI-8	DISCHARGE PRESSURE HIGH LIMIT ALARM	X	X	X
BI-9	HEATING COIL PUMP STATUS			X
BI-10	EXHAUST AIR DAMPER END SWITCH			
BI-11	OUTDOOR AIR DAMPER END SWITCH			
BI-12	EXHAUST FAN INLET PRESSURE HIGH LIMIT ALARM	X	X	X
BI-13	UPS LOW BATTERY ALARM	X	X	X
BI-14	UPS ON BATTERY ALARM	X	X	X
BO-1	EXHAUST FAN A START/STOP			X
BO-2	EXHAUST FAN B START/STOP			X
BO-3	SUPPLY FAN B START/STOP			X
BO-4	SUPPLY FAN A START/STOP			X
BO-5	HEATING COIL PUMP START-STOP			X
BO-6	OA HEAT EXCHANGER BYPASS DAMPER	X		X

SETUP ALL TRENDS IN 15 MINUTE INTERVALS AND STORE FOR MINIMUM 45 DAYS.
INCLUDE TRENDS FOR THE ASSOCIATED COMMAND AND/OR SETPOINT IN ADDITION
TO POINT LISTED ABOVE.
PROVIDE CHANGE OF VALUE AND INTERVAL TRENDS FOR BI AND BO POINTS.



GENERAL NOTES:

- ALL AHUS TO BE SINGLE POINT POWER FOR ALL 400/3 EQUIPMENT WITH AN ADDITIONAL 120/1 CIRCUIT FOR OUTLETS AND LIGHTS WITHIN UNITS.
- DAMPERS TO BE INTEGRAL WITH AHU; ACTUATORS BY MANUFACTURER.
- THRU THE LAN, VFC TO TRANSMIT TO BAS STATUS AND ALARMS OF ALL DATA AVAILABLE VFC SUPPLIER TO FURNISH INTEGRAL COMMUNICATION CARD. TCC TO MAP ALL OWNER REQUESTED INFORMATION POINTS. NOTE THAT START/STOP SIGNAL AND SPEED CONTROL ARE HARD WIRED TO DDC TO ENSURE OPERATION ON LOSS OF LAN.
- AIRFLOW PIEZOMETER INTEGRAL TO FAN, PROVIDED BY FAN MANUFACTURER, TRANSDUCER BY MANUFACTURER.
- LOCATE 3" FROM COOLING COIL INLET.
- VFC FURNISHED, INSTALLATION, WIRING, AND COMMISSIONED BY MANUFACTURER.
- SPRING RETURN OPEN UPON LOSS OF POWER.
- SPRING RETURN CLOSED UPON LOSS OF POWER.
- REFER TO PLANS FOR LOCATIONS.
- HARD WIRE DUCT STATIC SENSOR TO THE SAME DDC CONTROLLER WHICH CONTROLS THE FAN.

PLAN NOTES:

SEQUENCE OF OPERATION:

- ALL SETPOINTS TO BE ADJUSTABLE. SETPOINTS TO BE EXPOSED ON GRAPHIC DISPLAY OR HIDDEN BASED ON OWNER REQUEST.
- ENABLE:** AIR HANDLER IS STARTED FROM DDC PANEL OR FROM COMMAND OF FACILITY MANAGEMENT SYSTEM.

OCCUPIED MODE:

- AHU SHALL UTILIZE AN "OPTIMUM START" ALGORITHM FOR START-UP. THE CONTROL SYSTEM SHALL CALCULATE THE ZONE TEMPERATURES AND COMPARE WITH THE BUILDING SETPOINT. CONTROL SYSTEM SHALL CALCULATE THE TIME REQUIRED TO BRING THE ZONE UP TO SETPOINT TEMPERATURE AND START THE UNIT APPROPRIATELY.
- STARTUP:** ON UNIT STARTUP, SUPPLY AND EXHAUST FANS START. SUPPLY FAN SPEED CONTROL AND SUPPLY AIR TEMPERATURE CONTROL LOOPS ARE ACTIVE. EXHAUST AIR DAMPER IS FULLY OPEN, MIN AND MAX OUTDOOR AIR AND RELIEF DAMPERS FULLY CLOSED FOR 5 MIN. (ADJ) WITH EXHAUST FAN TRACKING CFM SETPOINT SET TO NULL (ADJ).
- NORMAL OPERATION:** AFTER STARTUP, MINIMUM OUTDOOR AIR DAMPER TO OPEN FULLY (ADJ) ON PROVED FAN OPERATION AND EXHAUST FAN CFM TRACKING OFFSET TO INCREASE TO SETPOINT.
- SUPPLY AIR TEMPERATURE:** THE SUPPLY AIR CONTROL LOOP SHALL PERFORM AS DESCRIBED BELOW:
 - ABOVE 70°F (ADJ) OAT:
 - MODULATE THE ECONOMIZER DAMPERS TO MAINTAIN MIXED AIR TEMPERATURE OF 52°F. MIXED AIR SETPOINT SHALL TRACK SUPPLY AIR TEMPERATURE SETPOINT MINUS 3°F (ADJ) TO ACCOUNT FOR TEMP RISE ACROSS FANS.
 - ECONOMIZER DAMPER SPAN IS AS FOLLOWS:

DAMPER	MIN % OPEN	MAX % OPEN
OAT	MIN	100
RAD	100	0
EAD	MIN	100
 - ONCE THE RELIEF AIR DAMPERS ARE 100% (ADJ) OPEN AND THE SUPPLY AIR TEMPERATURE RISES, THE COOLING COIL CONTROL VALVE SHALL BE MODULATED TO MAINTAIN SUPPLY AIR SETPOINT.
 - IF THERE IS A DROP IN SUPPLY AIR TEMPERATURE, THE COOLING COIL CONTROL VALVE SHALL BE MODULATED CLOSED AND IN SEQUENCE THE ECONOMIZER DAMPERS SHALL BE MODULATED TOWARD THE MINIMUM POSITION.
 - PREHEAT: ONCE THE MAXIMUM OUTDOOR AIR DAMPER IS AT THE MINIMUM POSITION AND THE PREHEAT TEMPERATURE REMAINS BELOW SET POINT, THE PREHEAT COIL CONTROL VALVE SHALL BE MODULATED OPEN TO MAINTAIN PREHEAT AIR TEMPERATURE OF 55°F. PREHEAT AIR SETPOINT SHALL TRACK SUPPLY AIR TEMPERATURE SETPOINT MINUS 5°F (ADJ) TO ACCOUNT FOR TEMP RISE ACROSS FANS AND TO KEEP PREHEAT SHUT UNLESS REQUIRED FOR EXTREME COLD OAT.
 - PREHEAT COIL PUMP: OPERATES BELOW 35°F OAT AND SHUT OFF ABOVE 38°F.
 - CONTROLS SHALL PREVENT SIMULTANEOUS OPERATION OF COOLING COIL AND HEATING COIL AND ECONOMIZER WHEN OUTDOOR AIR CONDITIONS EXIST TO USE FULL ECONOMIZER.
 - RESET SUPPLY AIR TEMPERATURE SETPOINT FROM 55°F TO 60°F (ADJ) AS OUTDOOR AIR TEMPERATURE VARIES FROM 70°F TO 30°F. IN THE EVENT THE EXHAUST AIR HUMIDITY EXCEEDS 55% RH THE UNIT SUPPLY TEMPERATURE SHALL BE RESET TO 55°F AND SHALL IDENTIFY THIS ON THE OWS GRAPHICS. AFTER 15 MINUTES THE SUPPLY TEMPERATURE SHALL BE RESET UP IN 2°F INCREMENTS EVERY 10 MINUTES.
 - ABOVE 70°F (ADJ) OAT: THE ECONOMIZER DAMPERS SHALL BE FIXED IN MINIMUM POSITION AND THE COOLING COIL CONTROL VALVE SHALL MODULATE TO MAINTAIN SUPPLY AIR TEMPERATURE SETPOINT.
- SAFETIES:** FURNISH SAFETIES AS FOLLOWS TO PROVIDE ALARM AND SHUT DOWN UNIT.
 - COOLING COIL INLET TO HAVE TWO (2) OR MORE SAFETY LOW LIMIT THERMOSTATS WIRED IN SERIES TO BREAK BELOW 38°F (ADJ). ON ALARM, CHILLED WATER CONTROL VALVES OPEN 100% (ADJ).
 - SUPPLY HIGH LIMIT STATIC PRESSURE SWITCH SET TO 3.0" W.G. (ADJ).
 - EXHAUST DUCT LOW LIMIT STATIC SENSOR SWITCH SET TO (+) 2.0" W.G. (ADJ).
 - IN THE EVENT OF SMOKE (AHU SUPPLY OR EXHAUST DUCT DETECTOR).
 - HEAT EXCHANGER FROST BYPASS: OA BYPASS DAMPER SHALL MODULATE TO MAINTAIN NO LESS THAN HEAT EXCHANGER LEAVING OAT TEMPERATURE OF 28.8°F (ADJ).
- SUPPLY FAN SPEED:** STATIC PRESSURE SENSOR LOCATED 90% OF DISTANCE THROUGH SUPPLY DUCT AND FAN DISCHARGE STATIC PRESSURE SENSOR LOCATED IN AHU DISCHARGE TO FUNCTION WITH SUPPLY FAN VFC CONTROL LOOP TO MODULATE SUPPLY FAN SPEED TO ACHIEVE DUCT STATIC PRESSURE SET AT 1.25" W.G. (ADJ), NOT TO EXCEED DISCHARGE PRESSURE OF 3.1" W.G. (ADJ). FINAL SETPOINT DETERMINED BY TAB CONTRACTOR AND APPROVED BY ENGINEER.
 - IF ZERO OR ONE (ADJ) CRITICAL TERMINAL BOXES INDICATE A 95% OR GREATER DAMPER POSITION, THE STATIC PRESSURE SETPOINT SHALL DECREASE AT A RATE OF 0.1" W.G. EVERY 5 MINUTES (ADJ).
 - IF TWO OR THREE (ADJ) CRITICAL TERMINAL BOXES IN THE SYSTEM INDICATE 95% OR GREATER OPEN DAMPER POSITION, HOLD THE STATIC PRESSURE AT CURRENT SETPOINT.
 - IF FOUR (ADJ) OR MORE CRITICAL TERMINAL BOXES IN THE SYSTEM INDICATE 95% OPEN DAMPER POSITION, THE SETPOINT SHALL INCREASE AT A RATE OF 0.1" W.G. EVERY 5 MINUTES.
 - DUCT STATIC PRESSURE SETPOINT SHALL HAVE A MINIMUM AND MAXIMUM RANGE OF SETPOINT +/- 0.3" W.G. (ADJ).
- EXHAUST FAN SPEED:** EXHAUST FAN VFC CONTROL LOOP TO BE CONTROLLED THROUGH VELOCITY PRESSURE SENSORS MEASURING AIR VOLUME THROUGH AIR MEASURING PIEZOMETER STATIONS IN SUPPLY AND EXHAUST FAN INLET CONES SO AS TO ALLOW FANS TO TRACK SUPPLY VOLUME WITH DIFFERENTIAL FOR BUILDING EXHAUST AND BUILDING PRESSURIZATION. THE RELIEF AIR DAMPERS MODULATE OPEN TO MAINTAIN SPACE DIFFERENTIAL PRESSURE NO GREATER THAN 0.02" W.G. RELATIVE TO THE OUTSIDE.
- FILTER SWITCH:** FILTER DIFFERENTIAL PRESSURE SWITCHES ALARM AS NOTED - PRE-FILTER: 1.0" W.G. (ADJ).

- COOLING COIL INLET TO HAVE TWO (2) OR MORE SAFETY LOW LIMIT THERMOSTATS WIRED IN SERIES TO BREAK BELOW 38°F (ADJ). ON ALARM, CHILLED WATER CONTROL VALVES OPEN 100% (ADJ).
 - SUPPLY HIGH LIMIT STATIC PRESSURE SWITCH SET TO 3.0" W.G. (ADJ).
 - EXHAUST DUCT LOW LIMIT STATIC SENSOR SWITCH SET TO (+) 2.0" W.G. (ADJ).
 - IN THE EVENT OF SMOKE (AHU SUPPLY OR EXHAUST DUCT DETECTOR).
 - HEAT EXCHANGER FROST BYPASS: OA BYPASS DAMPER SHALL MODULATE TO MAINTAIN NO LESS THAN HEAT EXCHANGER LEAVING OAT TEMPERATURE OF 28.8°F (ADJ).
- 5. SUPPLY FAN SPEED:** STATIC PRESSURE SENSOR LOCATED 90% OF DISTANCE THROUGH SUPPLY DUCT AND FAN DISCHARGE STATIC PRESSURE SENSOR LOCATED IN AHU DISCHARGE TO FUNCTION WITH SUPPLY FAN VFC CONTROL LOOP TO MODULATE SUPPLY FAN SPEED TO ACHIEVE DUCT STATIC PRESSURE SET AT 1.25" W.G. (ADJ), NOT TO EXCEED DISCHARGE PRESSURE OF 3.1" W.G. (ADJ). FINAL SETPOINT DETERMINED BY TAB CONTRACTOR AND APPROVED BY ENGINEER.
- IF ZERO OR ONE (ADJ) CRITICAL TERMINAL BOXES INDICATE A 95% OR GREATER DAMPER POSITION, THE STATIC PRESSURE SETPOINT SHALL DECREASE AT A RATE OF 0.1" W.G. EVERY 5 MINUTES (ADJ).
 - IF TWO OR THREE (ADJ) CRITICAL TERMINAL BOXES IN THE SYSTEM INDICATE 95% OR GREATER OPEN DAMPER POSITION, HOLD THE STATIC PRESSURE AT CURRENT SETPOINT.
 - IF FOUR (ADJ) OR MORE CRITICAL TERMINAL BOXES IN THE SYSTEM INDICATE 95% OPEN DAMPER POSITION, THE SETPOINT SHALL INCREASE AT A RATE OF 0.1" W.G. EVERY 5 MINUTES.
 - DUCT STATIC PRESSURE SETPOINT SHALL HAVE A MINIMUM AND MAXIMUM RANGE OF SETPOINT +/- 0.3" W.G. (ADJ).
- 6. EXHAUST FAN SPEED:** EXHAUST FAN VFC CONTROL LOOP TO BE CONTROLLED THROUGH VELOCITY PRESSURE SENSORS MEASURING AIR VOLUME THROUGH AIR MEASURING PIEZOMETER STATIONS IN SUPPLY AND EXHAUST FAN INLET CONES SO AS TO ALLOW FANS TO TRACK SUPPLY VOLUME WITH DIFFERENTIAL FOR BUILDING EXHAUST AND BUILDING PRESSURIZATION. THE RELIEF AIR DAMPERS MODULATE OPEN TO MAINTAIN SPACE DIFFERENTIAL PRESSURE NO GREATER THAN 0.02" W.G. RELATIVE TO THE OUTSIDE.
- 7. FILTER SWITCH:** FILTER DIFFERENTIAL PRESSURE SWITCHES ALARM AS NOTED - PRE-FILTER: 1.0" W.G. (ADJ).

POD EXHAUST SYSTEM MODE:

- UPON COMMAND FROM POD EXHAUST SYSTEM CONTROL PANEL (SHEET M-905) OR BAS, AHU SHALL PRIORITIZE SUPPLY FAN STATIC PRESSURE SETPOINT TO ACHIEVE COMMANDING ZONE TERMINAL BOX CFM SETPOINT, WHILE MAINTAINING PRIOR SAFETIES.
- IF TWO OR MORE ZONES ARE COMMANDED ON BY POD EXHAUST SYSTEM CONTROL PANEL, AHU SHALL STRIVE TO ACHIEVE SETPOINTS OF ALL COMMANDING ZONES TERMINAL BOX CFM SETPOINTS, WHILE MAINTAINING PRIOR SAFETIES.
- UPON DISABLE OF POD EXHAUST SYSTEM FROM CONTROL PANEL OR BAS, AHU SHALL RESUME PRIOR CONDITIONS.

#	Revision	Date
---	----------	------

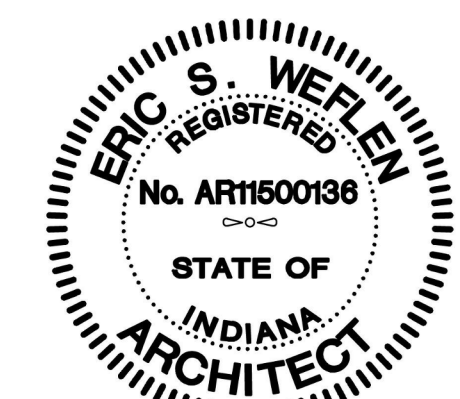
Project #: 17-700-078-2

Designed By: A.M.M

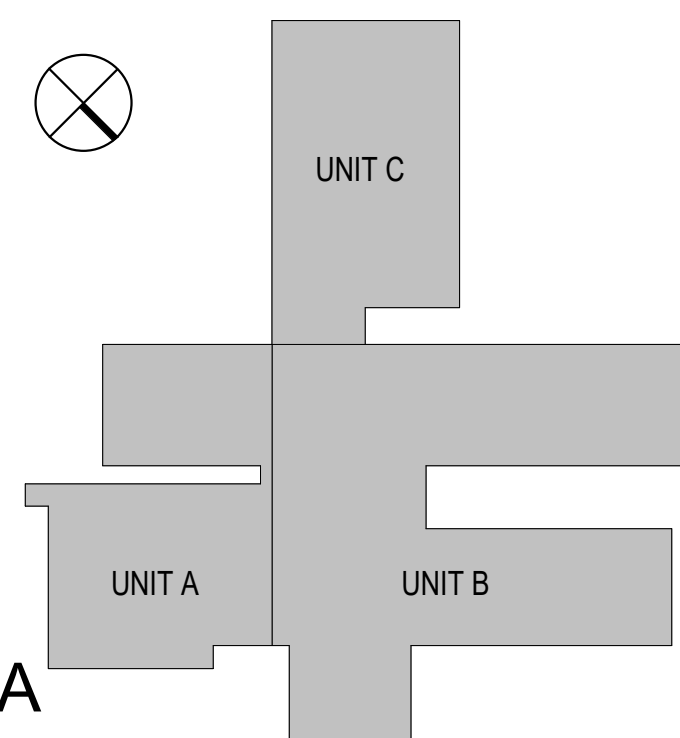
Drawn By: A.M.M

Checked By: D.E.B

Date: 04.12.2021

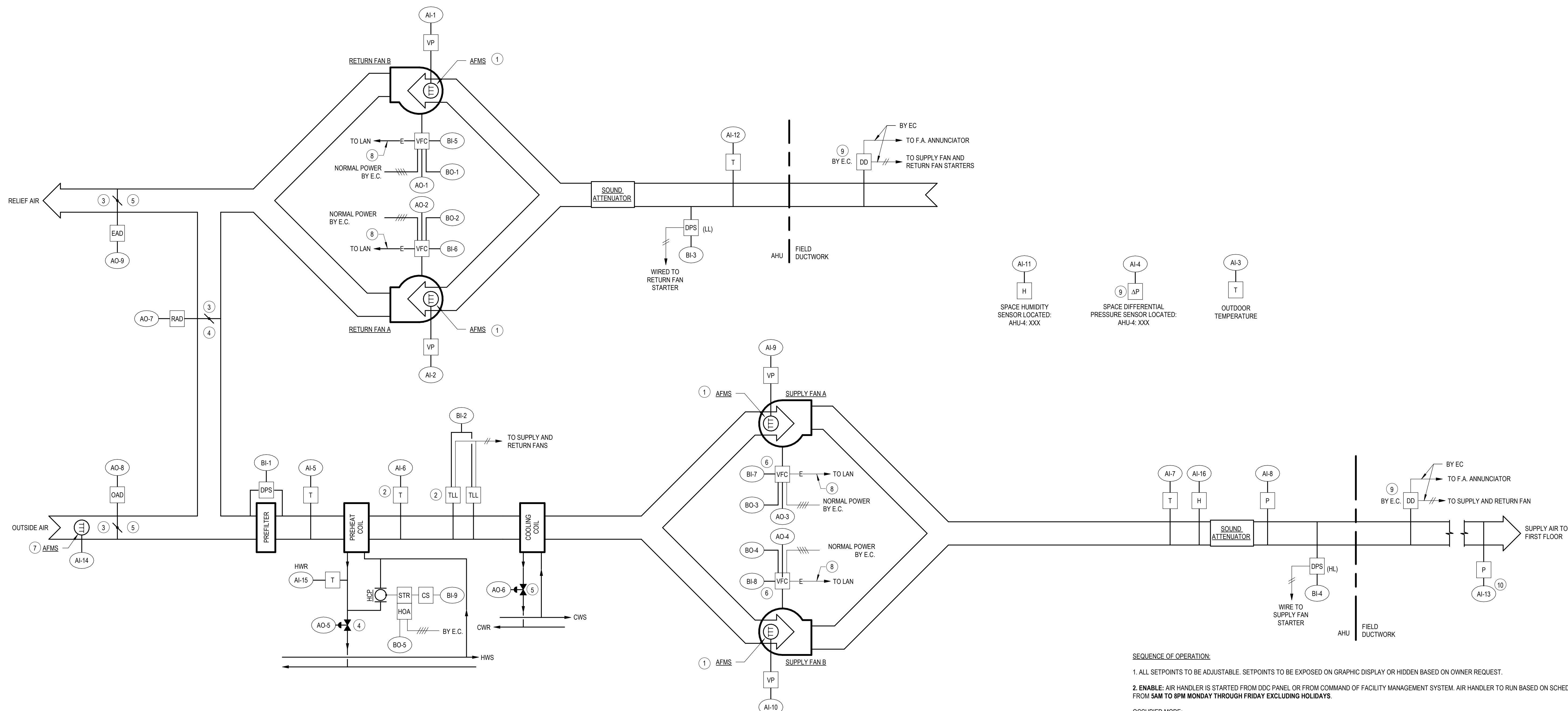


These Drawings and Specifications, and all copies thereof are and shall remain the property and copyright of the Architect. They shall be used only with respect to this Project and are not to be used on any other Project or Work without prior written permission from the Architect



AHU-4 CONTROL SCHEMATIC

M903



SEQUENCE OF OPERATION:

1. ALL SETPOINTS TO BE ADJUSTABLE. SETPOINTS TO BE EXPOSED ON GRAPHIC DISPLAY OR HIDDEN BASED ON OWNER REQUEST.

2. **ENABLE:** AIR HANDLER IS STARTED FROM DDC PANEL OR FROM COMMAND OF FACILITY MANAGEMENT SYSTEM. AIR HANDLER TO RUN BASED ON SCHEDULED OCCUPANCY. INITIALLY FROM **5AM TO 8PM MONDAY THROUGH FRIDAY EXCLUDING HOLIDAYS**.

OCCUPIED MODE:

1. AHU SHALL UTILIZE AN "OPTIMUM START" ALGORITHM FOR START-UP. THE CONTROL SYSTEM SHALL CALCULATE THE ZONE TEMPERATURES AND COMPARE WITH THE BUILDING SETPOINT. CONTROL SYSTEM SHALL CALCULATE THE TIME REQUIRED TO BRING THE ZONE UP TO SETPOINT TEMPERATURE AND START THE UNIT APPROPRIATELY.

2. **STARTUP:** ON UNIT STARTUP, SUPPLY AND RETURN FANS START. SUPPLY FAN SPEED CONTROL AND SUPPLY AIR TEMPERATURE CONTROL LOOPS ARE ACTIVE. RETURN AIR DAMPER IS FULLY OPEN, MIN AND MAX OUTDOOR AIR AND RELIEF DAMPERS FULLY CLOSED FOR **5 MIN (ADJ)** WITH RETURN FAN TRACKING CFM SETPOINT SET TO **NULL (ADJ)**.

2. **NORMAL OPERATION:** AFTER STARTUP, OUTDOOR AIR DAMPER TO OPEN **FULLY (ADJ)** ON PROVED FAN OPERATION AND RETURN FAN CFM TRACKING OFFSET TO INCREASE TO SETPOINT. ALARM IF MINIMUM OUTDOOR AIR FLOW CFM FALLS BELOW **OA SCHEDULED CFM** WHEN OUTDOOR AIR DAMPER IS AT MAXIMUM POSITION.

i. OA CFM SCHEDULE:
AHU-4: 2000 CFM

3. **SUPPLY AIR TEMPERATURE:** THE SUPPLY AIR CONTROL LOOP SHALL PERFORM AS DESCRIBED BELOW:

a. BELOW 70°F (ADJ) OAT:

i. MODULATE THE ECONOMIZER DAMPERS TO MAINTAIN MIXED AIR TEMPERATURE OF 52°F. MIXED AIR SETPOINT SHALL TRACK SUPPLY AIR TEMPERATURE SETPOINT MINUS 3°F (ADJ) TO ACCOUNT FOR TEMP RISE ACROSS FANS.

ii. ECONOMIZER DAMPER SPAN IS AS FOLLOWS:

DAMPER	MIN % OPEN	MAX % OPEN
OAD	MIN	100
RAD	100	0
EAD	MIN	100

iii. IF THE RELIEF AIR DAMPERS ARE **100% (ADJ)** OPEN AND THE SUPPLY AIR TEMPERATURE RISES, THE COOLING COIL CONTROL VALVE SHALL BE MODULATED TO MAINTAIN SUPPLY AIR SETPOINT.

iv. IF THERE IS A DROP IN SUPPLY AIR TEMPERATURE, THE COOLING COIL CONTROL VALVE SHALL BE MODULATED CLOSED AND IN SEQUENCE THE ECONOMIZER DAMPERS SHALL BE MODULATED TOWARD THE MINIMUM POSITION.

v. **PREHEAT:** ONCE THE OUTDOOR AIR DAMPER IS AT THE MINIMUM POSITION AND THE PREHEAT TEMPERATURE REMAINS BELOW SET POINT, THE PREHEAT COIL CONTROL VALVE SHALL BE MODULATED OPEN TO MAINTAIN PREHEAT AIR TEMPERATURE OF 55°F. PREHEAT AIR SETPOINT SHALL TRACK SUPPLY AIR TEMPERATURE SETPOINT MINUS 5°F (ADJ) TO ACCOUNT FOR TEMP RISE ACROSS FANS AND TO KEEP PREHEAT SHUT UNLESS REQUIRED FOR EXTREME COLD OAT.

vi. **PREHEAT COIL PUMP:** OPERATES BELOW 35°F OAT AND SHUT OFF ABOVE 38°F.

vii. CONTROLS SHALL PREVENT SIMULTANEOUS OPERATION OF COOLING COIL AND HEATING COIL AND ECONOMIZER WHEN OUTDOOR AIR CONDITIONS EXIST TO USE FULL ECONOMIZER.

viii. RESET SUPPLY AIR TEMPERATURE SETPOINT FROM **55°F TO 60°F (ADJ)** AS OUTDOOR AIR TEMPERATURE VARIES FROM **70°F TO 30°F**. IN THE EVENT THE RETURN AIR HUMIDITY EXCEEDS **55% RH** THE UNIT SUPPLY TEMPERATURE SHALL BE RESET TO **55°F** AND SHALL IDENTIFY THIS ON THE OWS GRAPHICS. AFTER 15 MINUTES THE SUPPLY TEMPERATURE SHALL BE RESET UP IN 2°F INCREMENTS EVERY 10 MINUTES.

b. ABOVE 70°F (ADJ) OAT: THE ECONOMIZER DAMPERS SHALL BE FIXED IN MINIMUM POSITION AND THE COOLING COIL CONTROL VALVE SHALL MODULATE TO MAINTAIN SUPPLY AIR TEMPERATURE SETPOINT.

4. **SAFETIES:** FURNISH SAFETIES AS FOLLOWS TO PROVIDE ALARM AND SHUT DOWN UNIT.

a. COOLING COIL INLET TO HAVE TWO (2) OR MORE SAFETY LOW LIMIT THERMOSTATS WIRED IN SERIES TO BREAK BELOW **38°F (ADJ)**. ON ALARM, CHILLED WATER CONTROL VALVES OPEN **100% (ADJ)**.

b. SUPPLY HIGH LIMIT STATIC PRESSURE SWITCH SET TO **3.0" W.G. (ADJ)**.

c. RETURN DUCT LOW LIMIT STATIC SENSOR SWITCH SET TO (c) **2.0" W.G. (ADJ)**.

d. IN THE EVENT OF SMOKE (AHU SUPPLY OR RETURN DUCT DETECTOR).

5. **SUPPLY FAN SPEED:** STATIC PRESSURE SENSOR LOCATED 90% OF DISTANCE THROUGH SUPPLY DUCT AND FAN DISCHARGE STATIC PRESSURE SENSOR LOCATED IN AHU DISCHARGE TO FUNCTION WITH SUPPLY FAN VFC CONTROL LOOP TO MODULATE SUPPLY FAN SPEED TO ACHIEVE DUCT STATIC PRESSURE SET AT **1.25" W.G. (ADJ)**, **NOT TO EXCEED DISCHARGE PRESSURE OF 3.9" W.G. (ADJ)**. **FINAL SETPOINT DETERMINED BY TAB CONTRACTOR AND APPROVED BY ENGINEER.**

A. IF **ZERO OR ONE (ADJ)** CRITICAL TERMINAL BOXES INDICATE A 95% OR GREATER DAMPER POSITION, THE STATIC PRESSURE SETPOINT SHALL DECREASE AT A RATE OF 0.1" W.G. EVERY 5 MINUTES (ADJ).

B. IF **TWO OR THREE (ADJ)** CRITICAL TERMINAL BOXES IN THE SYSTEM INDICATE 95% OR GREATER OPEN DAMPER POSITION, HOLD THE STATIC PRESSURE AT CURRENT SETPOINT.

C. IF **FOUR (ADJ)** OR MORE CRITICAL TERMINAL BOXES IN THE SYSTEM INDICATE 95% OPEN DAMPER POSITION, THE SETPOINT SHALL INCREASE AT A RATE OF 0.1" W.G. EVERY 5 MINUTES.

D. DUCT STATIC PRESSURE SETPOINT SHALL HAVE A MINIMUM AND MAXIMUM RANGE OF **SETPOINT +/- 0.3" WG (ADJ)**.

6. **RETURN FAN SPEED:** RETURN FAN VFC CONTROL LOOP TO BE CONTROLLED THROUGH VELOCITY PRESSURE SENSORS MEASURING AIR VOLUME THROUGH AIR MEASURING PIEZOMETER STATIONS IN SUPPLY AND RETURN FAN INLET CONES SO AS TO ALLOW FANS TO TRACK SUPPLY VOLUME WITH DIFFERENTIAL FOR BUILDING EXHAUST AND BUILDING PRESSURE. THE RELIEF AIR DAMPERS MODULATE OPEN TO MAINTAIN SPACE DIFFERENTIAL PRESSURE NO GREATER THAN 0.02" W.G. RELATIVE TO THE OUTSIDE.

7. **FILTER SWITCH:** FILTER DIFFERENTIAL PRESSURE SWITCHES ALARM AS NOTED - **PRE-FILTER: 1.0" WG (ADJ)**.

8. **SHUTDOWN:** ON AHU SHUTDOWN SUPPLY AND RETURN FANS STOP. OUTDOOR AND RELIEF DAMPERS CLOSE, AND RETURN DAMPER OPENS. MAINTAIN PREHEAT AIR SETPOINT OF **60°F (ADJ)** BY MODULATING PREHEAT COIL VALVE. LOSS OF A SINGLE FAN SHALL NOT SHUT DOWN THE UNIT. CLOSE COOLING COIL VALVE EXCEPT ON LOW LIMIT SAFETY SHUTDOWN.

UNOCCUPIED MODE:

1. **TRANSITION TO UNOCCUPIED:** UPON TRANSITION TO UNOCCUPIED CYCLE, SUPPLY AND RETURN FAN SHALL STOP ACCORDING TO SHUTDOWN SEQUENCE DESCRIBED ABOVE.

2. **SETBACK AND SETUP:** UNIT IS CYCLED ON AND SHALL FULLY RECIRCULATE WITH RETURN FAN TRACKING SET TO **NULL (ADJ)** TO MAINTAIN A MINIMUM AND MAXIMUM SPACE TEMPERATURE OF **60°F (ADJ)** AND **80°F (ADJ)**. TEMPERATURE MINIMUM AND MAXIMUM LEVELS TO FUNCTION BY POLLING CRITICAL SPACE TEMPERATURE SENSORS. USE OF A HYSTERESIS OF 2°F (ADJ) IS REQUIRED TO PREVENT SHORT CYCLING.

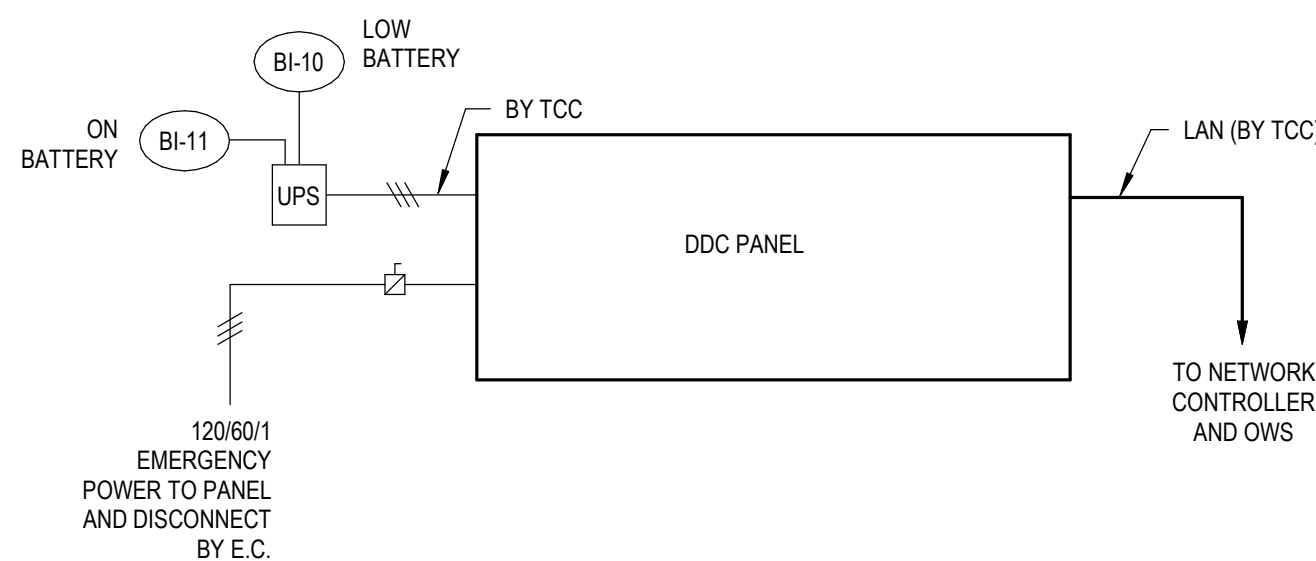
3. **HUMIDITY CONTROL:** UNIT IS CYCLED ON AND SHALL FULLY RECIRCULATE WITH RETURN FAN TRACKING SET TO **NULL (ADJ)** TO MAINTAIN SPACE HUMIDITY BELOW **60% RH (ADJ)** WHEN OUTSIDE TEMPERATURE EXCEEDS **70°F (ADJ)**. HUMIDIFICATION CONTROL CYCLE SHALL EMPLOY DEHUMIDIFICATION THROUGH COOLING COIL WITH SPACE REHEAT. USE OF A HYSTERESIS OF 10 % RH (ADJ) IS REQUIRED TO PREVENT SHORT CYCLING.

4. **TRANSITION TO OCCUPIED:** UNIT IS INDEXED TO OCCUPIED MODE BASED ON SCHEDULE OR TERMINAL UNIT SEQUENCE.

ID	DESCRIPTION	TREND	ALARM	GRAPHIC
AI-1	RETURN FAN B VOLUME	X		X
AI-2	RETURN FAN A VOLUME	X		X
AI-3	OUTDOOR TEMPERATURE (GLOBAL)	X		X
AI-4	SPACE DIFFERENTIAL PRESSURE	X		X
AI-5	MIXED AIR TEMPERATURE	X	X	X
AI-6	PREHEAT COIL DISCHARGE TEMPERATURE	X	X	X
AI-7	SUPPLY AIR TEMPERATURE	X		X
AI-8	SUPPLY AIR STATIC PRESSURE (AT UNIT)	X		X
AI-9	SUPPLY FAN A VOLUME	X		X
AI-10	SUPPLY FAN B VOLUME	X		X
AI-11	SPACE HUMIDITY	X	X	X
AI-12	RETURN AIR TEMPERATURE	X		X
AI-13	STATIC PRESSURE 90% THRU SYSTEM	X		X
AI-14	OUTDOOR AIRFLOW	X	X	X
AI-15	PREHEAT COIL HWR TEMPERATURE	X	X	X
AI-16	SUPPLY AIR HUMIDITY	X		X
AO-1	RETURN FAN B VFC	X		X
AO-2	RETURN FAN A VFC	X		X
AO-3	SUPPLY FAN A VFC	X		X
AO-4	SUPPLY FAN B VFC	X		X
AO-5	PREHEAT COIL VALVE	X		X
AO-6	COOLING COIL VALVE	X		X
AO-7	RETURN AIR DAMPER	X		X
AO-8	OUTDOOR AIR DAMPER	X		X
AO-9	RELIEF AIR DAMPER	X		X
BI-1	PREFILTER ALARM	X	X	X
BI-2	SAFETY LOW LIMIT STAT ALARM	X	X	X
BI-3	RETURN AIR LOW STATIC ALARM	X	X	X
BI-4	SUPPLY AIR HIGH STATIC ALARM	X	X	X
BI-5	RETURN FAN B STATUS	X	X	X
BI-6	RETURN FAN A STATUS	X	X	X
BI-7	SUPPLY FAN A STATUS	X	X	X
BI-8	SUPPLY FAN B STATUS	X	X	X
BI-9	HEATING COIL PUMP STATUS			X
BI-10	UPS LOW BATTERY ALARM	X	X	X
BI-11	UPS ON BATTERY ALARM	X	X	X
BO-1	RETURN FAN B START-STOP			X
BO-2	RETURN FAN A START-STOP			X
BO-3	SUPPLY FAN A START-STOP			X
BO-4	SUPPLY FAN B START-STOP			X
BO-5	HEATING COIL PUMP START-STOP			X

SETUP ALL TRENDS IN 15 MINUTE INTERVALS AND STORE FOR MINIMUM 45 DAYS.

INCLUDE TRENDS FOR THE ASSOCIATED COMMAND AND/OR SETPOINT IN ADDITION TO POINT LISTED ABOVE.
PROVIDE CHANGE OF VALUE AND INTERVAL TRENDS FOR BI AND BO POINTS.



GENERAL NOTES:

1. ALL AHU'S TO BE SINGLE POINT POWER FOR ALL 400V EQUIPMENT WITH AN ADDITIONAL 120V CIRCUIT FOR OUTLETS AND LIGHTS WITHIN UNITS.

PLAN NOTES:

- AIRFLOW PIEZOMETER INTEGRAL TO FAN, PROVIDED BY FAN MANUFACTURER, TRANSDUCER BY MANUFACTURER.
- LOCATE 3" FROM COOLING COIL INLET.
- DAMPERS TO BE INTEGRAL WITH AHU, ACTUATORS BY MANUFACTURER.
- SPRING RETURN OPEN UPON LOSS OF POWER.
- SPRING RETURN CLOSED UPON LOSS OF POWER.
- VARIABLE FREQUENCY CONTROLLER FURNISHED, INSTALLED, WIRED, AND COMMISSIONED BY MANUFACTURER.
- OUTDOOR AIRFLOW MEASURING STATION BY TCC.
- LAN TO OWS THRU THE LAN, VFC TO TRANSMIT TO BAS STATUS AND ALARMS OF ALL DATA AVAILABLE. SUPPLIER TO FURNISH INTEGRAL COMMUNICATION CARD. TCC TO MAP ALL OWNER REQUESTED INFORMATION POINTS. NOTE THAT START/STOP SIGNAL AND SPEED CONTROL ARE HARD WIRED TO DDC TO ENSURE OPERATION ON FAN LOSS.
- REFER TO PLANS FOR LOCATIONS.
- HARD WIRE DUCT STATIC SENSOR TO THE SAME DDC CONTROLLER WHICH CONTROLS THE FAN.

#	Revision	Date
---	----------	------

Project #: 17-700-078-2

Designed By: A.M.M

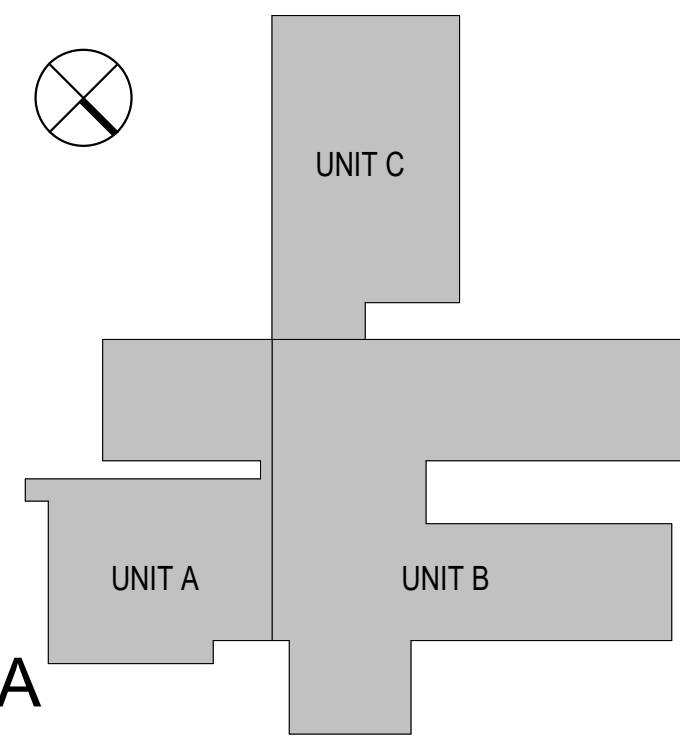
Drawn By: A.M.M

Checked By: D.E.B

Date: 04.12.2021

*Erik Weiler*

These Drawings and Specifications, and all copies thereof are and shall remain the property and copyright of the Architect. They shall be used only with respect to this Project and are not to be used on any other Project or Work without prior written permission from the Architect

DOAS-3 CONTROL
SCHEMATIC

M904

SEQUENCE OF OPERATION:

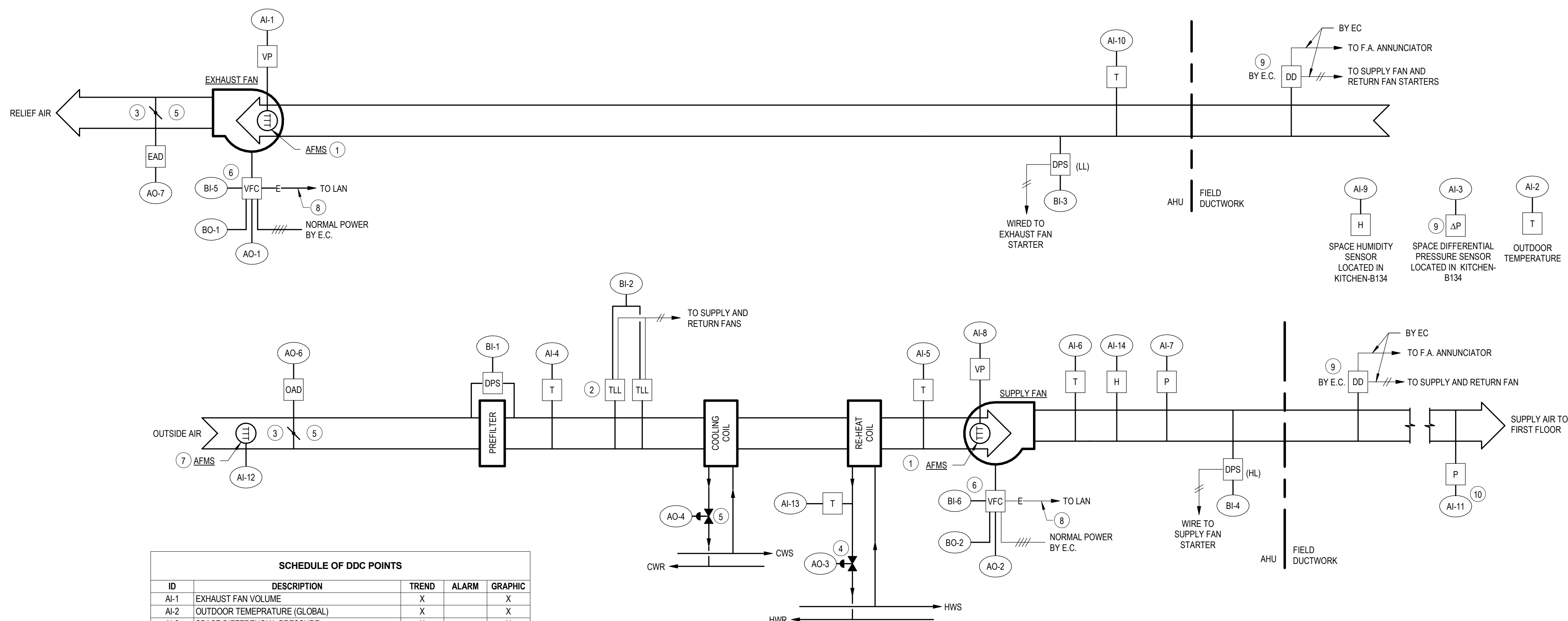
1. ALL SETPOINTS TO BE ADJUSTABLE. SETPOINTS TO BE EXPOSED ON GRAPHIC DISPLAY OR HIDDEN BASED ON OWNER REQUEST.
2. **ENABLE:** AIR HANDLER IS STARTED FROM DDC PANEL OR FROM COMMAND OF FACILITY MANAGEMENT SYSTEM.

OCCUPIED MODE:

1. AHU SHALL UTILIZE AN "OPTIMUM START" ALGORITHM FOR START-UP. THE CONTROL SYSTEM SHALL CALCULATE THE ZONE TEMPERATURES AND COMPARE WITH THE BUILDING SETPOINT. CONTROL SYSTEM SHALL CALCULATE THE TIME REQUIRED TO BRING THE ZONE UP TO SETPOINT TEMPERATURE AND START THE UNIT APPROPRIATELY.
2. **STARTUP:** ON UNIT STARTUP, SUPPLY AND EXHAUST FANS START. SUPPLY FAN SPEED CONTROL AND SUPPLY AIR TEMPERATURE CONTROL LOOPS ARE ACTIVE. EXHAUST AIR DAMPER IS FULLY OPEN. WITH EXHAUST FAN TRACKING CFM SETPOINT SET TO **NULL (ADJ)**.
3. **SUPPLY AIR TEMPERATURE:** THE SUPPLY AIR CONTROL LOOP SHALL PERFORM AS DESCRIBED BELOW:
 - a. BELOW 52°F (ADJ) OAT:
 - i. REHEAT COIL SHALL MODULATE TO MAINTAIN DISCHARGE AIR TEMPERATURE SETPOINT TO MAINTAIN SPACE SETPOINT.
 - b. BETWEEN 52°F AND 70°F (ADJ) OAT:
 - i. COOLING COIL CONTROL VALVE SHALL BE MODULATED TO MAINTAIN SUPPLY AIR SETPOINT OF 52°F.
 - iv. IF THERE IS A DROP IN SUPPLY AIR TEMPERATURE, THE COOLING COIL CONTROL VALVE SHALL BE MODULATED CLOSED.
 - v. REHEAT COIL SHALL MODULATE TO PROVIDE DISCHARGE AIR TEMPERATURE REQUIRED TO MEET SPACE SETPOINT.
 - vi. CONTROLS SHALL PREVENT SIMULTANEOUS OPERATION OF COOLING COIL AND HEATING COIL AND ECONOMIZER **WHEN** OUTDOOR AIR CONDITIONS EXIST TO USE FULL ECONOMIZER.
 - vi. RESET SUPPLY AIR TEMPERATURE SETPOINT FROM 55°F TO 60°F (ADJ) AS OUTDOOR AIR TEMPERATURE VARIES FROM 70°F TO 30°F. IN THE EVENT THE EXHAUST AIR HUMIDITY EXCEEDS 80% RH THE UNIT SUPPLY TEMPERATURE SHALL BE RESET TO 55°F AND SHALL IDENTIFY THIS ON THE OWS GRAPHICS. AFTER 15 MINUTES THE SUPPLY TEMPERATURE SHALL BE RESET UP IN 1/2°F INCREMENTS EVERY 10 MINUTES.
 - c. ABOVE 70°F (ADJ) OAT: THE COOLING COIL CONTROL VALVE SHALL MODULATE TO MAINTAIN SUPPLY AIR TEMPERATURE SETPOINT.
4. **SAFETIES:** FURNISH SAFETIES AS FOLLOWS TO PROVIDE ALARM AND SHUT DOWN UNIT.
 - a. COOLING COIL INLET TO HAVE TWO (2) OR MORE SAFETY LOW LIMIT THERMOSTATS WIRED IN SERIES TO BREAK BELOW 20°F (ADJ), ON ALARM, CHILLED WATER CONTROL VALVES OPEN 100% (ADJ).
 - b. SUPPLY HIGH LIMIT STATIC PRESSURE SWITCH SET TO 3.0" W.G. (ADJ).
 - c. EXHAUST DUCT LOW LIMIT STATIC SENSOR SWITCH SET TO (+) 2.0" W.G. (ADJ).
 - d. IN THE EVENT OF SMOKE (AHU SUPPLY OR EXHAUST DUCT DETECTOR).
- e. **WHEN SYSTEM DETECTS A CHANGE IN ZONE PRESSURE RELATIVE TO OUTDOORS, EXHAUST FAN SHALL REDUCE FAN SPEED TO MAINTAIN POSITIVE PRESSURIZATION.**
5. **SUPPLY FAN SPEED:** STATIC PRESSURE SENSOR LOCATED 90% OF DISTANCE THROUGH SUPPLY DUCT AND FAN DISCHARGE STATIC PRESSURE SENSOR LOCATED IN AHU DISCHARGE TO FUNCTION WITH SUPPLY FAN VFC CONTROL LOOP TO MODULATE SUPPLY FAN SPEED TO ACHIEVE DUCT STATIC PRESSURE SET AT 1.25" W.G. (ADJ), NOT TO EXCEED DISCHARGE PRESSURE OF 3.0" W.G. (ADJ), FINAL SETPOINT DETERMINED BY TAB CONTRACTOR AND APPROVED BY ENGINEER.
- a. DUCT STATIC PRESSURE SETPOINT SHALL HAVE A MINIMUM AND MAXIMUM RANGE OF SETPOINT +/- 0.3" W.G. (ADJ).
6. **EXHAUST FAN SPEED:** EXHAUST FAN VFC CONTROL LOOP TO BE CONTROLLED THROUGH VELOCITY PRESSURE SENSORS MEASURING AIR VOLUME THROUGH AIR MEASURING PIEZOMETER STATIONS IN SUPPLY AND EXHAUST FAN INLET CONES SO AS TO ALLOW FANS TO TRACK SUPPLY VOLUME WITH DIFFERENTIAL FOR BUILDING EXHAUST AND BUILDING PRESSURIZATION. THE RELIEF AIR DAMPERS MODULATE OPEN TO MAINTAIN SPACE DIFFERENTIAL PRESSURE NO GREATER THAN 0.25" W.G. RELATIVE TO THE OUTSIDE.
7. **FILTER SWITCH:** FILTER DIFFERENTIAL PRESSURE SWITCHES ALARM AS NOTED - **PRE-FILTER: 1.0" W.G. (ADJ)**.
8. **SHUTDOWN:** ON AHU SHUTDOWN SUPPLY AND EXHAUST FANS STOP. OUTDOOR AND RELIEF DAMPERS CLOSE. AND EXHAUST DAMPER OPENS. MAINTAIN REHEAT AIR SETPOINT OF 60°F (ADJ) BY MODULATING REHEAT COIL VALVE. LOSS OF A SINGLE FAN SHALL NOT SHUT DOWN THE UNIT. CLOSE COOLING COIL VALVE EXCEPT ON LOW LIMIT SAFETY SHUTDOWN.

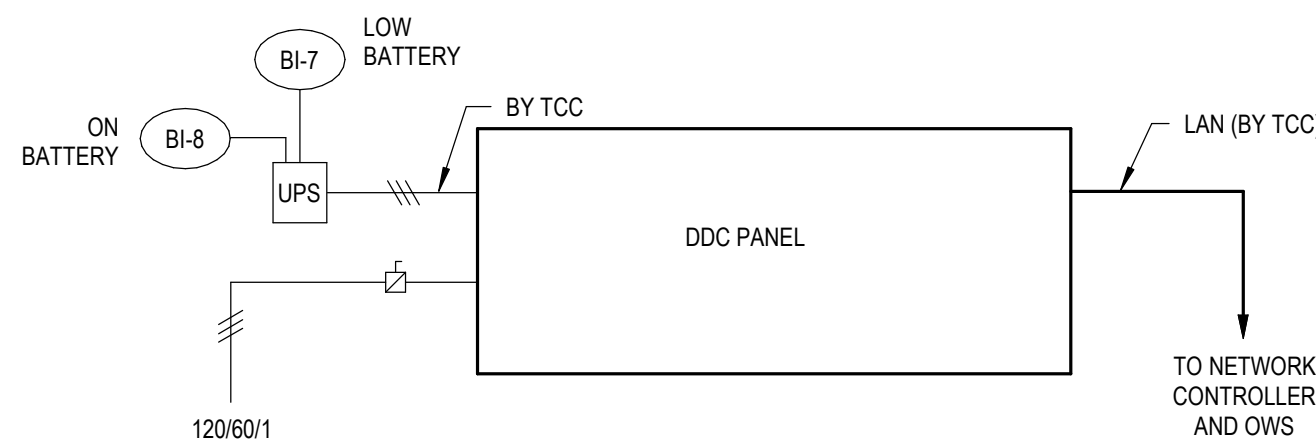
GENERAL NOTES:

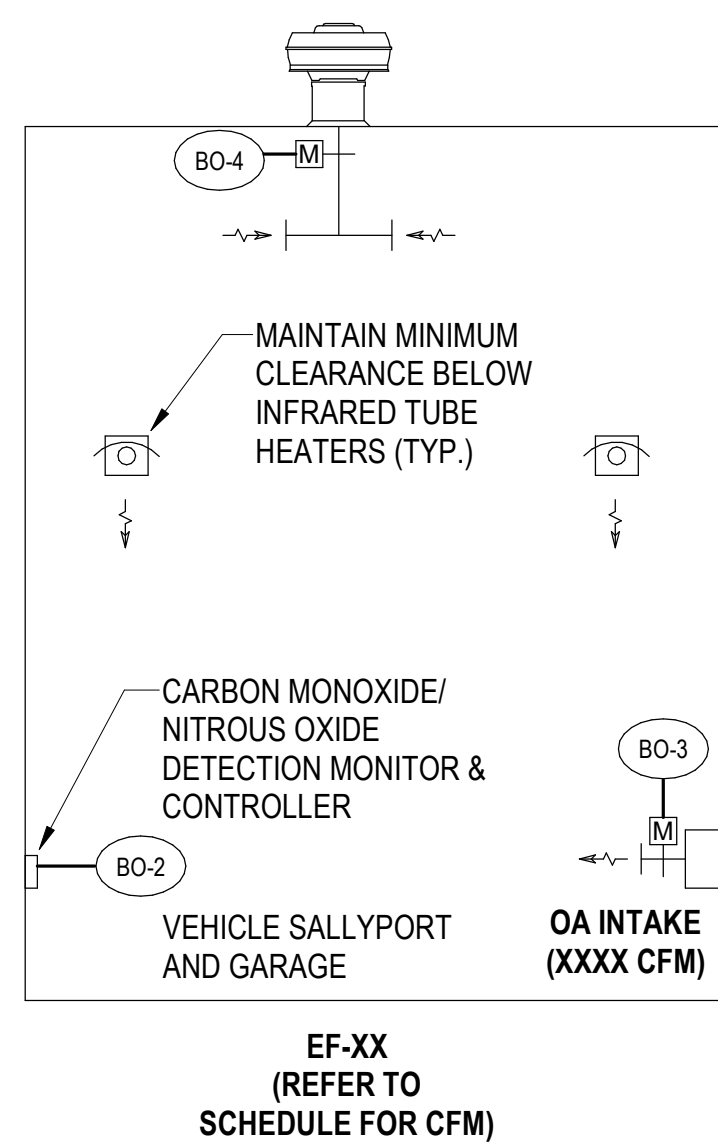
1. ALL AHUS TO BE SINGLE POINT POWER FOR ALL 460/3 EQUIPMENT WITH AN ADDITIONAL 120V1 CIRCUIT FOR OUTLETS AND LIGHTS WITHIN UNITS.
- PLAN NOTES
 1. AIRFLOW PIEZOMETER INTEGRAL TO FAN, PROVIDED BY FAN MANUFACTURER. TRANSDUCER BY MANUFACTURER.
 2. LOCATE 3" FROM COOLING COIL INLET.
 3. DAMPERS TO BE INTEGRAL WITH AHU. ACTUATORS BY MANUFACTURER.
 4. SPRING RETURN OPEN UPON LOSS OF POWER.
 5. SPRING RETURN CLOSED UPON LOSS OF POWER.
 6. VARIABLE FREQUENCY CONTROLLER FURNISHED, INSTALLED, WIRED, AND COMMISSIONED BY MANUFACTURER.
 7. OUTDOOR AIRFLOW MEASURING STATION BY TCC.
 8. LAN TO OWS. THRU THE LAN, VFC TO TRANSMIT TO BAS STATUS AND ALARMS OF ALL DATA AVAILABLE. SUPPLIER TO FURNISH INTEGRAL COMMUNICATION CARD. TCC TO MAP ALL OWNER REQUESTED INFORMATION POINTS. NOTE THAT START/STOP SIGNAL AND SPEED CONTROL ARE HARD WIRED TO DDC TO ENSURE OPERATION ON FAN LOSS.
 9. REFER TO PLANS FOR LOCATIONS.
 10. HARD WIRE DUCT STATIC SENSOR TO THE SAME DDC CONTROLLER WHICH CONTROLS THE FAN.



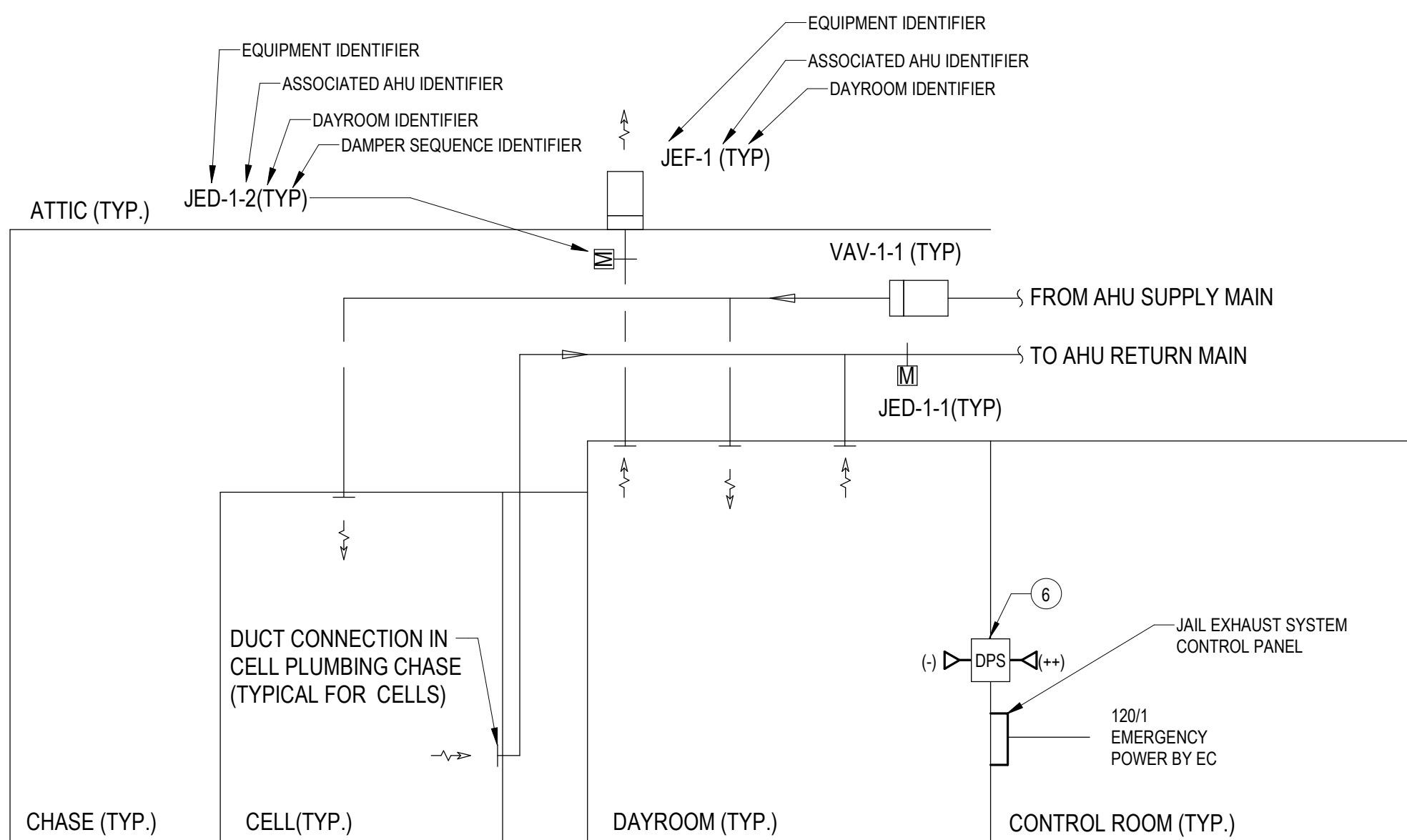
ID	DESCRIPTION	TREND	ALARM	GRAPHIC
AI-1	EXHAUST FAN VOLUME	X		X
AI-2	OUTDOOR TEMPERATURE (GLOBAL)	X		X
AI-3	SPACE DIFFERENTIAL PRESSURE	X		X
AI-4	MIXED AIR TEMPERATURE	X	X	X
AI-5	REHEAT COIL DISCHARGE TEMPERATURE	X	X	X
AI-6	SUPPLY AIR TEMPERATURE	X	X	X
AI-7	SUPPLY AIR STATIC PRESSURE (AT UNIT)	X		X
AI-8	SUPPLY FAN VOLUME	X		X
AI-9	SPACE HUMIDITY	X	X	X
AI-10	EXHAUST AIR TEMPERATURE	X		X
AI-11	STATIC PRESSURE 90% THRU SYSTEM	X		X
AI-12	OUTDOOR AIRFLOW	X	X	X
AI-13	RETURN WATER TEMPERATURE	X	X	X
AI-14	SUPPLY AIR HUMIDITY	X		X
AO-1	EXHAUST FAN VFC	X		X
AO-2	SUPPLY FAN VFC	X		X
AO-3	REHEAT COIL VALVE	X		X
AO-4	COOLING COIL VALVE	X		X
AO-6	OUTDOOR AIR DAMPER	X		X
AO-7	EXHAUST AIR DAMPER	X		X
BI-1	PREFILTER ALARM	X	X	X
BI-2	SAFETY LOW LIMIT STAT ALARM	X	X	X
BI-3	EXHAUST AIR LOW STATIC ALARM	X	X	X
BI-4	SUPPLY AIR HIGH STATIC ALARM	X	X	X
BI-5	EXHAUST FAN STATUS	X	X	X
BI-6	SUPPLY FAN STATUS	X	X	X
BI-7	UPS LOW BATTERY ALARM	X	X	X
BI-8	UPS ON BATTERY ALARM	X	X	X
BO-1	EXHAUST FAN START-STOP			X
BO-2	SUPPLY FAN START-STOP			X

SETUP ALL TRENDS IN 15 MINUTE INTERVALS AND STORE FOR MINIMUM 45 DAYS.

INCLUDE TRENDS FOR THE ASSOCIATED COMMAND AND/OR SETPOINT IN ADDITION TO POINT LISTED ABOVE.
PROVIDE CHANGE OF VALUE AND INTERVAL TRENDS FOR BI AND BO POINTS.1 DOAS-3 CONTROL SCHEMATIC
NOT TO SCALE



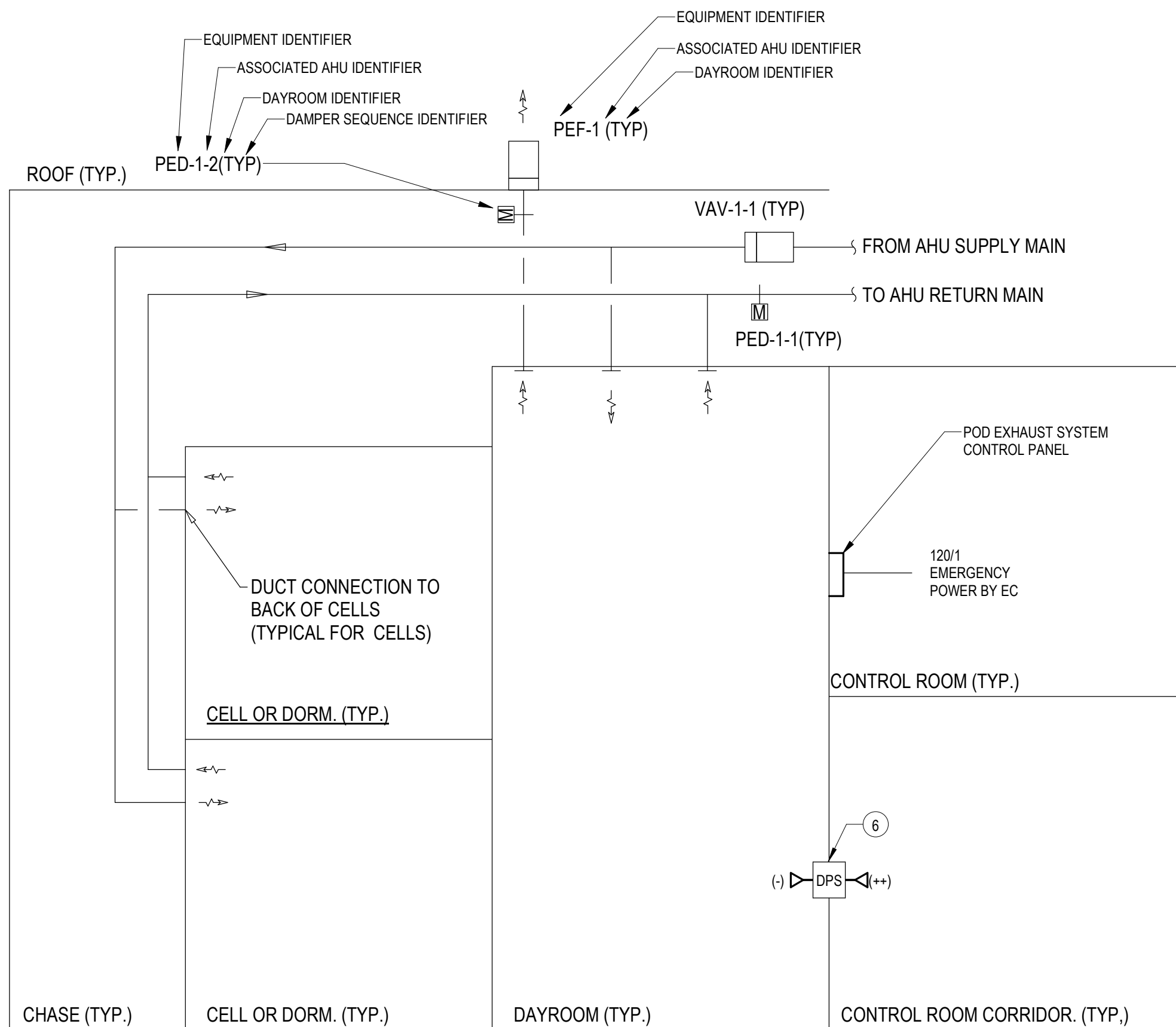
1 M - DIAGRAM - SALLYPORT AIRFLOW1
NOT TO SCALE



NOTE:

REFER TO FLOOR PLAN FOR MOTORIZED DAMPERS. PROVIDE VOLUME DAMPERS FOR ALL POINTS OF CONNECTION TO GRILLES, REGISTERS, DIFFUSERS, AND BACK OF CELLS/DORMS.

6 M - DIAGRAM - JAIL CELLS AIRFLOW
NOT TO SCALE



NOTE:

REFER TO FLOOR PLAN FOR MOTORIZED DAMPERS. PROVIDE VOLUME DAMPERS FOR ALL POINTS OF CONNECTION TO GRILLES, REGISTERS, DIFFUSERS, AND BACK OF CELLS/DORMS.

3 M - DIAGRAM - DAYROOM CELL AIRFLOW1
NOT TO SCALE

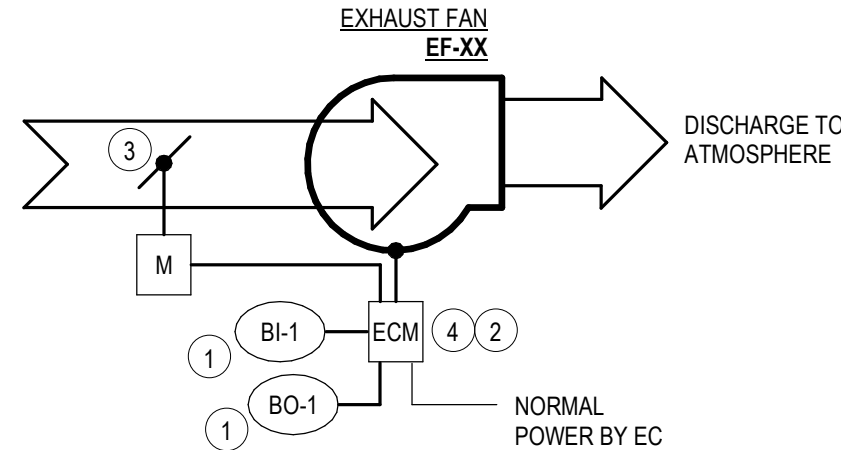
CONTROL SEQUENCES:

INFRARED TUBE HEATER CONTROL:

- HEATERS SHALL OPERATE TO MAINTAIN SPACE TEMPERATURE SET POINT OF 60 DEGREES (ADJ.) FROM STAND ALONE WALL THERMOSTAT.
- BAS SHALL MONITOR AND ALARM FOR HIGH/LOW TEMPERATURE ALARM OF 45 AND 95 DEGREES F (ADJ.)

SALLYPORT MONITORING & EXHAUST FAN CONTROL:

- BAS SHALL MONITOR ALARM FROM VEHICLE SALLYPORT MONITORING SYSTEMS. MONITOR SHALL BE FIELD PROGRAMMED TO MAINTAIN THE FOLLOWING LEVELS. UPON RECEIVING A CRITICAL ALARM FROM THE MONITORING PANEL, ENERGIZE THE ASSOCIATED OUTDOOR AIR INTAKE DAMPER, EXHAUST FAN DAMPER, AND EXHAUST FAN. MOTORIZED DAMPERS SHALL HAVE POWER TO OPEN AND SPRING POWER TO CLOSE.
 - CARBON MONOXIDE BELOW 25.0 PPM (ADJ.)
 - NITROGEN DIOXIDE BELOW 1.0 PPM (ADJ.)
- PURGE VENTILATION: EXHAUST FANS SHALL RUN 15 MINUTES AFTER ALARM CONDITIONS HAVE CLEARED.
- EXHAUST FAN SHALL RUN TO MAINTAIN MAXIMUM SPACE TEMPERATURE OF 85 DEGREES F (ADJ.). FAN SHALL RUN FOR A MINIMUM OF 5 MINUTES (ADJ.) AFTER SETPOINT IS REACHED.
- SAFETIES: FAN STATUS SHALL BE MONITORED THROUGH CURRENT SENSOR AND BE ALARMED TO THE BAS.



SCHEMATIC NOTES:

- WIRE BAS FAN STATUS AND FAN ENABLE/DISABLE TO THE NEAREST DDC SYSTEM PANEL.
- TCC TO PROVIDE CURRENT SENSOR AND CONTROL LOGIC FOR FAN FAULT DETERMINATION.
- CONTROL DAMPERS TO BE DUCT MOUNTED; DAMPERS AND ACTUATORS BY TCC.
- ELECTRO-COMMUTATED MOTOR WITH INTEGRAL SPEED CONTROLLER FURNISHED BY THE FAN MANUFACTURER. FAN SPEED IS MANUALLY SET BY TAB ADJUSTMENT OF THE ECM CONTROLLER.

SEQUENCE OF OPERATION:

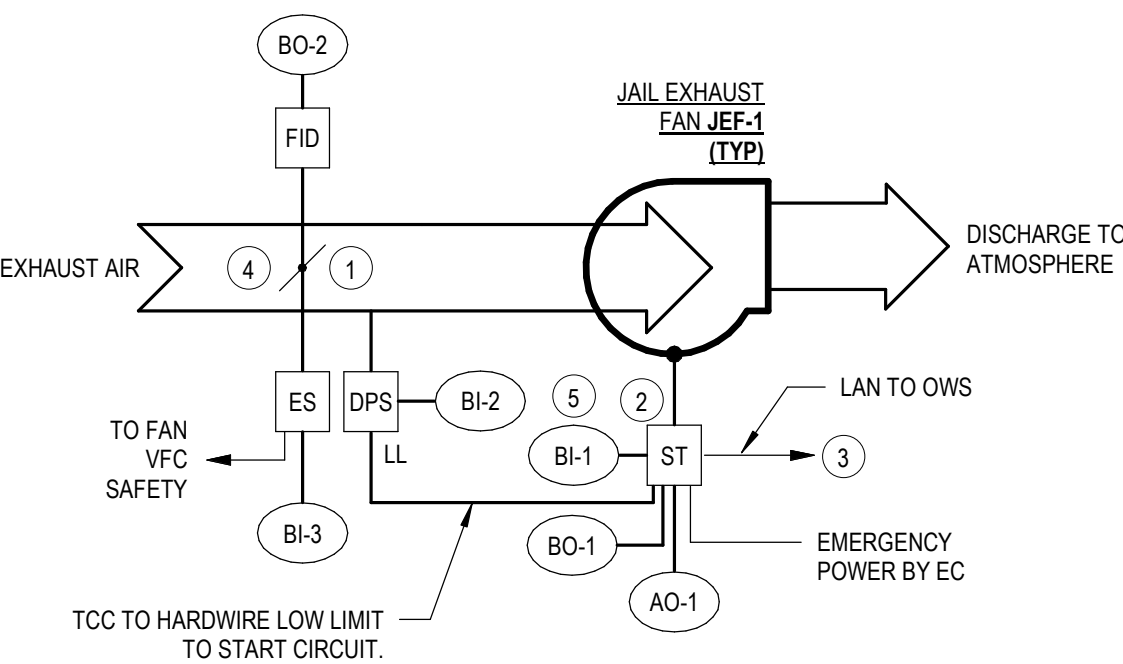
- NORMAL OPERATION:
 - ALL SETPOINTS TO BE ADJUSTABLE. SETPOINTS ARE TO BE EXPOSED ON THE GRAPHIC DISPLAY OR HIDDEN BASED ON OWNER REQUEST.
 - ENABLE: EXHAUST FAN IS STARTED FROM COMMAND OF FACILITY MANAGEMENT SYSTEM. FAN ISOLATION DAMPER OPENS. DDC SHALL CYCLE FAN ON AND OFF ACCORDING TO THE SALLYPORT AIRFLOW SEQUENCE (THIS DRAWING).
 - EXHAUST FAN HAS AN INTEGRAL SPEED CONTROLLER WHICH IS MANUALLY SET DURING TAB. ENABLING THE FAN TO RUN OPENS THE ISOLATION DAMPER. DISABLING THE FAN CLOSES THE ISOLATION DAMPER.

2 SALLYPORT EXHAUST FAN CONTROL SCHEMATIC
NOT TO SCALE

JAIL EXHAUST SYSTEM CONTROL SEQUENCES:

GENERAL SYSTEM CONTROL:

- JAIL EXHAUST SYSTEM CONTROL SEQUENCE WILL BEGIN WHEN THE SYSTEM IS ENGAGED MANUALLY AT THE CONTROL PANEL FROM THE ASSOCIATED CONTROL ROOM.
- JAIL EXHAUST SEQUENCE WILL CONTINUE UNTIL RESET FROM DDC.
- UPON ACTIVATION OF JAIL EXHAUST FAN:
 - ZONE MOTORIZED RETURN DAMPERS SHALL CLOSE.
 - ZONE VAV BOX DAMPER SHALL OPEN TO A SETPOINT CFM DETERMINED BY TABB CONTRACTOR TO MAINTAIN PROPER PRESSURIZATION OF NOT MORE THAN 0.05 INCHES OF WATER COLUMN (ADJ.) AS MEASURED USING SPACE DIFFERENTIAL PRESSURE SENSORS. INSTALLED IN EACH ZONE.
 - ASSOCIATED AHU SHALL PRIORITIZE ZONE VAV BOX THAT HAS BEEN INITIATED BY THE JAIL EXHAUST SYSTEM SEQUENCE TO ACHIEVE SETPOINT CFM. AHU SHALL RETURN TO NORMAL OPERATION UPON DEACTIVATION.
- EXHAUST FAN MOTORIZED DAMPER IS NORMALLY CLOSED, RETURN DUCT MOTORIZED DAMPERS ARE NORMALLY OPEN.
- JAIL EXHAUST SYSTEM SHALL HAVE A GRAPHIC DISPLAY PANEL INSTALLED IN CONTROL ROOM. CONTROL PANEL SHALL DISPLAY THE HOUSING AREA AND HAVE TOUCH CONTROLS TO ACTIVATE AND DEACTIVATE SYSTEM FOR EACH ZONE.

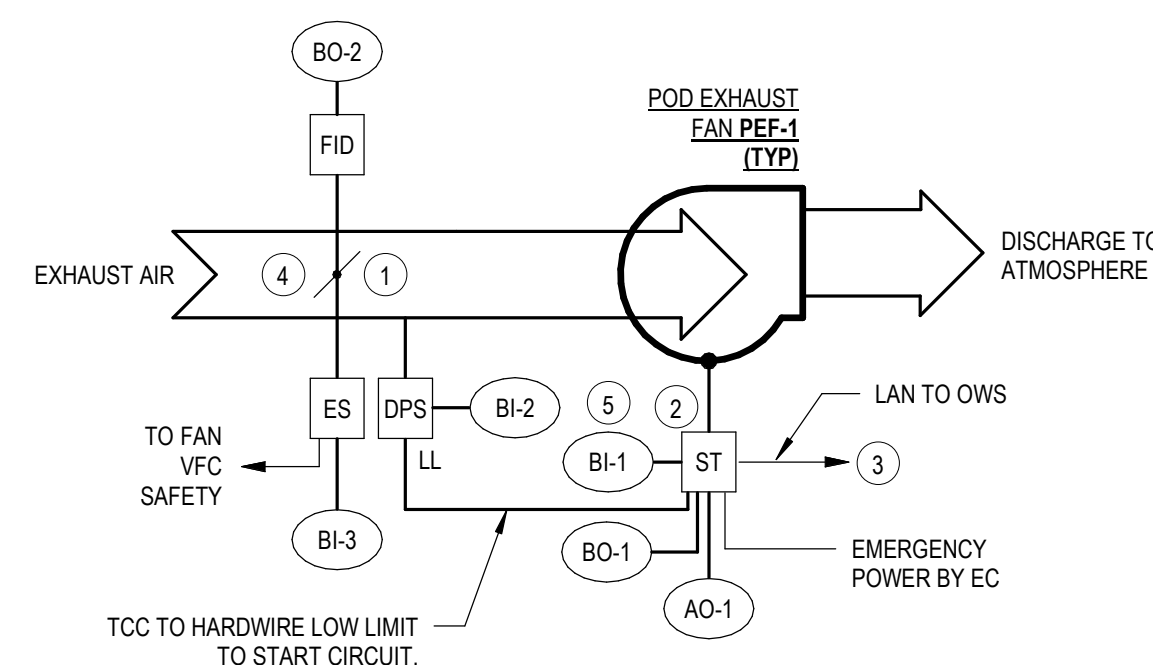


5 JAIL EXHAUST FAN SCHEMATIC
NOT TO SCALE

POD EXHAUST SYSTEM CONTROL SEQUENCES:

GENERAL SYSTEM CONTROL:

- POD EXHAUST SYSTEM CONTROL SEQUENCE WILL BEGIN WHEN THE SYSTEM IS ENGAGED MANUALLY AT THE CONTROL PANEL FROM THE ASSOCIATED CONTROL ROOM.
- POD EXHAUST SEQUENCE WILL CONTINUE UNTIL RESET FROM DDC.
- UPON ACTIVATION OF POD EXHAUST FAN:
 - ZONE MOTORIZED RETURN DAMPERS SHALL CLOSE.
 - ZONE VAV BOX DAMPER SHALL OPEN TO A SETPOINT CFM DETERMINED BY TABB CONTRACTOR TO MAINTAIN PROPER PRESSURIZATION OF NOT MORE THAN 0.05 INCHES OF WATER COLUMN (ADJ.) AS MEASURED USING SPACE DIFFERENTIAL PRESSURE SENSORS. INSTALLED IN EACH ZONE.
 - ASSOCIATED AHU SHALL PRIORITIZE ZONE VAV BOX THAT HAS BEEN INITIATED BY THE POD EXHAUST SYSTEM SEQUENCE TO ACHIEVE SETPOINT CFM. AHU SHALL RETURN TO NORMAL OPERATION UPON DEACTIVATION.
- EXHAUST FAN MOTORIZED DAMPER IS NORMALLY CLOSED, RETURN DUCT MOTORIZED DAMPERS ARE NORMALLY OPEN.
- POD EXHAUST SYSTEM SHALL HAVE A GRAPHIC DISPLAY PANEL INSTALLED IN EACH CONTROL ROOM. CONTROL PANEL SHALL DISPLAY THE POD AND HAVE TOUCH CONTROLS TO ACTIVATE AND DEACTIVATE SYSTEM FOR EACH ZONE.



4 POD EXHAUST FAN SCHEMATIC
NOT TO SCALE

SCHEDULE OF DDC POINTS				
ID	DESCRIPTION	TREND	ALARM	GRAPHIC
BI-1	FAN STATUS	X	X	X
BO-1	FAN START-STOP	X		X
BO-2	GAS MONITORING SYSTEM EXHAUST FAN ON/OFF	X		X
BO-3	OUTDOOR INTAKE MOTORIZED DAMPER	X		X
BO-4	EXHAUST FAN MOTORIZED DAMPER	X		X

SCHEDULE OF DDC POINTS				
ID	DESCRIPTION	TREND	ALARM	GRAPHIC
AO-1	EXHAUST FAN STARTER	X		X
AO-1	EXHAUST FAN STARTER	X		X
BI-1	EXHAUST FAN STATUS	X	X	X
BI-1	EXHAUST FAN STATUS	X	X	X
BI-2	EXHAUST FAN DIFFERENTIAL PRESSURE SWITCH	X	X	X
BI-2	EXHAUST FAN DIFFERENTIAL PRESSURE SWITCH	X	X	X
BI-3	EXHAUST FAN DAMPER END SWITCH	X	X	X
BI-3	EXHAUST FAN DAMPER END SWITCH	X	X	X
BO-1	EXHAUST FAN START-STOP	X		X
BO-1	EXHAUST FAN START-STOP	X		X
BO-2	EXHAUST FAN ISOLATION DAMPER	X		X
BO-2	EXHAUST FAN ISOLATION DAMPER	X		X

PLAN NOTES:

- SPRING RETURN CLOSED UPON LOSS OF POWER.
- STARTER FURNISHED BY MANUFACTURER, POWERED BY EC AND CONTROLLED BY TCC.
- LAN TO OWS. THRU THE LAN, VFC TO TRANSMIT TO BAS STATUS AND ALARMS OF ALL DATA AVAILABLE SUPPLIER TO FURNISH INTEGRAL COMMUNICATION CARD. TCC TO MAP ALL OWNER REQUESTED INFORMATION POINTS. NOTE THAT START/STOP SIGNAL AND SPEED CONTROL ARE HARD WIRED TO DDC TO ENSURE OPERATION ON FAN LOSS.
- CONTROL DAMPERS TO BE DUCT MOUNTED; DAMPERS AND ACTUATORS BY TCC.
- TCC TO PROVIDE CURRENT SENSING AND CONTROL LOGIC FOR FAN FAULT DETERMINATION.
- REFER TO FLOOR PLANS FOR LOCATION.

EXHAUST FAN SEQUENCE OF OPERATION:

- ALL SETPOINTS TO BE ADJUSTABLE. SETPOINTS TO BE EXPOSED ON GRAPHIC DISPLAY OR HIDDEN BASED ON OWNER REQUEST.
- NORMAL OPERATION MODE:
 - EXHAUST FAN IS STARTED FROM COMMAND OF FACILITY MANAGEMENT SYSTEM. EXHAUST FAN TO BE NORMALLY OFF.
 - EXHAUST FAN TO FOLLOW SEQUENCE OF OPERATIONS AS DEFINED IN THE JAIL EXHAUST SYSTEM AIRFLOW DIAGRAM (THIS PAGE).
- SAFETIES:
 - FURNISH SAFETY LOW LIMIT HARD WIRED TO FAN TO DISABLE FAN IF LOW LIMIT EXCEEDS 3.0" W.G. FURNISH A BINARY ALARM INPUT TO THE BAS.

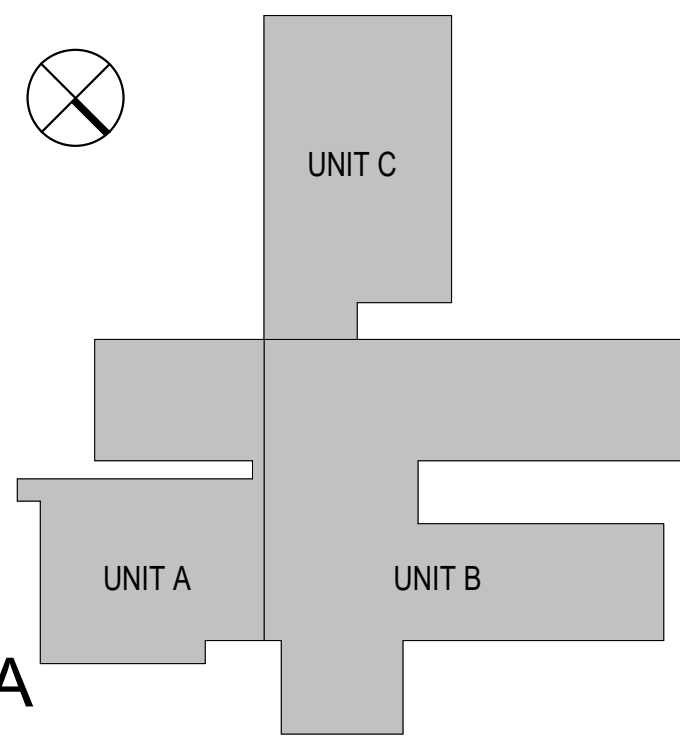
SCHEDULE OF DDC POINTS				
ID	DESCRIPTION	TREND	ALARM	GRAPHIC
AO-1	EXHAUST FAN STARTER	X		X
AO-1	EXHAUST FAN STARTER	X		X
BI-1	EXHAUST FAN STATUS	X	X	X
BI-1	EXHAUST FAN STATUS	X	X	X
BI-2	EXHAUST FAN DIFFERENTIAL PRESSURE SWITCH	X	X	X
BI-2	EXHAUST FAN DIFFERENTIAL PRESSURE SWITCH	X	X	X
BI-3	EXHAUST FAN DAMPER END SWITCH	X	X	X
BI-3	EXHAUST FAN DAMPER END SWITCH	X	X	X
BO-1	EXHAUST FAN START-STOP	X		X
BO-1	EXHAUST FAN START-STOP	X		X
BO-2	EXHAUST FAN ISOLATION DAMPER	X		X
BO-2	EXHAUST FAN ISOLATION DAMPER	X		X

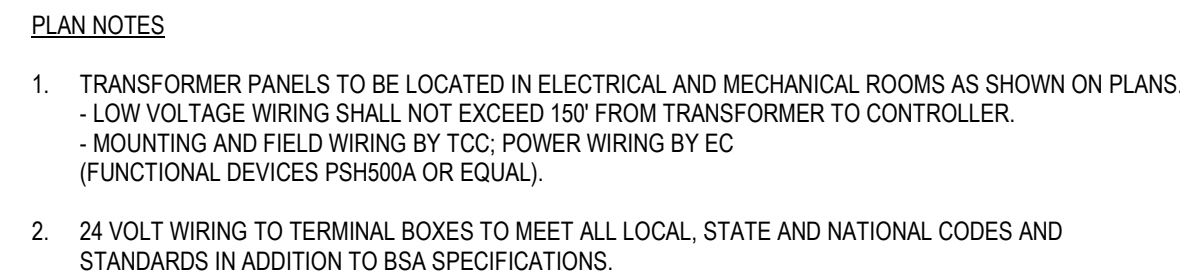
PLAN NOTES:

- SPRING RETURN CLOSED UPON LOSS OF POWER.
- STARTER FURNISHED BY MANUFACTURER, POWERED BY EC AND CONTROLLED BY TCC.
- LAN TO OWS. THRU THE LAN, VFC TO TRANSMIT TO BAS STATUS AND ALARMS OF ALL DATA AVAILABLE SUPPLIER TO FURNISH INTEGRAL COMMUNICATION CARD. TCC TO MAP ALL OWNER REQUESTED INFORMATION POINTS. NOTE THAT START/STOP SIGNAL AND SPEED CONTROL ARE HARD WIRED TO DDC TO ENSURE OPERATION ON FAN LOSS.
- CONTROL DAMPERS TO BE DUCT MOUNTED; DAMPERS AND ACTUATORS BY TCC.
- TCC TO PROVIDE CURRENT SENSING AND CONTROL LOGIC FOR FAN FAULT DETERMINATION.
- REFER TO FLOOR PLANS FOR LOCATION.

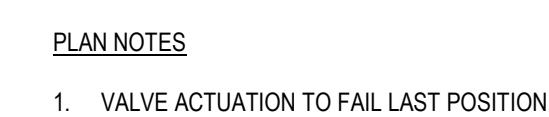
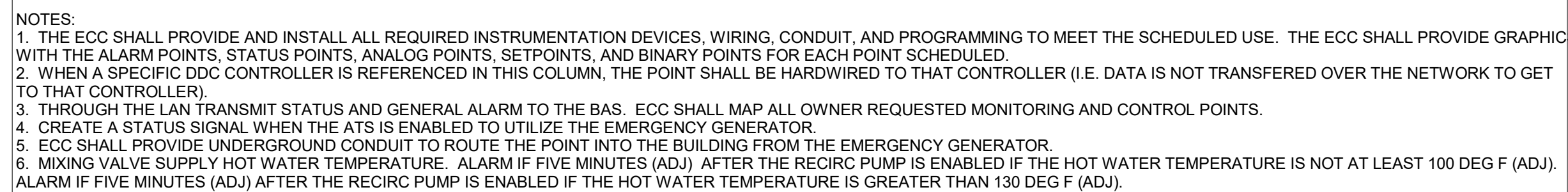
EXHAUST FAN SEQUENCE OF OPERATION:

- ALL SETPOINTS TO BE ADJUSTABLE. SETPOINTS TO BE EXPOSED ON GRAPHIC DISPLAY OR HIDDEN BASED ON OWNER REQUEST.
- NORMAL OPERATION MODE:
 - EXHAUST FAN IS STARTED FROM COMMAND OF FACILITY MANAGEMENT SYSTEM. EXHAUST FAN TO BE NORMALLY OFF.
 - EXHAUST FAN TO FOLLOW SEQUENCE OF OPERATIONS AS DEFINED IN THE POD EXHAUST SYSTEM AIRFLOW DIAGRAM (THIS PAGE).
- SAFETIES:
 - FURNISH SAFETY LOW LIMIT HARD WIRED TO FAN TO DISABLE FAN IF LOW LIMIT EXCEEDS 3.0" W.G. FURNISH A BINARY ALARM INPUT TO THE BAS.





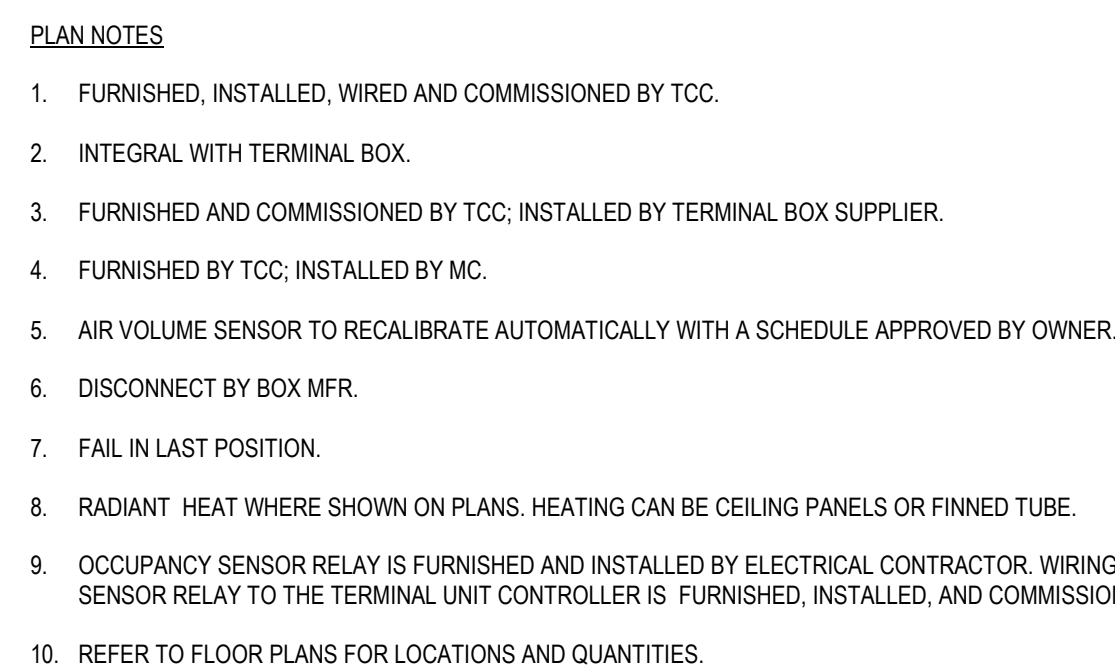
①



SEQUENCE OF OPERATION:

1. HRC-1 SHALL BE ENABLED TO RUN AT ALL TIMES.
2. WHEN THE OUTSIDE AIR TEMPERATURE IS ABOVE 50°F HRC-1 SHALL CONTROL TO MAINTAIN CONDENSER WATER (HOT WATER) AT BOILER SYSTEM TEMPERATURE SETPOINT.
3. UPON BEING ENABLED TO RUN, HRC SHALL OPEN ITS CHILLED WATER AND CONDENSER WATER ISOLATION VALVES. AFTER VALVES ARE FULL OPEN, HEAT RECOVERY CHILLER PUMPS SHALL BE COMMANDED TO RUN.

6 HEATING RECOVERY CHILLER CONTROL SCHEMATIC
NOT TO SCALE



②



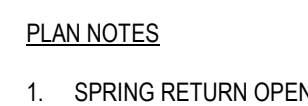
1. WIRE BAS FAN STATUS AND FAN ENABLE/DISABLE TO THE NEAREST DDC SYSTEM PANEL.
2. ECC TO PROVIDE CURRENT SENSOR AND CONTROL LOGIC FOR FAN FAULT DETERMINATION
3. GRAVITY BACKDRAFT DAMPER.
4. ELECTRO-COMMUTATED MOTOR WITH INTEGRAL SPEED CONTROLLER FURNISHED BY THE F MANUALLY SET BY TAB ADJUSTMENT OF THE ECM CONTROLLER.

SCHEDULE OF DDC POINTS				
ID	DESCRIPTION	TREND	ALARM	GRAPHIC
BI-1	FAN STATUS	X	X	X
BO-1	FAN START-STOP	X		X

SEQUENCE OF OPERATION

1. NORMAL OPERATION:
 - a. ALL SETPOINTS TO BE ADJUSTABLE. SETPOINTS ARE TO BE EXPOSED ON THE GRAPHIC DISPLAY OR HIDDEN BASED ON OWNER REQUEST.
 - b. ENABLE: EXHAUST FAN IS STARTED FROM COMMAND OF FACILITY MANAGEMENT SYSTEM.
 - c. EXHAUST FAN HAS AN INTEGRAL SPEED CONTROLLER WHICH IS MANUALLY SET DURING TAB. ENABLING THE FAN TO RUN OPENS THE ISOLATION DAMPER. DISABLING THE FAN CLOSES THE ISOLATION DAMPER.

③



SEQUENCE OF OPERATION

- OCCUPIED CYCLE**
1. BELOW 60° F/O THE UNIT FAN OPERATES CONTINUOUSLY AND ABOVE 62° F/O THE FAN IS INOPERATIVE AND THE VALVE IS CLOSED.
 2. BELOW 55° F/O THE UNIT COIL VALVE IS MODULATED TO ACHIEVE A SPACE TEMPERATURE SET AT 70° F.
 3. UNIT SHALL SEND ALARM FOR TEMPERATURE FALLING BELOW 50 DEGREES F (ADD.) OR RISING ABOVE 55 DEGREES F (ADD.).
- UNOCCUPIED CYCLE:**
1. UNOCCUPIED CYCLE OCCURS FROM 8:00 P.M. TO 5:00 A.M. MONDAY THROUGH FRIDAY AND ALL DAY SATURDAY, SUNDAY, AND HOLIDAYS.
 2. UNIT IS DISABLED ABOVE 45° F/O. THE UNIT SHALL CYCLE TO MAINTAIN A 50° F/O.
 3. BELOW 45° F/O, THE UNIT SHALL CYCLE TO MAINTAIN A 60° F/O. SPACE TEMPERATURE 45°-50° DEGS SHOULD BE PROGRAMMED INTO THE CONTROLLER AND BE ADJUSTABLE).

4

SEQUENCE OF OPERATION

- The diagram shows a central box labeled "DDC CONTROLLER". To its left is a box labeled "TCC" with a note "FURNISHED AND COMMISSIONED BY TCC - MOUNTED AND WIRED BY TCC.". A line connects the "TCC" box to the "DDC CONTROLLER" box. Below the "TCC" box are two labels: "SPRING RETURN OPEN." and "FOR MONITORING / TROUBLESHOOTING USE ONLY". To the right of the "DDC CONTROLLER" box is a section titled "OCCUPIED CYCLE:". It contains three numbered items:

 1. UNIT HEATING COIL VALVE IS MODULATED TO ACHIEVE A SPACE TEMPERATURE SET AT 65°F.
 2. UNIT COOLING COIL VALVE IS MODULATED TO ACHIEVE A SPACE AIR TEMPERATURE OF 78°F.
 3. IF SPACE TEMPERATURE DROPS BELOW 40°F FOR 5 MINUTES (ADJ.) THEN FULLY OPEN BOTH THE HEATING VALVE AND THE COOLING VALVE. SIGNAL AN ALARM AT THE OWS.

(5)